

JOHANNES GUTENBERG-UNIVERSITÄT, MAINZ

LECTURE NOTES ON

Foundations of Motivic Homotopy Theory

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Foreword and Notation

These notes originate from the lecture course “foundations of motivic homotopy theory” given in the winter term 2024/2025 at the Johannes Gutenberg-University Mainz by Prof. Dr. Tom Bachmann.

We will freely use the language of ∞ -categories as it provides a natural language to talk about homotopy theory in general. We write $\mathbf{Ani}$ for the ∞ -category of anima or spaces. We will usually write $\mathcal{C}, \mathcal{D}$ for ∞ -categories and $\mathcal{X}, \mathcal{Y}$ for ∞ -topoi.

Chapter 1

Introduction

The basic vision which led to the construction of motivic homotopy theory was to be able to do homotopy theory in algebraic topology. But why do we even want to do homotopy theory in algebraic geometry? To answer this question let us give a brief review of some constructions from algebraic topology which would be nice to have in algebraic geometry. As a first example let us take a look at topological K -theory. Given a topological space $X \in \mathbf{Ani}$ one can define the *complex K -theory space of X* KU . This space has the nice property that

$$\pi_0 KU(X) \cong KU_0(X),$$

the *complex K -theory group*. This group is basically given by the isomorphism classes of complex vector bundles on X where we define that given a short exact sequence of vector bundles

$$0 \longrightarrow V' \longrightarrow V \longrightarrow V'' \longrightarrow 0$$

then V should be the same as the Whitney-sum $V' \oplus V''$. $KU(-)$ in fact defines a functor $\mathbf{Top} \rightarrow \mathbf{Ani}$. Now the nice thing about algebraic topology this functor is in fact representable, that is we have a natural equivalence

$$KU(X) \simeq \mathrm{Map}_{\mathbf{Top}}(X, \mathbb{Z} \times BU).$$

Here $U = \mathrm{colim}_n (U_n)$, U_n the unitary groups, and BU is the classifying space of complex vector bundles.

Now, K -groups were originally defined by Alexander Grothendieck himself in algebraic geometry. Hence it would be quite helpful to have an analogous statement of representability in algebraic geometry, which is sadly not the case.

Another, though related, example is the following. In algebraic topology, the space of isomorphism classes of complex rank n vector bundles on a topological space X is equivalent as an anima to

$$\mathrm{Vect}(X)^\simeq \simeq \mathrm{Map}_{\mathbf{Top}}(X, BU_n).$$

Here BU_n is (tautologically enough) the classifying space of complex rank n vector bundles, but, as you probably know from a course on algebraic topology, $BU_n \simeq \mathrm{Gr}_n(\mathbb{C})$, the infinite complex grassmannian.

However the interesting part is know that we have an action of U_n on $\mathcal{C}^n$ compatible with the standard hermitian form. So in particular, we obtain an action

$$U_n \curvearrowright S^{2n-1} \subseteq \mathcal{C}^n.$$

This action has as a point stabilizer the subgroup U_{n-1} , so we obtain a homeomorphism $S^{2n-1} \cong U_n/U_{n-1}$ as homogenous spaces. In particular we get a fibre sequence

$$S^{2n-1} \longrightarrow BU_{n-1} \longrightarrow BU_n$$

Fixnotation,
in par-
ticular fr
simplicial
categories
and
Yoneda
and
Anima

What this fibre sequence tell us is, that the difference of rank n and rank $(n - 1)$ vector bundles is somewhat given by maps from X into S^{2n-1} . In particular, if X does not have any non contractible maps into S^{2n-1} , for example if X is a manifold of dimension $\dim(X) < 2n - 1$, and if we have a complex rank- $(n - 1)$ -vector bundle V on X such that $V \oplus \mathcal{C} \cong \mathcal{C}^n$, $\mathcal{C}$ the trivial bundle of rank 1, than $V \cong \mathcal{C}^{n-1}$. So this gives a great insight into the stability question about vector bundles by explicitly measuring the difficulty of answering this question (and in particular telling you that it is related to the homotopy groups of spheres, so really hard). More generally there is a well developed theory of obstruction theory in algebraic topology for lifts, like in the above example, where the obstructions are living in $H^i(X, \pi_{i+1}S^{2n-1})$.

Such a theory for obstructions would, once again, be really good to have in algebraic geometry. Sadly enough this is, again, not the case.

As a last point there is Brown's representability theorem, stating that every nice-enough functor, that is every generalized cohomology theory like KU , MU , $H\mathbb{Z}$ from the category $\mathbf{Ani}$ to the category of spectra, $\mathbf{Sp}$ is representable. This greatly simplifies the study of these cohomology theories, as spectra are better behaved and easier to understand then topological spaces.

So how can we manage to press all of this structure into algebraic geometry? The important idea is to change the category we are working with, and in our case, we have to change the category of spaces to the category of motivic spaces

Definition 1.0.1 (Morel, Voevodsky). let S be a (qcqs) scheme. Then we define the *category of motivic spaces* as

$$\mathrm{Spc}(S) := L_{\mathrm{Nis}+\mathbb{A}^1}\mathcal{P}(\mathrm{Sm}_S) \hookrightarrow \mathcal{P}(\mathrm{Sm}_S).$$

Here Sm_S is the 1-category of smooth, finitely presented schemes over S , $\mathcal{P}(-)$ is the ∞ -category of presheaves of spaces and $L_{\mathrm{Nis}+\mathbb{A}^1}$ is a certain localization functor. This means $\mathrm{Spc}(S)$ is the universal presentable ∞ -category generated by smooth S -schemes up to relations given by L_{Nis} and $L_{\mathbb{A}^1}$. In other words we get that a functor out of $\mathrm{Spc}(S)$ is the same thing as a cocontinuous functor out of $\mathcal{P}(\mathrm{Sm}_S)$ which invert those maps we are localizing at, which in turn is the same thing as any functor out of Sm_S inverting the maps we are localizing at. We now set out to give a quick explanation of what these maps we localize at are.

Lets start with $L_{\mathbb{A}^1}$. This is the localization generated by maps

$$X \times \mathbb{A}^1 \xrightarrow{\mathrm{pr}} X \in \mathrm{Sm}_S.$$

In particular, $X \xrightarrow{\iota_i} X \times \mathbb{A}^1$ is inverted, for i any point of the affine line.

Next we have L_{Nis} . For this we need some terminology. We call a diagramm in Sm_S

$$\begin{array}{ccc} W & \longrightarrow & V \\ \downarrow & & \downarrow \\ U & \longrightarrow & X \end{array}$$

a *distinguished Nisnevich square*, if it is a cartesian square such that $U \hookrightarrow X$ is an open immersion, $V \rightarrow X$ is étale and $(U \setminus W)_{\mathrm{red}} \xrightarrow{\cong} (X \setminus V)_{\mathrm{red}}$ is an isomorphism of schemes with the reduced structure sheaf¹.

¹There is no explicit need for taking the reduced structure on the complement. Any structure making $U \setminus W$ and $X \setminus V$ closed subschemes would work.

If we write $y_X \in \mathcal{P}(\mathrm{Sm}_S)$ for the image of X under the Yoneda-embedding, then L_{Nis} is generated by the maps

$$y_U \coprod_{y_W} y_V \rightarrow y_X \in \mathcal{P}(\mathrm{Sm}_S)$$

$$\emptyset \rightarrow y_\emptyset$$

We call $L_{\mathrm{Nis}+\mathbb{A}^1}$ the *motivic localization functor*. Be aware that this is **not** simply the composition of $L_{\mathbb{A}^1}$ and L_{Nis} .

Now that we “discussed” all of this technicalities, what does this even mean? Let us first start with a small philosophical remark by Tom. He thinks that the most relevant case is given by $S = \mathrm{Spec}(k)$ for k some field. However, even in this case motivic homotopy theory is impossible to understand and one does not get any significant simplifications by assuming $S = \mathrm{Spec}(k)$, so we will work in greater generality and explicitly mention if there can be done more by reducing to $S = \mathrm{Spec}(k)$.

After this remark, let us return to the problem at hand, explaining what $L_{\mathbb{A}^1}$ and L_{Nis} are good for. The easier thing to explain is $L_{\mathbb{A}^1}$. For this consider the following definition.

Definition 1.0.2. Two maps $f, g : X \rightarrow Y \in \mathrm{Sm}_S$ are called *naively $\mathbb{A}^1$ homotopic* if there exists $H : X \times \mathbb{A}^1 \rightarrow Y$ such that $H|_{X \times \{0\}} = f$ and $H|_{X \times \{1\}} = g$.

This definition is terrible to work with. Neither does this define an equivalence relation nor is it compatible with composition. But there is salvation. If f and g are naively $\mathbb{A}^1$ homotopy equivalent, then $L_{\mathbb{A}^1}(f) \sim L_{\mathbb{A}^1}(g)$, that is they have the same image in $\pi_0 \mathrm{Map}_{\mathrm{Spd}(S)}(L_{\mathbb{A}^1}X, L_{\mathbb{A}^1}Y)$. Indeed, we have a commutative diagram in Sm_S

$$\begin{array}{ccccc} X & & \xrightarrow{f} & & Y \\ & \searrow \iota_0 & & \nearrow H & \\ & & X \times \mathbb{A}^1 & \xrightarrow{H} & Y \\ & \nearrow \iota_1 & & \nwarrow g & \\ X & & \xrightarrow{g} & & Y \end{array}$$

Where the outer diagram Now applying $L_{\mathbb{A}^1}$ $L_{\mathbb{A}^1}(\iota_i)$ becomes an inverse equivalence for the projection $\mathrm{pr} : X \times \mathbb{A}^1 \rightarrow X$. In particular we obtain a commutative diagram fitting into the left of the diagram above

$$X \xrightarrow{L_{\mathbb{A}^1}(\iota_0)} X \times \mathbb{A}^1 \xrightarrow{L_{\mathbb{A}^1}(\mathrm{pr})} X,$$

$$\quad \quad \quad \text{id}$$

hence define a two cell $(L_{\mathbb{A}^1}(f), L_{\mathbb{A}^1}(g), \mathrm{id})$ witnessing the homotopy $L_{\mathbb{A}^1}(f) \sim L_{\mathbb{A}^1}(g)$.

This means in particular that $L_{\mathbb{A}^1}$ turns naive $\mathbb{A}^1$ homotopies into real ones.

There is also another interpretation of $L_{\mathbb{A}^1}$. For this we define the following cosimplicial object

$$\mathbb{A}^\bullet : \Delta \rightarrow \mathrm{Sm}_{\mathbb{Z}}$$

$$[n] \mapsto \mathrm{Spec}(Z[X_0, \dots, X_n] / \sum X_i = 1).$$

This is an imitation of the definition of the standard simplices considered as a cosimplicial topological space

$$\Delta^\bullet : \Delta \rightarrow \mathrm{Sm}_{\mathbb{Z}}$$

$$[n] \mapsto \Delta^n := \{(x_0, \dots, x_n) \mid \sum x_i = 1, x_i \geq 0 \forall i\}.$$

Now given $F \in \mathcal{P}(\mathbf{Sm}_S)$ we define the *singular Scheme corresponding to F* , $\text{Sing}F \in \mathcal{P}(\mathbf{Sm}_S)$ as the presheave

$$X \mapsto |F(X \times \mathbb{A}^\bullet)| := \text{colim}_{\Delta^{\text{op}}} (F(X \times A^n))$$

Where $|-| : \mathbf{sAn} \rightarrow \mathbf{An}$ is the geometric realization. This functor gives us a second interpretation of $L_{\mathbb{A}^1}$,

Theorem 1.0.3. *There is a natural equivalence*

$$L_{\mathbb{A}^1} \simeq \text{Sing}.$$

This theorem gives us quite an explicit description (and construction!) of $L_{\mathbb{A}^1}$, and we will prove it later in the course.

Let us proceed with the interpretation of LN . The first interpretation is the one coming directly from the definition of a Nisnevich square. It is that in the Nisnevich local world we can “glue” a scheme X out of the to constituent pieces U, V along their “intersection” W . Of course, this would seem more reasonable if $V \rightarrow X$ was an open immersion.

But there is also a second interpretation. For this we define a *Nisnevich coever* (also known as completely decomposed étale cover) as a family of morphisms $\{\alpha_i : X_i \rightarrow X\}_{i \in I}$ such that each α_i is étale and for all $x \in X$ there exists some $i \in I$ and $y \in X_i$ with $\alpha_i(y) = x$ and $\kappa(x) \xrightarrow{\cong} \kappa(y)$.

Example 1.0.4. *If*

$$\begin{array}{ccc} W & \longrightarrow & V \\ \downarrow & & \downarrow \\ U & \longrightarrow & X \end{array}$$

is a Nisnevich square then $\{U \rightarrow X, V \rightarrow X\}$ is a Nisnevich cover.

In fact, Nisnevich covers form a Grothendieck topology on $\mathbf{SM}_S$ and, as we discuss in this course, there is a notion of a sheave of spaces on a Grothendieck site.

Theorem 1.0.5. *A presheave $F \in \mathcal{P}(\mathbf{Sm}_S)$ is a sheave for the Nisnevich topology if and only if F satisfies the following two conditions*

- 1) *The canonical map $F(\emptyset) \simeq \text{Map}(y_0, F) \rightarrow \text{Map}(\emptyset, F) \cong *$ is an equivalence.*

Every Nisnevich square becomes a pullback square of anima.

In particular, L_{Nis} is nothing else then the sheafification functor for the Nisnevich topology. The only reason Tom knows for why the Nisnevich topology is used in motivic homotopy theory is that it works pretty well, so he does not know a heuristic reason for why we use this topology explicitly.

Now that we “understand” $L_{\mathbb{A}^1}$ and L_{Nis} , what is with $L_{\text{mot}} = L_{\mathbb{A}^1 + \text{Nis}}$? As we will see, L_{mot} is the transfinite alternating composition fo $L_{\mathbb{A}^1}$ and L_{Nis} , so motivic spaces are in some sense exactly the $\mathbb{A}^1$ -invariant Nisnevich sheaves.

In the beginning we mentioned that spectra are a great simplification in the life of algebraic topologists. What about motivic spectra, i.e. spectral in the world of motivic spaces? There is a notion of presentably symmetric monoidal ∞ -categories, $\mathbf{CAlg}(\mathbf{Pr}^L)$ which we will talk about in the second part of this course. Now let $\mathcal{C} \hookrightarrow \mathbf{CAlg}(\mathbf{Pr}^L)_{\mathbf{Spc}(S)_*/}$ be the full subcategory on those functors $F : \mathbf{Spc}(S) \rightarrow \mathcal{D}$, $\mathcal{D}$ a presentably symmetric monoidal ∞ -category, such that $F(\mathbb{P}^1) \in \mathcal{D}$ is invertible, i.e. there exists $X \in \mathcal{D}$ such that $F(\mathbb{P}^1) \otimes X \simeq 1_{\mathcal{D}}$. Then we have the following

Theorem 1.0.6 (Robalo). *The category $\mathcal{C}$ has an initial object denoted by $\mathrm{SH}(\emptyset, S)$. This is the category of motivic spectra or the stable motivic homotopy category.*

Let us now return to the topological constructions we talked about in the beginning. First of all, one can show that if S is a regular scheme, there exists a motivic spectrum $KGL \in \mathrm{SH}(S)$ such that for any $X \in \mathrm{SM}_S$ we have an equivalence

$$\mathrm{map}_{\mathrm{SH}(S)}(\Sigma_+^\infty X, KGL) \simeq K(X).$$

That is K -theory becomes representable in the stable motivic homotopy category! Furthermore we have

$$\Omega^\infty KGL \simeq \mathbb{Z} \times BGL$$

mimicking what we have seen in algebraic topology. If we restrict to $S = \mathrm{Spec}(k)$, k a field, and to $X \in \mathrm{Sm}_k$ affine, there exists an equivalence

$$\pi_0 \mathrm{Map}_{\mathrm{Spd}(k)}(X, BGL_n) \cong \mathrm{Vect}_n(X) / \cong.$$

and there is a fibre sequence

$$\mathbb{A}^n \setminus \{0\} \longrightarrow BGL_{n-1} \longrightarrow BGL_n.$$

So in fact, the stable motivic homotopy category fulfills all the wishes we stated in the beginning. However, the last statements are some really serious theorems, so will will most certainly not do them here in the course. However the goal is that afterwards, you yourself will be versed enough to read and understand the proof of these theorems.

Chapter 2

Sheaves of Spaces

In this chapter we will introduce the necessary technical framework for the definition of motivic spaces. The main source will, of course, be [Lur09].

2.1 ∞ -Topoi

In classical 1-category theory, a *topos* is a category which is equivalent to the sheaves of sets on some Grothendieck site. A topos has many desirable properties, as for example shown in [MM12], so much as that they behave a lot like the category of sets. In ∞ -category theory there is a generalized notion which we are now going to introduce.

Definition 2.1.1. A (in general large) ∞ -category $\mathcal{X} \in \widehat{\text{Cat}}_\infty$ is called an ∞ -*topos* if there exists a locally small ∞ -category $\mathcal{C} \in \text{Cat}_\infty$ and a left exact accessible localization $\mathcal{P}(\mathcal{C}) \rightarrow \mathcal{X}$.

Note that even if $\mathcal{C}$ is small, $\mathcal{X}$ does not need to be small.

Remark 2.1.2. By definition, a topos $\mathcal{X}$ is always presentable and the localization $\mathcal{P}(\mathcal{C}) \rightarrow \mathcal{X}$ preserves finite limits and has a fully faithful right adjoint.

The typical (trivial) example of an ∞ -topos is $\mathcal{X} = \text{Ani} = \mathcal{P}(*)$, or more generally $\mathcal{P}(\mathcal{C})$ for $\mathcal{C} \in \text{Cat}_\infty$.

Remark 2.1.3. There is also an intrinsic characterization of ∞ -topoi using *Giraud's axioms*, just as it is the case for 1-topoi. For a discussion of this see [Lur09, §6.1].

Similar to the 1-categorical case, ∞ -topoi have a few important properties which we are going to review now. However, most of these facts are quite hard to prove, so we will mostly only state them and refer the interested reader to [Lur09, Ch. 6] for precise statements and proofs.

The first important fact is that *coproducts are disjoint*. That means that given an ∞ -topos $\mathcal{X}$ and $A, B \in \mathcal{X}$, then the pushout square

$$\begin{array}{ccc} \emptyset & \longrightarrow & A \\ \downarrow & & \downarrow \\ B & \longrightarrow & A \amalg B \end{array}$$

is also a pullback square.

Proof. We choose a presentation $L : \mathcal{P}(\mathcal{C}) \rightarrow \mathcal{X}$. Let $A, B \in \mathcal{P}(\mathcal{C})$. Then, as the statement holds in Ani and as limits and colimits of presheaves are computed section wise, the

pushout square in $\mathcal{P}((\mathcal{C}))$

$$\begin{array}{ccc} \emptyset & \longrightarrow & A \\ \downarrow & & \downarrow \\ B & \longrightarrow & A \amalg B \end{array}$$

is also pullback square. Applying L and using that it is left adjoint, hence preserves colimits, and left exact, hence preserves pullbacks, we conclude using the essential surjectivity of L . $\square$

The second important fact is that *colimits are universal*, that is that colimits commute with base change. To be precise, let $f : A \rightarrow B \in \mathcal{X}$. Then the induced postcomposition functor $f_{\#} : \mathcal{X}_{/A} \rightarrow \mathcal{X}_{/B}$ has a right adjoint, the base change functor $f^* : \mathcal{X}_{/B} \rightarrow \mathcal{X}_{/A}$. The functor f^* preserves colimits.

Remark 2.1.4. Contrary to the first fact, this is nontrivial even for $\mathcal{X} = \mathbf{Ani}$.

Exercise 2.1.5. Reduce the general case to $\mathcal{X} = \mathbf{Ani}$.

The third important fact about ∞ -topoi is that they satisfy *descent*:

Let I be a (small) index ∞ -category and let $F : I \rightarrow \mathcal{X}$ be a diagram. Then the canonical map

$$\mathcal{X}_{/\mathrm{colim}_I(F)} \xrightarrow{\cong} \lim_I \left(\mathcal{X}_{/F(i)} \right),$$

defined by the conglomeration of the pullbacks $\mathcal{X}_{/\mathrm{colim}_I(F)} \rightarrow \mathcal{X}_{/F(i)}$, is an equivalence.

Remark 2.1.6. Again, this is highly non-trivial, even in the case $\mathcal{X} = \mathbf{Ani}$.

In particular, to construct an object $T \rightarrow \mathrm{colim}_I(F) \in \mathcal{X}$ is the same data as to exhibit $G \rightarrow F \in \mathrm{Fun}(I, \mathcal{X})$ compatible with restrictions, that is that for all $\alpha : i \rightarrow j \in I$ we have an equivalence

$$G(i) \cong G(j) \times_{F(j)} F(i).$$

Such natural transformation is more generally called *cartesian*.

Given such a cartesian transformation $G \rightarrow F \in \mathrm{Fun}(I, \mathcal{X})$ we obtain an object $T = \mathrm{colim}_I(G) \rightarrow \mathrm{colim}_I(F) \in \mathcal{X}$, and given an object $T \rightarrow \mathrm{colim}_I(F)$ we obtain a functor $G(i) := T \times_{\mathrm{colim}_I(F)} F(i)$ and a cartesian transformation $G \rightarrow F$ using the projection.

The forth and last important fact is that *groupoids are effective*. To explain this statement we need to go a little further.

Reminder 2.1.7. We denote by Δ_+ the 1-category of totally ordered, finite sets. Then Δ is a full subcategory of the category of Δ_+ . Further we write $\Delta_{(+)}^{\leq r} \subseteq \Delta_{(+)}$ for the full subcategory of (possibly empty) totally ordered sets of cardinality at most r .

Given an ∞ -category $\mathcal{C}$ a functor $U : \Delta^{\mathrm{op}} \rightarrow \mathcal{C}$ is called a simplicial object of $\mathcal{C}$, while $U_{\bullet} : \Delta_+^{\mathrm{op}}$ is called an augmented simplicial object of $\mathcal{C}$.

As usual we will write U_n for $U([n])$ or $U_{\bullet}([n])$ respectively. We further use the convention $[-1] = \emptyset$.

Definition 2.1.8. A simplicial object $U : \Delta^{\mathrm{op}} \rightarrow \mathcal{C}$ of an ∞ -category $\mathcal{C}$ is called a *groupoid* if for all subsets $S, T \subseteq [n] \in \Delta$ such that $S \cap T = \{s\}$ and $S \cup T = [n]$ the square

$$\begin{array}{ccc} U([n]) & \longrightarrow & U(S) \\ \downarrow & & \downarrow \\ U(T) & \longrightarrow & U(\{s\}) \end{array}$$

is a pullback square in $\mathcal{C}$.

Exercise 2.1.9. Show that a groupoid object in the category of sets recovers the usual notion of a groupoid category.

Lemma 2.1.10. *Let $\mathcal{C}$ be a complete ∞ -category, $U_\bullet : \Delta_+^{\text{op}} \rightarrow \mathcal{C}$ an augmented simplicial object in $\mathcal{C}$. Then $U_\bullet$ is right Kan extended from $(\Delta_+^{\leq 0})^{\text{op}}$ if and only if*

1) $U_\bullet$ is a groupoid.

2)

$$\begin{array}{ccc} U_1 & \longrightarrow & U_0 \\ \downarrow & & \downarrow \\ U_0 & \longrightarrow & U_{-1} \end{array}$$

is a pullback square in $\mathcal{C}$.

Proof. To compute the right Kan extension $\text{RKan}_F : \Delta_+ \rightarrow \mathcal{C}$ of a functor $F : \Delta_+^{\leq 0}$ at $[n]$ we can use the “local formula”, that is we take the limit over the diagram

$$(\Delta_+^{\leq 0})_{[n]/}^{\text{op}} \rightarrow \Delta_+^{\leq 0} \xrightarrow{F} \mathcal{C}$$

where the first functor is the forgetful functor and where we can identify the first category with

$$(\Delta_+^{\leq 0})_{[n]/}^{\text{op}} \cong \left((\Delta_+^{\leq 0})_{/[n]} \right)^{\text{op}} \cong \left\{ \begin{array}{ccc} [-1] & \longrightarrow & [0] \\ & \searrow \exists! & \downarrow \dots \downarrow \\ & & [n] \end{array} \right\}^{\text{op}}$$

It follows that limit is just the limit over $n + 1$ copies of $F([0]) \rightarrow F([-1])$, that is

$$\text{RKan}_F([n]) = F([0]) \times_{F([-1])} \dots \times_{F([-1])} F([0]) = F([0])^{\times_{F([-1])} (n+1)}.$$

To check that this implies the statement is an inductive argument which we leave to the reader. $\square$

Definition 2.1.11. Let $\mathcal{C}$ be an ∞ -category.

- We call an augmented simplicial object $U_\bullet : \Delta_+ \rightarrow \mathcal{C}$ a *Čech nerve* if it satisfies the above condition.
- A groupoid $U : \Delta \rightarrow \mathcal{C}$ is called *effective* if $\tilde{U}_\bullet : \Delta_+^{\text{op}} \rightarrow \mathcal{C}$ is a Čech nerve, where

$$\tilde{U}_n := \begin{cases} U_n & : n \geq 0 \\ |U| & : n = -1 \end{cases}$$

where the higher cells are either defined by U or by the universal property of the colimit.

In the 1-categorical setting an epimorphism $f : X \rightarrow Y \in \mathcal{C}$ for a 1-category is called *effective* if it exhibits Y as the coequaliser of the kernel pair of f , that is $X \times_Y X \xrightarrow[\text{pr}_1]{\text{pr}_2} Y$

$X \xrightarrow{f} Y$ is a coequaliser diagram, assuming the appearing limits and colimits exist in $\mathcal{C}$. This generalizes the idea that surjection of sets $f : X \rightarrow Y$ exhibit Y as X modulo some equivalence relation. We will see that the notion of an effective groupoid, or more precise of the Čech nerve, generalizes the notion of an effective morphism to the ∞ -categorical world. This notion is quite important, so we will discuss it a bit more.

2.2 More about Čech nerves

We want to define an analog of the image of a morphism in an ∞ -topos. For this we want to generalize the notion of a monomorphism and an epimorphism to ∞ -topoi.

Reminder 2.2.1. We call $S \in \mathbf{Ani}$ *n-truncated* if $\pi_i(S, s) = 0$ for all $n > 0$ and all $s \in S$.

Example 2.2.2. $X \in \mathbf{Ani}$ is *(-1)-truncated* if and only if $X = \emptyset$ or $X = *$.

Definition 2.2.3. Let $\mathcal{C} \in \mathbf{Cat}_\infty$ be an ∞ -category.

- We call an object $X \in \mathcal{C}$ *n-truncated* if the mapping space $\mathrm{Map}_{\mathcal{C}}(Y, X) \in \mathbf{Ani}$ is *n-truncated* for all $Y \in \mathcal{C}$.
- We call a morphism $f : X \rightarrow Y \in \mathcal{C}$ *n-truncated*, if it is *n-truncated* as an object in $\mathcal{C}_{/Y}$.
- A *(-1)-truncated* morphism is called a *monomorphism*.

This seems to be completely unrelated to the usual definition of a monomorphism. However, the following lemma shows that from an abstract-categorical point of view, this notion of a monomorphism is the correct one.

Lemma 2.2.4. Let $\mathcal{C}$ be an ∞ -category. Then $f : X \rightarrow Y$ is a monomorphism if and only if

$$\begin{array}{ccc} X & \xrightarrow{\mathrm{id}} & X \\ \downarrow \mathrm{id} & & \downarrow f \\ X & \xrightarrow{f} & Y \end{array}$$

is a cartesian square.

Proof. To show that $f : X \rightarrow Y \in \mathcal{C}$ is a monomorphism, we have to show that $f \in \mathcal{C}_{/Y}$ is *(-1)-truncated*. Using that the product in the slice category is the fibre product it suffices to show that the diagonal $\Delta : X \rightarrow X \times X \in \mathcal{C}$ is an equivalence. Furthermore, by the Yoneda-embedding, the functor $\mathrm{Map}_{\mathcal{C}}(Z, -)$ preserves and detects being cartesian for all $Z \in \mathcal{C}$, so we may assume $\mathcal{C} = \mathbf{Ani}$. But now the claim is an explicit computation: For $X \in \mathbf{Ani}$ we have that the diagonal is an equivalence if and only if $X = *$ or $X = \emptyset$. $\square$

Example 2.2.5. Let $\mathcal{C} = \mathbf{Ani}$ and $x \in X \in \mathcal{C}$. Then, for a monomorphism $f : X \rightarrow Y$ we have the following commutative diagram

$$\begin{array}{ccccc} * & \xrightarrow{x} & X & \xrightarrow{\mathrm{id}} & X \\ \downarrow & & \downarrow \mathrm{id} & & \downarrow f \\ * & \xrightarrow{x} & X & \xrightarrow{f} & Y \end{array}$$

where both squares are pullback squares. By the pasting law for pullbacks we see that the outer rectangle is also a pullback square. It follows that the fiber $X \times_Y \{y\}$, $y \in Y$, is either empty or contractible. Hence a monomorphism is the same thing as an inclusion of connected components.

Reminder 2.2.6. Let $\mathcal{C}$ be an ∞ -category, $X \in \mathcal{C}$. We define the ∞ -category $\mathrm{Sub}(X)$ as the subcategory $\mathrm{Sub}(X) := \{Y \rightarrow X \text{ mono}\} \hookrightarrow \mathcal{C}_{/X}$. Then every mapping space in $\mathrm{Sub}(X)$ is either empty or contractible, so it is a poset. We call this the *poset of subobjects of X*.

Remark 2.2.7. Let $\mathcal{C} = \mathcal{X}$ be an ∞ -topos and $f : X \rightarrow Y \in \mathcal{X}$. Notice that $\Delta_+^{\leq 0} \cong \Delta^1$ so f promotes to $f : (\Delta_+^{\leq 0})^{\text{op}} \rightarrow \mathcal{X}$. In particular we can consider the right Kan extension of f . We call this the *augmented Čech nerve generated by f* and denote this as $\check{C}(f)_+ : \Delta_+^{\text{op}} \rightarrow \mathcal{X}$. We denote the corresponding simplicial object $\check{C}(f) : \Delta^{\text{op}} \rightarrow \mathcal{X}$ and call it simply the *Čech nerve generated by f* .

We can describe the objects in this Čech nerve quite explicitly:

$$\begin{aligned}\check{C}(f)_0 &= X \\ \check{C}(f)_1 &= X \times_Y X \\ \check{C}(f)_2 &= X \times_Y X \times_Y X\end{aligned}$$

In particular, a groupoid $U : \Delta^{\text{op}} \rightarrow \mathcal{C}$ is effective if and only if it is equivalent to the Čech nerve generated by $U_0 \rightarrow |U|$.

Proposition 2.2.8. *Let $f : X \rightarrow Y \in \mathcal{X}$. Then the canonical map $|\check{C}(f)| \rightarrow X$, defined by the universal property of the colimit, is a monomorphism.*

Proof. We want to use the criterion of lemma 2.2.4, that is we want the canonical map $|\check{C}(f)| \simeq |\check{C}(f)| \times_{|\check{C}(f)|} |\check{C}(f)| \rightarrow |\check{C}(f)| \times_Y |\check{C}(f)|$ to be an equivalence. Before we show this, notice the following.

Let $(U \rightarrow Y) = U_\bullet$ be an augmented simplicial object. Then, using universality of colimits twice, we have

$$|U| \times_Y |U| \simeq \operatorname{colim}_{m,n \in \Delta^{\text{op}}} (U_n \times_Y U_m).$$

Now we use this twice, once to $\check{C}(f) \rightarrow Y$ to calculate $|\check{C}(f)| \times_Y |\check{C}(f)|$, and once to $\check{C}(f) \rightarrow |\check{C}(f)|$ to calculate $|\check{C}(f)|$. This gives us a morphism of bi-simplicial objects, given explicitly on the (n, m) stage by

$$\begin{array}{c} \overbrace{X \times_Y \dots \times_Y X}^n \times_{|\check{C}(f)|} \overbrace{X \times_Y \dots \times_Y X}^m \\ \downarrow \\ \underbrace{X \times_Y \dots \times_Y X}_n \times_Y \underbrace{X \times_Y \dots \times_Y X}_m \end{array}$$

As $\mathcal{X}$ is an ∞ -topos, groupoids are effective. In particular Čech nerves are effective, so $X \times_{|\check{C}(f)|} X \simeq X \times_Y X$, finishing the proof. $\square$

Definition 2.2.9. Let $\mathcal{X}$ be an ∞ -topos, $X \in \mathcal{X}$.

- We call $\operatorname{im}(f) := |\check{C}(f)| \hookrightarrow X \in \operatorname{Sub}(X)$ the *image of f* .
- $f : X \rightarrow Y$ is called an *epimorphism*, or *effective epimorphism*, or *covering* if $\operatorname{im}(f) \simeq X$ via the above map.

This definition resembles what we are used to from classical abelian categories, or more general regular categories, in that every morphism $f : X \rightarrow Y \in \mathcal{X}$ factors over its image via an epi-mono-factorization $f : X \twoheadrightarrow |\check{C}(f)| \hookrightarrow Y$.

Example 2.2.10. *If $f : X \rightarrow Y \in \mathcal{X}$, $\mathcal{X}$ an ∞ -topos, is a section, that is there exists some $s : Y \rightarrow X \in \mathcal{X}$ such that $f \circ s \simeq \operatorname{id}$, then f is an epimorphism.*

Exercise 2.2.11. Show that $f : X \rightarrow Y$ is an epimorphism if and only if $\pi_0(f) : \pi_0(X) \rightarrow \pi_0(Y)$ is surjective.

So we get some kind of duality: While a for f to be a monomorphism has to be an injection on π_0 and a bijection on π_i for all $i \geq 0$, which is a really string restriction,, for f to be an epimorphism it only has to be surjective on π_0 .

Remark 2.2.12. Let $\mathcal{X}$ be an ∞ -topos, and

$$\begin{array}{ccc} Y' & \longrightarrow & Y \\ \downarrow f' & & \downarrow f \\ X' & \longrightarrow & X \end{array}$$

be a cartesian diagram in $\mathcal{X}$. Then we have $\check{C}(f) \times_X X' \simeq \check{C}(f')$, so by universality of colimits $|\check{C}(f)| \times_X X' \simeq |\check{C}(f')|$. In particular it follows that epimorphisms are stable under basechange.

Exercise 2.2.13. (Equivalence is a local property) Let $\mathcal{X}$ be an ∞ -topos and $f : Y \rightarrow X \in \mathcal{X}$ be a morphism. Then show that if we have a cartesian diagram in $\mathcal{X}$

$$\begin{array}{ccc} Y' & \longrightarrow & Y \\ \downarrow f' & & \downarrow f \\ X' & \xrightarrow{g} & X \end{array}$$

such that g is an epimorphism and f' is equivalence, f is an equivalence.

Before we continuo on our quest of defining sheaves on ∞ -categories, let us discuss a last example of a Čech nerve.

Example 2.2.14. Let $X \in \mathbf{Ani}$ be connected. Choose a basepoint $b : * \rightarrow X$. Then b is an epimorphism and we have

$$\begin{aligned} \check{C}(b)_0 &= * \\ \check{C}(b)_1 &= \Omega_b X \\ \check{C}(b)_2 &= \Omega_b X \times \Omega_b X \text{ etc.} \end{aligned}$$

Note that $\Omega_b X$ is in fact a group, and that this groupoid encodes the path concatenation operation and the corresponding homotopies on $\Omega_b X$. In particular we learn

$$X \simeq |\check{C}(b)|$$

which is not really obvious why this should be true at all.

2.3 Sheaves and sheafification

In classical 1-category theory to define a sheave on a category, one needs a good notion of “coverings”. This leads to the definition of a Grothendieck topology. And, as it was to be expected, this notion can be generalized to ∞ -categories, and we will let ourself guide by what we are used to from classical sheave theory. As a notational side note, we (mostly) will write $C \in \mathcal{P}(\mathcal{C})$ for the image of C , y_C , under the Yoneda embedding $\mathcal{C} \rightarrow \mathcal{P}(\mathcal{C})$.

Definition 2.3.1. Let $\mathcal{C} \in \mathbf{Cat}_\infty$. We define a *sieve on $\mathcal{C}$* as a subobject of the terminal object $* \in \mathcal{P}(\mathcal{C})$. Further, for $X \in \mathcal{C}$, a *sieve on X* is a sieve on the slice category $\mathcal{C}_{/X}$.

Note that the slice category $\mathcal{C}_{/X}$ has a terminal object and hence $* = X \in \mathcal{P}(\mathcal{C}_{/X})$.

Remark 2.3.2. What does this mean? As $\mathcal{P}(\mathcal{C})$ is an ∞ -topos, it follows from the last section that $F \hookrightarrow * \in \mathcal{P}(\mathcal{C})$ is a subobject of $*$ if and only if for all $X \in \mathcal{C}$ we have functoriality $F(X) = \emptyset$ or $F(X) = *$. That is, F is specified by some family of objects $X \in \mathcal{C}$ such that if $F(X) = *$ and $Y \rightarrow X \in \mathcal{C}$ is a morphism, then $F(Y) = *$. We will write $X \in F$ if $F(X) = *$.

Definition 2.3.3. Let $\mathcal{C} \in \text{Cat}_\infty$. A *Grothendieck topology* τ on $\mathcal{C}$ consists of the specification of a collection of sieves on C for every $C \in \mathcal{C}$, called the *covering sieves*. They are required to satisfy the following conditions.

- 1) For $C \in \mathcal{C}$ the terminal object $C \hookrightarrow C \in \mathcal{P}(\mathcal{C}_{/C})$ is a covering sieve.
- 2) Given $f : C \rightarrow D \in \mathcal{C}$ and a covering sieve $F \hookrightarrow D$ of D then $f^*F \hookrightarrow f^*D = C$ is a covering sieve, where $f^* : \mathcal{P}(\mathcal{C}_{/D}) \rightarrow \mathcal{P}(\mathcal{C}_{/C})$ is the pullback functor.
- 3) Given $C \in \mathcal{C}$, a covering sieve $F \hookrightarrow C \in \mathcal{P}(\mathcal{C}_{/C})$ and any sieve $G \hookrightarrow C$ such that for all $f : D \rightarrow C \in \mathcal{C}$ with $f \in F$, the pullback sieve $f^*G \hookrightarrow D$ is a covering sieve, then $G \hookrightarrow C$ was a covering sieve to begin with.

With the notion of a Grothendieck topology we can mimic the classical definition of a sheave, see [MM12, §III.4], also for a good heuristic explanation for this definition.

Definition 2.3.4. Let $\mathcal{C} \in \text{Cat}_\infty$ be a (small) ∞ -category together with a Grothendieck topology τ . Then we define $\text{Shv}_\tau(\mathcal{C}) \subseteq \mathcal{P}(\mathcal{C})$ as the full subcategory of those presheaves $F \in \mathcal{P}(\mathcal{C})$ such that for all $C \in \mathcal{C}$ and all covering sieves $(j \hookrightarrow C) \in \tau$ we have that the map

$$F(C) \simeq \text{Map}_{\mathcal{P}(\mathcal{C})}(C, F) \rightarrow \text{Map}_{\mathcal{P}(\mathcal{C})}(j, F)$$

is an equivalence.

Example 2.3.5. Let $\mathcal{C}$ be an ∞ -category, $C \in \mathcal{C}$ and $X_1, X_2, \dots \rightarrow C \in \mathcal{C}$ be a family of morphisms. Write $f : \coprod_i y_{X_i} \rightarrow y_C$. Then $|\check{C}(f)| \hookrightarrow C \in \mathcal{P}(\mathcal{C})$ defines a sieve, called the *sieve generated by the family* $\{X_i \rightarrow C\}_{i \in I}$.

Exercise 2.3.6. In the context of the above example, show that a morphism the morphisms contained in the sieve generated by $\{X_i \rightarrow C\}_{i \in I}$ are exactly those who factor through some X_i . That is, a morphism $Y \rightarrow C \in \mathcal{C}$ is contained in the sieve generated by $\{X_i \rightarrow C\}_{i \in I}$ if and only if there exists some $i \in I$ and a map $Y \rightarrow X_i$ rendering the following diagram commutative

$$\begin{array}{ccc} & & X_i \\ & \nearrow & \downarrow \\ Y & \longrightarrow & C \end{array}$$

Using this exercise we get a more classical description of the sheave condition, at least for sieves generated by a family of morphisms. So first assuming that the sieve generated by $\{X_i \rightarrow C\}_{i \in I}$ is a covering sieve of C we get the following chain of equivalences.

$$F(C) \simeq \text{Map}_{\mathcal{P}(\mathcal{C})}(|\check{C}(f)|, F) \simeq \lim_{\Delta} \left(\text{Map}_{\mathcal{P}(\mathcal{C})}(\check{C}(f), F) \right)$$

now assuming further that $\mathcal{C}$ has fiber products we obtain

$$F(C) \simeq \lim \left(\prod_i F(X_i) \rightleftharpoons \prod_{i,j} F(X_i \times_C X_j) \rightleftharpoons \prod_{i,j,k} F(X_i \times_C X_j \times_C X_k) \dots \right)$$

One of the most important operations in sheave theory is the sheafification of a presheave. Let τ be a Grothendieck topology on some ∞ -category $\mathcal{C}$. Then, with “some” work it is possible to define a functor, the plus construction,

$$(-)^+ : \mathcal{P}(\mathcal{C}) \rightarrow \mathcal{P}(\mathcal{C})$$

$$F \mapsto \left(X \mapsto F^+(X) \simeq \operatorname{colim}_{(C \hookrightarrow X) \in \tau} \left(\operatorname{Map}_{\mathcal{P}(\mathcal{C})}(C, F) \right) \right)$$

Remark 2.3.7. Note that for two covering sieves $C, D \hookrightarrow X \in \mathcal{P}(\mathcal{C})$ we have that there intersection $C \cap D \hookrightarrow X$ is again a covering sieve. In particular the colimit above is filtered.

Theorem 2.3.8. *Let τ be a Grothendieck topology on some ∞ -category $\mathcal{C}$. Then the ∞ -category $\operatorname{Shv}_\tau(\mathcal{C})$ is an ∞ -topos. The left adjoint of $\operatorname{Shv}_\tau(\mathcal{C}) \hookrightarrow \mathcal{P}(\mathcal{C})$ is left exact and obtained as the transfinite composition of $(-)^+$.*

Remark 2.3.9. For presheave of sets, that is 0-truncated spaces, it is a classical fact that the plus construction has to be applied twice, once to obtain a separated presheave, and a second time to obtain a sheave. For groupoids of sets, so 1-truncated spaces, the plus construction has to be applied three times, once to make the automorphism group of sections into separated presheaves, a second time to turn them into sheaves and a third time to ensure that descent is satisfied. This theme continuous, for a n -truncated space, the plus construction has to be applied $n + 2$ times, see [Lur09, Ch. 6.5.3].

Proof sketch. First one shows that if $G \in \operatorname{Shv}_\tau(\mathcal{C})$ and $F \in \mathcal{P}(\mathcal{C})$ then

$$\operatorname{Map}_{\mathcal{P}(\mathcal{C})}(F, G) \simeq \operatorname{Map}_{\mathcal{P}(\mathcal{C})}(F^+, G).$$

Afterwards one shows that there exists a regular cardinal κ independent $F \in \mathcal{P}(\mathcal{C})$ such that the κ -transfinite composition

$$F \longrightarrow F^+ \longrightarrow F^{++} \longrightarrow \dots \longrightarrow F' =: L_\tau F$$

is a τ -sheave.

One then defines L_τ as the κ -transfinite composition of $(-)^+$. This is then left adjoint to the inclusion $\operatorname{Shv}_\tau(\mathcal{C}) \hookrightarrow \mathcal{P}(\mathcal{C})$. Now, since limits and filtered colimits commute with finite limits in ∞ -topoi, we see that $(-)^+$ preserves finite limits and hence so does L_τ . $\square$

For a more rigorous discussion see [Lur09, Prop. 6.2.2.7]. We will sometimes write a_τ instead of L_τ .

We now proceed with a special construction of sheaves, the so called homotopy sheaves, a generalization of the homotopy groups.

Definition 2.3.10. Let $\mathcal{X}$ be an ∞ -topos. We call a 0-truncated object $X \in \mathcal{X}_{\leq 0}$ a *sheave of sets*.

Example 2.3.11. *If $\mathcal{X} \simeq \operatorname{Shv}_\tau(\mathcal{C})$ for some ∞ -site $(\mathcal{C}, \tau)$, then $\mathcal{X}_{\leq 0} \simeq \operatorname{Shv}_\tau(\mathcal{C}, \operatorname{Set})$, kind of justifying the naming.*

Reminder 2.3.12. • An ∞ -category is *cotensored* over spaces if for $X \in \operatorname{Ani}$ and $Y \in \mathcal{X}$ there exist an object $Y^X \in \mathcal{X}$ satisfying that for all $Z \in \mathcal{X}$ we have $\operatorname{Map}_{\mathcal{X}}(Z, Y^X) \simeq \operatorname{Map}_{\mathcal{X}}(Z \times X, Y)$ with $Z \times X := \operatorname{colim}_X(Z)$.

- If $\mathcal{C}$ is a presentable ∞ -category then $\mathcal{C}$ is cotensored. Further, for all $n \geq 0$, the inclusion $\mathcal{C}_{\leq n} \hookrightarrow \mathcal{C}$ preserves limits and is accessible. It from the adjoint functor theorem that there exists a left adjoint $\tau_{\leq n} : \mathcal{C} \rightarrow \mathcal{C}_{\leq 0}$.

Definition 2.3.13. Let $\mathcal{X}$ be an ∞ -topos, $X \in \mathcal{X}$. Then we define the *zeroth-homotopy sheave* $\pi_0(X) := \tau_{\leq 0}X \in \mathcal{X}_{\leq 0}$.

Let us see how this definition plays out in our favorite example of sheaves.

Example 2.3.14. Let $\mathcal{X} \simeq \mathrm{Shv}_\tau(\mathcal{C})$ for some ∞ -site $(\mathcal{C}, \tau)$. Let $X \in \mathcal{X}$. Write $X' \in \mathcal{P}(\mathcal{X})$ for the presheave defined by $C \mapsto \pi_0 X(C)$. The, for $F \in \mathrm{Shv}_\tau(\mathcal{C})_{\leq 0}$ we have using the adjunction from the previous remark

$$\mathrm{Map}_{\mathrm{Shv}_\tau(\mathcal{C})}(X, F) \simeq \mathrm{Map}_{\mathrm{Shv}_\tau(\mathcal{C})_{\leq 0}}(X', F).$$

In particular $\pi_0 X \simeq a_\tau X'$.

So the zeroth homotopy sheave is just the sheafification of the presheave which is section-wise the zeroth homotopy set, making it a “good” generalization. However, in algebraic geometry it is important to have all homotopy groups at hand, so next we want to generalize those too.

Definition 2.3.15. Let $\mathcal{X}$ be an ∞ -topos and choose a basepoint $* \rightarrow S^n \in \mathrm{Ani}$. As $\mathcal{X}$ is presentable, $\mathcal{X}$ is in particular cotensored and we can form X^{S^n} for $X \in \mathcal{X}$. Our choice of basepoint defines a map $X^{S^n} \rightarrow X^* \simeq X$, defining an object in $\mathcal{X}_X$.

- We then define the *n-th homotopy sheave* as the 0-truncation

$$\pi_n X := \tau_{\leq 0}(X^{S^n} \rightarrow X) \in (\mathcal{X}_X)_{\leq 0}$$

- Let $b : * \rightarrow X$ be a choice of basepoint for X . Then we define the *pointed n-th homotopy sheave*

$$\pi_n(X, b) := b^* \pi_n X \in \mathcal{X}_{\leq 0}$$

Remark 2.3.16. The second part of the definition can also be generalized in that a morphism $b : Y \rightarrow X \in \mathcal{X}$ defines $b^* \pi_n(X) \in (\mathcal{X}_Y)_{\leq 0}$. In particular $\pi_n X$ records all possible choices of “basepoints”.

In algebraic topology, $\pi_n X$ is a group for $n \geq 1$, commutative for $n \geq 2$. This behaviour too is covered by our more general definition.

Remark 2.3.17. We have a pushout diagram in Ani

$$\begin{array}{ccc} * & \longrightarrow & S^n \\ \downarrow & & \downarrow \\ S^n & \longrightarrow & S^n \vee S^n \end{array}$$

which in terms induces a pullback

$$\begin{array}{ccc} X^{S^n \vee S^n} & \longrightarrow & X^{S^n} \\ \downarrow & & \downarrow \\ X^{S^n} & \longrightarrow & X \end{array}$$

So that $X^{S^n \vee S^n} \simeq X^{S^n} \times_X X^{S^n} \in \mathcal{X}_X$. As $\tau_{\leq n}$ preserves finite products, see [Lur09, Lemma 6.5.1.2], we obtain a pairing

$$\pi_n X \times \pi_n X \simeq \pi_0(X^{S^n \vee S^n}) \rightarrow \pi_0(X^n) \simeq \pi_n X \in (\mathcal{X}_X)_{\leq 0}$$

where we used that $\tau_{\leq 0}(X \rightarrow X) = * \in (\mathcal{X}_X)_{\leq 0}$. One can show that this in fact promotes $\pi_n X \in (\mathcal{X}_X)_{\leq 0}$ to a sheave of groups.

Exercise 2.3.18. Let $X \in \mathbf{Ani}$. Show that $\pi_1(X) \in (\mathbf{Ani}/X)_{\leq 0} \simeq \mathbf{Fun}(\tau_{\leq 1}X, \mathbf{Set})$ and describe $\pi_1(X)$ explicitly.

Remark 2.3.19. Just as in algebraic topology, given a fibre sequence $X \rightarrow Y \rightarrow Z \in \mathcal{X}_*$ in an ∞ -topos $\mathcal{X}$ will induce a long exact sequence

$$\dots \rightarrow \pi_n(X, *) \rightarrow \pi_n(Y, *) \rightarrow \pi_n(Z, *) \rightarrow \pi_{n-1}(X, *) \rightarrow \dots$$

See [Lur09, Remark 6.5.1.5].

Now that we can talk about homotopy groups we would like that our objects are basically defined by their homotopy groups, as it is the case in classical algebraic topology (for nice spaces). As we will see, in the ∞ -topoi world, this is not true in general, leading us to the definitions of connectivity and postnikov-, and hyper-complete topoi.

Lemma 2.3.20. *Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a left exact colimit preserving functor of presentable ∞ -categories. The F commutes with $\tau_{\leq k}$*

Proof sketch. The idea of the proof is that a morphism $X \rightarrow Y$ is n -truncated if and only if the relative diagonal $X \rightarrow X \times_Y X$ is $n - 1$ -truncated, see [Lur09, Lemma 5.5.6.15]. This implies by an inductive argument, see [Lur09, Proposition 5.5.6.16], that a left exact functor preserves n -truncated objects, which in turn implies that F can be restricted to a functor of truncated objects. Hence there is an (a priori not necessary commutative) diagram

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{F} & \mathcal{D} \\ \downarrow \tau_{\leq n} & & \downarrow \tau_{\leq n} \\ \mathcal{C}_{\leq n} & \xrightarrow{F} & \mathcal{D}_{\leq n} \end{array}$$

Now a left exact colimit preserving functors of presentable ∞ -categories admits a left adjoint by the adjoint functor theorem [Lur09, Corollary 5.5.2.9] and hence the commutativity of the diagram follows from the adjoint diagram. For a full proof see [Lur09, Proposition 5.5.6.28]. $\square$

Lemma 2.3.21. *Let $\mathcal{X}$ be an ∞ -topos and $p : X \rightarrow \tau_{\leq k}X \in \mathcal{X}$ be the unit of the adjunction, $k \in \mathbb{Z}_{\geq -1}$. Then we have an equivalence*

$$\pi_n(X) \simeq p^* \pi_n(\tau_{\leq k}X)$$

for all $n \leq k$.

Proof. Consider the adjunction

$$L : \mathcal{P}(\mathcal{C}) \xrightleftharpoons[\perp]{} \mathcal{X} : \iota$$

By the previous lemma L commutes with π_n , so it suffices to prove the result for $\mathcal{X} = \mathcal{P}(\mathcal{C})$. But now everything can be computed section wise so we are reduced to the case of $\mathcal{X} = \mathbf{Ani}$, which is now just the classical result from algebraic topology. $\square$

Definition 2.3.22. Let $\mathcal{X}$ be an ∞ -topos, $f : X \rightarrow Y \in \mathcal{X}$ be a morphism and $0 \leq n \leq \infty$. We call f *n -connective* if f is an epimorphism and $\pi_k(f) = *$ for all $0 \leq k \leq n$. We call an object $X \in \mathcal{X}$ *n -connective* if the unique morphism $X \rightarrow *$ is so.

That this is coherent with the classical definitions shows the next example.

Example 2.3.23. Let $\mathcal{X} = \mathbf{Ani}$.

- An $n = 0$ connective space is just a non-empty space without any further assumptions. Now suppose that $n > 0$. Then we have $\pi_0(X) \simeq *$ and further $\pi_n(X \rightarrow *) \in (\mathcal{X}/_*)_{\leq 0} \simeq \mathbf{Ani}_{\leq 0}$. Under this identification we have an equivalence $\pi_n(X \rightarrow *) \simeq \pi_n(X, b)$ for all basepoints $b \in X$. In particular $\pi_i(X, b) = *$ for all $i < n$ and all basepoints $b \in X$.
- A morphism $f : X \rightarrow Y \in \mathbf{Ani}$ is n connective if $\pi_0(f) : \pi_0(X) \rightarrow \pi_0(Y)$ is surjective and for all basepoints $b \in Y$ the fiber F_b defined as the pullback

$$\begin{array}{ccc} F_b & \longrightarrow & X \\ \downarrow & & \downarrow f \\ * & \xrightarrow{b} & Y \end{array}$$

is n connective. This follows from the long exact sequence of homotopy groups.

Remark 2.3.24. Here are some facts about n -connective morphisms in an ∞ -topos X . For proofs of these statements see [Lur09, §6.5.1].

- $X \in \mathcal{X}$ is n connective if and only if $\tau_{\leq n-1} X \simeq *$.
- A morphism $f \in \mathbf{Arr}(\mathcal{X})$ is n connective if and only if f is m -connective for all $m \leq n$.
- A morphism $f \in \mathbf{Arr}(\mathcal{X})$ is n connective for all $n \geq 0$ if and only if f is ∞ -connective.
- n -connective morphisms are stable under transfinite composition, arbitrary colimits and pullbacks.
- If a morphism $f \in \mathbf{Arr}(\mathcal{X})$ is an equivalence then f is ∞ connective. The converse however is in general not true.

The last point is kind of a thorn in the side, and we will fix this by considering hyper-complete topoi. But before we do so, we quickly return to epimorphisms as they can be characterized in terms of homotopy sheaves.

Lemma 2.3.25. *Let $\mathcal{X}$ be an ∞ -topos and $X \in \mathcal{X}$ be $(n+1)$ -connective $n \geq 0$. Write $f : X \rightarrow * \in \mathcal{X}$ for the canonical morphism. Then the pullback functor induces an equivalence of truncated ∞ -topoi*

$$f^* : \mathcal{X}_{\leq n-1} \xrightarrow{\simeq} (\mathcal{X}/_X)_{\leq n-1}.$$

Proof sketch. We will only show fully-faithfulness of f^* . For this first remember that for $T \in \mathcal{X}$ we have $f^*(T) = T \times X$ considered as an object in $\mathcal{X}/_X$. Now f^* has a left adjoint $f_{\#}$, giving us $f_{\#}f^*(X) = T \times X \in \mathcal{X}$. We learn

$$\mathrm{Map}_{\mathcal{X}/_X}(f^*(T), f^*(U)) \simeq \mathrm{Map}_{\mathcal{X}}(f_{\#}f^*(T), U) \simeq \mathrm{Map}_{\mathcal{X}}(T \times X, U).$$

If U is $(n-1)$ -truncated we can use the inclusion-truncation adjunction

$$\mathrm{Map}_{\mathcal{X}/_X}(f^*(T), f^*(U)) \simeq \mathrm{Map}_{\mathcal{X}_{\leq n-1}}(\tau_{\leq n-1}(T \times X), U).$$

It is an exercise for the reader to show that

$$\tau_{\leq n-1}(T \times X) \simeq \tau_{\leq n-1}(\tau_{\leq n-1}T \times \tau_{\leq n-1}X).$$

But now $\tau_{\leq n-1}X \simeq *$ by assumption so that

$$\mathrm{Map}_{\mathcal{X}/_X}(f^*(T), f^*(U)) \simeq \mathrm{Map}_{\mathcal{X}_{\leq n-1}}(\tau_{\leq n-1}(T), U) \simeq \mathrm{Map}_{\mathcal{X}}(T, U).$$

Assuming $T \in \mathcal{X}_{\leq n-1}$ and remembering that f^* preserves truncated objects finishes the claim, of course after realizing that this chain of equivalences is indeed the same as f^* . To see this, notice that in the span diagram

$$\begin{array}{ccc} T \times X & \longrightarrow & T \\ \downarrow & & \\ \tau_{\leq n}(T \times X) & & \end{array}$$

the horizontal map is the counit, given by $\text{id}_X \times f$, while the vertical map is given by functoriality. But, after applying $\text{Map}_{\mathcal{X}}(-, U)$, this span consists of exactly those morphisms appearing in the chain of equivalences above.

Note that this part of the proof would also have worked for n truncated objects. However the harder part of showing essential surjectivity does in fact require to restrict to $n - 1$ truncated objects, see [Lur09, 7.2.1.1] for the full proof. $\square$

Proposition 2.3.26. *Let $\mathcal{X}$ be an ∞ -topos. Then $f : X \rightarrow Y \in \mathcal{X}$ is an epimorphism if and only if $\pi_0(f) : \pi_0(X) \rightarrow \pi_0(Y) \in \mathcal{X}_{\leq 0}$ is an epimorphism.*

Proof. First assume that $f : X \rightarrow Y$ is an epimorphism so in particular $|\check{C}(f)| \simeq Y$. As $\tau_{\leq 0} : \mathcal{X} \rightarrow \mathcal{X}_{\leq 0}$ preserves colimits, and since we are 0-truncated, we learn that $\pi_0(Y)$ is a the coequalizer

$$\pi_0(X \times_Y X) \rightrightarrows \pi_0(X) \longrightarrow \pi_0(Y)$$

and hence a quotient object.

Now assume f induces an epimorphism on π_0 . Consider the factorization of f over its image which we will denote by $\overline{X} := |\check{C}(f)|$, that is $X \twoheadrightarrow \overline{X} \hookrightarrow Y$. Further consider the unit $Y \rightarrow \pi_0(Y)$. Now, simply by construction, this last morphism $Y \rightarrow \pi_0(Y) \in \mathcal{X}_{/\pi_0(Y)}$ is 1-connective. But $\overline{X} \hookrightarrow Y$ is a monomorphism so in particular (-1) -truncated. By the previous lemma there is an equivalence

$$(\mathcal{X}_{/\pi_0(Y)})_{\leq -1} \simeq (\mathcal{X}_{/Y})_{\leq -1}.$$

In particular there exists an object $Z \rightarrow \pi_0(Y) \in (\mathcal{X}_{/\pi_0(Y)})_{\leq -1}$ such that the square in the following diagram is a pullback

$$\begin{array}{ccccc} X & \twoheadrightarrow & \overline{X} & \hookrightarrow & Y \\ & & \downarrow & & \downarrow \\ & & Z & \hookrightarrow & \pi_0(Y) \end{array}$$

Applying π_0 we obtain

$$\begin{array}{ccccc} \pi_0(X) & \twoheadrightarrow & \pi_0(\overline{X}) & & \\ & & \downarrow & & \\ & & Z \simeq \pi_0(Z) & \hookrightarrow & \pi_0(Y) \end{array}$$

(A curved arrow points from $\pi_0(X)$ to $\pi_0(Y)$.)

This implies $Z \simeq \pi_0(Y)$ and hence, by stability of equivalences under pullbacks $\overline{X} \simeq Y$. $\square$

2.4 Hypercompleteness

We are now finally ready to give the definition of hypercompleteness.

Definition 2.4.1. Let $\mathcal{X}$ be an ∞ -topos. An object $X \in \mathcal{X}$ is called *hypercomplete* if the functor $\mathrm{Map}_{\mathcal{X}}(-, X)$ inverts ∞ -connective maps. A topos $\mathcal{X}$ is called *hypercomplete* if all $X \in \mathcal{X}$ are hypercomplete.

Example 2.4.2. The ∞ -topos of spaces Ani is hypercomplete and thus so is $\mathcal{P}(\mathcal{C})$ for some ∞ -category $\mathcal{C}$.

We want to know whether a given ∞ -topos is hypercomplete. For this we will define the stronger notion of Postnikov-complete ∞ -topoi and show that such topoi can be characterized by a certain notion of dimension.

Definition 2.4.3. Let $\mathcal{X}$ be an ∞ -topos, $X \in \mathcal{X}$. Then we have a *Postnikov-tower* of X , that is a functor $X_{\bullet} : (\mathbb{N} \cup \{\infty\})^{\mathrm{op}} \rightarrow \mathcal{X}$, written diagrammatically as

$$X = X_{\infty} \longrightarrow \dots \longrightarrow X_2 \longrightarrow X_1 \longrightarrow X_0$$

exhibiting each X_n as the n -truncation of X . We call $X \in \mathcal{X}$ *Postnikov-complete* if the canonical map defined by the Postnikov-tower $X \rightarrow \lim_n (X_n)$ is an equivalence. A ∞ -topos is *Postnikov-complete* if ever $X \in \mathcal{X}$ is so.

The next lemma shows that Postnikov-completeness implies hypercompleteness

Lemma 2.4.4. Let $\mathcal{X}$ be an ∞ -topos. If $X \in \mathcal{X}_{\leq n}$ for some $n < \infty$, then X is hypercomplete.

Proof. Let $f : A \rightarrow B \in \mathcal{X}$ be an ∞ -connective map. Then we have the following chain of equivalences

$$\mathrm{Map}_{\mathcal{X}}(A, X) \simeq \mathrm{Map}_{\mathcal{X}_{/B}}(A, X \times B) \simeq \mathrm{Map}_{(\mathcal{X}_{/B})_{\leq n}}(\tau_{\leq n}^B A, X \times B) \simeq \mathrm{Map}_{\mathcal{X}}(B, X)$$

Here the first equivalence comes from the upper-lower-star adjunction, viewing A as an object in the slice topos via f and then pushing it into $\mathcal{X}$ by f_*A . The second equivalence stems from the assumption $X \in \mathcal{X}_{\leq n}$, so in particular $X \times B \in (\mathcal{X}_{/B})_{\leq n}$. The third one is just the truncation-inclusion adjunction¹. The last equivalence stems from the fact the $\tau_{\leq n}^B A \simeq B$ as $f : A \rightarrow B$ is ∞ -connective, so $\tau_{\leq n}^B A$ is terminal in the slice category, so equivalent to B via f . Then it is just undoing all adjunctions.

That (the inverse of) this chain of equivalences coincides with the precomposition by f is clear as the fourth equivalence is induced by f . $\square$

Remark 2.4.5. Be aware that although hypercomplete objects are closed under limits, basically by definition, in a general ∞ -topos $\mathcal{X}$ the limit $\lim_n (X_n)$ does not need to be Postnikov-complete. In particular, not all objects of all (hypercomplete) ∞ -topoi are Postnikov-complete.

However, this remark does not apply for a class of ∞ -topoi we will now introduce.

Definition 2.4.6. Let $\mathcal{X}$ be an ∞ -topos.

- $\mathcal{X}$ has *homotopy dimension* $\leq n$ if for all n -connective object $U \in \mathcal{X}$ there exists a map $* \rightarrow U$, that is a section of the canonical map $U \rightarrow *$ which we call a *global section*.

¹Here the τ^B just means to take the truncation in the slice category.

- $\mathcal{X}$ is *locally of finite homotopy dimension* if there exists a collection of objects $\{U_\alpha\}_{\alpha \in A}$ such that the U_α generate $\mathcal{X}$ under colimits and each slice topos $\mathcal{X}_{/U_\alpha}$ has homotopy dimension $\leq n_\alpha$.

Example 2.4.7. *Ani has homotopy dimension ≤ 0 and is locally of finite homotopy dimension.*

Exercise 2.4.8. $\text{Ani}_{/S_1}$ is not of homotopy dimension ≤ 0 .

Remark 2.4.9. Let $\mathcal{X}$ be an ∞ -topos of homotopy dimension $\leq n$. Then, if $X \in \mathcal{X}$ is $(n+d)$ -connective, the mapping space $\text{Map}_{\mathcal{X}}(*, X)$ is d -connective. This can be seen as follows:

$d = 0$ Here it is just the definition, remembering that for an object to be 0-connective means that it is non-empty.

$d = 1$ Being 1-connective means that any two points can be joined by a path. So let $a, b : * \rightarrow X$ be two points of X . Consider the pullback diagram

$$\begin{array}{ccc} L & \longrightarrow & * \\ \downarrow & & \downarrow b \\ * & \xrightarrow{a} & X \end{array}$$

Using the long exact homotopy sequence it follows that L is $(n+d-1 = n)$ -connective. In particular, as $\mathcal{X}$ has homotopy dimension $\leq n$, there exists a map $* \rightarrow L$ and hence a commutative diagram

$$\begin{array}{ccccc} & & a & & \\ & \searrow & & \searrow & \\ * & \longrightarrow & L & \rightrightarrows & X \\ & \swarrow & & \swarrow & \\ & & b & & \end{array}$$

showing the claim in this case.

$d \geq 2$ just repeat the above argument for $\Omega^{d-1}X$.

Proposition 2.4.10. *Let $\mathcal{X}$ be an ∞ -topos of locally finite homotopy dimension. Then all objects $X \in \mathcal{X}$ are Postnikov-complete. In particular $\mathcal{X}$ is hypercomplete.*

Proof. We have to check that for all $X \in \mathcal{X}$ the canonical map $X \rightarrow \lim_n (X_n)$ is an equivalence. By Yoneda, it suffices to check that post-composition by the canonical map induces an equivalence between mapping spaces $\text{Map}_{\mathcal{X}}(Y, X) \rightarrow \text{Map}_{\mathcal{X}}(Y, \lim_n (X_n))$. But by assumption $\mathcal{X}$ is generated under colimits by objects U such that the slice topos $\mathcal{X}_{/U}$ is of finite homotopy dimension. As the class of test-objects for which a map is an equivalence is closed under colimits, it suffices to prove the claim for the global sections of the slice topos $\text{Map}_{\mathcal{X}}(U, X) \rightarrow \text{Map}_{\mathcal{X}}(U, \lim_n (X_n))$.

Now realize that

$$\text{Map}_{\mathcal{X}}\left(U, \lim_n (X_n)\right) \simeq \lim_n (\text{Map}_{\mathcal{X}}(U, X_n)),$$

and further that, by the long exact homotopy sequence, $X \rightarrow X_{\leq n}$ is $(n+1)$ -connective. Now, by an elaboration of the previous remark (exercise for you!) $\text{Map}_{\mathcal{X}}(U, -)$ converts r -connective maps into $(r-d)$ -connective maps and hence

$$\text{Map}_{\mathcal{X}}(U, X) \rightarrow \text{Map}_{\mathcal{X}}(U, X_n)$$

is $(n+1-d)$ -connective. This now implies the claim, as in Ani if we have a tower $X_\bullet : (\mathbb{N} \cup \{\infty\})^{\text{op}} \rightarrow \text{Ani}$ such that for every m there exists an n_0 such that $\pi_m(X_\infty) \simeq \pi_m(X_n)$ for all $n \geq n_0$, then this tower is a homotopy limit tower, i.e. $X_\infty \simeq \lim_n (X_n)$. $\square$

Chapter 3

The Nisnevich topology

Now that we have the purely technical part out of the way, let us consider the application of this formalism most important for our goal, the Nisnevich topos.

Definition 3.0.1. let S be a scheme. A family of étale maps $\{U_\alpha \xrightarrow{f_\alpha} S\}_{\alpha \in A}$ is called a *Nisnevich covering* if the maps are jointly surjective on k points for every field k .

Remark 3.0.2. To state this in another way, a family as above is a Nisnevich cover if for all points $s \in S$ there exists some $\alpha \in A$ and some $x \in U_\alpha$ such that $f_\alpha(x) = s$ and $\kappa(x) \cong \kappa(s)$, where $\kappa(S)$ is the residue field at s .

The Nisnevich coverings defined above define a Grothendieck-pretopology. We can turn this into a Grothendieck topology in usual way and leave it to you to check that this in fact satisfies all the axioms.

Definition 3.0.3. We write $\mathcal{E}t_S$ for the category of étale S -schemes. For a scheme $X \in \mathcal{E}t_S$ we call a $C \hookrightarrow X \in \mathcal{P}(\mathcal{E}t_S)$ a covering sieve if it is generated by a Nisnevich covering, that is if there exists a Nisnevich covering $\{f_\alpha : U_\alpha \rightarrow S\}_{\alpha \in A}$ such that $U_\alpha \in C$ for all α . We write $(\mathcal{E}t_S, \tau_{\text{Nis}})$ for the *Nisnevich site of S* and $S_{\text{Nis}} := \text{Shv}_{\text{Nis}}(\mathcal{E}t_S)$ for the *Nisnevich ∞ -topos of S* .

That is, the Nisnevich ∞ -topos is the localization of $\mathcal{P}(\mathcal{E}t_S)$ at the Nisnevich covering sieves.

Exercise 3.0.4. Show that the Nisnevich covering sieves defined above do satisfy the axioms of a Grothendieck topology.

3.1 Formalities

In this section we discuss some of the formal properties of the Nisnevich topos. We start with the base-change functor.

Let $S' \rightarrow S \in \text{Sch}$ be a morphism of schemes. As usual we obtain the *base-change functor*

$$f^* : \mathcal{E}t_S \rightarrow \mathcal{E}t_{S'}, X \mapsto X \times_S S'.$$

To obtain a base-change functor for the Nisnevich topos, we compose the above functor with the Yoneda-embedding $y : \mathcal{E}t_{S'} \rightarrow \mathcal{P}(\mathcal{E}t_{S'})$ and consider the left Kan extend of $y \circ f$ along $y : \mathcal{E}t_S \rightarrow \mathcal{P}(\mathcal{E}t_S)$, giving us a functor, which we call by the same name

$$f^* : \mathcal{P}(\mathcal{E}t_S) \rightarrow \mathcal{P}(\mathcal{E}t_{S'}).$$

We can explicitly describe what this functor does on objects as follows.

Let $F \in \mathcal{P}(\acute{\text{Et}}_S)$ be a presheaf and $X \in \mathcal{P}(\acute{\text{Et}}_{S'})$ be an étale S' -scheme. Then, using the colimit formula for left Kan extensions, we learn

$$(f^*F)(X) = \operatorname{colim}_{I^{\text{op}}} (F(Y))$$

where $I = X \downarrow f^*$ is the comma-category

$$I = \left\{ \begin{array}{ccc} X & \longrightarrow & f^*Y \\ & \searrow & \downarrow \\ & & S' \end{array} \right\}$$

Using $f : S' \rightarrow S$ we see that X has a canonical étale S -scheme structure. Further using the étale morphism given by the fibre product $f^*Y \rightarrow Y$ the index category I is equivalent to

$$(\acute{\text{Et}}_S)_{X/} = \left\{ \begin{array}{ccc} X & \longrightarrow & Y \\ & \searrow & \downarrow \\ & & S \end{array} \right\}$$

so that

$$(f^*F)(X) = \operatorname{colim}_{Y \in (\acute{\text{Et}}_S)_{X/}^{\text{op}}} (F(Y)).$$

As $(\acute{\text{Et}}_S)_{X/}^{\text{op}}$ is filtered, see for example [Tam12, II, 1.4], it follows that $f^* : \mathcal{P}(\acute{\text{Et}}_S) \rightarrow \mathcal{P}(\acute{\text{Et}}_{S'})$ preserves finite limits, as those are computed section wise.

The pullback $f^* : \acute{\text{Et}}_S \rightarrow \acute{\text{Et}}_{S'}$ further preserve Nisnevich coverings, that is if $\{U_\alpha \rightarrow U\}_\alpha \in \acute{\text{Et}}_S$ is a Nisnevich cover, so is $\{f^*(U_\alpha) \rightarrow f^*(U)\}_\alpha \in \acute{\text{Et}}_{S'}$. In particular, f^* induces a morphism of pretopologies. We want to argue that it even induces a morphism of sites and hence of ∞ -topoi.

Let $C \hookrightarrow X \in \mathcal{P}(\acute{\text{Et}}_S)$ be a covering sieve, that is it contains a pretopological sieve, or a sieve generated by a covering family $\{U_\alpha \rightarrow X\}_\alpha$. Remind yourself that a sieve generated by a covering family is the same as the geometric realization of the Čech-nerve, so that we obtain a commutative diagram

$$\begin{array}{ccc} | \check{C}(\coprod_\alpha U_\alpha \rightarrow X) | & & \\ \downarrow & \searrow & \\ C & \longrightarrow & X \end{array}$$

Now applying f^* and using that this functor preserves all colimits and finite limits, in particular fiberproducts, we see that

$$\begin{array}{ccc} f^* | \check{C}(\coprod_\alpha U_\alpha \rightarrow X) | \cong | \check{C}(\coprod_\alpha f^*U_\alpha \rightarrow f^*X) | & & \\ \downarrow & \searrow & \\ f^*C & \longrightarrow & f^*X \end{array}$$

It follows from the discussion above that f^*C is again a covering sieve, that is $f^* : \mathcal{P}(\acute{\text{Et}}_S) \rightarrow \mathcal{P}(\acute{\text{Et}}_{S'})$ preserves covering sieves and hence induces a morphism of ∞ -topoi, which we call once more f^* , making the following diagram commute

$$\begin{array}{ccc} \mathcal{P}(\acute{\text{Et}}_S) & \xrightarrow{\text{L}_{\text{Nis}}} & \text{Shv}_{\text{Nis}}(\acute{\text{Et}}_S) \\ \downarrow f^* & & \downarrow f^* \\ \mathcal{P}(\acute{\text{Et}}_{S'}) & \xrightarrow{\text{L}_{\text{Nis}}} & \text{Shv}_{\text{Nis}}(\acute{\text{Et}}_{S'}) \end{array}$$

To be precise, $f^* : \mathrm{Shv}_{\mathrm{Nis}}(\acute{\mathrm{Et}}_S) \rightarrow \mathrm{Shv}_{\mathrm{Nis}}(\acute{\mathrm{Et}}'_{S'})$ is given by the composition $L_{\mathrm{Nis}} \circ f^* \circ \iota$ where $\iota : \mathrm{Shv}_{\mathrm{Nis}}(\acute{\mathrm{Et}}_S) \rightarrow \mathrm{Shv}_{\mathrm{Nis}}(\acute{\mathrm{Et}}'_{S'})$ is the inclusion and the f^* is the presheaf pullback functor. In particular, the sheaf f^* preserves colimits and finite limits as L_{Nis} , the f^* for presheaves and ι do so.

Remark 3.1.1. Notice that $f^* : \mathcal{P}(\acute{\mathrm{Et}}_S) \rightarrow \mathcal{P}(\acute{\mathrm{Et}}'_{S'})$ does in general not preserve Nisnevich sheaves, that is the diagram

$$\begin{array}{ccc} \mathcal{P}(\acute{\mathrm{Et}}_S) & \longleftarrow & \mathrm{Shv}_{\mathrm{Nis}}(\acute{\mathrm{Et}}_S) \\ \downarrow f^* & & \downarrow f^* \\ \mathcal{P}(\acute{\mathrm{Et}}'_{S'}) & \longleftarrow & \mathrm{Shv}_{\mathrm{Nis}}(\acute{\mathrm{Et}}'_{S'}) \end{array}$$

does in general not commute.

Next we want to consider the stalks in the Nisnevich topology. To do so we need to short interludes.

Interlude: Henselian rings

In étale cohomology the stalks of a sheaf correspond to strict henselisations. In the Nisnevich topology, we will see that they correspond to henselisations. So it should come in handy to have an idea of the properties of henselian rings, which is what we are going to do in this interlude. However, we will not prove anything and instead refer the reader to classical literature like [Ray06] or simply the stacks project [Sta25, Tag 04GE].

We begin with the classical definition of henselian rings, which is also one of the most irrelevant characterizations (at least for our purposes).

Definition 3.1.2. A local ring $(A, \mathfrak{m})$ with residue field k is called an *henselian local ring* if for every monic polynomial $f \in A[T]$ and every root $a_0 \in k$ of $\bar{f} \in k[T]$ such that $\bar{f}'(a_0) \neq 0 \in k$, there exists some $a \in A$ such that $\bar{a} = a_0$ and $f(a) = 0$.

Remark 3.1.3. More informally, an henselian local ring is a local ring such that every simple root of the image of a monic polynomial under the natural projection $A[T] \rightarrow k[T]$ has a lift.

Proposition 3.1.4 ((Fun-)Facts about henselian rings). *Let A be a commutative ring and $\mathfrak{p} \in \mathrm{Spec}(A)$ a prime ideal.*

- i) Consider the full subcategory $\mathcal{C} \subseteq \mathrm{Alg}_A$ consisting of those A -algebras which are étale¹ and such that the canonical map*

$$\kappa(\mathfrak{p}) \cong A \otimes_A \kappa(\mathfrak{p}) \mapsto B \otimes_A \kappa(\mathfrak{p})$$

is an equivalence, where $\kappa(\mathfrak{p})$ is the residue field at $\mathfrak{p} \in \mathrm{Spec}(A)$. Then $\mathcal{C}$ is filtered and the colimit $A_{\mathfrak{p}}^h := \mathrm{colim}_{\mathcal{C}}(B)$ is a henselian local ring. We call $A_{\mathfrak{p}}^h$ the henselisation of A in $\mathfrak{p}$.

Analogously, given an scheme $X \in \mathrm{Sch}$ we define the henselisation of X in $x \in X$ as $X_x^h := \mathrm{Spec}(\mathcal{O}_{X,x}^h)$.

- ii) A is henselian local if and only if for any finite A -algebra B there is an isomorphism $B \cong \prod_{i=1}^n B_i \in \mathrm{Alg}_A$ such that every B_i is local.*

¹That is a ring B with a map $A \rightarrow B$ such that the corresponding map $\mathrm{Spec}(B) \rightarrow \mathrm{Spec}(A)$ is étale.

- iii) If A is henselian local and $A \rightarrow B$ is quasi-finite² then $B \cong B_1 \times B_2 \in \text{Alg}_A$ where $A \rightarrow B_1$ is finite and $B_2 \otimes_A k = 0$, that is the fiber in B_2 of the closed point is empty.
- iv) The functor $- \otimes_A k : \text{Alg}_A \rightarrow \text{Alg}_B$ induces an equivalence of categories

$$- \otimes_A k : \{A \rightarrow B \text{ finite étale}\} \longrightarrow \{k \rightarrow l \text{ finite étale}\}.$$

- v) Any complete local ring is henselian.
- vi) 0-dimensional local rings are henselian.

Interlude: Pro-Objects

The idea of the pro-completion of an ∞ -category is that we add formally all cofiltered limits, as made precise by the following definition.

Definition 3.1.5. Let $\mathcal{C} \in \text{Cat}_\infty$ be an ∞ -category. We define the *pro-completion* of $\mathcal{C}$ as

$$\text{Pro}(\mathcal{C}) := \text{Ind}(\mathcal{C}^{\text{op}})^{\text{op}}.$$

Remark 3.1.6. More informally, for an ∞ -category $\mathcal{C}$ the objects of $\text{Pro}(\mathcal{C})$ is a cofiltered diagram³ in $\mathcal{C}$

$$\text{“}\lim_{i \in I}\text{”}(c_i) := \dots \longrightarrow c_3 \longrightarrow c_2 \longrightarrow c_1.$$

Further the mapping spaces are given as

$$\text{Map}_{\text{Pro}(\mathcal{C})}(c_\bullet, d_\bullet) \simeq \lim_{j \in J} \text{colim}_{i \in I} (\text{Map}_{\mathcal{C}}(c_i, d_j))$$

In the case of schemes filtered colimits do not need to exist in general. However, we have the following theorem by Grothendieck:

Theorem 3.1.7. Let S be a scheme and $\mathcal{C} \subseteq \text{Pro}(\text{Sch}_S)$ the full subcategory of pro-objects $X_\bullet$ such that each $X_i \rightarrow S$ is locally of finite presentation and all transition maps $X_i \rightarrow X_j$ are affine. Then the functor

$$\begin{aligned} \lim : \mathcal{C} &\rightarrow \text{Sch}_S \\ X_\bullet &\mapsto \lim_{i \in I} (X_i) \end{aligned}$$

is well defined and fully faithful.

Proof. Exercise. (Hint: Start with the affine case.)

□

add proof

What can we take from this? Let $X \in \text{Sch}_S$ be an S -scheme which can be written as a cofiltered limit of S -schemes locally of finite presentation with affine transition maps. Let $F \in \mathcal{P}(\text{Sch}_S^{\text{lfp}})$, where the subscript means “locally of finite presentation”. Then $F(X)$ makes sense:

Choose a representation $X = \lim_{i \in I} (X_i)$ and set $F(X) := \text{colim}_{i \in I} (F(X_i))$. The full faithfulness in the theorem guarantees that this expression is well defined and independent of choices.

Now that we talked about all these technicalities we can go on to define stalks in the Nisnevich topology.

²For a definition of quasi-finite see [Sta25, Tag 02MK].

³here we depict an sequential diagram. of course not every cofiltered diagram is sequential.

Definition 3.1.8. Let $X \in \mathcal{E}t_S$ be an étale S -scheme and let $x \in X$ be a point. Then X_x^h is a cofiltered limit of the form discussed in the previous interlude, so for all $F \in \mathcal{P}(\mathcal{E}t_S)$ the expression $F(X_x^h) \in \mathbf{Ani}$ is well defined. We call this *the stalk of F at $x \in X$* .

One crucial point in the classical theory of sheaves is that the stalk of a presheaf and its sheafification are isomorphic. The next lemma shows that this is also true in the Nisnevich ∞ -topos.

Lemma 3.1.9. *Let $X \in \mathcal{E}t_S$ be an étale S -scheme, let $x \in X$ be a point and $F \in \mathcal{P}(\mathcal{E}t_S)$. Then there exists a natural equivalence*

$$F(X_x^h) \simeq (L_{\mathbf{Nis}}F)(X_x^h).$$

Proof. Assume for the moment that $S = X_x^h$. Let $C \hookrightarrow X_x^h \in \mathcal{P}(\mathcal{E}t_{X_x^h})$ be a Nisnevich covering sieve. Then, by definition C contains a Nisnevich covering of X_x^h , so there exists some étale morphism of schemes $p : Y \rightarrow X_x^h$ and some $y \in Y$ such that $p(y) = x$ and $\kappa(y) \cong \kappa(x)$, using that $x \in X_x^h$ is the closed point.

But then p admits a section. Indeed, by 3.1.4 iii) Y splits as a disjoint union $Y_1 \coprod Y_2$ such that Y_1 misses x while Y_2 is finite over X_x^h . By 3.1.4 ii) we have another splitting $Y_2 \cong Y_3 \coprod Y_4$, where Y_4 is some disjoint union of local rings and $Y_3 \cong X_x^h$.

It follows that X_x^h is contained in the covering sieve $C \hookrightarrow X_x^h$. So in particular $C \cong X_x^h$ and therefore the topology is the trivial topology where X_x^h is the only covering sieve. But this implies

$$(F^+)(X_x^h) = \operatorname{colim}_{C \hookrightarrow X_x^h \text{ covering}} (F(C)) = F(X_x^h).$$

Now, as $L_{\mathbf{Nis}}F$ is the transfinite iteration of the plus construction $F^{++\dots}$, it follows that in the special case $S = X_x^h$

$$L_{\mathbf{Nis}}F(X_x^h) \simeq F(X_x^h). \quad (**)$$

For the general case consider the canonical morphism $f : X_x^h \rightarrow S$. Then we have an equivalence

$$(f^*F)(X_x^h) \simeq \operatorname{colim}_{U \in ((\mathcal{E}t_S)_{X_x^h})^{\text{op}}} (F(U)) \simeq F(X_x^h) \quad (*)$$

where the last equivalence exists because $X_x^h \in \mathbf{Pro}(\mathcal{E}t_S)$. By the definition of f^* for sheaves we have a natural equivalence

$$L_{\mathbf{Nis}} \circ \iota \circ f^* \circ L_{\mathbf{Nis}} \simeq L_{\mathbf{Nis}} \circ \iota \circ L_{\mathbf{Nis}} \circ f^* = L_{\mathbf{Nis}} \circ f^* : \mathcal{P}(\mathcal{E}t_S) \rightarrow \mathbf{Shv}_{\mathbf{Nis}}(\mathcal{E}t_{X_x^h}). \quad (***)$$

Hence we obtain the following chain of equivalences, where we suppressed the inclusions for readability,

$$\begin{aligned} (L_{\mathbf{Nis}}F)(X_x^h) &\simeq (f^*L_{\mathbf{Nis}}F)(X_x^h) && \text{(by } (*)) \\ &\simeq (L_{\mathbf{Nis}}f^*L_{\mathbf{Nis}}F)(X_x^h) && \text{(by } (**)) \\ &\simeq (L_{\mathbf{Nis}}f^*F)(X_x^h) && \text{(by } (***)) \\ &\simeq (f^*F)(X_x^h) && \text{(by } (**)) \\ &\simeq F(X_x^h). && \text{(by } (*)) \end{aligned}$$

This finishes the claim. $\square$

In classical sheaf theory a lot of properties can be checked local, like a morphism being an equivalence. The same holds true in our situation.

Check
this again!

Lemma 3.1.10. *A morphism of sheaves of sets $\alpha : F \rightarrow G \in \mathrm{Shv}_{\mathrm{Nis}}(\acute{\mathrm{E}}t_S)_{\leq 0}$ is an equivalence if and only if $\alpha(X_x^h)$ is an equivalence for all $x \in X \in \acute{\mathrm{E}}t_S$*

Proof. This is a standard proof. We will only show that injectivity of α at the stalks implies injectivity α and leave the rest as an exercise. To this end, let $a, b \in F(S)$ be such that $\alpha(a) = \alpha(b)$. By assumption we have for all $s \in S$ that $a|_{S_s^h} = b|_{S_s^h}$. By the definition of the stalk this is equivalent to an affine $U_s \in \acute{\mathrm{E}}t_S$ and a factorization

$$\begin{array}{ccc} s & \xrightarrow{\quad} & S \\ & \searrow & \uparrow \\ & & U_s \end{array}$$

such that $a|_U = b|_U$. Choosing such a U_s for every $s \in S$ defines a Nisnevich covering $\{U_s \rightarrow S\}_{s \in S}$ and hence $a = b$. $\square$

This lemma seems unsatisfying as we considered only sheaves of sets. However, combining this statement with the theory of homotopy sheaves we obtain the following

Theorem 3.1.11. *Let $\alpha : F \rightarrow G \in \mathcal{P}(\acute{\mathrm{E}}t_S)$ be a morphism of presheaves. Then $L_{\mathrm{Nis}}(\alpha) \in \mathrm{Shv}_{\mathrm{Nis}}(\acute{\mathrm{E}}t_S)$ is ∞ -connective if and only if $\alpha(X_x^h)$ is an equivalence for all $x \in X \in \acute{\mathrm{E}}t_S$.*

Proof. α as a morphism of sheaves of spaces being ∞ -connective is equivalent to the statement that $\pi_n(\alpha)$ is an isomorphism as a morphism of sheaves of sets for all $n \in \mathbb{N}_0$. By the previous lemma, this is equivalent to $\pi_n(\alpha)(X_x^h) = \pi_n(\alpha(X_x^h))$ being an isomorphism for all $n \in \mathbb{N}_0$ and all $x \in X \in \acute{\mathrm{E}}t_S$. This is in turn equivalent to $\alpha(X_x^h)$ being an equivalence by Whitehead's theorem, finishing the claim. $\square$

We will later on see that the Nisnevich ∞ -topos is Postnikov-complete so that in particular every ∞ -connective morphism is an equivalence. Hence, in the Nisnevich ∞ -topos, one can indeed check equivalences stalk-wise.

3.2 Distinguished squares

We will now turn to an alternative description of the Nisnevich topology, which we already used in the introduction. Our goal in the end will be to give a more concrete description of Nisnevich sheaves.

Definition 3.2.1. Consider the following commutative square of schemes.

$$\begin{array}{ccc} W & \longrightarrow & V \\ \downarrow & & \downarrow p \\ U & \xrightarrow{j} & X \end{array}$$

Such a diagram is called a *distinguished Nisnevich square* if the following four conditions hold.

- the square is cartesian
- $j : U \hookrightarrow X$ is an open immersion.
- p is étale. Let $Z \hookrightarrow X$ be the reduced closed complement of U . Then the étale pullback map $V_Z \rightarrow Z$, where we also take the reduced closed subscheme structure on V_Z , is an isomorphism of schemes.

Remark 3.2.2. As you can check yourself, every distinguished square defines a Nisnevich covering $\{U, V \rightarrow X\}$.

Proposition 3.2.3. *Assume we are given a distinguished Nisnevich square for $X \in \acute{\text{Et}}_S$. Then applying the Yoneda embedding turns this distinguished square into a cocartesian square in $\text{Shv}_{\text{Nis}}(\acute{\text{Et}}_S)$.*

Proof. By definition $U \amalg V \rightarrow X$ is surjective on k -points for all fields k , and hence an epimorphism as a map of schemes. As equivalences in an ∞ -topos are local it suffices to proof that the map $U \amalg_W V \rightarrow X$ is an equivalence after pulling back the diagram along $U \amalg V \rightarrow X$. Further using universality of colimits in an ∞ -topos, it suffices to check this after pulling back to U and V separately.

If we pull the diagram back along $U \hookrightarrow X$ we obtain

$$\begin{array}{ccc} W & \xrightarrow{\text{id}} & W \\ \downarrow & & \downarrow \\ U & \xrightarrow{\text{id}} & U \end{array}$$

which is clearly a cocartesian diagram.

Next we consider the pullback of the distinguished square along $V \rightarrow X$ we obtain

$$\begin{array}{ccc} W \times_X V & \longrightarrow & V \times_X V \\ \downarrow & & \downarrow \\ W & \longrightarrow & V \end{array}$$

To show that this diagram is a cocartesian diagram we extend this as

$$\begin{array}{ccccc} W & \xrightarrow{\Delta} & W \times_X V & \longrightarrow & W \\ \downarrow & & \downarrow & & \downarrow \\ V & \xrightarrow{\Delta} & V \times_X V & \longrightarrow & V \end{array}$$

as the outer rectangle is again clearly cocartesian it suffices to check that the left square is cocartesian. However, notice that this diagram is again a Nisnevich distinguished square. So it seems that we bit our tail, but this time, the situation is much better. Namely we have $V \hookrightarrow V \times_X V$ as V and $V \times_X V$ are both étale over S , hence $V \xrightarrow{\Delta} V \times_X V$ is étale and as such an open immersion.

So in particular we can repeat the argument of the proof, but this time both maps are open immersions, so that both pullback maps are just of the type $U \hookrightarrow X$ from before, showing that

$$\begin{array}{ccc} W & \xrightarrow{\Delta} & W \times_X V \\ \downarrow & & \downarrow \\ V & \xrightarrow{\Delta} & V \times_X V \end{array}$$

is in fact cocartesian. □

Let now $F \in \text{Shv}_{\text{Nis}}(\acute{\text{Et}}_S)$ be an Nisnevich sheave. Then F satisfies the following two properties.

- 1) The above proposition tells us that applying F to a given distinguished square

$$\begin{array}{ccc} W & \longrightarrow & V \\ \downarrow & & \downarrow p \\ U & \xrightarrow{j} & X \end{array}$$

turns this into the cartesian diagram

$$\begin{array}{ccc} F(X) & \longrightarrow & F(U) \\ \downarrow & & \downarrow \\ F(V) & \longrightarrow & F(W). \end{array}$$

2) By the sheaf property

$$F(\emptyset) = *$$

where $\emptyset$ is the empty cover and $*$ is the terminal object of $\mathbf{Ani}$.

A great theorem of Fabien Morel and Vladimir Voevodsky states that these two properties classify Nisnevich sheaves, at least under some mild conditions on S and on $\mathbf{Ét}_S$.

Theorem 3.2.4. *Let $S \in \mathbf{Sch}^{\text{qcqs}}$ be a quasi-compact, quasi-separated scheme. Let $F \in \mathcal{P}(\mathbf{Ét}_S^{\text{fp}})$ be a presheaf on finitely presented S -schemes⁴. Then F is a Nisnevich sheaf if and only if F satisfies the conditions 1) and 2).*

Remark 3.2.5. The finiteness assumption on the étale S -schemes is to prevent pathologies like $X = \coprod_{\mathbb{Z}} S$ in which case $F(X) = \prod_{\mathbb{Z}} F(S)$ which can not be deduced from the condition 1). However we can remove this assumption by adding a third condition of F being a Zariski sheaf, that is that F satisfies Zariski descent.

Our next goal is to prove this theorem. However, this is quite hard and we will need some time to develop the theory necessary to do so.

3.3 Splitting sequences

We start by discussing a general technique which enables us to do inductive arguments by proving a statement over a finite sequence of closed subschemes.

Lemma 3.3.1. *Let $X \in \mathbf{Sch}^{\text{qcqs}}$ be a qcqs-schemes. Let $Y \xrightarrow{p} X$ be a Nisnevich cover (take the disjoint union of the family). Then there exists a finite sequence*

$$\emptyset = Z_n \hookrightarrow Z_{n-1} \hookrightarrow \cdots \hookrightarrow Z_0 = X$$

of finitely presented, closed subschemes such that p admits a section over the complement $Z_i \setminus Z_{i+1}$.

Proof. We will only proof the lemma for X noetherian. For the general case see [BH21, Lemma A.1]. If $X = \emptyset$ is the empty scheme the statement is trivial so we may assume X non-empty.

As X is non-empty and noetherian it has a minimal point, i.e. a point with no generalization. Pick such a point $\eta \in X$. Then $\mathcal{O}_{X,\eta}$ is a local ring of dimension 0 so in particular henselian. Thus $p|_{Y \times_X \text{Spec}(\mathcal{O}_{X,\eta})}$ admits a section. In particular, this section exists in some open neighborhood U of η . No define $Z_1 := X \setminus U$ and repeat the argument. by noetherian induction, this process terminates after finitely many steps. $\square$

Corollary 3.3.2. *Let X be a qcqs-scheme. Any Nisnevich cover $\{U_\alpha \rightarrow X\}_{\alpha \in A}$ can be refined by a finite cover consisting of schemes finitely presented over X . In particular the Nisnevich topology is noetherian.*

⁴Note that in being a finitely presented étale S -scheme for S qcqs is the same as being quasi-compact.

Proof. By the definition of a Nisnevich cover, the structure map $U_\alpha \rightarrow X$ is étale and hence locally of finite presentation. By covering each U_α by affine open sets we obtain a refinement by finitely presented X -schemes, so we will from now on assume that $U_\alpha \rightarrow X$ is finitely presented for all $\alpha \in A$.

To see that finitely many α suffice let us apply the lemma to $p : \coprod_\alpha U_\alpha \rightarrow X$. We obtain a finite increasing sequence of finitely presented, closed subschemes $(Z_i)_{i \in I}$ of X together with sections $s_i : Z_i \setminus Z_{i+1} \rightarrow \coprod_\alpha U_\alpha$ of the Nisnevich cover p . In particular $\{s_i^{-1}(U_\alpha)\}_\alpha$ forms an open cover of $Z_i \setminus Z_{i+1}$. But $Z_{i+1} \hookrightarrow Z_i$ is finitely presented by the lemma and it is a fact that a closed immersion into a quasi-compact scheme is finitely presented if and only if its complement is quasi-compact. As there are finitely many Z_i , it follows that there exist a finite subset $A' \subseteq A$ such that $\{U_\alpha \rightarrow X\}_{\alpha \in A'}$ is a Nisnevich cover of X . $\square$

Remark 3.3.3. Except from the fact that the Nisnevich topology is noetherian for a qcqs scheme S we further obtain an equivalence of sites and hence of ∞ -topoi

$$\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Ét}_S) \xrightarrow{\simeq} \mathrm{Shv}_{\mathrm{Nis}, \mathrm{fin}}(\mathrm{Ét}_S^{\mathrm{fp}}).$$

Now we turn to the second ingredient we need for the proof of theorem 3.2.4.

3.4 Generated topologies

Definition 3.4.1. Let $\mathcal{C} \in \mathrm{Cat}_\infty$ be a small ∞ -category and τ an rule assigning for every $X \in \mathcal{C}$ a family of sieves $\tau(X) := \{C \hookrightarrow X \in \mathcal{P}(\mathcal{C})\}$. We call τ a *quasi-topology* if for all morphisms $f : X \rightarrow Y \in \mathcal{C}$, the basechange along f of the sieves for Y are contained in the sieves for X , that is $f^*\tau(Y) \subseteq \tau(X)$.

Remark 3.4.2. Given any such rule as in the above definition, there exists a smallest topology $\bar{\tau}$ containing τ . This is simply due to the fact that the assignment of all sieves is a topology, the discrete topology where everything is open, and topologies are stable under intersection. We call $\bar{\tau}$ the topology generated by τ . As the name already suggests, quasi-topologies are quite close to being topologies in that it suffices to check the sheaf-conditions for all sieves in the quasi-topology. This is made precise by the next lemma.

Lemma 3.4.3. *Let $\mathcal{C} \in \mathrm{Cat}_\infty$ be a small ∞ -category and τ be a quasi-topology on $\mathcal{C}$. Let $\bar{\tau}$ be the topology on $\mathcal{C}$ generated by τ . Then, a presheaf $F \in \mathcal{P}(\mathcal{C})$ is a sheaf in the $\bar{\tau}$ -topology, that is $F \in \mathrm{Shv}_{\bar{\tau}}(\mathcal{C})$ if and only if $F(X) \simeq \mathrm{Map}_{\mathcal{P}(\mathcal{C})}(C, F)$ for all $X \in \mathcal{C}$ and all $C \in \tau(X)$.*

Proof. The necessity of the condition is clear as $\tau \subseteq \bar{\tau}$ and $F(X) \simeq \mathrm{Map}_{\mathcal{P}(\mathcal{C})}(C, F)$ is simply the sheaf condition.

For the other direction, let us write the subcategory of τ -“sheaves”

$$\mathrm{Shv}_\tau(\mathcal{C}) := \left\{ F \in \mathcal{P}(\mathcal{C}) \mid F(X) \simeq \mathrm{Map}_{\mathcal{P}(\mathcal{C})}(C, F) \ \forall X \in \mathcal{C}, C \in \tau(X) \right\}.$$

Further let τ' be the quasi-topology consisting of all sieves $C \hookrightarrow X \in \mathcal{P}(\mathcal{C})$ such that $F(f^*C) \simeq F(Y)$ for all $f : Y \rightarrow X \in \mathcal{C}$ and all τ -sheaves $F \in \mathrm{Shv}_\tau(\mathcal{C})$. So this is the quasi-topology consisting of all sieves that are inverted by all sheaves which invert the sieves in τ . So in particular, any τ -sheaf is also a τ' -sheaf, so $\mathrm{Shv}_\tau(\mathcal{C}) \subseteq \mathrm{Shv}_{\tau'}(\mathcal{C})$ and τ' contains the trivial covering sieves.

We want to prove now that τ' is a topology, in which case $\bar{\tau} \subseteq \tau'$ by definition of $\bar{\tau}$. It follows $\mathrm{Shv}_{\tau'}(\mathcal{C}) \subseteq \mathrm{Shv}_{\bar{\tau}}(\mathcal{C})$, proving the claim.

So we have to check the axioms of a Grothendieck topology for τ' , see definition 2.3.3. The Stability under basechange is clear from the construction, as is the containment of the trivial sieve. We thus only have to check that for $S \in \tau'(X)$ and R a sieve on X such that $g^*R \in \tau'(X')$ for all $g : X' \rightarrow X \in S$, then already $R \in \tau'(X)$.

So let $f : Y \rightarrow X \in \mathcal{C}$ be any map to X and let $F \in \text{Shv}_\tau(\mathcal{C})$. We have to show that $F(Y) \rightarrow \text{Map}_{\mathcal{P}(\mathcal{C})}(f^*R, F)$ is an equivalence. This map fits into a commutative square

$$\begin{array}{ccc} \text{Map}_{\mathcal{P}(\mathcal{C})}(Y, F) & \xrightarrow{1)} & \text{Map}_{\mathcal{P}(\mathcal{C})}(f^*S, F) \\ \downarrow & & \downarrow 2) \\ \text{Map}_{\mathcal{P}(\mathcal{C})}(f^*R, F) & \xrightarrow{3)} & \text{Map}_{\mathcal{P}(\mathcal{C})}(f^*R \times_Y f^*S, F). \end{array}$$

Hence it suffices to show that 1), 2) and 3) are equivalences. 1) is true as by assumption S was a covering sieve.

For 2) remember that the left Kan-extension of the Yoneda embedding along the Yoneda embedding is the identity functor⁵, so we can write

$$f^*S \simeq \text{colim}_{Z \rightarrow f^*S} (Z)$$

with $Z \in \mathcal{C}$. But $\mathcal{P}(\mathcal{C})$ is a topos, so by universality of colimits we obtain

$$f^*R \times_Y f^*S \simeq \text{colim}_{Z \rightarrow f^*S} (f^*R \times_Y Z).$$

By assumption, $f^*R \times_Y Z \in \tau'(Z)$ for all $Z \rightarrow f^*S$, implying that the canonical map $f^*R \times_Y Z \hookrightarrow Z \in \mathcal{P}(\mathcal{C})$ induces an equivalence

$$\text{Map}_{\mathcal{P}(\mathcal{C})}(Z, F) \simeq \text{Map}_{\mathcal{P}(\mathcal{C})}(f^*R \times_Y Z, F).$$

for all $F \in \text{Shv}_\tau(\mathcal{C})$. Thus we have a chain of equivalences

$$\begin{aligned} \text{Map}_{\mathcal{P}(\mathcal{C})}(f^*S, F) &\simeq \lim_{Z \rightarrow f^*S} \left(\text{Map}_{\mathcal{P}(\mathcal{C})}(Z, F) \right) \\ &\simeq \lim_{Z \rightarrow f^*S} \left(\text{Map}_{\mathcal{P}(\mathcal{C})}(f^*R \times_Y Z, F) \right) \\ &\simeq \text{Map}_{\mathcal{P}(\mathcal{C})}(f^*R \times_Y f^*S, F). \end{aligned}$$

That this is the canonical map is clear as we only decomposed f^*S and applied the canonical morphism locally.

3) is analogous to 2). □

Lemma 3.4.4. *Let $S \in \text{Sch}^{\text{qcqs}}$, τ_{dist} the quasi-topology on $\mathcal{P}(\text{Ét}_S^{\text{qcqs}})$ generated by the distinguished squares $\{U, V \rightarrow X\}$. Then $\bar{\tau}_{\text{dist}} = \tau_{\text{Nis}}$.*

Proof. Let $X \in \text{Ét}_S^{\text{qcqs}}$ and let $U \rightarrow X$ be a Nisnevich cover. Let $\emptyset = Z_k \hookrightarrow Z_{k-1} \hookrightarrow \dots \hookrightarrow Z_0 = X$ be a splitting sequence. We will prove inductively on i that $U|_{X \setminus Z_i} \rightarrow X \setminus Z_i$ is a $\bar{\tau}_{\text{dist}}$ -covering. For $i = 0$ there is nothing, so assume the claim is proven for $i - 1$. By the properties of a splitting sequence, there exists a splitting $s : Z_{i-1} \setminus Z_i \rightarrow U_{Z_{i-1} \setminus Z_i} =: U_i$. Notice that s is étale, as it is the section of an étale map, and such sections are always étale by the cancellation lemma for étale maps. In particular, s is an open map and hence $V := U_i \setminus s(Z_{i-1} \setminus Z_i)$ is a closed subset of U_i . Consider the following diagram

$$\begin{array}{ccccc} s(Z_{i-1} \setminus Z_i) & \longrightarrow & U|_{X \setminus Z_i} & \longleftarrow & ? \\ \downarrow & & \downarrow & & \downarrow \\ Z_{i-1} \setminus Z_i & \longrightarrow & X \setminus Z_i & \longleftarrow & X \setminus Z_{i-1} \end{array}$$

⁵This is also known as the Yoneda-density theorem.

where both squares are pullback squares. Now $s(Z_{i-1} \setminus Z_i) \subseteq U|_{X \setminus Z_i}$ is open, $X \setminus Z_{i-1} \subseteq X \setminus Z_i$ is open, $V \subseteq U_i$ is closed, $Z_{i-1} \setminus Z_i \subseteq X \setminus Z_i$ is closed and $V \subseteq U|_{X \setminus Z_i}$ is closed. It follows that both squares are distinguished squares, so to check that $U|_{X \setminus Z_i} \rightarrow X \setminus Z_i$ is a $\bar{\tau}_{\text{dist}}$ -covering, it suffices to check after pullback to $X \setminus Z_{i-1}$, where the claim is true by induction, and to $U|_{X \setminus Z_i} \setminus V$ which admits a section, so the claim is also clear. $\square$

Proof of theorem 3.2.4. By what we did already it suffices to proof that if $F \in \mathcal{P}(\dot{\text{Et}}_S^{\text{fp}})$ satisfies condition 1) and 2), then $F \in \text{Shv}_{\tau_{\text{dist}}}(\dot{\text{Et}}_S^{\text{qcs}})$.
Let Q be the distinguished square

$$\begin{array}{ccc} W & \longrightarrow & V \\ \downarrow & & \downarrow \\ U & \longrightarrow & X. \end{array}$$

Using the yoneda embedding, we can consider the pushout

$$\begin{array}{ccc} W & \longrightarrow & V \\ \downarrow & & \downarrow \\ U & \longrightarrow & K \end{array}$$

$\mathcal{P}(\dot{\text{Et}}_S^{\text{qcs}})$. We further set $\check{C} := \check{C}(U \amalg V \rightarrow X)$.
Consider now the pullback

$$\begin{array}{ccc} \check{C} \times_X K & \xrightarrow{c)} & K \\ \downarrow d) & & \downarrow b) \\ \check{C} & \xrightarrow{a)} & X \end{array}$$

The statement is equivalent to showing that the arrow $a)$ becomes an equivalence after applying F . We claim that this is true for the other three.

- b) This is an equivalence by condition 1) of the assumption on F .
- c) By universality of colimits, this arrow is a pushout of

$$\begin{array}{ccc} \check{C} \times_X V & \longrightarrow & V \\ \uparrow & & \uparrow \\ \check{C} \times_X W & \longrightarrow & W \\ \downarrow & & \downarrow \\ \check{C} \times_X U & \longrightarrow & U \end{array}$$

so it suffices to proof that F inverts these maps individually. But this is true as $U \amalg V \rightarrow X$ acquires a section after being pulled back to U, V or W .

- d) We write $Q_{ij} := Q \times_X U^{\times i} \times_X V^{\times j}$ and $K_{Q_{ij}} := K \times_X Q_{ij}$. Then the map $d)$ is a colimit of maps of the form $K_{Q_{ij}} \rightarrow U^{\text{times } i} \times V^{\times j}$, all of which are inverted by F due to our assumptions.

$\square$

As an application we want to show that the pullback of a Nisnevich sheaf is again a Nisnevich sheaf, as one is used to from classical algebraic geometry.

Proposition 3.4.5. *Let $f : X \rightarrow Y \in \text{Sch}^{\text{qcqs}}$ such that X is a cofiltered limit of schemes in $\text{Ét}_Y^{\text{qcqs}}$ with affine transition maps. Let $F \in \text{Shv}_{\text{Nis}}(\text{Ét}_Y^{\text{qcqs}})$. Then $f^*F \in \mathcal{P}(\text{Ét}_X^{\text{qcqs}})$ is again a Nisnevich sheaf.*

Proof. We verify the conditions 1) and 2) of theorem 3.2.4.

First of all we have

$$(f^*F)(\emptyset) \simeq \text{colim}_{\{I\}^{\text{op}}} (F(T)) \simeq F(\emptyset) \simeq *,$$

where the second equivalence is due to the fact that the indexing category

$$I^{\text{op}} = \left\{ \begin{array}{ccc} \emptyset & \longrightarrow & T \\ & & \downarrow \text{ét} \\ & & Y \end{array} \right\}^{\text{op}}$$

has a terminal object (note the op!)

Next let

$$\begin{array}{ccc} W & \longrightarrow & V \\ \downarrow & & \downarrow \\ U & \longrightarrow & T \end{array}$$

be a distinguished square in $\text{Ét}_X^{\text{qcqs}}$. Write $X = \lim iX_i$. We now use the fact that if $S \rightarrow X$ is finitely presented then there exist some index i_0 and an étale scheme $S_0 \rightarrow X_{i_0}$ such that $S_0 \times_{X_{i_0}} X \cong S$.

It follows that there exists a distinguished square

$$\begin{array}{ccc} W_o & \longrightarrow & V_o \\ \downarrow & & \downarrow \\ U_o & \longrightarrow & T_o \end{array}$$

in $\text{Ét}_{X_{i_0}}^{\text{qcqs}}$ such that pulling this square back to X recovers the original square. Now we calculate

$$\begin{aligned} (f^*F)(T) &\simeq \text{colim}_{i > i_0} \left(F(T_0 \times_{X_{i_0}} X_i) \right) \\ &\simeq \text{colim}_{i > i_0} \left(F(U_i) \times_{F(W_i)} F(V_i) \right) \\ &\simeq (\text{colim}_{i > i_0} (F(U_i))) \times_{(\text{colim}_{i > i_0} (F(W_i)))} (\text{colim}_{i > i_0} (F(V_i))) \\ &\simeq (f^*F)(U) \times_{(f^*F)(W)} (f^*F)(V). \end{aligned}$$

Here we used that we know that the pullback of Nisnevich sheaves along étale maps are Nisnevich sheaves and universality of colimits. $\square$

3.5 Homotopy dimension

In this section we discuss the homotopy dimension of the Nisnevich ∞ -topos.

Theorem 3.5.1. *Let $X \in \text{Sch}^{\text{qcqs}}$ be of dimension $\dim(X) \leq d$. Then $\text{Shv}_{\text{Nis}}(\text{Ét}_X^{\text{fp}})$ has homotopy dimension $\leq d$.*

Remark 3.5.2. If $Y \in \text{Ét}_X^{\text{fp}}$ there is an equivalence $\text{Shv}_{\text{Nis}}(\text{Ét}_Y^{\text{fp}}) \simeq \text{Shv}_{\text{Nis}}(\text{Ét}_X^{\text{fp}})/Y$ and $\dim(Y) \leq \dim(X)$. It follows that $\text{Shv}_{\text{Nis}}(\text{Ét}_X^{\text{fp}})$ is of locally finite homotopy dimension $\leq d$ and so is hypercomplete.

Proof. We only proof the theorem for noetherian schemes, the general case can be found in [CM21, § 3.3].

If $\dim(X) = 0$ then $X \cong \coprod_{i=1}^n X_i$, with X_i connected, as noetherian schemes have only finitely many minimal points. It follows $\mathrm{Shv}_{\mathrm{Nis}}(\acute{\mathrm{E}}\mathrm{t}_X^{\mathrm{fp}}) \simeq \prod_i \mathrm{Shv}_{\mathrm{Nis}}(\acute{\mathrm{E}}\mathrm{t}_{X_i}^{\mathrm{fp}})$, so we may assume that X is connected. In particular, X is henselian local and hence n -connective, so F is n -conneective⁶.

For the general case, let $F \in \mathrm{Shv}_{\mathrm{Nis}}(\acute{\mathrm{E}}\mathrm{t}_X^{\mathrm{fp}})$ and let $1 \leq \dim(X) \leq d$. We want to show that $F(X) \neq \emptyset$. For this let $\emptyset \neq Z \hookrightarrow X$ be a closed immersion such that $F(X \setminus Z) \neq \emptyset$, e.g. $Z = X$. We claim that there exists a closed subscheme $Z' \hookrightarrow Z$ with $Z' \neq Z$ such that $F(X \setminus Z') \neq \emptyset$. The claim follows then from noetherian induction.

Let $\eta \in Z$ be a minimal point. Let $\{U_\alpha\}$ be a system of étale neighborhoods of $\eta \in X$, that is maps $U_\alpha \rightarrow X$ such that $U_\alpha \times_X \{\eta\} \xrightarrow{\sim} \{\eta\}$. Note that $\lim_\alpha U_\alpha \simeq X_\eta^h$, so $\lim_\alpha U_\alpha \setminus \bar{\eta} \simeq X_\eta^h \setminus \eta$, which has dimension $< d$. Thus

$$\mathrm{colim}_\alpha (\pi_0(F(U_\alpha))) \simeq \pi_0(F(\lim_\alpha U_\alpha)) = * = \pi_0(F(\lim_\alpha U_\alpha \setminus \bar{\eta})) = \mathrm{colim}_\alpha (\pi_0(F(U_\alpha \setminus \bar{\eta}))).$$

Pick now $s \in F(X \setminus Z)$. By the above calculation there exists some α and some $t \in F(U_\alpha)$ such that

$$s|_{U_\alpha \setminus \bar{\eta}} \sim t|_{U_\alpha \setminus \bar{\eta}}.$$

Consider the pullback

$$\begin{array}{ccc} ? & \xrightarrow{f} & Z \\ \downarrow & & \downarrow \\ U_\alpha & \longrightarrow & X \end{array}$$

As $U_\alpha \rightarrow X$ hits η , f is an isomorphism over $Z \setminus Z'$ for some proper subscheme $Z' \hookrightarrow Z$ containing η . Set now $U := U_\alpha \times_X (X \setminus Z')$. We obtain a distinguished square

$$\begin{array}{ccc} W & \longrightarrow & U \\ \downarrow & & \downarrow \\ X \setminus Z & \longrightarrow & X \setminus Z' \end{array}$$

such that $W \subseteq U_\alpha \setminus \bar{\eta}$. It follows that $s|_W \sim t|_W$. But this implies $F(X \setminus Z') \neq \emptyset$. $\square$

3.6 Summary

Let us summarize the most important facts about the Nisnevich ∞ -topos that we have (more or less) proven.

Let X be a noetherian scheme of finite dimension.

- There is an equivalence $X_{\mathrm{Nis}} := \mathrm{Shv}_{\mathrm{Nis}}(\acute{\mathrm{E}}\mathrm{t}_X) \xrightarrow{\sim} \mathrm{Shv}_{\mathrm{Nis}}(\acute{\mathrm{E}}\mathrm{t}_X^{\mathrm{fp}})$.
- A presheaf $F \in \mathcal{P}(\acute{\mathrm{E}}\mathrm{t}_X)$ is a Nisnevich sheaf if and only if $F(\emptyset) = *$ and F turns distinguished squares into pullback squares.
- A morphism $\alpha \dots F \rightarrow G \in \mathcal{P}(\acute{\mathrm{E}}\mathrm{t}_X^{\mathrm{fp}})$ becomes an equivalence after Nisnevich localization of and only if for all $y \in Y \in \acute{\mathrm{E}}\mathrm{t}_X^{\mathrm{fp}}$

$$\alpha(Y_y^h) : F(Y_y^h) \xrightarrow{\sim} G(Y_y^h).$$

⁶ $(\pi_i F)(X) = \pi_i(F(X))$.

- X_{Nis} has finite homotopy dimension, hence is hypercomplete, hence is Postnikov complete.

This finishes the abstract technical part of this lecture. We will now proceed to study motivic spaces, the bread and butter of motivic homotopy theory, in more detail.

Chapter 4

Motivic Spaces

The main object of interest in motivic homotopy theory is given by the following

Definition 4.0.1. Let S be a scheme. Denote by $W_{\text{mot}} \subseteq \text{Ar}(\mathcal{P}(\text{Sm}_S^{\text{fp}}))$ the strongly saturated class generated by

- 1) The Nisnevich sieves,
- 2) the collection of all projections $\{X \times \mathbb{A}^1 \rightarrow X\}_{X \in \text{Sm}_S^{\text{fp}}}$

We call the associated localization functor L_{mot} and $\text{Spc}(S)$ the associated Bousfield localization. We call the objects of $\text{Spc}(S)$ *motivic spaces*.

To ensure the existence of the localization functor and the Bousfield localization, we needed the collection W_{mot} to be of small generation¹. That is the reason why we only considered finitely presented smooth S -schemes. However, by the discussions of the previous chapter, as we are considering Nisnevich sieves, we could also have omitted the finitely presented condition.

As, due to set theoretic reasons, it is easier to work with Sm_S^{fp} , we will from now on always mean this category, even if we are writing Sm_S . As a further convention we will call the localization functor for the saturated class generated by 1) L_{Nis} and the one for the class generated by 2) $L_{\mathbb{A}^1}$. Note, however, that in the older literature L_{mot} is often called $L_{\mathbb{A}^1}$. The goal of this chapter is to get a better understanding of motivic spaces by understanding the motivic localization functor in more detail. We will do so by first studying the Nisnevich localization and the $\mathbb{A}^1$ -localization.

4.1 Nisnevich localization

Lemma 4.1.1. Let $X \in \text{Sm}_S$ be a smooth S -scheme and $F \in \mathcal{P}(\text{Sm}_S)$. Let $\mathcal{P}(\text{Sm}_S) \rightarrow \mathcal{P}(\text{Ét}_X^{\text{fp}})$ denote the restriction functor $F \mapsto F|_{X_{\text{Nis}}}$. Then

$$(L_{\text{Nis}} F)|_{X_{\text{Nis}}} \simeq L_{\text{Nis}}(F|_{X_{\text{Nis}}}).$$

Proof. It suffices to prove that the restriction functor preserves colimits, Nisnevich-local objects and generating Nisnevich equivalences. As colimits are computed section-wise, the first point is obvious.

Now let $U \rightarrow V \in \text{Ét}_X$ be a Nisnevich covering. Then $G \in \mathcal{P}(\text{Ét}_X^{\text{fp}})$ is a Nisnevich sheaf if and only if it satisfies the sheaf condition $G(V) \simeq \text{Tot}(G(U^{\times_V n}))$. But this is exactly the same condition that a presheaf $F \in \mathcal{P}(\text{Sm}_S)$ has to satisfy to be a Nisnevich sheaf.

¹See [Lur25, 04MD].

Lastly let $\alpha : C \hookrightarrow Y \in \mathbf{Sm}_S$ be a Nisnevich local covering. But then $\alpha|_{X_{\text{Nis}}}$ is still a monomorphism so it suffices to prove that $\alpha|_{X_{\text{Nis}}}$ is Nisnevich locally split. To show this, let $W_\bullet \rightarrow Y$ be a Nisnevich covering generating C and consider $T \xrightarrow{\beta} Y$ for some $T \in \mathcal{E}t_X^{\text{fp}}$. Consider the following Diagram

$$\begin{array}{ccccc} T & \xrightarrow{\beta} & Y & \xleftarrow{\alpha} & C \\ \uparrow & \lrcorner & \uparrow & \nearrow & \\ W'_\bullet & \longrightarrow & W_\bullet & & \end{array}$$

Here, the dotted arrow is due to $W_\bullet \rightarrow Y$ generating the sieve $\alpha : C \hookrightarrow Y$. In particular β lives in C and $W'_\bullet \rightarrow T$ is a Nisnevich covering, as needed. $\square$

The next proposition describes the maps that are inverted by the Nisnevich localization functor.

Proposition 4.1.2. *Let S be a qcqs scheme. Then $W_{\text{Nis}} \subseteq \text{Ar}(\mathcal{P}(\mathbf{Sm}_S))$ is generated by Nisnevich squares and the unique map $\emptyset \rightarrow y_\emptyset$.*

Proof. Write $\tilde{W}_{\text{Nis}}$ for the class generated by Nisnevich squares and the map $\emptyset \rightarrow y_\emptyset$. Then we have $\tilde{W}_{\text{Nis}} \subseteq W_{\text{Nis}}$ due to the lemma above. To prove the other inclusion, we will show that every $\tilde{W}_{\text{Nis}}$ -local object is already W_{Nis} -local.

So assume that F is local for $\tilde{W}_{\text{Nis}}$. Then $F|_{X_{\text{Nis}}}$ is a Nisnevich sheaf for all $X \in \mathbf{Sm}_S$. As the restriction is right adjoint to the inclusion, so we get that in particular, F is local for all morphisms in W_{Nis} . $\square$

Corollary 4.1.3. *Let S be a qcqs scheme. Then the fully faithful inclusion $\text{Shv}_{\text{Nis}}(\mathbf{Sm}_S) \hookrightarrow \mathcal{P}(\mathbf{Sm}_S)$ preserves filtered colimits. Equivalently, for every $X \in \mathbf{Sm}_S$ the corresponding representable functor $y_X \in \text{Shv}_{\text{Nis}}(\mathbf{Sm}_S)$ is compact.*

Proof. The sheaf condition is expressed in terms of finite limits in anima, to be precise, to turn Nisnevich squares into pullback squares in anima. But in spaces, finite limits commute with filtered colimits, giving the statement. $\square$

4.2 $\mathbb{A}^1$ -localization

Next we want to study the $\mathbb{A}^1$ -localization. For this we imitate what is one does in algebraic topology.

Definition 4.2.1. For $X \in \Delta$, set $\mathbb{Z}(X) := \mathbb{Z}[\{x \in X\}] / (\sum_{x \in X} x = 1)$. For $\alpha : X \rightarrow Y \in \Delta$, we define $\mathbb{Z}(\alpha) : \mathbb{Z}(Y) \rightarrow \mathbb{Z}(X)$, $y \mapsto \sum_{\alpha(x)=y} x$. This defines a functor $\mathbb{Z} : \Delta^{\text{op}} \rightarrow \text{Ring}$. We now define a cosimplicial scheme, the *standard cosimplicial affine scheme*, as the composition $\mathbb{A}_{\mathbb{Z}} = \mathbb{A} : \Delta \xrightarrow{\mathbb{Z}(-)^{\text{op}}} \text{Ring}^{\text{op}} \xrightarrow{\text{Spec}} \mathbf{Sm}$, explicitly

$$\begin{array}{ccccc} \mathbb{A}^{[0]} & \xrightarrow{\quad} & \mathbb{A}^{[1]} & \xleftarrow{\quad} & \mathbb{A}^{[2]} & \dots \\ & \xleftarrow{\quad} & & \xrightarrow{\quad} & & \\ & \xrightarrow{\quad} & & \xleftarrow{\quad} & & \end{array}$$

Notice that $\mathbb{A}^{[0]} = \text{Spec}(\mathbb{Z}[X_0]/(X_0-1)) \cong \mathbb{A}^0$, and $\mathbb{A}^{[1]} = \text{Spec}(\mathbb{Z}[X_0, X_1]/(X_0+X_1-1)) \cong \mathbb{A}^1$. More generally, $\mathbb{A}^{[n]} \simeq \mathbb{A}^n$.

Now let X be a scheme and consider the composition

$$\begin{aligned} (-) \times \mathbb{A}^{[-]} : \mathbf{Sm}_S \times \Delta &\xrightarrow{\text{id} \times \mathbb{A}^{[-]}} \mathbf{Sm}_S \times \mathbf{Sm}_S \xrightarrow{\times} \mathbf{Sm}_S \\ (X, [n]) &\mapsto X \times \mathbb{A}^{[n]}. \end{aligned}$$

We define the *simplicial Sing-functor* as the composition

$$\mathrm{Sing}_\bullet : \mathcal{P}(\mathrm{Sm}_S) \xrightarrow{((-) \times \mathbb{A}^{[-]})^*} \mathcal{P}(\mathrm{Sm}_S \times \Delta) \simeq \mathrm{Fun}(\Delta^{\mathrm{op}}, \mathcal{P}(\mathrm{Sm}_S)),$$

where the equivalence is given by the Fun-product adjunction.

Lastly, we define the *singular simplicial presheave* as the composition as the geometric realization of the simplicial Sing-functor, that is

$$\mathrm{Sing} : \mathcal{P}(\mathrm{Sm}_S) \xrightarrow{(-) \times \mathbb{A}^{[-]}} \mathcal{P}(\mathrm{Sm}_S \times \Delta) \simeq \mathrm{Fun}(\Delta^{\mathrm{op}}, \mathcal{P}(\mathrm{Sm}_S)) \xrightarrow{\mathrm{colim}} \mathcal{P}(\mathrm{Sm}_S).$$

Explicitly, on objects, we have $(\mathrm{Sing}(F))(X) \simeq \left| F\left(X \times \mathbb{A}^{[-]}\right) \right|$, where $|-| = \mathrm{colim}_{\Delta^{\mathrm{op}}} (-)$ denotes the geometric realization.

We now have the abstract notions of the $\mathbb{A}^1$ -localization and the Sing functor. To relate these two and to understand, why they can be seen as a algebraic geometry version of the concept of homotopy equivalence, we introduce two different versions of homotopies, simplicial homotopies, and naïve $\mathbb{A}^1$ -homotopies. We start with the former.

Remark 4.2.2. Let $\mathcal{C}$ be a ∞ -category with finite disjoint unions. Then $\mathcal{C}$ is tensored over the category Fin of finite sets. In particular there is a functor

$$\begin{aligned} \mathrm{Fin} \times \mathcal{C} &\rightarrow \mathcal{C} \\ (X, C) &\mapsto \amalg_X C. \end{aligned}$$

To define this explicitly, this is more delicate than it may seem. This can be upgraded to a functor

$$\mathrm{Fun}(\Delta^{\mathrm{op}}, \mathrm{Fin}) \times \mathrm{Fun}(\Delta^{\mathrm{op}}, \mathcal{C}) \rightarrow \mathrm{Fun}(\Delta^{\mathrm{op}}, \mathcal{C})$$

by applying the construction above section wise.

In particular, for $[n] \in \Delta$, $S_\bullet \in \mathrm{Fun}(\Delta^{\mathrm{op}}, \mathcal{C})$ the expression $\Delta^n \times S_\bullet \in \mathrm{Fun}(\Delta^{\mathrm{op}}, \mathcal{C})$ does make sense.

Definition 4.2.3. A *simplicial homotopy* between two morphisms $f_0, f_1 : F \rightarrow G \in \mathrm{Fun}(\Delta^{\mathrm{op}}, \mathcal{C})$ is a morphism $H : [1] \times F \rightarrow G$ with $H|_i \simeq f_i$ for $i = 0, 1$.

The following lemma will be the first step in understanding the singular scheme functor.

Lemma 4.2.4. *Let $\mathcal{C}$ be an ∞ -category with finite products and finite disjoint unions. Suppose that products in $\mathcal{C}$ preserve colimits in each variable. Then the geometric realization $|-| : \mathrm{Fun}(\Delta^{\mathrm{op}}, \mathcal{C}) \rightarrow \mathcal{C}$ preserves finite products.*

Proof. First of all note that for $c \in \mathcal{C}$, $n \in \mathbb{N}$, and $F \in \mathrm{Fun}(\Delta^{\mathrm{op}}, \mathcal{C})$ we have an equivalence due to Yoneda

$$\mathrm{Map}_{\mathcal{C}_\Delta}(\Delta^n \times c, F) \simeq \mathrm{Map}_{\mathcal{C}_\Delta}(c, F_n).$$

Thus $\mathrm{Fun}(\Delta^{\mathrm{op}}, \mathcal{C})$ is generated under colimits by objects of the form $\Delta^n \times c := \Delta^n \times y_c$. Note further that

$$\mathrm{Map}_{\mathcal{C}}(|\Delta^n \times c|, d) \simeq \mathrm{Map}_{\mathcal{C}_\Delta}(\Delta^n \times c, d) \simeq \mathrm{Map}_{\mathcal{C}}(c, d).$$

Hence, by Yoneda, $|\Delta^n \times c| \xrightarrow{\simeq} |c| \simeq c$.

Fix $c \in \mathcal{C}$ and consider the functor

$$\begin{aligned} \mathrm{Ani}_\Delta &\rightarrow \mathcal{C} \\ X &\mapsto |X \times c| \end{aligned}$$

given by the colimit preserving extension of $\text{Ani} \rightarrow \mathcal{C}, X \mapsto \text{colim}_X^*$. By the above discussion, this functor inverts all morphisms of the form $\Delta^n \rightarrow \Delta^0 \simeq *$. In particular, the functor factors through the Bousfield localization of Ani_Δ on the class of morphisms W generated by the morphisms $\Delta^n \rightarrow \Delta^0$. But $\text{Ani}_\Delta[W^{-1}] \simeq \text{Ani}$, so that we can conclude $|X \times c| \simeq c$ whenever $|X| \simeq *$ is a contractible simplicial anima. In particular

$$|(\Delta^n \times c) \times (\Delta^m \times d)| \simeq |(\Delta^n \times \Delta^m) \times (c \times d)| \simeq |c \times d| \simeq c \times d.$$

Here we used that $|\Delta^n \times \Delta^m| \simeq *$.

Thus, the natural comparison morphism $(\eta : | - | \circ \times \Rightarrow \times \circ | - |) : \mathcal{C}_\Delta \times \mathcal{C}_\Delta \rightarrow \mathcal{C}$ is an equivalence on the image of the Yoneda embedding. As these generate $\mathcal{C}_\Delta$ under colimits, we conclude that η is point wise an equivalence. But being a natural equivalence can be checked point wise, so we are done. $\square$

Corollary 4.2.5. *Assume that $f, g : F \rightarrow G \in \mathcal{C}_\Delta$ a simplicial homotopic with simplicial homotopy given by $H : \Delta^1 \times F \rightarrow G$. Then $|f| \simeq |g|$.*

Proof. That f and g are simplicial homotopic can be described by commutativity of the following diagram

$$\begin{array}{ccc} \Delta^0 \times F & \xrightarrow{f} & G \\ \searrow 0 & & \uparrow H \\ & \Delta^1 \times F & \\ \nearrow 1 & & \\ \Delta^0 \times F & \xrightarrow{g} & G \end{array}$$

Applying the geometric realization and using Lemma 4.2.4, we obtain the following commutative diagram

$$\begin{array}{ccc} |F| & \xrightarrow{|f|} & |G| \\ \searrow \text{id} & & \uparrow |H| \\ & |F| & \\ \nearrow \text{id} & & \\ |F| & \xrightarrow{|g|} & |G| \end{array}$$

which was exactly the claim. $\square$

We now turn $\mathbb{A}^1$ -homotopies.

Definition 4.2.6. Let $f, g : X \rightarrow Y \in \mathcal{P}(\text{Sm}_S)$ be two morphisms. We call f and g *naïvely $\mathbb{A}^1$ -homotopic*, if there exists a morphism $H : \mathbb{A}^1 \times X \rightarrow Y$, such that $H|_0 = f$ and $H|_1 = g$. Here $(-)_i : S \rightarrow \mathbb{A}^1 \in \mathcal{P}(\text{Sm}_S)$ means the inclusion of S at i .

Lemma 4.2.7. *The $\mathbb{A}^1$ -localization functor preserves finite products.*

Proof. We have to show that if $f, g \in W_{\mathbb{A}^1}$, then we also have $f \times g \in W_{\mathbb{A}^1}$. Indeed, if we denote by $l_X : X \rightarrow L_{\mathbb{A}^1}X$ and by $l_Y : Y \rightarrow L_{\mathbb{A}^1}Y$ the respective localization morphisms, that is the components of the unit, and if the claim holds, then $l_X \times l_Y : X \times Y \rightarrow L_{\mathbb{A}^1}X \times L_{\mathbb{A}^1}Y$ is an $\mathbb{A}^1$ -equivalence and the right hand side is $\mathbb{A}^1$ -local, hence $L_{\mathbb{A}^1}(X \times Y) \simeq L_{\mathbb{A}^1}X \times L_{\mathbb{A}^1}Y$.

Fix $X \in \text{Sm}_S$. The class $\mathcal{C}_1 := \{h \in \mathcal{P}(\text{Sm}_S)^{[1]} | h \times (X \times \mathbb{A}^1 \rightarrow X) \in W_{\mathbb{A}^1}\}$ is strongly saturated and contains $Y \times \mathbb{A}^1 \rightarrow Y$ for all smooth S -schemes Y . The second claim follows

Did I already use this notation?

from the following factorization

$$\begin{array}{ccc} (Y \times \mathbb{A}^1) \times (X \times \mathbb{A}^1) & \longrightarrow & Y \times X \\ \cong \downarrow & & \uparrow \\ ((Y \times X) \times \mathbb{A}^1) \times \mathbb{A}^1 & \longrightarrow & (Y \times X) \times \mathbb{A}^1 \end{array}$$

where all morphisms of the lower path are obviously in $W_{\mathbb{A}^1}$. It follows that $W_{\mathbb{A}^1} \subseteq \mathcal{C}_1$.

Consider now $C_2 := \{h \in \mathcal{P}(\mathrm{Sm}_S)^{[1]} \mid h \times f \in W_{\mathbb{A}^1} \ \forall f \in W_{\mathbb{A}^1}\}$. This is again strongly saturated and contains $(X \times \mathbb{A}^1 \rightarrow X)$ by the discussion of C_1 . In particular $W_{\mathbb{A}^1} \subseteq C_2$ finishing the claim. $\square$

Corollary 4.2.8. *If $f, g : X \rightarrow Y \in \mathcal{P}(\mathrm{Sm}_S)$ are naively $\mathbb{A}^1$ -homotopic, then $L_{\mathbb{A}^1}f \simeq L_{\mathbb{A}^1}g$.*

We now relate the Sing-functor and the $\mathbb{A}^1$ -localization.

Lemma 4.2.9. *If $f, g : X \rightarrow Y \in \mathcal{P}(\mathrm{Sm}_S)$ are naively $\mathbb{A}^1$ -homotopic, then $\mathrm{Sing}_\bullet(f)$ and $\mathrm{Sing}_\bullet(g)$ are simplicially homotopic. In particular, $\mathrm{Sing}(f) \simeq \mathrm{Sing}(g)$.*

Proof. By construction the $\mathrm{Sing}_\bullet$ -functor preserves products. Hence it suffices to construct the dashed line in

$$\begin{array}{ccccc} \Delta^0 & & \Delta^0 & \xrightarrow{\quad} & \mathrm{Sing}_\bullet(*) \\ \downarrow \iota_0 & & \downarrow \iota_1 & & \downarrow \iota_0 \\ \Delta^1 & \xrightarrow{\quad} & \mathrm{Sing}_\bullet(\mathbb{A}^1) & & \end{array}$$

(Note: The diagram shows curved arrows from the top Δ^0 to the top $\mathrm{Sing}_\bullet(*)$ and from the top Δ^0 to the bottom $\mathrm{Sing}_\bullet(\mathbb{A}^1)$, and a dashed arrow from Δ^1 to $\mathrm{Sing}_\bullet(\mathbb{A}^1)$.)

rendering the diagram commutative in $\mathcal{P}(\mathrm{Sm}_S)_\Delta$. Here, the two curved arrows are the canonical equivalences. Indeed, if we have a commutative diagram as above, we obtain a commutative diagram

$$\begin{array}{ccccccc} X \times \Delta^0 & \xrightarrow{\simeq} & X \times \mathrm{Sing}_\bullet(*) & \xrightarrow{\simeq} & \mathrm{Sing}_\bullet(X) & & \\ \downarrow \iota_0 & & \downarrow \iota_0 & & \downarrow \iota_0 & \searrow \mathrm{Sing}_\bullet(f) & \\ X \times \Delta^1 & \longrightarrow & X \times \mathrm{Sing}_\bullet(\mathbb{A}^1) & \xrightarrow{\simeq} & \mathrm{Sing}_\bullet(X \times \mathbb{A}^1) & \xrightarrow{\mathrm{Sing}_\bullet(H)} & T \\ \uparrow \iota_1 & & \uparrow \iota_1 & & \uparrow \iota_1 & \nearrow \mathrm{Sing}_\bullet(g) & \\ X \times \Delta^0 & \xrightarrow{\simeq} & X \times \mathrm{Sing}_\bullet(*) & \xrightarrow{\simeq} & \mathrm{Sing}_\bullet(X) & & \end{array}$$

implying the claim. To see the existence of the dashed morphism above, notice that

$$\begin{aligned} \mathrm{Map}_{\mathcal{P}(\mathrm{Sm}_S)_\Delta}(\Delta^1, \mathrm{Sing}_\bullet(\mathbb{A}^1)) &\simeq \mathrm{Map}_{\mathcal{P}(\mathrm{Sm}_S)}(*, \mathrm{Sing}_1(\mathbb{A}^1)) \\ &\simeq \mathrm{Map}_{\mathcal{P}(\mathrm{Sm}_S)}(\mathbb{A}^{[1]}, \mathbb{A}^1) \\ &\simeq \mathrm{Hom}_{\mathrm{Sm}_S}(\mathbb{A}^1, \mathbb{A}^1). \end{aligned}$$

Here the first equivalence is given by the Yoneda lemma, the second is the explicit definition of $\mathrm{Sing}_1(\mathbb{A}^1)$, and in the last equivalence that the Yoneda embedding is fully faithful, and that $\mathbb{A}^{[1]} \cong \mathbb{A}^1$, see Definition 4.2.1.

The right hand side now contains $[\mathrm{id}] \in \mathrm{id} : \mathbb{A}^1 \rightarrow \mathbb{A}^1$. It is easily verified that the corresponding morphism has the desired properties. Indeed, under the string of equivalences, the morphism in $\mathrm{Map}_{\mathcal{P}(\mathrm{Sm}_S)_\Delta}(\Delta^1, \mathrm{Sing}_\bullet(\mathbb{A}^1))$ corresponding to the identity of $\mathbb{A}^1$,

is mapped in $\text{Hom}_{\text{Sm}_S}(\mathbb{A}^{[1]}, \mathbb{A}^1)$ to the morphism induced by the ring homomorphism² $\mathbb{Z}[X] \rightarrow \mathbb{Z}[X_0, X_1]/(X_0 + X_1 - 1)$ sending X to X_0 . Furthermore, the morphism

$$\iota_0^* : \text{Map}_{\mathcal{P}(\text{Sm}_S)_\Delta}(\Delta^1, \text{Sing}_\bullet(\mathbb{A}^1)) \rightarrow \text{Map}_{\mathcal{P}(\text{Sm}_S)_\Delta}(\Delta^0, \text{Sing}_\bullet(\mathbb{A}^1))$$

is sent under the Yoneda lemma to the morphism $\mathbb{A}^{[1]} \rightarrow \mathbb{A}^{[0]}$ corresponding to the morphism of rings $\mathbb{Z}[X_0, X_1]/(X_0 + X_1 - 1) \rightarrow \mathbb{Z}[X]/(X - 1)$ sending $X_0 \mapsto 0$ and $X_1 \mapsto X$. Hence, $\iota_0^*[\text{id}]$ corresponds to the morphism of rings $\mathbb{Z}[X] \rightarrow \mathbb{Z} \cong \mathbb{Z}[X]/(X - 1)$ sending X to 0. The discussion for ι_1 is similar.

The in particular part now follows from Corollary 4.2.5. $\square$

Proposition 4.2.10. *There is an equivalence of endofunctors*

$$\text{Sing} \simeq L_{\mathbb{A}^1} : \mathcal{P}(\text{Sm}_S) \rightarrow \mathcal{P}(\text{Sm}_S).$$

Proof. By Lemma 4.2.9 the Sing-functor inverts naïve $\mathbb{A}^1$ -homotopies. Furthermore, by universality of colimits, the Sing-functor further preserves colimits. It follows that the Sing-functor inverts all of $W_{\mathbb{A}^1}$, the strong saturated class of morphisms generated by the projections $X \times \mathbb{A}^1 \rightarrow X$, also known as $\mathbb{A}^1$ -equivalences.

Thus, applying, Sing to the unit $\text{id} \Rightarrow L_{\mathbb{A}^1}$ induces an equivalence $\text{Sing} \simeq \text{Sing}(L_{\mathbb{A}^1})$. However, if $X \in \mathcal{P}(\text{Sm}_S)$ is $\mathbb{A}^1$ invariant, then $\text{Sing}_\bullet(X) \simeq X$ is the constant simplicial diagram. Thus $\text{Sing}(X) \simeq X$ for all $\mathbb{A}^1$ -invariant presheaves, hence $\text{Sing}L_{\mathbb{A}^1} \simeq L_{\mathbb{A}^1}$, finishing the claim. $\square$

4.3 Motivic localization

Having discussed Nisnevich and $\mathbb{A}^1$ localizations, we are now going to use the information about both of them to deduce properties of the motivic localization.

Lemma 4.3.1. *There is a natural equivalence*

$$L_{\text{mot}} \simeq \text{colim}_{n \leq \kappa} ((L_{\mathbb{A}^1} \circ L_{\text{Nis}})^{\circ n}) : \mathcal{P}(\text{Sm}_S) \rightarrow \text{Spc}(S)$$

of the motivic localization with the κ -transfinite composition of $L_{\mathbb{A}^1}$ and L_{Nis} where κ is some regular cardinal.

Proof. Since the inclusion $\text{Shv}_{\text{Nis}}(\text{Sm}_S) \hookrightarrow \mathcal{P}(\text{Sm}_S)$ is accessible, it preserves κ -filtered colimits for some regular cardinal κ . Let F be the κ -transfinite composition of L_{mot} and L_{Nis} . Then, using cofinality of the even/odd subdiagrams, we have that FP is an $\mathbb{A}^1$ -invariant Nisnevich sheaf for any $P \in \mathcal{P}(\text{Sm}_S)$. Indeed, as the inclusion of Nisnevich sheaves is κ -accessible, using the subdiagram where the last localization is the Nisnevich localization, we see that the colimit in presheaves is already the colimit in Nisnevich sheaves, so FP is Nisnevich local. On the other hand, $\mathbb{A}^1$ -local presheaves are closed under all filtered colimits, using Proposition 4.2.10 and that products and sifted colimits commute with filtered colimits. Thus FP is also $\mathbb{A}^1$ -local. $\square$

Remark 4.3.2. In fact, if S is qcqs, $\kappa = \aleph_0$ already suffices. This follows from $\text{Spc}(S)$ being closed under filtered colimits for S qcqs, see Proposition 4.3.4, so that in the above proof we can replace κ with $\aleph_0$.

Corollary 4.3.3. *The motivic localization functor preserves finite products.*

²Be wary of the variance!

Proof. This follows from Lemma 4.3.1, using Lemma 4.2.7, the fact that the Nisnevich site is a topos, so that in particular L_{Nis} is a left exact-left adjoint, and the fact, that κ -filtered limits commute with finite products inside presheaves. $\square$

Proposition 4.3.4. *Let S be a qcqs scheme. The inclusion $\text{Spc}(S) \rightarrow \mathcal{P}(\text{Sm}_S)$ is closed under filtered colimits. Equivalently, for $X \in \text{Sm}_S$, the object $L_{\text{mot}}X$ is compact in $\text{Spc}(S)$.*

Proof. We can write $\text{Spc}(S) = L_{\mathbb{A}^1}(\mathcal{P}(\text{Sm}_S)) \cap L_{\text{Nis}}(\mathcal{P}(\text{Sm}_S))$, where the intersection is taken as subcategories of $\mathcal{P}(\text{Sm}_S)$. Now the claim is true for the first factor by Proposition 4.2.10, while it is true for the Nisnevich localization, as the Nisnevich topology is generated by distinguished Nisnevich squares, which are in particular finite covers. Hence, the sheaf-condition for Nisnevich square is a finite limit condition, which in particular commutes with filtered colimits. $\square$

Remark 4.3.5. Notice that $\text{Spc}(S)$ is *not* an ∞ -topos. This is due to the fact, that $L_{\mathbb{A}^1}$ is not left exact. Indeed, assume $L_{\mathbb{A}^1}$ would be left exact. Then it would be localization with respect to the Grothendieck topology containing coverings of the form $X \hookrightarrow \mathbb{A}_X^1$. Consider now the pullback

$$\begin{array}{ccc} \emptyset & \longrightarrow & X \\ \downarrow & \lrcorner & \downarrow 1 \\ X & \hookrightarrow_0 & \mathbb{A}^1 \end{array}$$

As coverings in Grothendieck topologies are stable under pullback, the morphism $\emptyset \rightarrow X$ would be a covering, so that $\emptyset \simeq L_{\mathbb{A}^1}\emptyset \simeq L_{\mathbb{A}^1}X$. Thus the only $\mathbb{A}^1$ -invariant presheaf would be the empty presheaf. However, if $S = \text{Spec}k$ is the spectrum of a reduced ring, e.g. a field, then $\mathbb{G}_m$ is already $\mathbb{A}^1$ -local, see . But the presheaf represented by $\mathbb{G}_m$ is surley not the empty presheaf. Hence, $L_{\mathbb{A}^1}$ can not be a left exact localization.

Refernce
exercises.

This ends our discussion of the general discussion about motivic spaces. our next goal is to explain some of the fundamental theorems in motivic homotopy theory.

Chapter 5

The Homotopy Purity Theorem

In differential topology, one important theorem is the “tubular neighborhood theorem” which states, that for any smooth manifold M and any smooth submanifold Z , there exists a tubular neighborhood in M , essentially given by the normal bundle of Z in M . The homotopy purity theorem of Morel and Voevodsky is an analogue of this theorem in motivic homotopy theory. The importance of the theorem might not be obvious at once, however, it has multiple important implications, for example, it shows that motivic cohomology satisfies the purity isomorphism, that is, that the motivic cohomology of a closed smooth subscheme of codimension c is isomorphic to the motivic cohomology of the ambient scheme supported in Z , shifted by c . See [Tin] for more details.

Theorem 5.0.1. *Let $S \in \text{Sch}$ be a base scheme, and let $X, Z \in \text{Sm}_S$ such that $Z \hookrightarrow X$ is a closed immersion. Then there is a canonical equivalence*

$$X/(X \setminus Z) \simeq N/(N \setminus 0_Z) \in \text{Spc}(S),$$

where $N := N_{Z/X}$ is the normal bundle of the immersion $Z \hookrightarrow X$, and 0_Z is the zero section. Remember that, as a closed subscheme, Z induces an exact sequence $J \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_Z$. As J/J^2 is a locally free $\mathcal{O}_Z$ -module, it defines a vector bundle on Z .

Remark 5.0.2. Suppose that Z is locally closed in X . Let $U \hookrightarrow X$ be an open neighborhood, such that Z is closed in U . Then $X/(X \setminus Z) \simeq U/(U \setminus Z)$ see . This is even more generally true for any étale neighborhood of Z , so that the left hand side in the homotopy purity theorem depends only on the “local behavior around Z ”.

reference
exercises

As furthermore the formation of the normal bundle compatible with flat base change, that is for any flat neighborhood $U \rightarrow X$ of Z there is an isomorphism $N_{Z/U} \cong N_{Z/X}$. Thus, also the right hand side only depends on the “local behavior around Z ”.

Before we begin the discussion of the proof of Theorem 5.0.1 we seize the momentum and define one of the most important objects in motivic homotopy theory.

Definition 5.0.3. Let $X \in \text{Sm}_S$ be a smooth scheme, and $\mathcal{E}$ a vector bundle on X . We define the *Thom space* of $\mathcal{E}$ as the cofiber

$$\text{Th}(\mathcal{E}) := \mathcal{E}/(\mathcal{E} \setminus 0_X) = \text{cofib}(X \xrightarrow{0_X} \mathcal{E}).$$

Example 5.0.4. Assume $\mathcal{E} \cong \mathbb{A}^n \times X$ is trivial. Then $\text{Th}(\mathcal{E}) \simeq (\mathbb{A}^n/(\mathbb{A}^n \setminus 0) \wedge X_+)$. For a general $\mathcal{E}$, there is an equivalence $\text{Th}(\mathcal{E}) \simeq \text{Th}(\mathcal{E}^\vee)$ between the Thom space of $\mathcal{E}$ and the Thom space of its dual.

5.1 Heuristic explanation

We start with a special case to motivate why the theorem can be believed to be true. To be precise, we assume for the moment $S = \operatorname{Spec} k$ to be the spectrum of a perfect field, and that $Z = \{z\}$ is a closed point in X .

Notice first that the residue field $\kappa(z)$ of the henselian local ring $\mathcal{O}_{X,z}^h$ is a finite separable extension of the ground field k . Indeed, finiteness follows from X being smooth, while separability follows from k being perfect. Thus, by the primitive element theorem, we can write $\kappa(z) \cong k[T]/p(T)$ with $p \in k[T]$ a separable polynomial.

Now, by the henselian property Definition 3.1.2, there already exists an $t \in \mathcal{O}_{X,z}^h$ such that $P(t) = 0$, where we write P for a lift of p . But this implies that we obtain a canonical inclusion $\kappa(z) \cong k[T]/(P) \hookrightarrow \mathcal{O}_{X,z}^h$, inducing a factorization of the identity

$$\kappa(z) \hookrightarrow \mathcal{O}_{X,z}^h \twoheadrightarrow \mathcal{O}_{X,z}^h/\mathfrak{m} \cong \kappa(z).$$

Geometrically, this translates into a retraction

$$\operatorname{Spec} \mathcal{O}_{X,z}^h \xrightarrow{r} z.$$

By the definition of $\operatorname{Spec} \mathcal{O}_{X,z}^h$, the structure morphism to $\operatorname{Spec} k$ factors over all étale neighborhoods of $\{z\}$. Thus we can appeal to continuity to find an étale neighborhood U of z in X , together with a retraction, by abuse of notation also called $r : U \rightarrow z$, such that the following diagram commutes

$$\begin{array}{ccc} z & \xrightarrow{\quad r \quad} & \operatorname{Spec} \mathcal{O}_{X,z}^h \\ \text{id} \downarrow & & \downarrow \\ z & \xrightarrow{\quad r \quad} & U \\ \text{id} \downarrow & & \downarrow \\ z & \longrightarrow & X \end{array}$$

Thus, up to replacing X by an étale neighborhood, we may assume that $z \rightarrow X$ admits a smooth retraction.

Now assume further that X has dimension d , so that in particular the normal bundle has dimension $\dim N_{z/X} = d$, as z is a closed point. Under this assumption, the local ring $\mathcal{O}_{X,z}$ is a regular local ring of dimension d , so that the maximal ideal $\mathfrak{m}_{z/X}$ is generated by a regular sequence $(f_1, \dots, f_d)$. This yields a morphism

$$(f_\bullet, r) : X \rightarrow \mathbb{A}^d \times_{\operatorname{Spec} k} z = \mathbb{A}_z^d,$$

so that, by construction, $z \mapsto 0$.

Consider now the Kähler differentials

$$\Omega_{X/k,z}^1 = \mathfrak{m}_{X,z}/\mathfrak{m}_{X,z}^2 \cong k\{df_1, \dots, df_d\} \cong \Omega_{\mathbb{A}_z^d/k,z}^1.$$

Thus, by the algebraic inverse function theorem, there exists an open neighborhood U of z in X , such that $(f_\bullet, r)|_U$ is étale. But then

$$X/(X \setminus z) \simeq U/(U \setminus z) \simeq \mathbb{A}_z^d/(\mathbb{A}_z^d \setminus z) \simeq \operatorname{Th}(N_{z/X}).$$

Here the first two equivalences follow from Nisnevich excision, while the last equivalence uses Remark 5.0.2.

The problem with this proof is, that it depends on a lot of choices, and it is not at all clear, how these choices can be glued, given that Z is no longer just a closed point. So we need to deploy a different method of proof.

5.2 Deformation to the normal cone

In this section we discuss the procedure of the deformation to the normal cone, see [Ful98, Chapter 5].

Definition 5.2.1. Let $Z \hookrightarrow X \in \text{Sm}_S$ be a closed immersion of smooth S -schemes. We define the *deformation space* as

$$D_Z X := \text{Bl}_{Z \times 0}(X \times \mathbb{A}^1) \setminus \text{Bl}_{Z \times 0}(X \times \mathbb{A}^1).$$

The deformation space comes equipped with a natural projection map to $\mathbb{A}_S^1$.

Remark 5.2.2. To understand how the deformation space looks like, let us discuss how it looks like away from 0 and at 0.

Away from 0 As we only blow up Z at 0, we see that over $\mathbb{A}^1 \setminus 0$ the deformation space is simply $X \times (\mathbb{A}^1 \setminus 0)$.

At 0 Consider the following two intertwined pullback diagrams relating all schemes appearing in the definition of the deformation space

$$\begin{array}{ccccc} \mathbb{P}(N_{Z/(X \times \mathbb{A}^1)}) & \xrightarrow{\quad} & \text{Bl}_{Z \times 0}(X \times \mathbb{A}^1) & & \\ \downarrow & \swarrow & \downarrow & \nearrow & \\ & \mathbb{P}(N_{Z/X}) & \xrightarrow{\quad} & \text{Bl}_{Z \times 0}(X \times 0) & \\ \downarrow & \downarrow & \downarrow & \downarrow & \\ Z \times 0 & \xrightarrow{\text{id}} & Z \times 0 & \hookrightarrow & X \times 0 \hookrightarrow X \times \mathbb{A}^1 \end{array}$$

Here the outer square is also a pullback square.

Using that $N_{Z/(X \times \mathbb{A}^1)} \cong N_{Z/X} \oplus \mathcal{O}$, we see that over 0, the deformation space can (heuristically) be described as

$$((X \times \mathbb{A}^1 \setminus Z \times 0)_0 \cup \mathbb{P}(N_{Z/X} \oplus \mathcal{O})) \setminus ((X \setminus Z) \cup \mathbb{P}(N_{Z/X})) = \mathbb{P}(N_{Z/X} \oplus \mathcal{O}) \setminus \mathbb{P}(N_{Z/X}) = N_{Z/X}$$

where we embed $\mathbb{P}(N_{Z/X}) \hookrightarrow \mathbb{P}(N_{Z/X} \oplus \mathcal{O})$ as the hyperplane at ∞ .

So in summary, we can imagine $D_Z X$ as a continuous deformation of X to the normal bundle of Z over X , parametrized by $t \in \mathbb{A}_S^1$. This procedure is called the *deformation to the normal cone*.

Remark 5.2.3. In algebraic geometry, a “cone” is a scheme together with an action by $\mathbb{A}^1$. For example, every vector bundle is a cone in this notation.

5.3 Smooth closed pairs

The next ingredient we will need for the proof of the purity theorem is the notion of smooth closed pairs.

Definition 5.3.1. A *smooth closed pair* is a tuple (X, Z) where $X, Z \in \text{Sm}_S$ are two smooth S -schemes, together with a closed immersion $Z \hookrightarrow X$. A morphism of smooth closed pairs $(X, Z) \rightarrow (Y, W)$ is a morphism of schemes $X \rightarrow Y$, such that $f(Z) \subseteq W$, and a given isomorphism $Z \xrightarrow{\cong} X \times_Y W$.

Lemma 5.3.2. Let P be a class of smooth closed pairs. Suppose the following holds for all smooth closed pairs (X, Z) .

- (1) If $\{U_\bullet \rightarrow X\}$ is a Zariski cover such that $(U_{i_1} \times_X \dots \times_X U_{i_k}, Z \times_X U_{i_1} \times_X \dots \times_X U_{i_k}) \in P$, then already $(X, Z) \in P$.
- (2) If $(Y, W) \rightarrow (X, Z)$ is a morphism of smooth pairs, such that $Y \rightarrow X$ is étale and induces an isomorphism $W \xrightarrow{\cong} Z$, then $(Y, W) \in P$ if and only if $(X, Z) \in P$.
- (3) $(\mathbb{A}_Z^n, Z) \in P$ for all $n \geq 0$.

Then P already contains all smooth closed pairs.

Proof. Let (X, Z) be a general smooth pair and $z \in Z$. By Lemma 5.3.3 there exists an open neighborhood U_z of z in X , and étale neighborhood X' of $U_z \cap Z$ in U admitting a smooth retraction $X' \rightarrow U_z \cap Z$. Thus we can cover X by the opens $\{U_z | z \in Z\} \cup \{X \setminus Z\}$. Using condition (1), we can replace X by iterated intersections of the sets in this cover, and hence reduce to one of two cases.

In the first case, $Z = \emptyset$. But then, using that $\emptyset \rightarrow X$ is étale, it follows from condition (2), that $(X, \emptyset)$ is contained in P if and only if $(\emptyset, \emptyset)$ is contained in P . But $\mathbb{A}_\emptyset^n \cong \emptyset$, and hence $(\emptyset, \emptyset) \cong (\mathbb{A}_\emptyset^n, \emptyset)$ is in P by condition (3).

In the other case we can replace X by (some open subscheme of) X' and hence assume the existence of a smooth retraction. As $N_{Z/X}^\vee$ is a vector bundle, there exists for every $z \in Z$ an open neighborhood, such that $N_{Z/X}^\vee$ is a free $\mathcal{O}_Z$ module of rank c . By condition (1) and using Lemma 5.3.4 we may further reduce to the case $Z = Z(f_1, \dots, f_c)$, and hence that the conormal sheaf is given by $N_{Z/X}^\vee \cong \mathcal{O}_Z\{df_1, \dots, df_c\}$.

Consider now the map $(f_1, \dots, f_c, s) : X \rightarrow \mathbb{A}_Z^c$ over Z , where $s : X \rightarrow Z$ is the smooth retraction. This map is étale at every point of Z . Indeed, as $Z \hookrightarrow X$ is a smooth closed immersion, there exists a short exact sequence

$$0 \longrightarrow N_{Z/X}^\vee \longrightarrow s^* \Omega_{X/S}^1 \longrightarrow \Omega_{Z/S}^1 \longrightarrow 0.$$

Notice the exactness on the left is due to smoothness of Z . By our reduction, $\Omega_{Z/S}^1$ is locally free of rank c , in particular, the above sequence splits and we obtain an isomorphism at every $z \in Z$

$$\Omega_{X/S, z}^1 \cong (s^* \Omega_{X/Z}^1)_z \cong N_{Z/X, z}^\vee \oplus \Omega_{Z/S, z}^1 \cong \Omega_{\mathbb{A}_Z^c/Z, z}^1.$$

In particular, there exists an open neighborhood U of Z in X . So, by (2), we may replace X with U . But by (2) again, we may further replace U by $\mathbb{A}_Z^c$, which is in P by condition (3). $\square$

Lemma 5.3.3. *Let (X, Z) be a smooth closed pair, and let $z \in Z$. Then there exists an open neighborhood U of z in X and an étale neighborhood U' of $U \cap Z$ in U , such that there exists a smooth retraction $U' \rightarrow U \cap Z$, see also the following diagram:*

$$\begin{array}{ccc} U' & \xrightarrow{\text{ét}} & U \\ & \searrow & \nearrow \\ & U \cap Z & \end{array}$$

Proof. We start by constructing the open neighborhood U of Z in X together with an étale morphism $\varphi : U \rightarrow \mathbb{A}_S^n$ such that $Z = \varphi^{-1}(\mathbb{A}_S^d)$, where we consider $\mathbb{A}_S^d \subseteq \mathbb{A}_S^n$ as a linear subspace. Consider the short exact sequence induced by the smooth closed immersion $Z \hookrightarrow X$

$$0 \longrightarrow N_{Z/X}^\vee \longrightarrow \Omega_{X/S}^1|_Z \longrightarrow \Omega_{Z/S}^1 \longrightarrow 0.$$

As $N_{Z/X}^\vee$ is a vector bundle, replacing X with an suitable open subset, there are $f_i \in I_Z$ such that $N_{Z/X}^\vee \simeq \mathcal{O}_Z\{df_1, \dots, df_c\}$. In particular, the above sequence splits, and we obtain an isomorphism

$$\Omega_{X/S}^1|_Z \cong \mathcal{O}_Z\{df_1, \dots, df_c, dg_1, \dots, dg_e\}$$

for some $g_j \in \mathcal{O}_Z$. Given this basis, by going even further locally, we may assume

$$\Omega_{X/Z}^1 \cong \mathcal{O}_X\{df_1, \dots, df_c, dg_1, \dots, dg_e\}.$$

Then, the map $\varphi = (f_\bullet, g_\bullet) : X \rightarrow \mathbb{A}_S^{c+e}$ has the required properties. Étaleness is clear as the morphism was constructed using a basis of the Kähler differentials, and we have $Z = \varphi^{-1}(0 \times \mathbb{A}_S^e)$, also by construction.

Now consider the pullback

$$\begin{array}{ccccc} & & Z & & \\ & \swarrow & & \searrow & \\ & & X' & \xrightarrow{\quad} & X \\ & \downarrow & \lrcorner & & \downarrow \varphi \\ \text{id} \swarrow & & \mathbb{A}_Z^c & \xrightarrow{\alpha} & \mathbb{A}_S^{c+d} \\ & & \downarrow \text{pr}_Z & & \\ & & Z & & \end{array}$$

Here, $\alpha = \text{id}_{\mathbb{A}_Z^c} \times \varphi|_Z$. Then the morphism $X' \rightarrow \mathbb{A}_Z^c$ is étale, and α is so too, as seen by the pullback diagram

$$\begin{array}{ccc} Z & \xrightarrow{\quad} & X \\ \varphi|_Z \downarrow & \lrcorner & \downarrow \varphi \\ \mathbb{A}_Z^e & \xrightarrow{\quad} & \mathbb{A}_S^{c+e} \end{array}$$

Thereby, X' is an étale neighborhood of Z in X which admits a smooth retraction, as the projection $\mathbb{A}_Z^e \rightarrow Z$ is smooth. $\square$

Lemma 5.3.4. *Let (X, Z) be a closed smooth pair where Z is defined by an ideal sheaf I_Z . Assume we are given sections $f_1, \dots, f_c \in I_Z$ such that $I_Z/I_Z^2 \cong \mathcal{O}_Z\{df_1, \dots, df_c\}$. Then there exists an open neighborhood U of Z in X such that $I_Z|_U \cong \langle f_1, \dots, f_c \rangle$.*

Proof. Working locally, we may assume that $X = \text{Spec} A$ is affine. Set $I = I_Z$ and $J = \langle f_1, \dots, f_c \rangle \subseteq I$, which we consider as Ideals in A . Then, since Z , being affine, hence quasi compact, and smooth over S , is finitely presented, J is finitely generated. We then have

$$I \cdot I/J \cong (I^2 + J)/J \cong J,$$

where the second isomorphism is due to the assumption on I/I^2 . As I is finitely generated, we obtain from Nakayama's lemma some $r \in A$ with the property that $r \equiv 1 \pmod I$ such that $r \cdot I/J = 0$. Set $U = D(r)$. Then $I|_U = J|_U$ and since $r \notin I$ we have $Z \subseteq D(r)$ implying the claim. $\square$

5.4 Proof of purity

The last ingredient we need for the proof of the purity theorem Theorem 5.0.1 is the following

Definition 5.4.1. A morphism $(X', Z') \rightarrow (X, Z)$ of smooth closed pairs is called *weakly excisive*, if the following diagram of presheaves is motivically cocartesian

$$\begin{array}{ccc} Z' & \longrightarrow & X'/(X' \setminus Z') \\ \downarrow & & \downarrow \\ Z & \longrightarrow & X/(X \setminus Z) \end{array}$$

Lemma 5.4.2. Let $(X'', Z'') \xrightarrow{g} (X', Z') \xrightarrow{f} (X, Z)$ be morphisms of smooth closed pairs.

- (1) If g is weakly excisive, then f is weakly excisive if and only if $f \circ g$ is.
- (2) If $f \circ g$ and f are both weakly excisive and $Z' \rightarrow Z$ is a motivic equivalence, then g is weakly excisive.
- (3) If $\{U_\bullet \rightarrow X\}$ is a Nisnevich cover and $f_{U_{i_1} \times_X \dots \times_X U_{i_k}}$ is weakly for all indices, then f is weakly excisive.

Proof. We recommend doing these as exercises, but for the convenience of the reader, we include the proofs here.

For (1), Writing out the definitions we see that the claim is equivalent to the statement that in the following diagram, the lower square is motivically cocartesian if and only if the outer rectangle is so

$$\begin{array}{ccc} Z'' & \longrightarrow & X''/(X'' \setminus Z'') \\ g \downarrow & \lrcorner & \downarrow \\ Z' & \longrightarrow & X'/(X' \setminus Z') \\ f \downarrow & & \downarrow \\ Z & \longrightarrow & X/(X \setminus Z) \end{array}$$

But this is just the pasting law for pullbacks.

For (2), by assumption the upper square and the outer rectangle in the above diagram are motivically cocartesian squares. Furthermore, f is a motivic equivalence. As the lower square is a cocartesian diagram, the induced morphism $X'/(X' \setminus Z') \rightarrow X/(X \setminus Z)$ is a motivic equivalence. Thus the upper square is equivalent to the outer square, which is motivically cocartesian by assumption, applying the claim.

For (3)

□

todo

We now have everything we need to prove purity. In fact, we prove the following, stronger statement.

Theorem 5.4.3. For any smooth pair (X, Z) the following cospan consists of weakly excisive morphisms

$$(X, Z) \rightarrow (D_Z X, Z \times \mathbb{A}^1) \leftarrow (N_{Z/X}, Z)$$

where we identified $Z \hookrightarrow N_{Z/X}$ with its image under the zero section. In particular, $X/(X \setminus Z) \xrightarrow{\cong} D_Z X/(D_Z X \setminus (Z \times \mathbb{A}^1)) \xleftarrow{\cong} N_{Z/X}/(N_{Z/X} \setminus Z) \in \mathrm{Spc}(S)$.

Proof. The “in particular” part follows from the motivically cocartesian diagram

$$\begin{array}{ccc} Z & \longrightarrow & X/(X \setminus Z) \\ \downarrow & \lrcorner & \downarrow \\ Z \times \mathbb{A}^1 & \longrightarrow & D_Z X/(D_Z X \setminus (Z \times \mathbb{A}^1)) \\ \uparrow & \lrcorner & \uparrow \\ Z & \longrightarrow & N_{Z/X}/(N_{Z/X} \setminus Z) \end{array}$$

the fact that both left vertical morphisms are motivic equivalences, and that pushout preserves equivalences.

So all that is left is the statement itself. Denote by P the collection of all smooth closed pairs for which the theorem holds. We want to use Lemma 5.3.2 to show that P actually contains all smooth local pairs. Thus, we have to show that P satisfies conditions (1), (2), and (3) of Lemma 5.3.2. Condition (1), being in P is Zariski local, follows directly from Lemma 5.4.2 (3), as every Zariski cover is in particular a Nisnevich cover.

Condition (2), being in P is invariant under étale morphisms, follows from Lemma 5.4.2 (2) and (3). Indeed, given an étale morphism of smooth closed pairs $(X, Z) \rightarrow (X', Z')$, consider the following diagram

$$\begin{array}{ccc} (X', Z') & \longrightarrow & (D_{Z'} X', Z' \times \mathbb{A}^1) \\ \downarrow & & \downarrow \\ (X, Z) & \longrightarrow & (D_Z X, Z \times \mathbb{A}^1) \end{array}$$

Here, both vertical maps are weakly excisive. Thus, by Lemma 5.4.2 (2) and (3), the top morphism is weakly excisive if and only if the lower morphism is weakly excisive. The other square is exactly the same argument.

For condition (3), we have to show that $(X, Z) = (\mathbb{A}^n, Z)$ is in P . One explicitly calculate

$$(X, Z) = (\mathbb{A}_Z^n, Z) \xrightarrow{\iota_0} D \cong (X \times \mathbb{A}^1, Z \times \mathbb{A}^1) \xleftarrow{\iota_1} (N_{Z/X}, Z) \cong (\mathbb{A}_Z^n, Z) = (X, Z)$$

Maybe try to implement this

where ι_i is the inclusion at i . The inclusions composed with the retraction are both motivic equivalences. We want to show that the following diagram is cocartesian

$$\begin{array}{ccc} Z & \longrightarrow & X/(X \setminus Z) \\ \downarrow & & \downarrow \\ Z \times \mathbb{A}^1 & \longrightarrow & (X \times \mathbb{A}^1)/((X \times \mathbb{A}^1) \setminus (Z \times \mathbb{A}^1)) \end{array}$$

as the left vertical morphism is an equivalence, this is motivically a pushout if and only if the right morphism is an equivalence. But there is an equivalence

$$(X \times \mathbb{A}^1) \cong (Z \times \mathbb{A}^1) \cong (X \setminus Z) \times \mathbb{A}^1,$$

finishing the proof. □

Chapter 6

Connectivity theorems

In this chapter we are going to discuss Morels connectivity theorems for motivic spaces and spectra over a field.

6.1 Connectivity and motivic spaces

To talk about connectivity of motivic spaces, we need to introduce homotopy sheaves for motivic spaces. We already know how to define them for ∞ -topoi, see Definition 2.3.15. As $\mathrm{Spc}(S)$ for some base scheme S is not a topos, see Remark 4.3.5, we cannot directly use this definition. There is a remedy though, as $\mathrm{Spc}(S)$ is a subcategory of the Nisnevich topos over S .

Definition 6.1.1. Let $X \in \mathrm{Spc}(S)$ and consider the inclusion $\iota : \mathrm{Spc}(S) \hookrightarrow \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S)$. We then define the *homotopy sheaves of X* as

$$\pi_i(X) := \pi_i^{\mathrm{Nis}}(\iota X).$$

Given a basepoint $b : * \simeq S \rightarrow X$ we define the *homotopy sheaves of X at b* as

$$\pi_i(X, b) := \pi_i^{\mathrm{Nis}}(\iota X, \iota b).$$

We call $X \in \mathrm{Spc}(S)$ *(n -)connected*, *n -truncated*, etc. if it holds for ιX .

Remark 6.1.2. Explicitly, we have

$$\pi_0(X) = \mathrm{L}_{\mathrm{Nis}}(Y \mapsto [Y, X]_{\mathrm{Spc}(S)})$$

etc.

Notice that for $X \in \mathrm{Spc}(S)$ it is not necessarily true that $\pi_0(X)$ is $\mathbb{A}^1$ -invariant.

Remark 6.1.3. It is often easier to work with general presheaves. When we do so, we call $P \in \mathcal{P}(\mathrm{Sm}_S)$ *Nisnevich locally n -connected*, if this is true for $\mathrm{L}_{\mathrm{Nis}}P \in \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S)$ is. We often suppress the Nisnevich and simply say locally connected.

Then connectivity theorems will give a statement about the change in connectivity of a presheaf if one motivically localizes the given presheaf. A first incarnation of this is the following

Proposition 6.1.4. *Let $X \in \mathcal{P}(\mathrm{Sm}_S)$ be locally connected. Then $\mathrm{L}_{\mathrm{mot}}X \in \mathrm{Spc}(S)$ is connected.*

Proof. It suffices to prove that if $X \in \mathcal{P}(\mathrm{Sm}_S)$ is locally connected, then $L_{\mathbb{A}^1}X$ is also locally connected. Indeed, if this is the case, by definition of locally connectedness Remark 6.1.3, the presheaf $(L_{\mathrm{Nis}} \circ L_{\mathbb{A}^1})^{\circ n}X$ is locally connected for all n , and by Lemma 4.3.1 and the fact, that colimits in $\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S)$ preserve connected objects implies the claim. We will compute the values of the homotopy sheaves on all stalks. So let $y \in Y \in \mathrm{Sm}_S$. We compute

$$\begin{aligned} \pi_0(L_{\mathbb{A}^1}X)(Y_y^h) &\cong \pi_0(L_{\mathbb{A}^1}X(Y_y^h)) \\ &\cong \pi_0|X(\mathbb{A}^\bullet \times Y_y^h)| \\ &\cong * \end{aligned}$$

Here we used that $\pi_0|X(\mathbb{A}^\bullet \times Y_y^h)|$ is a discrete set, which is hence a quotient of $\pi_0(X(Y_y^h))$. But X was assumed to be connected, hence $\pi_0(X(Y_y^h)) \cong *$. $\square$

6.2 Sheaves of spectra

The connectivity theorem we want to prove does not make a statement about motivic spaces themselves. Instead, it will give an answer to how much the connectivity will change when we change into the stabilization of motivic spaces. To define this category, we first have to talk about the stabilization of topoi.

Definition 6.2.1. let $\mathcal{X}$ be a topos. We define the *sheaves of spectra on $\mathcal{X}$* as the stabilization of $\mathcal{X}$, that is

$$\mathrm{Sp}(\mathcal{X}) \simeq \lim(\dots \mathcal{X}_* \xrightarrow{\Omega} \mathcal{X}_* \xrightarrow{\Omega} \mathcal{X}_*).$$

Explicitly, an object in $\mathrm{Sp}(\mathcal{X})$ is a sequence $X^0, X^1, X^2, \dots \in \mathcal{X}_*$ together with equivalences $\omega^i : X^i \xrightarrow{\sim} \Omega X^{i+1}$.

As each X^i is pointed, we will usually suppress the base point in the homotopy sheaves and simply write $\pi_i(X)$. Notice that $\pi_i(\Omega X^n, *) \cong \pi_i(X^n)$ for all n . Thus the following definition is justified.

Definition 6.2.2. Let $X \in \mathrm{Sp}(\mathcal{X})$ and let $i \in \mathbb{Z}$. We define the *stable homotopy sheaves of X* as

$$\pi_i(X) := \pi_{i+n}(X^n)$$

for n great enough.

Remark 6.2.3. All stable homotopy sheaves are sheaves of abelian groups. Indeed, this follows from the fact that $X^i \simeq \Omega^2 X^{i+2}$.

Lemma 6.2.4. *Let $\mathcal{X}$ be hypercomplete. Then $f : X \rightarrow Y \in \mathrm{Sp}(\mathcal{X})$ is an equivalence if and only if $\pi_i(f) : \pi_i(X) \rightarrow \pi_i(Y)$ is an isomorphism of abelian sheaves for all $i \in \mathbb{Z}$.*

That is, we have more homotopy sheaves, to check, but we do not need to check for all the base points anymore.

Proof. Since $\pi_i(-)$ is a functor for all i , the “only if” part is clear. For the “if” part we use that f is an equivalence if and only if $f^i : X^i \rightarrow Y^i \in \mathcal{X}_*$ is an equivalence for all $i \geq 0$. Now let $i \geq 1$. By assumption, $\pi_*(f^i) = \pi_*(f^i, *)$ is an equivalence, where $*$ is the canonical base point. So, using hypercompleteness, we learn that f^i induces an equivalence $(f^i)_{\geq 1} : (X^i)_{\geq 1} \rightarrow (Y^i)_{\geq 1}$ of connective covers. Indeed, for connective spaces, it suffices to consider only one singular base point, as all the homotopy sheaves for different base points are isomorphic to chosen one.

Now $\Omega : \mathcal{X}_* \rightarrow \mathcal{X}_*$ is a functor, hence $\Omega(f^i)_{\geq 1}$ is an equivalence. As $\Omega \circ \tau_{\geq 1} \simeq \Omega$, and by the definition of sheaves of spectra, we obtain

$$\Omega(f^i)_{\geq 1} \simeq \Omega(f^i) \simeq f^{i-1}.$$

As $i \geq 1$ was arbitrary, this implies the claim. $\square$

We restrict now to the special case of sheaf topoi, that is, assume we are given a category $\mathcal{C}$ with a Grothendieck topology, and consider $\mathcal{X} = \mathrm{Shv}_\tau(\mathcal{C})$. We denote the localization, the left adjoint to the inclusion, by $L_\tau : \mathcal{P}(\mathcal{C}) \rightarrow \mathrm{Shv}_\tau(\mathcal{C})$. Stabilizing the inclusion, we obtain an inclusion

$$\mathrm{Sp}(\mathcal{X}) \hookrightarrow \mathrm{Sp}(\mathcal{P}(\mathcal{C})) \simeq \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathrm{Sp}),$$

and a description of $\mathrm{Sp}(\mathcal{X})$ as the localizing subcategory of $\mathrm{Sp}(\mathcal{P}(\mathcal{C}))$ consisting of those sequences $(X^0, X^1, \dots)$ such that $X^i \in \mathcal{X}$ for all i . In particular, using the adjoint functor theorem, this admits a left adjoint, which, by abuse of notation, we also call L_τ . In fact, one can show

$$L_\tau(X^0, X^1, \dots) \simeq (L_\tau X^0, L_\tau X^1, \dots),$$

see

Exercise

6.3 Gabber's lemma and the S^1 -stable connectivity theorem

6.3.1 Gabber's lemma

The main ingredient in the proof of the connectivity theorem, which we have yet to state, is a presentation lemma by Gabber. As the proof of Gabber's lemma is hard and technical, we postpone it Section 6.6, while only stating it in this section.

Theorem 6.3.1 (Gabber's Lemma). *Let k be a field, $X \in \mathrm{Sm}_k$ a smooth scheme, $Z \hookrightarrow X$ a not-necessarily closed subscheme which is everywhere of positive codimension. Let $z \in Z$. Then there exists*

- (1) *an open neighborhood $X' \subseteq X$ of z ;*
- (2) *an étale morphism $\varphi : X' \rightarrow \mathbb{A}_S^1$, where S is a smooth affine scheme over k .*

They further fulfill the following properties.

- (a) *The composition $Z' := Z \cap X' \hookrightarrow X' \xrightarrow{\varphi} \mathbb{A}_S^1 \rightarrow S$ is finite.*
- (b) *The restriction $\varphi|_{Z'} : Z' \rightarrow \mathbb{A}_S^1$ is a closed immersion.*
- (c) *Z' is saturated under φ , i.e. $Z' \xrightarrow{\cong} \varphi^{-1}(\varphi(Z'))$.*

Theorem 6.3.2 (Homotopy sheaves are unramified). *Let $E \in \mathrm{Sp}(\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_k))$ be $\mathbb{A}^1$ invariant, $X \in \mathrm{Sm}_k$ a smooth k scheme, and $x \in X$ a point. Then*

- (i) $\pi_i(E(X_x)) \hookrightarrow \pi_i(E(k(X_x)))$, where $k(X_x)$ denotes the generic point, that is, the generic point of the connected component of X containing x .
- (ii) $\pi_i(E)(X_x^h) \cong \pi_i(E(X_x^h)) \hookrightarrow \pi_i(E(k(X_x^h))) \cong \pi_i(E)(X_x^h)$.

Proof. We will only proof this if k is infinite. For the finite case, see for example [Bac24, Proposition 2.9].

Choose a presentation $X_x^h \cong \lim U_\alpha$ by étale neighborhoods of x . Denote by $x_\alpha \in U_\alpha$ the image of x . Then we can also write $X_x^h = \lim U_{\alpha, x_\alpha}$. Assume now that we have already proven (i). Then, from the commutative diagram

$$\begin{array}{ccc} \pi_i(E(X_x^h)) & \xrightarrow{\cong} & \operatorname{colim} \pi_i(E(U_{\alpha, x_\alpha})) \\ \downarrow & & \downarrow (i) \\ \pi_i(E(k(X_x^h))) & \xrightarrow{\cong} & \operatorname{colim} \pi_i(E(k(U_{\alpha, x_\alpha}))) \end{array}$$

where we know that the right vertical morphism is an injection by (i), it follows that the left vertical morphism is an injection too. Thus it suffices to prove (i).

Fix now some $i \in \mathbb{Z}$. Let $U \subseteq X$ be an open neighborhood of x , and let $Z \hookrightarrow X$ be a closed subscheme containing x . We claim that there exists an open neighborhood $U' \subseteq U \subseteq X$ of x such that¹

$$\pi_i(E(U')) \hookrightarrow \pi_i(E(U' \setminus Z)).$$

This implies the theorem. Indeed, let $\Lambda := \{(U, Z) | U, Z \text{ as above}\}$ and view this as a partially ordered set by declaring $(U, Z) \leq (U', Z')$ if $U \supseteq U'$ and $Z \subseteq Z'$. Obviously, Λ is filtered, since finite intersections of opens are open, and finite unions of closed are closed. We then have

$$\begin{aligned} \lim_{\Lambda} U &\cong X_x, \\ \lim_{\Lambda} (U \setminus Z) &\cong k(X_x). \end{aligned}$$

But by the claim, the family $\Lambda' := \{(U', Z) | \pi_i(E(U')) \hookrightarrow \pi_i(E(U' \setminus Z))\}$ is cofinal, and hence $\lim_{\Lambda'} U' = X_x$. We conclude by the commutative diagram

$$\begin{array}{ccc} \pi_i(E(X_x)) & \xrightarrow{\quad} & \pi_i(E(k(X_x))) \\ \cong \downarrow & & \downarrow \cong \\ \operatorname{colim}_{\Lambda'} (\pi_i(E(U'))) & \hookrightarrow & \operatorname{colim}_{\Lambda'} (\pi_i(E(k(U')))) \end{array}$$

Here the lower horizontal morphism is injective by the claim, and as filtered colimits in abelian groups are exact.

Thus it remains to prove the claim. To do so, we apply Gabber's lemma Theorem 6.3.1 to the triple $(U, Z \cap U, x)$. We obtain a commutative diagram

$$\begin{array}{ccc} U' & \xrightarrow{\text{étale}} & \mathbb{A}_S^1 \\ \uparrow & \nearrow & \downarrow \\ Z' & \xrightarrow{\text{finite}} & S \end{array}$$

Let s denote the image of $x \in Z'$ under the lower horizontal morphism. As S is infinite, being an affine smooth scheme over an infinite field, $\mathbb{A}_S^1(s)$ has infinitely many points too. Hence, Z' being finite over S , we obtain a section

$$\begin{array}{ccc} \mathbb{A}_S^1 \setminus Z' & & \\ \downarrow & \nearrow & \\ S & \longleftarrow & s \end{array}$$

Thus there exists some open neighborhood $S' \subseteq S$ of s to which the above section can be extended. Indeed, it suffices to give a function $\sigma \in \mathcal{O}(S) \cong \operatorname{Map}(S, \mathbb{A}_S^1)$ such that $\sigma^{-1}(Z')$

¹By $U \setminus Z$ we of course mean $U \setminus (U \cap Z)$.

does not meet the yet to define open neighborhood S' . As closed immersions are closed under pullback, $\sigma^{-1}(Z')$ is again closed, hence $S' := S \setminus \sigma^{-1}(Z')$ is an appropriate open neighborhood, provided $s \notin \sigma^{-1}(Z)$. That is, given the section $z \rightarrow \mathbb{A}_S^1 \setminus Z'$ from above, if we can extend this to a section $S \rightarrow \mathbb{A}_S^1$, we find the open neighborhood we wished for. But this we can do. In fact, S is affine and hence there is a surjection $\text{Map}(S, \mathbb{A}_S^1) \cong \mathcal{O}_S(S) \twoheadrightarrow \mathcal{O}_S(s)$.

Replacing S by S' we get a new commutative diagram

$$\begin{array}{ccccc} U' & \longrightarrow & \mathbb{A}_{S'}^1 & \longleftarrow & \mathbb{A}_{S'}^1 \setminus Z' \\ \uparrow & \nearrow & \downarrow & \searrow & \uparrow \\ Z' & \xrightarrow{\text{finite}} & S' & \xrightarrow{\sigma} & \mathbb{A}_{S'}^1 \setminus Z' \end{array}$$

Notice now that

$$\begin{array}{ccc} U' \setminus Z' & \hookrightarrow & U' \\ \downarrow & & \downarrow \\ \mathbb{A}_{S'}^1 \setminus Z' & \hookrightarrow & \mathbb{A}_{S'}^1 \end{array}$$

is an Nisnevich distinguished square, due to Gabber's lemma. Hence there is an equivalence by Theorem 5.0.1

$$U'/(U' \setminus Z') \simeq \mathbb{A}_{S'}^1/(\mathbb{A}_{S'}^1 \setminus Z') \in \text{Shv}_{\text{Nis}}(\text{Sm}_k).$$

The cofiber sequences induce long exact sequences by applying $\pi_i(E(-))$

$$\begin{array}{ccccc} \pi_i(E(U' \setminus Z')) & \xleftarrow{d} & \pi_i(E(U')) & \xleftarrow{c} & \pi_i(E(U'/(U' \setminus Z'))) \\ \uparrow & & \uparrow & & \uparrow \cong \\ \pi_i(E(\mathbb{A}_{S'}^1 \setminus Z')) & \xleftarrow{a} & \pi_i(E(\mathbb{A}_{S'}^1)) & \xleftarrow{b} & \pi_i(E(\mathbb{A}_{S'}^1/(\mathbb{A}_{S'}^1 \setminus Z'))) \\ & \searrow \sigma^* & \uparrow \cong & & \\ & & \pi_i(E(S')) & & \end{array}$$

Here we used that E is $\mathbb{A}^1$ invariant, and the equivalence $U'/(U' \setminus Z') \simeq \mathbb{A}_{S'}^1/(\mathbb{A}_{S'}^1 \setminus Z')$ from above.

Now $\sigma^* \circ a$ is an isomorphism, thus a is injective. By exactness, b is the zero map, and hence, by commutativity and the right vertical isomorphism, so is c . By exactness again, d is injective, which was exactly the claim. $\square$

6.3.2 The Dold-Kan equivalence

Before we can prove the stable connectivity, we give a short recap of the Dold-Kan correspondence.

let $\mathcal{C}$ be an abelian category. Then there exists a functor, see [Wei94, Definition 8.3.6]

$$N_* : \text{Fun}(\Delta^{\text{op}}, \mathcal{C}) \rightarrow \text{Ch}(\mathcal{C})_{\geq 0}.$$

Explicitly, this functor sends a $\mathcal{C}$ valued simplicial object X to the chain complex given in degree $n \geq 0$ by

$$N_n(X) := \bigcap_{i=0}^{n-1} \ker(d_i) \subseteq X_n$$

where $d_i : X_n \rightarrow X_{n-1}$ is the i -th face map. The differential $d : N_n(X) \rightarrow N_{n-1}(X)$ is given by

$$d = (-1)^n d_n.$$

This functor is called the *normalized chain complex of X* and is an equivalence of categories, see [Wei94, Theorem 8.4.1]. This is known as the Dold-Kan equivalence. But even more is true. We can embed the n -th degree of the normalized complex of X into the n -simplices of X and then consider the dual construction, that is we have a commutative diagram

$$\begin{array}{ccc} N_n(X) & \hookrightarrow & X_n \\ & \searrow \cong & \downarrow \\ & & \operatorname{coker}(\bigoplus_{i=0}^{n-1} X_{n-1} \xrightarrow{\oplus s_i} X_n) \end{array}$$

One can show that the composition is an isomorphism, as indicated. In particular, $N_n(X)$ is a summand of X_n .

But the dual construction makes sense for any additive, idem-complete 1-categories. In particular, N_* can be defined for any such category, and it induces an equivalence. For more informations on this, see [Lur17a]Section 1.2.3..

Now suppose, $\mathcal{C}$ is a stable, idempotent-complete category, and $X_\bullet : \Delta^{\text{op}} \rightarrow \mathcal{C}$ is a simplicial object. Then we can consider the image of this functor under taking homotopy categories. We obtain a simplicial object $hX : \Delta^{\text{op}} \rightarrow h\mathcal{C}$, which by the aboce discussion corresponds to a chain complex

$$N_* hX \in \operatorname{Ch}_{\geq 0}(h\mathcal{C})$$

Proposition 6.3.3. *There exists an equivalence*

$$\operatorname{cofib}(\operatorname{colim}_{\Delta_{\leq n-1}^{\text{op}}} X \rightarrow \operatorname{colim}_{\Delta_{\leq n}^{\text{op}}} X) \simeq \Sigma^n N_n hX$$

as a summand of $\Sigma^n X_n$.

Proof. This is [Lur17a, Remark 1.2.4.3]. □

Corollary 6.3.4. *Let $X_\bullet : \Delta^{\text{op}} \rightarrow \operatorname{Sp}$ be a simplicial spectrum, such that X_i is $(-i)$ -connective for all i . Then $|X_\bullet|$ is connective.*

Proof. We first prove that $\operatorname{colim}_{\Delta_{\leq n}^{\text{op}}} X =: C_n$ is connective. We start the induction at $n = -1$, in which case $C_{-1} \simeq 0$ is trivially connected. For the induction step, consider the cofiber sequence

$$C_{n-1} \rightarrow C_n \rightarrow \Sigma^n N_n hX$$

where we used Proposition 6.3.3 to identify the cofiber with a summand of $\Sigma^n X_n$. But $\Sigma^n X_n$ is connective by assumption, hence so is the cofiber. Using induction and the induced long exact sequence on homotopy groups, we see that C_n is connective. But then, $|X_\bullet| \simeq \operatorname{colim} C_n$ is connective as well. Indeed, the sphere spectrum $\mathbb{1} \in \operatorname{Sp}$ being compact, homotopy groups commute with filtered colimits. □

6.3.3 Stable connectivity

In this section, we want to prove the following connectivity theorem.

Theorem 6.3.5. *Let k be a field, $E \in \operatorname{Fun}(\operatorname{Sm}_S^{\text{op}}, \operatorname{Sp})$ a Nisnevich locally connected spectrum. Then also $L_{\text{mot}} E$ is Nisnevich locally connected.*

Here L_{mot} is the left adjoint of the composition

$$\{\mathbb{A}^1\text{-invariant Nisnevich sheaves of spectra}\} \hookrightarrow \text{Sp}(\text{Shv}_{\text{Nis}}(\text{Sm}_k) \hookrightarrow \text{Sp}(\mathcal{P}(\text{Sm}_k) \simeq \text{Fun}(\text{Sm}_k^{\text{op}}, \text{Sp}).$$

We need to show one last lemma before we can give the proof of the stable connectivity theorem.

Lemma 6.3.6. *Let $E \in \text{Fun}(\text{Sm}_k^{\text{op}}, \text{Sp})$ be a Nisnevich sheaf. Then $L_{\text{mot}}E \simeq L_{\mathbb{A}^1}E \simeq \text{Sing}E$.*

Proof. As in the proof of Proposition 4.2.10, $E \rightarrow \text{Sing}E$ is an $\mathbb{A}^1$ -equivalence to an $\mathbb{A}^1$ -local object, hence defines $L_{\mathbb{A}^1}^1$. It therefor remains to prove that $L_{\mathbb{A}^1}E$ is a Nisnevich sheaf. By Theorem 3.2.4, we have to show that

- $(L_{\mathbb{A}^1}E)(\emptyset) = 0$,
- for every Nisnevich distinguished square $U \hookrightarrow X \xleftarrow{V}$ the following square is cartesian

$$\begin{array}{ccc} (L_{\mathbb{A}^1}E)(X) & \longrightarrow & (L_{\mathbb{A}^1}E)(U) \\ \downarrow & & \downarrow \\ (L_{\mathbb{A}^1}E)(V) & \longrightarrow & (L_{\mathbb{A}^1}E)(W) \end{array}$$

Here we write $W = U \times_X V$ for the pullback.

As Sp is a stable category, the initial and terminal object are equivalent, i.e. $* \simeq \emptyset \simeq 0$, and the square is a cartesian if and only if it is cocartesian. So we may instead check that the corresponding colimit conditions are satisfied. But this is clear from the definition of Sing as a colimit, see Definition 4.2.1. Hence $L_{\mathbb{A}^1}$ preserves Nisnevich sheaves of spectra. $\square$

Remark 6.3.7. In particular $L_{\text{mot}} \simeq L_{\mathbb{A}^1} \circ L_{\text{Nis}} \in \text{End}(\text{Fun}(\text{Sm}_k^{\text{op}}, \text{Sp}))$. Furthermore, the proof even shows that the inclusion

$$\text{Sp}(\text{Shv}_{\text{Nis}}(\text{Sm}_k)) \hookrightarrow \text{Fun}(\text{Sm}_k^{\text{op}}, \text{Sp})$$

preserves colimits.

Proof of Theorem 6.3.5. Let $E \in \text{Fun}(\text{Sm}_S^{\text{op}}, \text{Sp})$ a Nisnevich locally connected spectrum. We want to show that $\pi_i(L_{\text{mot}}E) \simeq 0$ for all $i < 0$. By hypercompleteness, it suffices to check the vanishing on stalks, and since $\pi_i(L_{\text{mot}}E)$ is unramified, see Theorem 6.3.2, it suffices to prove that $\pi_i(L_{\text{mot}}E(K)) \cong 0$ for all separable field extensions K/k , not necessarily finite or algebraic. But by Lemma 6.3.6, we have

$$(L_{\text{mot}}E)(K) \simeq (L_{\mathbb{A}^1} \circ L_{\text{Nis}}E)(K) \simeq |L_{\text{Nis}}E(K \times \mathbb{A}^\bullet)|.$$

But $E(K \times \mathbb{A}^n)$ is $(-n)$ -connective since K is a Nisnevich local ring, and as $\dim_{\text{Nis}}(\mathbb{A}_K^n) = \dim(\mathbb{A}_K^n = n)$, where $\dim_{\text{Nis}}$ denotes the homotopy dimension of the Nisnevich topos. So we conclude by Corollary 6.3.4. $\square$

6.4 The homotopy t -structure

Using the stable connectivity theorem, we are going to introduce a powerful tool on the category of $\mathbb{A}^1$ -invariant Nisnevich sheaves of spectra, the so-called *homotopy t -structure*. We start with a short reminder on t -structures.

6.4.1 t -Structure

Definition 6.4.1. Let $\mathcal{C}$ be a stable category. A t -structure on $\mathcal{C}$ consists of two full subcategories $\mathcal{C}_{\geq 0}, \mathcal{C}_{\leq 0} \hookrightarrow \mathcal{C}$ such that

- (i) For $X \in \mathcal{C}_{\geq 0}$ and $Y \in \mathcal{C}_{\leq 0}$ we have $\mathrm{Map}_{\mathcal{C}}(\Sigma X, Y) \simeq *$.
- (ii) The suspension functor restricts to an endofunctor of $\mathcal{C}_{\geq 0}$, while the inverse of the suspension functor restricts to an endofunctor of $\mathcal{C}_{\leq 0}$, that is $\Sigma \mathcal{C}_{\geq 0} \subseteq \mathcal{C}_{\geq 0}$ and $\Sigma^{-1} \mathcal{C}_{\leq 0} \subseteq \mathcal{C}_{\leq 0}$.
- (iii) For every $X \in \mathcal{C}$ there exists a cofiber sequence

$$A \rightarrow X \rightarrow B$$

such that $A \in \mathcal{C}_{\geq 0}$ and $\Sigma B \in \mathcal{C}_{\leq 0}$.

We follow the usual conventions and write $\mathcal{C}_{\geq n} := \Sigma^n \mathcal{C}_{\geq 0}$ and $\mathcal{C}_{\leq n} := \Sigma^n \mathcal{C}_{\leq 0}$. We call objects in $\mathcal{C}_{\geq 0}$ *connective*, and objects in $\mathcal{C}_{\leq 0}$ *coconnective*.

6.4.2 Fun-facts about t -structure

In this short section we gather some important facts about t -structures. The main reference for this is the original paper by Beilinson-Bernstein-Deligne-Gabber, see also [Lur17a, Section 1.2.1] for a modern point of view.

- We have that the connective objects in a t -structure are exactly the left orthogonal objects to the (-1) -coconnective ones, i.e. $\mathcal{C}_{\geq 0} \simeq \{X \in \mathcal{C} \mid \mathrm{Map}_{\mathcal{C}}(X, Y) \simeq 0 \forall Y \in \mathcal{C}_{\leq -1}\}$. Similarly $\mathcal{C}_{\leq 0} \simeq \{Y \in \mathcal{C} \mid \mathrm{Map}_{\mathcal{C}}(X, Y) \simeq 0 \forall X \in \mathcal{C}_{\geq 1}\}$.
- The inclusion $\mathcal{C}_{\geq 0} \hookrightarrow \mathcal{C}$ admits a right adjoint $\tau_{\geq 0}$, while the inclusion $\mathcal{C}_{\leq 0} \hookrightarrow \mathcal{C}$ admits a left adjoint $\tau_{\leq 0}$, and similarly for all other $n \in \mathbb{Z}$.
- For a fixed $X \in \mathcal{C}$, the composition

$$\tau_{\geq 0} X \rightarrow X \rightarrow \tau_{\leq -1} X$$

is canonically a cofiber sequence. In particular, the objects A and B from Definition 6.4.1 (iii) are uniquely determined as $A \simeq \tau_{\geq 0} X$ and $B \simeq \tau_{\leq -1} X$.

- The heart of a t -structure $\mathcal{C}^{\heartsuit} := \mathcal{C}_{\geq 0} \cap \mathcal{C}_{\leq 0}$ is an abelian 1-category.

Example 6.4.2. For $\mathcal{C} = \mathrm{Sp}$ we obtain a t -structure by defining

$$\begin{aligned} \mathrm{Sp}_{\geq 0} &:= \{E \mid \pi_i E = 0 \forall i < 0\}, \\ \mathrm{Sp}_{\leq 0} &:= \{E \mid \pi_i E = 0 \forall i > 0\}. \end{aligned}$$

In this case the zeroth homotopy group induces an equivalence $\pi_0 : \mathrm{Sp}^{\heartsuit} \xrightarrow{\cong} \mathrm{Ab}$ of the heart with the category of abelian groups.

It is worth pointing out that $E \in \mathrm{Sp}_{\leq -1}$ if and only if $\Omega^{\infty} E \simeq *$.

We now want to extend this example to the stabilization of a topos.

Proposition 6.4.3. Let $\mathcal{X}$ be a topos. Then the stabilization $\mathrm{Sp}(\mathcal{X})$ admits a t -structure with coconnective part given by

$$\mathrm{Sp}(\mathcal{X})_{\leq -1} := \{E \mid \Omega^{\infty} E \simeq *\}.$$

If $\mathcal{X}$ is further hypercomplete, we obtain a more explicit description of the t -structure as

$$\begin{aligned} \mathrm{Sp}(\mathcal{X})_{\geq 0} &:= \{E \mid \underline{\pi}_i(E) = 0 \forall i < 0\}, \\ \mathrm{Sp}(\mathcal{X})_{\leq 0} &:= \{E \mid \underline{\pi}_i(E) = 0 \forall i > 0\}. \end{aligned}$$

Proof. Formal arguments, which we elect not to elaborate on, show that the inclusion

$$\mathrm{Sp}(\mathcal{X})_{\leq -1} \hookrightarrow \mathrm{Sp}(\mathcal{X})$$

is accessible and closed under limits. Thus, by the adjoint functor theorem and presentability of all involved categories, there exists a left adjoint $X \mapsto X_{\leq -1}$.

We first show that precomposition with the unit of the above adjunction induces an injective morphism

$$[X_{\leq -1}X, \Sigma B] \hookrightarrow [X, \Sigma B]$$

for all $B \in \mathrm{Sp}(\mathcal{X})_{\leq -1}$.

Indeed, let $\eta : X_{\leq -1} \rightarrow \Sigma B$ be arbitrary. Consider the cofiber sequence

$$B \rightarrow A \rightarrow X_{\leq -1} \xrightarrow{\eta} \Sigma B.$$

Assume that the composition $X \rightarrow X_{\leq -1} \xrightarrow{\eta} \Sigma B$, with the first morphism the unit, is null. By the universal property, we obtain a factorization of the unit through A . But $\mathrm{Sp}(\mathcal{X})_{\leq -1}$ is closed under limits, hence $A \in \mathrm{Sp}(\mathcal{X})_{\leq -1}$. Hence we obtain a further factorization

$$X \longrightarrow X_{\leq -1} \xrightarrow{\quad} A \xrightarrow{\quad} X_{\leq -1} \xrightarrow{\eta} \Sigma B$$

$\searrow \text{id}$

It follows that already $\eta = 0$.

Now let $X \in \mathrm{Sp}(\mathcal{X})$ and set $F := \mathrm{fib}(X \rightarrow X_{\leq -1})$. Then for $B \in \mathrm{Sp}(\mathcal{X})_{\leq -1}$ consider the long exact sequence

$$[X_{\leq -1}, B] \xrightarrow{\cong} [X, B] \longrightarrow [F, B] \longrightarrow [X_{\leq -1}, \Sigma B] \hookrightarrow [X, \Sigma B]$$

Here the left most morphism is an equivalence by adjunction, while the right most morphism is injective by the discussion above. Thus, by exactness, $[F, B] = 0$. Using that

$$\pi_i \mathrm{Map}_{\mathcal{C}}(F, B) \cong \pi_0 \Omega^i \mathrm{Map}_{\mathcal{C}}(F, B) \cong \pi_0 \mathrm{Map}_{\mathcal{C}}(F, \Omega^i B)$$

and that $\omega^i B = \Sigma^{-i} B \in \mathrm{Sp}(\mathcal{X})_{\leq -1}$, we conclude $\mathrm{Map}_{\mathcal{C}}(F, B) \simeq *$.

We now define the connective part as

$$\mathrm{Sp}(\mathcal{X})_{\geq 0} := \{X \mid \mathrm{Map}_{\mathcal{C}}(X, B) \simeq * \forall B \in \mathrm{Sp}_{\leq -1}\}.$$

The verification of the axioms of a t -structure is a good exercise and is left for the reader. □

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6.4.3 The Homotopy t -structure

We now use the above discussion for topoi to descent the t -structure on the stabilization of the Nisnevich topos to motivic spaces.

Definition 6.4.4. Let k be a field. Let $\mathrm{SH}^{S^1}(k) \hookrightarrow \mathrm{Sp}(\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_k))$ denote the full subcategory of $\mathbb{A}^1$ -invariant sheaves of spectra. This is called the *category of motivic S^1 -spectra*.

Notice that by definition $\mathrm{SH}^{S^1}(k) \simeq \mathrm{Sp}(\mathrm{Spc}(k))$.

Proposition 6.4.5. *For k a field, the category of motivic S^1 -spectra admits a t -structure defined by*

$$\begin{aligned} \mathrm{SH}^{S^1}(k)_{\geq 0} &:= \{E \mid \pi_i(E) = 0 \forall i < 0\}, \\ \mathrm{SH}^{S^1}(k)_{\leq 0} &:= \{E \mid \pi_i(E) = 0 \forall i > 0\}. \end{aligned}$$

The functor $\pi_0(-) : \mathrm{SH}^{S^1}(k)^{\heartsuit} \rightarrow \mathrm{Ab}(\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_k)_{\leq 0})$ is fully faithful.

Its essential image consists of the so-called strictly $\mathbb{A}^1$ -invariant sheaves, those sheaves $F \in \mathrm{Ab}(\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_k)_{\leq 0})$ such that the Nisnevich cohomology $H_{\mathrm{Nis}}^i(-, F)$ with values in F is $\mathbb{A}^1$ -invariant for all i .

Proof. Consider the induced t -structure on the stabilization of the Nisnevich topos, see Proposition 6.4.3. We want to show, that this descends to a t -structure of motivic S^1 -spectra. So let $E \in \mathrm{Sp}(\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_k))$ be an $\mathbb{A}^1$ -invariant Nisnevich sheaf of spectra. As the E is $\mathbb{A}^1$ -invariant, the counit $E_{\geq 0} \rightarrow L_{\mathrm{mot}}E \in \mathrm{Sp}(\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_k))$ factors essentially uniquely as $E_{\geq 0} \rightarrow L_{\mathrm{mot}}(E_{\geq 0}) \rightarrow E$. Since L_{mot} preserves connective objects by Theorem 6.3.5, $L_{\mathrm{mot}}(E_{\geq 0})$ is again connective, and hence the induced morphism $L_{\mathrm{mot}}(E_{\geq 0}) \rightarrow E$ factors itself essentially uniquely over $E_{\geq 0}$. In summary, we have the following commutative diagram

$$\begin{array}{ccc} E_{\geq 0} & \xrightarrow{\quad} & E \\ \downarrow & \searrow \exists! & \uparrow \\ L_{\mathrm{mot}}(E_{\geq 0}) & \dashrightarrow \exists! & E_{\geq 0} \end{array}$$

It follows from the universal property of $E_{\geq 0}$ as the terminal connective object with a morphism to E , that the composition $E_{\geq 0} \rightarrow L_{\mathrm{mot}}(E_{\geq 0}) \rightarrow E_{\geq 0}$ is homotopic to the identity. In particular, $E_{\geq 0}$ is a retract of an $\mathbb{A}^1$ -invariant object and hence itself $\mathbb{A}^1$ -invariant².

As further $\mathbb{A}^1$ -local sheaves of spectra are closed under colimits by the explicit description of $L_{\mathbb{A}^1}$, see Proposition 4.2.10, the cofiber $E_{\leq -1} := \mathrm{cof}(E_{\geq 0} \rightarrow E)$ is also $\mathbb{A}^1$ -invariant. Thus the t -structure on the Nisnevich topos descends to a t -structure on motivic S^1 -spectra. The explicit description of the t -structure follows from the Nisnevich topos being hypercomplete, see Theorem 3.5.1 in combination with Proposition 6.4.3.

By the description of the heart in Proposition 6.4.3, we further get a fully faithful factorization

$$\begin{array}{ccc} \mathrm{SH}^{S^1}(k)^{\heartsuit} & \hookrightarrow & \mathrm{Ab}(\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_k)_{\leq 0}) \\ \downarrow & & \downarrow \\ \mathrm{SH}^{S^1}(k) & \hookrightarrow & \mathrm{Sp}(\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_k)) \end{array}$$

If now $F \in \mathrm{Ab}(\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_k)_{\leq 0})$, and if we denote by $HF \in \mathrm{Sp}(\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_k))$ the corresponding spectrum, then

$$\pi_i(HF(X)) \cong H_{\mathrm{Nis}}^{-i}(X, F)$$

for any smooth scheme $X \in \mathrm{Sm}_k$. Thus HF is $\mathbb{A}^1$ invariant if and only if F is strictly $\mathbb{A}^1$ -invariant. $\square$

6.5 Unstable connectivity

In this section we shortly review the (much harder to prove) unstable version of the connectivity theorem Theorem 6.3.5. We will not prove this in full and refer to [Bac24] for a complete treatment of the subject.

²Local objects are always stable under retract by the classification of local objects as those X for which $\mathrm{Map}(-, X)$ detects weak equivalences.

We fix for this section a perfect field k

Definition 6.5.1. A Nisnevich sheaf of groups $G \in \text{Grp}(\text{Shv}_{\text{Nis}}(\text{Sm}_k)_{\leq 0})$ is called *strongly $\mathbb{A}^1$ -invariant* if $H^i(-, G)$ is $\mathbb{A}^1$ -invariant for $i \in \{0, 1\}$.

Remark 6.5.2. A Nisnevich sheaf of groups G is $\mathbb{A}^1$ -invariant if and only if the delooping $BG \in \text{Shv}_{\text{Nis}}(\text{Sm}_k)$ is $\mathbb{A}^1$ -invariant.

reference
exercises

As it turns out, for abelian sheaves, the notion of strictly and strongly $\mathbb{A}^1$ -invariance are equivalent.

Theorem 6.5.3 (Morel). *If an abelian Nisnevich sheaf $A \in \text{Ab}(\text{Shv}_{\text{Nis}}(\text{Sm}_k)_{\leq 0})$ is strongly $\mathbb{A}^1$ -invariant, then A is strictly $\mathbb{A}^1$ -invariant.*

Furthermore, there is a rich supply on strongly $\mathbb{A}^1$ invariant sheaves.

Theorem 6.5.4 (Morel). *Let $X \in \text{Spc}(k)_*$ be a pointed motivic space. Then $\pi_1(X)$ is a strongly $\mathbb{A}^1$ -invariant sheaf.*

Both these theorems are extremely hard to prove. Thus we will only state them, and use them as a black box and instead discuss some of their consequences. For the proofs, see [Bac24].

Corollary 6.5.5. *Let $X \in \text{Spc}(k)_*$ be a pointed motivic space. Then $\pi_i(X)$ is strictly $\mathbb{A}^1$ -invariant for all $i \geq 2$.*

Proof. We have an equivalence $\pi_i(X) \cong \pi_1(\Omega^{i-1})$. Now the right hand side is strongly $\mathbb{A}^1$ -invariant by Theorem 6.5.4. As $\pi_i(X)$ is abelian for $i \geq 2$, they are also strictly $\mathbb{A}^1$ -invariant by Theorem 6.5.3. $\square$

Lemma 6.5.6. *Let $F \rightarrow E \rightarrow B \in \text{Shv}_{\text{Nis}}(\text{Sm}_k)$ be a fiber sequence such that B is connected and $\mathbb{A}^1$ -invariant. Then F is $\mathbb{A}^1$ invariant if and only if E is $\mathbb{A}^1$ -invariant.*

Proof. For simplicity we write $B'(X) := B(X \times \mathbb{A}^1)$ and consider this as a sheaf, similarly for E and F . Using that $\underline{\text{Hom}}(\mathbb{A}^1, -)$ preserves limits, with $\underline{\text{Hom}}$ the internal mapping sheaf in $\text{Shv}_{\text{Nis}}(\text{Sm}_k)$, and that $\underline{\text{Hom}}(\mathbb{A}^1, E) \simeq E'$, we obtain a morphism of fiber sequences

$$\begin{array}{ccccc} F' & \longrightarrow & E' & \longrightarrow & B' \\ \uparrow & & \uparrow & & \uparrow \simeq \\ F & \longrightarrow & E & \longrightarrow & B \end{array}$$

As the right most morphism is an equivalence, we can view the morphism $E \rightarrow E'$ as a morphism in $\text{Shv}_{\text{Nis}}(\text{Sm}_k)_{/B}$. As the above diagram is a morphism of fiber sequences, applying the functor $(* \rightarrow B)^* : \text{Shv}_{\text{Nis}}(\text{Sm}_k)_{/B} \rightarrow \text{Shv}_{\text{Nis}}(\text{Sm}_k)_{/*}$ to $E \rightarrow E'$ results in $F \rightarrow F'$. The claim now follows from the morphism $* \rightarrow B$ being an epimorphism, as B is connected, and since pulling back along an epimorphism reflects equivalences. $\square$

Proposition 6.5.7. *Let $E \in \text{Shv}_{\text{Nis}}(\text{Sm}_k)$ be a connected Nisnevich sheaf of spectra. Then E is $\mathbb{A}^1$ invariant if and only if $\pi_i(E)$ is strongly $\mathbb{A}^1$ -invariant for all $i \geq 1$.*

Proof. The “only if” follows similar to the proof of Corollary 6.5.5. For the “if” direction we show by induction that $\leq_n$ is $\mathbb{A}^1$ -invariant. By hypercompleteness, $E \simeq \lim_n E_{\leq n}$. As $\mathbb{A}^1$ -invariant objects, as the local objects for a given Bousfield-localization, are closed under limits, so is E .

For $n = 0$ we have $E_{\leq 0} \simeq *$ as E is connected, so the claim is true.

For $n > 0$ we consider the fiber sequence

$$K(\pi_{n+1}(E), n+1) \rightarrow E_{\leq n+1} \rightarrow E_{\leq n}.$$

As $E_{\leq n}$ is connected and $\mathbb{A}^1$ -invariant by induction, we conclude with Lemma 6.5.6. $\square$

Corollary 6.5.8. *Let $X \in \mathrm{Spc}(k)$ be connected. Then the truncation $\tau_{\leq n}^{\mathrm{Nis}} X$ is $\mathbb{A}^1$ -invariant for every $n \geq 0$ and hence descends to an endofunctor of connected motivic spaces.*

Proof. This follows directly from Proposition 6.5.7, as $\pi_i(\tau_{\leq n}^{\mathrm{Nis}} X) \simeq \pi_i(X)$ for $i \leq n$ and 0 else, and the $\mathbb{A}^1$ -invariance of X . $\square$

Due to this corollary we will from now on simply write $\tau_{\leq n} X$ or $X_{\leq n} \in \mathrm{Spc}(k)$.

Corollary 6.5.9. *Let $X \in \mathrm{Spc}(k)_*$ be a pointed, connected motivic space. Then the connected cover $\tau_{\geq n}^{\mathrm{Nis}} X$ is $\mathbb{A}^1$ -invariant for every $n \geq 1$ and hence descends to an endofunctor of connected motivic spaces.*

Proof. This follows directly from Proposition 6.5.7, as $\pi_i(\tau_{\geq n}^{\mathrm{Nis}} X) \simeq \pi_i(X)$ for $i \geq n$ and 0 else. $\square$

Due to this corollary we will from now on simply write $\tau_{\geq n} X$ or $X_{\geq n} \in \mathrm{Spc}(k)$.

We can now prove the unstable connectivity.

Proposition 6.5.10. *Let $X \in \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_k)$ be r -connective. Then the motivic localization $L_{\mathrm{mot}} X$ is also r -connective.*

Proof. The proof proceeds by induction, the case $r = 0$ being clear. Indeed, if X is 0-connective, that is non-empty, then the universal morphism $X \rightarrow L_{\mathrm{mot}} X$ shows that $L_{\mathrm{mot}} X$ cannot be empty. The case $r = 1$ was dealt with above in Proposition 6.1.4.

If $r > 1$, then $L_{\mathrm{mot}} X$ is connected by the $r = 1$ case, and $(L_{\mathrm{mot}} X)_{\geq r}$ is $\mathbb{A}^1$ -invariant by Corollary 6.5.9. Similar to the proof of Proposition 6.4.5, we obtain a commutative diagram

$$\begin{array}{ccc}
 X & \xrightarrow{\quad} & L_{\mathrm{mot}} X \\
 \downarrow & \searrow \exists! & \nearrow \\
 & (L_{\mathrm{mot}} X)_{\geq r} & \\
 \downarrow & \nearrow \exists! & \nearrow \mathrm{id} \\
 L_{\mathrm{mot}} X & &
 \end{array}$$

Here the upper left diagonal morphism exist because X is r -connective, while the lower diagonal morphism exists because $(L_{\mathrm{mot}} X)_{\geq r}$ is $\mathbb{A}^1$ -invariant.

Hence $L_{\mathrm{mot}} X$ is a retract of an r -connective sheaf, and thereby itself connective. $\square$

6.6 Proof of Gabber's lemma

The main ingredient for the proofs above was Gabber's lemma Theorem 6.3.1. In this section we want to give a proof of this theorem. We will not prove it in full generality, but only a slightly weaker form, and also only over an infinite field k . This will anyways suffice for our applications. What we will prove is the following

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Theorem 6.6.1 (weak Gabber's Lemma). *Let k be an infinite field, $X \in \mathrm{Sm}_k$ a smooth scheme, $Z \hookrightarrow X$ a not-necessarily closed subscheme which is everywhere of positive codimension. Let $z \in Z$. Then there exists*

- (1) an étale neighborhood $U \subseteq X$ of z ;
- (2) an étale morphism $\varphi : X' \rightarrow \mathbb{A}_S^1$, where S is a smooth affine scheme over k .

They further fulfill the following properties.

- (a) The composition $Z' := Z \cap U \hookrightarrow U \xrightarrow{\varphi} \mathbb{A}_S^1 \rightarrow S$ is finite.

(b) The restriction $\varphi|_{Z'} : Z' \rightarrow \mathbb{A}_S^1$ is a closed immersion.

(c) Z' is saturated under φ , i.e. $Z' \xrightarrow{\cong} \varphi^{-1}(\varphi(Z'))$.

We are only interested in the local behavior around z , hence we may assume that X is an affine smooth connected scheme over k of dimension d . As smooth implies of finite type by definition, we obtain a closed embedding $X \hookrightarrow \mathbb{A}^N$ for some N big enough. We want to construct U étale over $\mathbb{A}_S^1$ for some smooth affine S . We will in fact make it étale over $\mathbb{A}^d$. To do so, we start by making it smooth over $\mathbb{A}^{d-1}$ and then later add an additional factor making it étale.

Consider the variety of all linear maps $\mathbb{A}^N \rightarrow \mathbb{A}^{d-1}$. This space is defined by $P = \mathbb{A}^{N \times (d-1)}$, being parametrized by $N \times (d-1)$ matrices. The space P has two nice properties

- Given two open, non-empty subsets $V_1, V_2 \subseteq P$, then their intersection $V_1 \cap V_2$ is again non-empty and open, i.e. P is irreducible. This follows from $\mathbb{A}^{N \times (d-1)}$ being an integral scheme.
- If $V \subseteq P$ is open and non-empty, then it admits a rational point, that is, $V(k)$ is non-empty. This uses that k is infinite.

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Thus to show that there exists a morphism satisfying properties $P_1, \dots, P_n$, then it suffices to check that every subspace in P of morphisms satisfying P_i contains an open and non-empty subspace in P .

We start by defining the following two sets

- $V_1 := \{\varphi : \mathbb{A}^N \rightarrow \mathbb{A}^{d-1} | \varphi|_Z \text{ is quasi-finite at } z\}$;
- $V_2 := \{\varphi : \mathbb{A}^N \rightarrow \mathbb{A}^{d-1} | \varphi|_X \text{ is smooth at } z\}$.

We want to show that both these subsets are open and non-empty. To simplify, notice that, as the étale and the quasi-finite loci in the definitions of V_1 and V_2 are open, we may replace z with a closed specialization $\bar{z}$ of z .

Lemma 6.6.2. *The subset $V_1 \subseteq P$ is open and non-empty.*

Proof. We start by showing that V_1 is open. For this, consider the universal projection

$$\begin{array}{ccccc} & & \mathbb{A}^{d-1} \times P & & \\ & \nearrow \varphi^{\text{un}} & \uparrow & \searrow & \\ Z \times P & \hookrightarrow & \mathbb{A}^N \times P & \twoheadrightarrow & P \end{array}$$

Here the morphism $\mathbb{A}^N \times P \rightarrow \mathbb{A}^{d-1} \times P$ is given by evaluation on the first component and identity on the second, that is, $\mathbb{A}^N \times P \ni (x, \varphi) \mapsto (\varphi(x), \varphi)$. This explains the name universal projection: basechanging the above diagram along a specified morphism $\varphi : * \rightarrow P$ gives back the specified morphism $\varphi : \mathbb{A}^N \rightarrow \mathbb{A}^{d-1}$.

Consider now the quasi-finite locus $U \subseteq Z \times P$ of φ^{un} . This is open, see [Sta25, Tag 01TI]. Thus $(\bar{z} \times P) \cap U$ is open in $\bar{z} \times P$. As $\bar{z}$ is closed, it corresponds to a finite field extension of k . In particular, $\bar{z} \rightarrow \text{Spec } k$ is a finite flat morphism. It follows that $\bar{z} \times P \rightarrow P$ is also finitely generated and flat, and thereby open. The image of $(\bar{z} \times P) \cap U$ is therefore open in U and consists of those linear maps $\varphi \in P$ which are quasi-finite at $\bar{z}$, which is exactly V_1 . Thus V_1 is an open subset of P .

Now we need to show that this subset is non-empty. For this we use the following fact: Let $W \subseteq \mathbb{A}^N$ be a closed subscheme of dimension $r > 0$. Let $w \in W$ be a closed point. Then there exists a hyperplane $H \subseteq \mathbb{A}^N$ such that $\dim(H + w) \cap W < r$.

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Assume first that $\bar{z}$ has positive dimension. If we then apply the fact to the pair $(Z, \bar{z})$, we obtain a hyperplane H_1 . If $\dim(Z \cap H_1) > 0$, we can apply the fact again to the tuple $((Z \cap H_1) + \bar{z}, \bar{z})$, thereby obtaining another hyperplane H_2 . Iterating this construction, we obtain a family of hyperplanes $(H_i)_{1 \leq i \leq n}$ for some $0 \leq n < d-1$, where we now also allow $\bar{z}$ to have dimension 0 again, in which case this is the empty family. Fill up this family with arbitrary hyperplanes in general position, until we have a family of cardinality $d-1$.

Let now $\varphi_i : \mathbb{A}^N \rightarrow \mathbb{A}^1$ be the projections onto these hyperplanes, that is, $\varphi_i : \mathbb{A}^N \rightarrow \mathbb{A}^1$ is a linear map with kernel H_i . Write $\varphi : \mathbb{A}^N \rightarrow \mathbb{A}^{d-1}$ for the product of these maps. Then

$$\varphi^{-1}(\varphi(\bar{z})) \cap Z = Z \cap \bigcap_i (H_i + \bar{z})$$

has dimension 0 by construction, thus $\bar{z}$ is isolated in its fiber and therefor quasi-finite at $\bar{z}$, showing that V_1 is non-empty. $\square$

Lemma 6.6.3. *The subset $V_2 \subseteq P$ is open and non-empty.*

Proof. We again start by showing that V_2 is open. Indeed, since X and $\mathbb{A}^{d-1}$ are smooth over $\text{Spec} k$, that φ is smooth at $\bar{z}$ is equivalent to the differential on the cotangent spaces being injective. This condition can be expressed by the non-vanishing of certain minors of the Jacobian, which is a polynomial inequality, hence in particular Zariski open.

Thus it suffices to show that V_2 is non-empty. To do so, we may base change to the algebraic closure $\bar{k}$ of k . Indeed, $V_2 \times_{\text{Spec} k} \text{Spec} \bar{k}$ admits a canonical morphism to V_2 , so if the base change is non-empty, neither is V_2 .

Base changing might introduce another problem though, $\bar{z}$ might not be rational over $\bar{k}$, instead $\bar{z} \times_{\text{Spec} k} \text{Spec} \bar{k}$ might consist of multiple points. In fact, the number of points will be given by the separability degree of $\kappa(\bar{z})$ over k . So we need to prove smoothness at several points. However, being smooth at any of these points is again an open condition by what we have already proven, so we may consider one point at a time. Thus we may assume $\bar{z}$ is a rational point.

Now consider the cotangent complex $\Omega_{\mathbb{A}^N/\bar{k}, \bar{z}}^1 \cong \bar{k}^N$. As $X \hookrightarrow \mathbb{A}^N$ is a closed immersion, by the second fundamental sequence we obtain a surjection

$$\Omega_{\mathbb{A}^N/\bar{k}, \bar{z}}^1 \rightarrow \Omega_{X/\bar{k}, \bar{z}}^1.$$

Pick coordinates $v_1, \dots, v_d \in k^N$ generating $\Omega_{X/\bar{k}, \bar{z}}^1$ under the above surjection. Define

$$\varphi : \mathbb{A}^N \rightarrow \mathbb{A}^{d-1},$$

given row-wise by v_1 to v_{d-1} . Then, by construction, the differential $d\varphi^* : \Omega_{\mathbb{A}^{d-1}/\bar{k}, \bar{z}}^1 \rightarrow \Omega_{\mathbb{A}^N/\bar{k}, \bar{z}}^1$ satisfies $d\varphi^*(e_i) = v_i$, and hence, the composition $\Omega_{\mathbb{A}^{d-1}/\bar{k}, \bar{z}}^1 \xrightarrow{d\varphi^*} \Omega_{\mathbb{A}^N/\bar{k}, \bar{z}}^1 \twoheadrightarrow \Omega_{X/\bar{k}, \bar{z}}^1$ is injective, showing that $\varphi|_X : X \rightarrow \mathbb{A}^{d-1}$ is smooth at $\bar{z}$, hence $\varphi \in V_2$, finishing the claim. $\square$

Let us summarize what we did so far. We have now found morphisms

$$\begin{array}{ccccc} z \in Z & \hookrightarrow & X & \hookrightarrow & \mathbb{A}^N \\ & & & & \downarrow \varphi \\ & & & & \mathbb{A}^{d-1} \end{array}$$

such that the composition $Z \rightarrow \mathbb{A}^{d-1}$ is quasi finite at z , while the composition $X \rightarrow \mathbb{A}^{d-1}$ is smooth at z . We write $s \in \mathbb{A}^{d-1}$ for the image of z under the above composition.

find a reference why this is always possible over an infinite field

We now use the next fact:

In the setting from above, there exists an open neighborhood U of z in X and a morphism $f : U \rightarrow \mathbb{A}^1$ such that

- $z \rightarrow U_s := U \times_{\mathbb{A}^{d-1}} \{s\} \xrightarrow{f} \mathbb{A}_s^1$ is universally injective;
- $(f, \varphi|_U) : U \rightarrow \mathbb{A}^1 \times \mathbb{A}^{d-1}$ is étale at z .

Using this fact there exists an open neighborhood $U' \subseteq U$ of z such that $U' \rightarrow \mathbb{A}^d$ is in fact étale. Set $\tilde{S} = (\mathbb{A}^{d-1})_s^h$ and consider the following pullback

$$\begin{array}{ccccccc}
 z & \xrightarrow{\quad} & \tilde{Z} \cap U' & \hookrightarrow & U' & \xrightarrow{(f, \varphi)} & \mathbb{A}^1 \times \mathbb{A}^{d-1} \\
 & & & & & & \downarrow \text{pr} \\
 & & & & & & \mathbb{A}^{d-1} \\
 & \searrow & & & & & \swarrow \\
 & & s & & & & \tilde{S}
 \end{array}
 \quad
 \begin{array}{ccccc}
 z \in \tilde{Z} & \hookrightarrow & \tilde{U} & \xrightarrow{(f, \varphi)} & \mathbb{A}^1 \times \tilde{S} \\
 & & \downarrow \text{ét} & & \downarrow \text{pr} \\
 & & \tilde{S} & & \tilde{S}
 \end{array}$$

q.f. at z

Where we basechanged the left diagram via the morphism $\tilde{S} \rightarrow \mathbb{A}^{d-1}$. The lower dashed morphism is obtained from the fact that the henselisation does not change the residue field, and hence the point lifts to $\tilde{S}$. This in turn induces the upper dashed morphism, that is, z lifts to $\tilde{Z}$. The idea is now to solve the problem in the henselized world, and then use a filtered colimit argument to reduce back to the case we are interested in.

To solve the problem on the right hand side, notice that, since $\tilde{S}$ is henselian, by Proposition 3.1.4 (iii)

$$\tilde{Z} \cong Z_1 \amalg \dots \amalg Z_r \amalg W$$

where the Z_i are local and finite over $\tilde{S}$, while W is not quasi-finite at any point over $\tilde{S}$. In particular, as quasi-finite morphisms are stable under basechange, see [Sta25, Tag 01TM], we must have $z \in Z_i$ for some i , say $i = 1$.

Let now $\tilde{\tilde{U}} := \tilde{U} \setminus (Z_2 \amalg \dots \amalg Z_r \amalg W)$ be the open complement of $Z_2 \amalg \dots \amalg Z_r \amalg W$ in $\tilde{U}$. Then $z \in \tilde{\tilde{U}}$ and the morphism $\tilde{\tilde{U}} \rightarrow \mathbb{A}_S^1$ is étale as the restriction of an étale morphism to an open subscheme, while $\tilde{\tilde{U}} \cap \tilde{Z} = Z_1 \rightarrow \tilde{S}$ is finite. Thus, we have achieved almost everything we want, at least in the henselian world. The only thing left is to show that $\tilde{Z} \rightarrow \mathbb{A}_S^1$ is in fact a closed immersion, that is, we must show that this morphism induces a surjection $\mathcal{O}(\mathbb{A}_S^1) \rightarrow \mathcal{O}(Z_1)$. As Z is finite over $\tilde{S}$, by Nakayama's lemma it suffices to check this in the special fiber, that is, after basechanging to s . In this case, we have the following diagram

$$\begin{array}{ccc}
 Z & \longleftarrow & F \\
 \downarrow & & \downarrow \\
 \tilde{\tilde{U}} & \longleftarrow & \tilde{\tilde{U}}_s \\
 \downarrow & & \downarrow \\
 \mathbb{A}_S^1 & \longleftarrow & \mathbb{A}_s^1 \\
 \downarrow & & \downarrow \\
 \tilde{S} & \longleftarrow & s
 \end{array}$$

Then, as Z_1 is finite over $\tilde{S}$, its fiber must contain only the closed points of Z_1 . But Z_1 was local, hence there is only one closed point, and we conclude that $|F| = \{z\}$ as topological

spaces. Also, as $F \hookrightarrow \tilde{U}_s$ and $\tilde{U}_s \rightarrow \mathbb{A}_s^1$ are both unramified, so is their composition $F \rightarrow \mathbb{A}_s^1$. The morphism is furthermore universally injective, as the basechange of a universally injective morphism, and hence a closed immersion.

Let us summarize again. We have achieved the following diagram

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$$\begin{array}{ccc}
 Z_1 & \hookrightarrow & \tilde{U} \\
 \searrow & & \downarrow \\
 & & \mathbb{A}_{\tilde{S}}^1 \\
 \searrow \text{finite} & & \downarrow \\
 & & \tilde{S}
 \end{array}$$

But now, $\tilde{S} = \lim V$, where the limit is indexed over all étale neighborhoods of s in $\mathbb{A}^{d-1}$. As all schemes in the above diagram are finitely presented over $\tilde{S}$, and the properties of the morphisms are well behaved, the diagram as well as all the properties of the involved morphisms are already realized at some finite stage of the indexing category. That is there exists some étale neighborhood V_1 of s in $\mathbb{A}^{d-1}$ such that the following diagram commutes

$$\begin{array}{ccccc}
 & & \xrightarrow{\quad} & & \\
 Z_{U_0} & \hookrightarrow & U_0 & \xrightarrow{\quad} & \mathbb{A}_{V_1}^1 \\
 \uparrow & & \searrow & & \downarrow \\
 z & & & \xrightarrow{\text{fin}} & X_{V_1} \longrightarrow V_1 \\
 & \searrow & & \downarrow & \downarrow \\
 & & Z & \hookrightarrow & X \longrightarrow \mathbb{A}^d
 \end{array}$$

This finishes the claim by setting $U = U_0$ and $S = V_1$.

Chapter 7

$\mathbb{P}^1$ -Spectra

Let S be a qcqs basescheme. In algebraic geometry, there is more the one notion of a sphere, and in the motivic world, there are even more possibilities. For one we have the $S^1 \in \mathrm{Spc}(S)$ as the constant presheave on the topological sphere. But then there is also $\mathbb{G}_m$ and $\mathcal{P}^1$, both of which could be interpreted as algebraic analogs of the topological sphere. So there is no reason that, if we want to build a stable motivic homotopy theory, why we should only invert the topological sphere. In fact, as $\mathbb{P}^1 \simeq S^1 \otimes \mathbb{G}_m \in \mathrm{Spc}(S)$, inverting the projective plain seems to be a good idea, as in this case, we automatically invert all the sphere mentioned above. This leads to the notion of $\mathbb{P}^1$ -spectra.

7.1 $\mathbb{P}^1$ -spectra and S^1 -spectra

To invert $\mathbb{P}^1$, we note that the functor $(-)\otimes\mathbb{P}^1 : \mathrm{Spc}(S)_* \rightarrow \mathrm{Spc}(S)_*$ admits a right adjoint $\Omega_{\mathbb{P}^1}$.

Definition 7.1.1. We define the *motivic stable homotopy category* as the limit

$$\mathrm{SH}(S) := \lim(\dots \xrightarrow{\Omega_{\mathbb{P}^1}} \mathrm{Spc}(S)_\bullet \xrightarrow{\Omega_{\mathbb{P}^1}} \mathrm{Spc}(S)).$$

That is, a motivic spectrum $E \in \mathrm{SH}(S)$ is a sequence $(E^0, E^1, \dots)$ of pointed motivic spaces $E^i \in \mathrm{Spc}(S)_*$ together with equivalences $\omega^i : E^i \xrightarrow{\simeq} \Omega_{\mathbb{P}^1} E^{i+1}$.

Remark 7.1.2. Notice that since, limits in Cat over a diagram of presentable categories with right adjoints are equivalent to the colimit in Pr^L , $\mathrm{SH}(S)$ is again presentable, and we obtain an adjunction

$$\Sigma^\infty : \mathrm{Spc}(S)_* \xrightleftharpoons[\perp]{} \mathrm{SH}(S) : \Omega^\infty$$

where the right adjoint is given as the projection onto the last entry in the limit diagram, hence obviously preserves limits and is accessible, inducing the left adjoint by the adjoint functor theorem.

details

We now want to make precise what said above that “inverting $\mathbb{P}^1$ inverts all mentioned spheres”. As already mentioned, $\mathbb{P}^1 \simeq S^1 \otimes \mathbb{G}_m$. This follows from the cocartesian square of schemes

$$\begin{array}{ccc} \mathbb{G}_m & \xrightarrow{t \mapsto t} & \mathbb{A}^1 \\ t \mapsto t^{-1} \downarrow & & \downarrow \\ \mathbb{A}^1 & \longrightarrow & \mathbb{P}^1 \end{array}$$

considered in $\mathrm{Spc}(S)$. In particular $(-)\otimes\mathbb{P}^1 \simeq (-)\otimes S^1 \circ (-)\otimes\mathbb{G}_m \simeq (-)\otimes\mathbb{G}_m \circ (-)\otimes S^1$. In particular, we obtain an equivalence of the corresponding right adjoints $\Omega_{\mathbb{P}^1} \simeq \Omega_{S^1} \circ \Omega_{\mathbb{G}_m} \simeq$

$\Omega_{\mathbb{G}_m} \circ \Omega_{S^1}$. Consider now the following $\mathbb{N}^{\text{op}} \times \mathbb{N}^{\text{op}}$ indexed diagram

$$\begin{array}{ccccc}
& \text{Spc}(S)_* & \xleftarrow{\Omega_{S^1}} & \text{Spc}(S)_* & \xleftarrow{\Omega_{S^1}} & \text{Spc}(S)_* \\
\Omega_{\mathbb{G}_m} \uparrow & & & \Omega_{\mathbb{G}_m} \uparrow & & \\
& \text{Spc}(S)_* & \xleftarrow{\Omega_{S^1}} & \text{Spc}(S)_* & \xleftarrow{\Omega_{S^1}} & \dots \\
\Omega_{\mathbb{G}_m} \uparrow & & & \Omega_{\mathbb{G}_m} \uparrow & & \\
& \vdots & & \vdots & &
\end{array}$$

We may compute the limit of this diagram along any cofinal subdiagram. Taking the diagonal, we see that the limit of the above diagram is given by $\text{SH}(S)$. However, calculating the whole limit and using that limits commute, so that we can first take the horizontal limits, we see, that $\text{SH}(S)$ is equivalent to the limit

$$\lim(\dots \text{SH}^{S^1} \xrightarrow{\Omega_{\mathbb{G}_m}} \text{SH}^{S^1} \xrightarrow{\Omega_{\mathbb{G}_m}} \text{SH}^{S^1})$$

where $\Omega_{\mathbb{G}_m}$ is the induced functor on the limit, which is on objects explicitly given by sending an S^1 -spectrum $E^* \in \text{SH}^{S^1}(S)$ to the S^1 -spectrum $\Omega_{\mathbb{G}_m} E^*$ given component wise by $(\Omega_{\mathbb{G}_m} E)^*$.

7.2 The $\mathbb{G}_m$ -loop functor over a field

Let us study the $\Omega_{\mathbb{G}_m}$ functor in a bit more detail. For this, the following lemma will be helpful.

Lemma 7.2.1. *Let $A, B \in \text{Spc}(S)_*$ be two motivic spaces. Then there exists a stable splitting*

$$\Sigma^\infty(A \times B) \simeq \Sigma^\infty A \oplus \Sigma^\infty B \oplus \Sigma^\infty(A \wedge B).$$

Proof. Let $A, B \in \text{Spc}(k)_*$ be two pointed motivic spaces. We have the following morphisms

$$\begin{array}{ccccc}
& & & & A \\
& & \nearrow \text{id} \vee * & & \uparrow \\
A \vee B & \longrightarrow & A \times B & & \\
& \searrow * \vee \text{id} & & \searrow & \\
& & & & B
\end{array}$$

Notice, that $A \vee B$ is the coproduct in pointed motivic spaces, and that Σ^∞ preserves coproducts as a left adjoint. Thus, by stabilizing, that is applying Σ^∞ , we obtain composition

$$\Sigma^\infty(A \vee B) \simeq \Sigma^\infty A \oplus \Sigma^\infty B \rightarrow \Sigma^\infty(A \times B) \rightarrow \Sigma^\infty A \oplus \Sigma^\infty B.$$

For the last morphism we used the universal property of the biproduct, to add the two maps to $\Sigma^\infty A$ and $\Sigma^\infty B$. Thus, by construction, the composit above beomes the identity, and we have found a splitting of the inclusion $\Sigma^\infty A \oplus \Sigma^\infty B \rightarrow \Sigma^\infty(A \times B)$. Applying this splitting to the stabilization of the cofiber sequence

$$A \vee B \longrightarrow A \times B \longrightarrow A \wedge B$$

gives the claim. $\square$

Lemma 7.2.2. *Let $S = \mathrm{Spec} k$ be the spectrum of a field. Then $\Omega_{\mathbb{G}_m}$ commutes with truncations $\tau_{\leq 0}$, and connective covers $\tau_{\geq 0}$, that is $\Omega_{\mathbb{G}_m}$ is t -exact.*

Proof. Let $E \in \mathrm{SH}^{S^1}(k)$ be an S^1 -spectrum. Applying $\Omega_{\mathbb{G}_m}$ to the fiber sequence defining connected covers and truncations, and using that $\Omega_{\mathbb{G}_m}$ preserves limits as a right adjoint, we obtain the fiber sequence

$$\Omega_{\mathbb{G}_m}(E_{\geq 0}) \rightarrow \Omega_{\mathbb{G}_m} E \rightarrow \Omega_{\mathbb{G}_m}(E_{\leq -1}).$$

By the uniqueness of the connected-coconnective decomposition in a t -structure, it suffices to prove that

- (a) $\Omega_{\mathbb{G}_m}(E_{\geq 0}) \in \mathrm{SH}^{S^1}(k)_{\geq 0}$ is connective;
- (b) $\Omega_{\mathbb{G}_m}(E_{\leq -1}) \in \mathrm{SH}^{S^1}(k)_{\leq -1}$ is (-1) -coconnective.

For (b), let $E \in \mathrm{SH}^{S^1}(k)_{\leq -1}$, that is $\Omega^\infty E \simeq *$. But, as $\Omega_{\mathbb{G}_m}$ on $\mathrm{SH}^{S^1}(k)$ is applied component wise

$$\Omega_{\mathbb{G}_m} E \simeq (\underbrace{\Omega_{\mathbb{G}_m} E^0}_{\simeq *}, \Omega_{\mathbb{G}_m} E^1, \dots).$$

Here we used that $\Omega_{\mathbb{G}_m}$, as a right adjoint, preserves limits. Hence $\Omega^\infty(\Omega_{\mathbb{G}_m} E) \simeq \Omega_{\mathbb{G}_m} E^0 \simeq *$, that is $\Omega_{\mathbb{G}_m} E \in \mathrm{SH}^{S^1}(k)_{\leq -1}$.

We now turn to (a), that is we want to show that if $E \in \mathrm{SH}^{S^1}(k)_{\geq 0}$, then $\pi_i(\Omega_{\mathbb{G}_m} E) = 0$ for all $i < 0$. By unramifiedness, it suffices to check that $\pi_i((\Omega_{\mathbb{G}_m} E)(K)) = 0$ for all finitely generated separable field extensions K/k and for all $i < 0$. We have the following chain of isomorphisms

$$\pi_i((\Omega_{\mathbb{G}_m} E)(K)) \cong \pi_0((\Omega_{\mathbb{G}_m} E^{-i})(K)) \cong \pi_0 E^{-i}(\mathbb{G}_m \wedge \mathrm{Spec} K)$$

Here, the first equivalence is given by the fact that $\pi_i(-) \cong \pi_0(\Omega_{S^1}(-))$, while the second isomorphism is induced by the Yoneda lemma and the following chain of equivalences

$$\begin{aligned} \Omega_{\mathbb{G}_m} E^{-i}(\mathrm{Spec} K) &\simeq \mathrm{Map}_{\mathrm{Spd}(k)_*}(\mathrm{Spec} K_+, \Omega_{\mathbb{G}_m} E^{-i}) \\ &\simeq \mathrm{Map}_{\mathrm{Spd}(k)_*}(\mathbb{G}_m \wedge \mathrm{Spec} K_+, E^{-i}) \\ &\simeq E^{-i}(\mathbb{G}_m \wedge \mathrm{Spec} K). \end{aligned}$$

In particular, we can rewrite, using the $(\Sigma^\infty, \Omega^\infty)$

$$\begin{aligned} \pi_i((\Omega_{\mathbb{G}_m} E)(K)) &\cong [\Sigma^\infty(\mathbb{G}_m \wedge \mathrm{Spec} K_+), \Sigma^{-i} E] \subseteq [\Sigma^\infty(\mathbb{G}_m \times \mathrm{Spec} K_+), \Sigma^{-i} E] \\ &\cong \pi_i E(\mathbb{G}_m \times \mathrm{Spec} K_+). \end{aligned}$$

For the inclusion we used 7.2.1.

Now, since $\dim(\mathbb{G}_{m,K})$ is an affine scheme of dimension 1, $\pi_i E(\mathbb{G}_m \times \mathrm{Spec} K_+) = 0$ for all $i < -1$ using 3.5.1. Thus all that is left to understand is the first negative homotopy group. For this consider the cofiber sequence

$$E_{\geq 1} \longrightarrow E \longrightarrow E_{=0}$$

Notice that the cofiber is concentrated in degree 0, as E was assumed to be connective. Using that $\pi_i(\Omega_{\mathbb{G}_m}(E_{\geq 1})) = 0$ for all $i \leq -1$, as we took the connective cover and $\Omega_{\mathbb{G}_m}$ can at most drop connectivity by one, we see from the induced long exact sequence on homotopy groups that

$$\pi_{-1}(\Omega_{\mathbb{G}_m} E) \simeq \pi_{-1}(\Omega_{\mathbb{G}_m}(E_{=0})).$$

reference

As $E_{=0}$ is in the heart, it suffices to prove that if we consider $A \in \mathrm{SH}^{S^1}(k)^\heartsuit \hookrightarrow \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_k, \mathrm{Ab})$, then

$$H_{\mathrm{Nis}}^1(\mathbb{G}_m \wedge \mathrm{Spec} K_+, A) = 0.$$

Using the suspension isomorphism, this is equivalent to

$$H_{\mathrm{Nis}}^2(\Sigma(\mathbb{G}_m \wedge \mathrm{Spec} K_+), A) = 0.$$

But $\Sigma(\mathbb{G}_m \wedge \mathrm{Spec} K_+) \simeq S^1 \wedge \mathbb{G}_m \wedge \mathrm{Spec} K_+ \simeq \mathbb{P}^1 \wedge \mathrm{Spec} K_+$, and $\mathbb{P}^1$ has dimension 1, implying the claim again by 3.5.1. $\square$

7.3 The homotopy t -structure

Using that $\Omega_{\mathbb{G}_m}$ is t -exact, we can now equip $\mathrm{SH}(k)$ with a t -structure, by descending the one from $\mathrm{SH}^{S^1}(k)$. For this, let $E \in \mathrm{SH}(k) = \lim(\dots \xrightarrow{\Omega_{\mathbb{G}_m}} \mathrm{SH}^{S^1}(k) \xrightarrow{\Omega_{\mathbb{G}_m}} \mathrm{SH}^{S^1}(k))$ and think of this as a sequence $(E^0, E^1, \dots)$ of S^1 -spectra with given equivalences $\omega^i : E^i \xrightarrow{\simeq} \Omega_{\mathbb{G}_m} E^{i+1}$. Notice that we can extend the sequence to the left by setting $E^{-i} := \Omega_{\mathbb{G}_m}^i E^0$ for all $i < 0$.

We now set $\pi_i(E)_j := \pi_i(E^j)$.

Definition 7.3.1. Let k be a field. We define the subcategory of connective and coconnective motivic spectra as the full subcategories spanned by

$$\begin{aligned} \mathrm{SH}(k)_{\geq 0} &:= \{E \mid \pi_i(E)_* = 0 \forall i < 0\}, \\ \mathrm{SH}(k)_{\leq 0} &:= \{E \mid \pi_i(E)_* = 0 \forall i > 0\}. \end{aligned}$$

As we will see in a second, they constitute the connective and coconnective part of a t -structure. But to describe the heart of the t -structure, we need to define a new category.

Definition 7.3.2. Let k be a field. We define the *category of homotopy modules* $\mathrm{HI}_*(k)$ as the 1-category where the objects are sequences $(A_i)_i \in \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_k, \mathrm{Ab})^{\mathbb{Z}}$ together with equivalences $\Omega_{\mathbb{G}_m} A_{i+1} \cong A_i$ and such that each A_i is strictly $\mathbb{A}^1$ -invariant. The morphisms are morphisms of sequences which are compatible with the isomorphisms.

Here we use that $\Omega_{|\mathbb{G}_m}$ is t -exact and hence restricts to the heart $\mathrm{SH}^{S^1}(k)^\heartsuit$. In fact, one can characterize $\Omega_{\mathbb{G}_m} A \in \mathrm{SH}^{S^1}(k)^\heartsuit$ for some $A \in \mathrm{SH}^{S^1}(k)^\heartsuit$ as the sheaf associating to every $X \in \mathrm{Sm}_k$ the direct summand of $A(X \times \mathbb{G}_m)$, which is the complement of the retract

$$A(X) \xrightarrow{p^*} A(X \times \mathbb{G}_m) \xrightarrow{i^*} A(X)$$

Remark 7.3.3. Traditionally, $\Omega_{\mathbb{G}_m} A$ was called the *retraction* of A and denoted by $\Omega_{\mathbb{G}_m} A := A_{-1}$. Furthermore, one defines inductively $A_{-(n+1)} := (A_{-n})_{-1}$. This curious notation stems from the fact that for a homotopy module $A_* \in \mathrm{HI}_*(k)$, then $(A_{n+1})_{-1} \cong A_n$.

Theorem 7.3.4. The two full subcategories $\mathrm{SH}(k)_{\geq 0}$ and $\mathrm{SH}(k)_{\leq 0}$ of $\mathrm{SH}(k)$ from Definition 7.3.1 constitute the connective and coconnective part of a t -structure, and $\pi_0(-)_*$ upgrades to an equivalence $\mathrm{SH}(k)^\heartsuit \simeq \mathrm{HI}_*(k)$.

Proof. As $\Omega_{\mathbb{G}_m}$ is t -exact by Lemma 7.2.2 we obtain $\pi_0(\Omega_{\mathbb{G}_m} E) \cong \Omega_{\mathbb{G}_m}(\pi_0(E))$. Thus, $\pi_0(-)_*$ upgrades to a functor $\mathrm{SH}(k) \rightarrow \mathrm{HI}_*(k)$. However, we have

$$\begin{aligned} \mathrm{SH}(k)^\heartsuit &:= \mathrm{SH}(k)_{\geq 0} \cap \mathrm{SH}(k)_{\leq 0} \simeq \lim(\dots \xrightarrow{\Omega_{\mathbb{G}_m}} \mathrm{SH}(k)^\heartsuit \xrightarrow{\Omega_{\mathbb{G}_m}} \mathrm{SH}(k)^\heartsuit) \\ &\simeq \mathrm{HI}_*(k), \end{aligned}$$

where the last morphism is induced by $\pi_0(-) : \mathrm{SH}^{S^1}(k)^\heartsuit \simeq \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_k, \mathrm{Ab})$. In particular, $\pi_0(-)_* : \mathrm{SH}(k) \rightarrow \mathrm{HI}_*(k)$ restricts to an equivalence on the heart.

Now, for $E = (E^0, E^1, \dots) \in \mathrm{SH}(k)$, where each $E^i \in \mathrm{SH}^{S^1}(k)$ is an S^1 -spectrum, together with its equivalences $\omega^i : E^i \xrightarrow{\simeq} \Omega_{\mathbb{G}_m} E^{i+1}$, we obtain a new $\mathbb{P}^1$ -spectrum $\bar{E} = (E_{\geq 0}^0, E_{\geq 0}^1)$ with equivalences $(\omega^i)_{\geq 0} : E_{\geq 0}^i \xrightarrow{\simeq} (\Omega_{\mathbb{G}_m} E^{i+1})_{\geq 0} \simeq \Omega_{\mathbb{G}_m}(E_{\geq 0}^{i+1})$. This is exactly the functor induced by applying component wise the connective cover functor and using the t -exactness of $\Omega_{\mathbb{G}_m}$.

We obtain a cofiber sequence

$$\bar{E} \rightarrow E \rightarrow \bar{\bar{E}}.$$

But this cofiber sequence can be calculated component wise, so that $\bar{\bar{E}} \in \mathrm{SH}(k)_{\leq -1}$. This shows the existence of the cofiber sequence from Definition 6.4.1 (iii). As clearly by the above descriptions $\Sigma \mathrm{SH}(k)_{\geq 0} \subseteq \mathrm{SH}(k)_{\geq 0}$ and $\Sigma^{-1} \mathrm{SH}(k)_{\leq 0} \subseteq \mathrm{SH}(k)_{\leq 0}$ it remains to show that for $E \in \mathrm{SH}(k)_{\geq 0}$ and $F \in \mathrm{SH}(k)_{\leq -1}$ that $\mathrm{Map}_{\mathrm{SH}(k)}(E, F) \simeq *$. However, using the description of $\mathrm{SH}(k)$ as a limit, we obtain

$$\mathrm{Map}_{\mathrm{SH}(k)}(E, F) \simeq \lim(\dots \xrightarrow{\Omega_{\mathbb{G}_m}} \mathrm{Map}_{\mathrm{SH}^{S^1}}(E^1, F^1) \xrightarrow{\Omega_{\mathbb{G}_m}} \mathrm{Map}_{\mathrm{SH}^{S^1}}(E^0, F^0)).$$

As all $E^i \in \mathrm{SH}^{S^1}(k)_{\geq 0}$ are connective and all $F^i \in \mathrm{SH}^{S^1}(k)_{\leq -1}$ are coconnective, each entry in the above sequential limit is contractible. Hence the limit is contractible, fishing the claim. $\square$

Remark 7.3.5. Even though this may for some look like a good, accessible theory, it is already quite complicated to calculate only the zeroth homotopy group of the unit $\pi_0(\Sigma^\infty S^0)$.

7.4 Category of spans

Our next goal is to equip $\mathrm{SH}(k)$ with the structure of a tensor product, and further to show, that it satisfies a universal property as such. For this, we first want to explain what the correct notion of a “symmetric monoidal category” is. The most important notion for this will be the those of a “category of spans”. This acts, in a specific sense which we will made precise below, as a blueprint for commutative monoids.

Remark 7.4.1. Remember that the partially ordered sets $[n] \in \mathrm{Cat}$ define a cosimplicial category $[\bullet] : \Delta \rightarrow \mathrm{Cat}$. This yields a simplicial anima

$$\begin{aligned} \mathrm{Cat} &\rightarrow \mathrm{Fun}(\Delta^{\mathrm{op}}, \mathrm{Ani}), \\ \mathcal{C} &\mapsto (n \mapsto \mathrm{Map}_{\mathrm{Cat}}([n], \mathcal{C})). \end{aligned}$$

The above functor is known as the *Rezk-nerve*. It resembles quite closely the classical nerve construction from discrete categories to simplicial sets, as the following theorem shows. We explain the content of the theorem below.

Theorem 7.4.2. *This functor is fully faithful with essential image consisting of the complete Segal spaces, those simplicial anima $X_\bullet \in \mathcal{P}(\Delta)$ such that*

(1) *the morphisms*

$$[1] \amalg_{[0]} [1] \amalg_{[0]} \dots \amalg_{[0]} [1] \rightarrow [n]$$

induce equivalences

$$X_n \xrightarrow{\simeq} X_1 \times_{X_0} X_1 \times_{X_0} \dots \times_{X_0} X_1.$$

This is known as the “Segal condition”.

reference

- (2) the zeroth degenerate morphism induces an isomorphism to the space of equivalences from Definition 7.4.3, i.e. there exists a commutative diagram

$$\begin{array}{ccc} X_{\text{equiv}} & \hookrightarrow & X_1 \\ \uparrow \simeq & \nearrow s_0 & \\ X_0 & & \end{array}$$

This is known as “completeness”.

Proof. This is [JT07]. □

So, what is going on? To understand the Segal condition better, at least on a heuristic level, let us see how this appears naturally on the image of the Rezk-nerve. Consider $\mathcal{C} \in \text{Cat}$ and let $X_\bullet := \text{Map}_{\text{Cat}}([\bullet], \mathcal{C})$ for the image of the $\mathcal{C}$ under the Rezk-nerve. Then we have the following

- $X_0 = \text{Map}_{\text{Cat}}([0], \mathcal{C}) \simeq \text{Map}_{\text{Cat}}(*, \mathcal{C}) \simeq \mathcal{C}^\simeq$ are exactly the objects of $\mathcal{C}$ together with the possible ways on how we can identify those objects.
- $X_1 = \text{Map}_{\text{Cat}}([1], \mathcal{C}) \text{Ar}(\mathcal{C})$ are exactly the morphisms in $\mathcal{C}$. Notice in particular that there is a pullback diagram

$$\begin{array}{ccc} \text{Map}_{\mathcal{C}}(c, d) & \longrightarrow & \text{Map}_{\text{Cat}}([1], \mathcal{C}) \\ \downarrow & \lrcorner & \downarrow (\text{ev}_0, \text{ev}_1) \\ * & \xrightarrow{(c, d)} & \mathcal{C}^\simeq \times \mathcal{C}^\simeq \end{array}$$

- $X_2 = \text{Map}_{\text{Cat}}([2], \mathcal{C})$ is the set of all possible composites in $\mathcal{C}$. Given $y \in \mathcal{C}$, the space of all morphisms factoring through y is given as

$$\begin{array}{ccc} P & \longrightarrow & \text{Map}_{\text{Cat}}([1], \mathcal{C}) \times \text{Map}_{\text{Cat}}([1], \mathcal{C}) \\ \downarrow & \lrcorner & \downarrow \text{ev}_0 \times \text{ev}_1 \\ * & \xrightarrow{(y, y)} & \mathcal{C}^\simeq \times \mathcal{C}^\simeq \end{array}$$

By [CCNW26, Lemma 1.3.18], this is the same as the pullback $\text{Map}_{\text{Cat}}([1], \mathcal{C}) \times_{\text{ev}_0, \{y\}, \text{ev}_1} \text{Map}_{\text{Cat}}([1], \mathcal{C})$. Taking the colimit over $\mathcal{C}^\simeq$ and using universality of colimits in anima, we see that the space of all compositions is exactly

$$\text{Map}_{\text{Cat}}([1], \mathcal{C}) \times_{\mathcal{C}^\simeq} \text{Map}_{\text{Cat}}([1], \mathcal{C}) \simeq X_1 \times_{X_0} X_1.$$

Thus we see that the Segal conditions encodes precisely, that an n -simplex in the simplicial anima Map is uniquely determined by giving n composable morphisms, or equivalently, that categories have the extension property for the spin inclusion, see [Lur25, Example 1.5.7.7].

Let us now dissect the completeness property. For this, assume we are given any simplicial anima $X_\bullet \in \mathcal{P}(\Delta)$ satisfying the Segal condition. Then we can define a discrete category $hX_\bullet$ with objects $\pi_0 X_0$ and such, that the morphism set between $a, b \in \pi_0 X_0$ is given by

$$\text{Hom}_{hX_\bullet}(a, b) := \pi_0(\{a\} \times_{X_0, d^1} X_1 \times_{d^0, X_0} \{b\}).$$

Composition is then induced from the span

$$X_1 \times_{X_0} \times X_1 \xleftarrow{\simeq} X_2 \xrightarrow{d^1} X_1.$$

Definition 7.4.3. Let $X_\bullet \in \mathcal{P}(\Delta)$ be a Segal space, that is a simplicial anima satisfying the Segal condition. We define the *space of equivalences* X_{equiv} of $X_\bullet$ as the pullback

$$\begin{array}{ccc} X_{\text{equiv}} & \xrightarrow{\quad} & X_1 \\ \downarrow & & \downarrow \pi_0 \\ \text{Map}_{\text{Cat}}(BS^1, hX_\bullet) & \hookrightarrow & \text{Map}_{\text{Cat}}([1], hX_\bullet) \end{array}$$

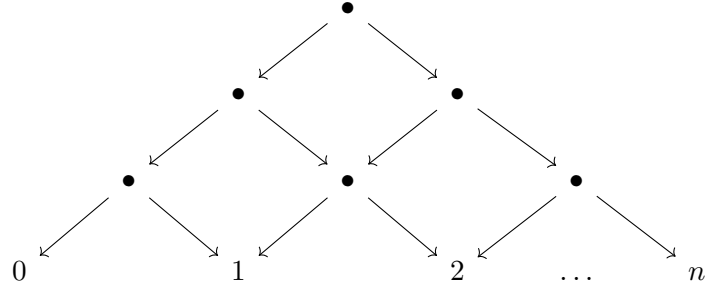
That is, an object in X_{equiv} is a 1-simplex in the Segal space $X_\bullet$ which induces an isomorphism on the homotopy category.

With this discussion in mind, a complete Segal space is a simplicial anima for which we can define a homotopy category, for which giving an n -simplex is equivalent to giving n -composable 1-simplices, and for which every equivalence is equivalent to an identity morphism, i.e. a morphism in the image of the degenerate morphism $s^0 : X_0 \rightarrow X_1$, something which is most certainly true for categories. The interesting part about Theorem 7.4.2 is, that this completely characterizes categories inside the the category of simplicial Anima.

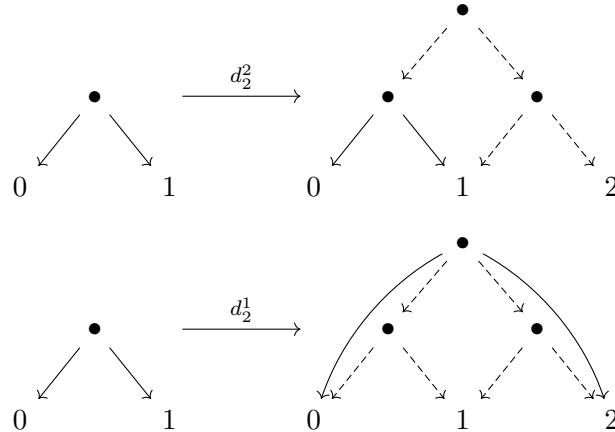
Remark 7.4.4. Notice that we are being lax here regarding set theoretical difficulties. What the described above only applies, in this form, to small categories. One could choose universes to circumvent this problem, but we elect to not discuss this subtleties in more detail, effectively ignoring these problems.

In our goal to define symmetric monoidal categories, we will now describe an explicit example of a a simplicial anima, which in the end will define us the category of spans. Our main reference for this is [HHLN23]. The explicit construction has to attributed to Barwick, see [Bar17].

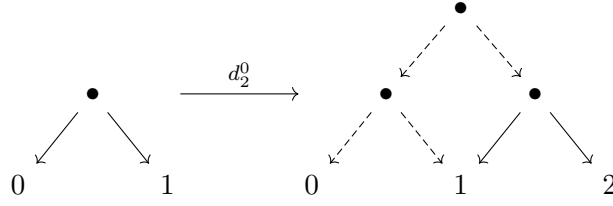
Define now a cosimplicial discrete category $[\tilde{\bullet}] : \text{Fun}(\Delta \rightarrow \text{Cat})$ as follows. For the value at $n \in \Delta$, we set the partially ordered set



We obtain the face and degeneracy morphisms by the respective inclusion and collapsing of spans. For example, the morphisms $d_2^{0,1,2} : 1 \rightarrow 2$ are given by the inclusion at the solid lines



and



The degeneracy morphisms are similarly given by collapsing order preserving some spans. Let now $\mathcal{C}$ be a category with pullbacks. We define $\text{Span}(\mathcal{C})_\bullet \in \mathcal{P}(\Delta)$ as the sub-simplicial anima

$$\text{Span}(\mathcal{C})_\bullet \hookrightarrow \text{Map}_{\text{Cat}}([\tilde{n}], \mathcal{C})$$

consisting of those diagrams where all squares are pullbacks. One can check that this defines a complete Segal anima, see [HHLN23, Theorem 2.13], and hence, by Theorem 7.4.2, a category $\text{Span}(\mathcal{C})$. What does this category look like? This can be inferred from Theorem 7.4.2. We will not explain, how one derives these formulas, but only state their explicit forms.

- We first have that the objects of $\text{Span}(\mathcal{C})$ are given as $\text{ob}(\text{Span}(\mathcal{C})) = \text{Span}(\mathcal{C})_0 \simeq \mathcal{C}^\simeq = \text{Ob}(\mathcal{C})$.
- Morphisms from x to y in $\text{Span}(\mathcal{C})$ are defined via the pullback

$$\begin{array}{ccc} \text{Map}_{\text{Span}(\mathcal{C})}(x, y) & \longrightarrow & \text{Span}(\mathcal{C})_1 \\ \downarrow & & \downarrow (d^1, d^0) \\ * & \xrightarrow{(x, y)} & \mathcal{C}^\simeq \times \mathcal{C}^\simeq \end{array}$$

That is, a morphism from x to y is given by a third object T together with two morphisms $x \leftarrow T \rightarrow y$.

- Composition is given by the span $\text{Span}(\mathcal{C})_1 \times_{\text{Span}(\mathcal{C})_0} \text{Span}(\mathcal{C})_1 \xleftarrow{\simeq} \text{Span}(\mathcal{C})_2 \xrightarrow{d^1} \text{Span}(\mathcal{C})_1$. Explicitly

$$(y \leftarrow T \rightarrow z) \circ (x \leftarrow U \rightarrow y) = (x \leftarrow T \times_y U \rightarrow z).$$

That is, the composition is defined via the pullback.

We are in particular interested in one incarnation of the span category, the one where $\mathcal{C} = \text{Fin}$ is the category of finite sets. In this case, $\text{Span}(\text{Fin})$ is a $(2, 1)$ -category, whose objects are given by finite sets, whose morphism 1-groupoid $\text{Map}_{\text{Span}(\text{Fin})}(X, Y)$ has as objects spans $(X \leftarrow T \rightarrow Y)$, and as morphisms isomorphisms $T \rightarrow T'$ compatible with the legs to X and Y , and whose composition is given by pullback.

Example 7.4.5. We as endomorphisms of the one-point finite set $\text{Map}_{\text{Span}(\text{Fin})}(*, *) \simeq \text{Fin}^\simeq \simeq \coprod_n B\Sigma_n$.

The category $\text{Span}(\text{Fin})$ has a few very convenient properties. For example, it has finite coproducts, and the coproduct object is just the coproduct object of the underlying finite sets, see [Cno26, Lemma 12.2.9]. Furthermore, there is an equivalence

$$\text{Span}(\text{Fin})^{\text{op}} \simeq \text{Span}(\text{Fin}),$$

given effectively by interchanging the legs, see [HHLN23, Lemma 2.14]. In particular, $\text{Span}(\text{Fin})$ also has finite products, and they agree with the finite coproducts, a phenomenon which is usually dubbed “semiadditivity”, see [Cno26, Definition 3.2.15]. Another

nice property is the fact, that we can embed Fin^{op} inside $\text{Span}(\text{Fin})$ as the backwards morphisms, that is, there is a functor

$$\text{Fin}^{\text{op}} \hookrightarrow \text{Span}(\text{Fin})$$

which is the identity on objects and sends a morphism $X \rightarrow Y \in \text{Fin}$ to the span $(Y \leftarrow X \rightrightarrows X)$, see [HHLN23, Proposition 2.15].

7.5 Symmetric monoidal categories

We will now use the semiadditivity property of $\text{Span}(\text{Fin})$ to define symmetric monoidal categories. The idea is, that the $\text{Span}(\text{Fin})$ behaves like the “universal commutative monoid on one object”, this one object being the one point set $*$. We will explain this slogan in more detail below.

Definition 7.5.1. Let $\mathcal{C}$ be a category admitting all finite products. We then define the *category* $\text{CMon}(\mathcal{C})$ of *commutative monoids in $\mathcal{C}$* as the category of product preserving functors from $\text{Span}(\text{Fin})$ to $\mathcal{C}$, i.e.

$$\text{CMon}(\mathcal{C}) := \text{Fun}^\times(\text{Span}(\text{Fin}), \mathcal{C}).$$

So, a commutative monoid in $\mathcal{C}$ is $X^\otimes : \text{Span}(\text{Fin}) \rightarrow \mathcal{C}$, such that $X^\otimes(A \amalg B) \xrightarrow{\cong} X^\otimes(A) \times X^\otimes(B)$ and $X^\otimes(\emptyset) \simeq *$.

Let us discuss why this encodes heuristically the intuitive notion of a commutative monoid as an object $X \in \mathcal{C}$ together with an unital, associative, and commutative multiplication. So let $X^\otimes$ be a commutative monoid in some category $\mathcal{C}$. Then $X^\otimes$ picks out a special element X , given as the value of $X^\otimes$ on the one point set $*$. what makes this element special is, that by the product preserving property of $X^\otimes$, we have that $X^\otimes(A) \simeq X^\otimes(\amalg_A *) \simeq \prod_A X^\otimes(*) = \prod_A X$. Furthermore we have the codiagonal

$$\nabla : * \amalg * \rightarrow * \in \text{Span}(\text{Fin}),$$

which, in this case, is the unique morphism into the terminal object. The codiagonal induces a morphism

$$m : X \times X \simeq X^\otimes(* \amalg *) \xrightarrow{X^\otimes(\nabla)} X^\otimes(*) = X.$$

Notice that there is a commutative diagram

$$\begin{array}{ccc} * \amalg * & \xrightarrow{\text{switch}} & * \amalg * \\ & \searrow \nabla & \downarrow \nabla \\ & & * \end{array}$$

where switch is the morphism permuting both factors. The existence of this morphism uses semiadditivity.

Commutativity of this diagram in $\text{Span}(\text{Fin})$ now induces a commutative diagram

$$\begin{array}{ccc} X \times X & \xrightarrow{\text{switch}} & X \times X \\ & \searrow m & \downarrow m \\ & & X \end{array}$$

showing that the morphism m is symmetric in both components. Similarly, one shows that m is associative and unital, where the unit is given by the value of $X^\otimes$ on the empty set. Thus, m is a unital, associative and commutative multiplication, which is, by design, is furthermore coherent.

Remark 7.5.2. We note that this is not the definition of commutative monoids from [Lur17b, Definition 2.4.2.1]. However, as it turns out, both definitions are equivalent, see [BH21, Appendix C.1].

Definition 7.5.3. A *symmetric monoidal category* $\mathcal{C}$ is a commutative monoid in the category Cat . They assemble into the category of symmetric monoidal categories $\text{CMon}(\text{Cat})$.

Proposition 7.5.4. Let $\mathcal{C} \in \text{Cat}$ be a category admitting finite products. Then there exists a unique symmetric monoidal structure on $\mathcal{C}$ such that $a \otimes b \simeq a \times b$.

Proof. This is [Lur17b, Proposition 2.1.4.5], see also [Cno26, Section 16.3]. $\square$

Proof. Let $\mathcal{C} \in \text{Pr}^{\text{L}}$ be a pointed category. Then there exists a canonical “smash product” symmetric monoidal structure, i.e. $a \otimes b \simeq (a \times b)/(a \vee b)$. $\square$

Proof. This is [Lur17b, Remark 4.8.2.14], see also [Cno26, Definition 17.3.2]. $\square$

Having symmetric monoidal categories, we want to define commutative algebras inside a symmetric monoidal category. For this, let $\mathcal{C}^{\otimes}$ be a symmetric monoidal category and denote by $\int \mathcal{C}^{\otimes} \rightarrow \text{Span}(\text{Fin})$ the cocartesian fibration classifying the functor $\mathcal{C}^{\otimes}$.

Definition 7.5.5. Let $\mathcal{C}^{\otimes}$ be a symmetric monoidal category. We define the *category* $\text{CAlg}(\mathcal{C})$ of commutative algebras in $\mathcal{C}$ as the full subcategory

$$\text{CAlg}(\mathcal{C}) \hookrightarrow \text{Sect}(\int \mathcal{C}^{\otimes}) := \text{Fun}_{\text{Span}(\text{Fin})}(\text{Span}(\text{Fin}), \int \mathcal{C}^{\otimes})$$

of those sections which are cocartesian over backwards morphism, that is, those section $A : \text{Span}(\text{Fin}) \rightarrow \int \mathcal{C}^{\otimes}$ of the canonical projection $\int \mathcal{C}^{\otimes}$, which, when restricted along $\text{Fin}^{\text{op}} \hookrightarrow \text{Span}(\text{Fin})$ are cocartesian sections.

Being a section cocartesian over backward morphisms can be essentially be summarized as being a section $A : \text{Span}(\text{Fin}) \rightarrow \int \mathcal{C}^{\otimes}$ of the canonical projection $\int \mathcal{C}^{\otimes}$, such that

$$A(Y \xleftarrow{X} \rightrightarrows X) : A(Y) \xrightarrow{\simeq} A(X) \in \mathcal{C}^{\otimes}(X).$$

Indeed, it is a cocartesian morphism inside the same fiber.

To understand the meaning of this definition, first recall that the objects of $\int \mathcal{C}^{\otimes}$ are those tuples (S, E) where $S \in \text{Span}(\text{Fin})$ and $E \in \mathcal{C}^{\otimes}(S) \simeq \coprod_S \mathcal{C}$, where we write $\mathcal{C} := \mathcal{C}^{\otimes}(*)$. The morphisms between two tuples (S, E) and (S', E') are given by

$$\text{Map}_{\int \mathcal{C}^{\otimes}}((S, E), (S', E')) := \{(f, \alpha) | f : S \rightarrow S' \in \text{Span}(\text{Fin}), \alpha : \mathcal{C}^{\otimes}(f)(E) \rightarrow E'\}.$$

A section $A : \text{Span}(\text{Fin}) \rightarrow \int \mathcal{C}^{\otimes}$ now assigns to every $S \in \text{Span}(\text{Fin})$ an object $A(S) \in \mathcal{C}^{\otimes}(S)$. Under the equivalence $\mathcal{C}^{\otimes}(S) \simeq \coprod_S \mathcal{C}$, this further corresponds to a sequence $\{A(S)_i\}_{i \in S}$. The section A further assigns to every $f : S \rightarrow S'$ a morphism $A(f) : \mathcal{C}^{\otimes}(f)A(S) \rightarrow A(S') \in \mathcal{C}^{\otimes}(S')$, and infinitely many compatibilities.

Let $S \in \text{Span}(\text{Fin})$ be a finite set. Then, the condition to be cocartesian over backward morphisms, applied to $(* \xleftarrow{S} \rightrightarrows S)$ shows that $A(S)_i \simeq A(*)$ for all i , that is, $\{A(S)_i\}_{i \in S} \simeq \{A(*)\}_{i \in S} = \{A\}_{i \in S}$. The key point is now, that the morphism induced by the fold morphism $\nabla : * \amalg * \rightarrow *$ is a multiplication, indeed

$$m = A(\nabla : * \amalg *) : A \otimes A \simeq \mathcal{C}^{\otimes}(\nabla)(A, A) \rightarrow A.$$

One proves similar that the multiplication for $\mathcal{C}^{\otimes}$, that this multiplication is unital, associative, and commutative in a coherent manner.

Example 7.5.6. One can show that the monoidal unit $\mathbb{1}(S) := \mathcal{C}^\otimes(\emptyset \hookrightarrow S)(*)$ always defines a commutative algebra. For a reason, see [Cno26, Paragraph beneath Proposition 18.1.14].

Example 7.5.7. Another example comes from the theory of presentable categories. Let $\mathcal{C}, \mathcal{D} \in \mathbf{Pr}^{\mathbf{L}}$. We “define” a functor $F \in \mathcal{P}((\mathbf{Pr}^{\mathbf{L}})^{\mathbf{op}})$ by setting

$$\mathcal{E} \mapsto F(\mathcal{E}) \subseteq \mathrm{Map}_{\mathrm{Cat}}(\mathcal{C} \times \mathcal{D}, \mathcal{E})$$

given by those functors which preserve colimits in each variable. This is representable by a presentable category $\mathcal{C} \otimes \mathcal{D} \in \mathbf{Pr}^{\mathbf{L}}$ called the Lurie-tensor product.

Consider now the product preserving functor $\mathrm{Cat}^\otimes : \mathrm{Span}(\mathrm{Fin}) \rightarrow \hat{\mathrm{Cat}}$ classifying the cartesian monoidal structure on the category of categories, see Proposition 7.5.4. Define a subcategory $X \hookrightarrow \int \mathcal{C}^\otimes \rightarrow \mathrm{Span}(\mathrm{Fin})$ as follows:

- The objects are tuple $\{(\mathcal{C}_1, \dots, \mathcal{C}_n) | \mathcal{C}_i \in \mathbf{Pr}^{\mathbf{L}} \ \forall i\}$.
- The morphism spaces $\mathrm{Map}_X((S, \mathcal{C}_\bullet), (T, \mathcal{D}_\bullet)) \hookrightarrow \mathrm{Map}_{\int \mathcal{C}^\otimes}((S, \mathcal{C}_\bullet), (T, \mathcal{D}_\bullet))$ consisting of the connected components of those functors which preserve colimits in each variable separately as appropriate, that is

$$\mathrm{Map}_{\mathbf{Pr}^{\mathbf{L}}}(\mathcal{C}_1 \otimes \mathcal{C}_2, \mathcal{D}) \simeq \mathrm{Map}_X((\ast \amalg \ast, (\mathcal{C}_1, \mathcal{C}_2)), (\ast, \mathcal{D})).$$

It turns out that the composition $X \rightarrow \mathrm{Span}(\mathrm{Fin})$ defines a cocartesian fibration and hence correspondence to a symmetric monoidal structure on $\mathbf{Pr}^{\mathbf{L}}$.

One can show that the functor $\mathrm{CAlg}(\mathbf{Pr}^{\mathbf{L}}) \rightarrow \mathrm{CMon}(\mathrm{Cat})$, induced by postcomposition with the inclusion $\mathbf{Pr}^{\mathbf{L}} \rightarrow \mathrm{Cat}$, is fully faithful with essential image given by presentably symmetric monoidal categories, those symmetric monoidal categories $\mathcal{C}^\otimes$ such that the underlying category $\mathcal{C} := \mathcal{C}^\otimes(\ast)$ is presentable and such that the tensor product $\otimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$ preserves colimits in each variable.

Furthermore, one can show that Ani is the tensor unit of $\otimes$ in $\mathbf{Pr}^{\mathbf{L}}$. The algebra structure on Ani coming from Example 7.5.6 is the cartesian monoidal structure.

For proofs and details, see [Lur17b, Section 4.8].

We will from now on often abbreviate a symmetric monoidal category simply by writing its underlying category, leaving the tensor structure implicate.

Definition 7.5.8. A symmetric monoidal functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is a morphism in $\mathrm{CMon}(\mathrm{Cat})$, that is a natural transformation

$$F^\otimes : \mathcal{C}^\otimes \rightarrow \mathcal{D} \in \mathrm{Fun}^\times(\mathrm{Span}(\mathrm{Fin}), \mathrm{Cat}).$$

Alternatively, a symmetric monoidal functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is a morphism of cocartesian fibrations $F : \int \mathcal{C}^\otimes \rightarrow \int \mathcal{D}^\otimes$ over $\mathrm{Span}(\mathrm{Fin})$.

A lax symmetric monoidal functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is a morphism $F : \int \mathcal{C}^\otimes \rightarrow \int \mathcal{D}^\otimes$ over $\mathrm{Span}(\mathrm{Fin})$ which sends cocartesian lifts of backwards morphisms to cocartesian lifts, i.e. F is cocartesian over backwards morphism.

Example 7.5.9. Let $\ast \in \mathrm{Cat}$ be the terminal category. Then the canonical functor $\ast : \mathrm{Span}(\mathrm{Fin}) \rightarrow \mathrm{Cat}$ defines a symmetric monoidal structure on $\ast \in \mathrm{Cat}$. As $\int \ast \simeq \mathrm{Span}(\mathrm{Fin})$ we get that the corresponding cocartesian fibration is given by $\int \ast \simeq \mathrm{Span}(\mathrm{Fin}) \xrightarrow{\mathrm{id}} \mathrm{Span}(\mathrm{Fin})$, so that a lax symmetric monoidal functor $\ast \rightarrow \mathcal{C}$ is the same thing as a commutative algebra in $\mathcal{C}$.

7.6 Relative adjunctions and symmetric monoidal localization

To describe the universal property of $\text{SH}(S)$ as a symmetric monoidal category, we need to talk about symmetric monoidal localization, which in turn needs the notion of a relative adjunction.

Definition 7.6.1. Consider the following commutative diagram in Cat

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{F} & \mathcal{D} \\ & \searrow p \quad \swarrow q & \\ & \mathcal{E} & \end{array}$$

Consider further a functor $G : \mathcal{D} \rightarrow \mathcal{C}$, as well as a natural transformation $\varepsilon : FG \rightarrow \text{id}_{\mathcal{D}}$. We say that ε *exhibits G as a right adjoint of F relative to $\mathcal{E}$* , or *exhibits G as a relative right adjoint to F* , if ε is the counit of an absolute adjunction, and ε induces a natural equivalence

$$q\varepsilon : pG \xrightarrow{\cong} q.$$

Lemma 7.6.2. Consider the setting from Definition 7.6.1. Assume further, that p and q be cocartesian fibrations. Then

- (1) F admits a relative right adjoint if and only if F preserves cocartesian edges, and F admits fiberwise a right adjoint, that is, for all $e \in \mathcal{E}$, the induced functor $F_e : \mathcal{C}_e \rightarrow \mathcal{D}_e$ admits a right adjoint.
- (1) G admits a relative left adjoint if and only if G admits fiberwise a left adjoint, and for all morphisms $f : e \rightarrow e' \in \mathcal{E}$ the natural transformation $F_{e'}f_* \rightarrow f_*F_e$ is an equivalence. Here the natural map is given by the composition

$$\begin{array}{ccc} F_{e'}f_* & \dashrightarrow & f_*F_e \\ u_{e'} \downarrow & & \uparrow c_e \\ F_{e'}f_*G_eF_e & \longrightarrow & F_{e'}G_{e'}f_*F_e \end{array}$$

where the left vertical morphism is the unit at e' , the lower horizontal morphism is the lax naturality of G as a morphism of cocartesian fibrations, and the right vertical morphism is the counit at e .

Proof. This is [Lur17b, Proposition 7.3.2.6 and Proposition 7.3.2.4]. $\square$

Corollary 7.6.3. Let $F^\otimes : \mathcal{C}^\otimes \rightarrow \mathcal{D}^\otimes$ be a symmetric monoidal functor and suppose that the functor $F : \mathcal{C} \rightarrow \mathcal{D}$ induced by $F^\otimes$ on underlying categories, admits a right adjoint G . Then G admits a canonical lax symmetric monoidal structure.

Proof. The claim can be translated into the statement, that $\int F : \int \mathcal{C}^\otimes \rightarrow \int \mathcal{D}^\otimes$ admits a right adjoint relative to $\text{Span}(\text{Fin})$. By Lemma 7.6.2, we only need to check that for all finite sets $S \in \text{Span}(\text{Fin})$, the functor induced on fibers $F_S : \mathcal{C}_S \rightarrow \mathcal{D}_S$ admits a right adjoint. But the fibers are equivalent to the S -fold product, that is to $F^S : \mathcal{C}^S \rightarrow \mathcal{D}^S$. This admits a right adjoint G^S by assumption. $\square$

We can now introduce symmetric monoidal localizations.

Proposition 7.6.4. Let $\mathcal{C}$ be a presentable symmetric monoidal category and $W \in \text{Ar}(\mathcal{C})$ be a class of morphisms of small generation with strong saturation¹ $\bar{W}$. Suppose that for

¹That is, $\bar{W}$ is the closure of W under two out of three and small colimits in $\text{Ar}(\mathcal{C}) = \text{Fun}(\Delta^1, W)$, and contains all equivalences of $\mathcal{C}$.

all $\alpha : Y \rightarrow Z \in W$ and for all $X \in \mathcal{C}$ we also have that $\text{id}_X \otimes \alpha : X \otimes Y \rightarrow X \otimes Z \in \bar{W}$. Then $\mathcal{C}[W^{-1}]$ inherits a canonical symmetric monoidal structure such that the localization $\mathcal{C} \rightarrow \mathcal{C}[W^{-1}]$ becomes symmetric monoidal.

Proof. We denote by $L \int \mathcal{C}^\otimes \hookrightarrow \int \mathcal{C}^\otimes$ the full subcategory generated by the objects $(S, (E_s)_{s \in S})$ such that for all $s \in S$ we have $E_s \in LC := \mathcal{C}[W^{-1}] \subseteq \mathcal{C}$. We want to show that the composition $L \int \mathcal{C}^\otimes \hookrightarrow \int \mathcal{C}^\otimes \rightarrow \text{Span}(\text{Fin})$ is a cocartesian fibration.

Let us first discuss an example, consider $\nabla : \{1, 2\} \cong * \amalg * \rightarrow *$, and consider the object in the fiber given by $(\{1, 2\}, (E_1, E_2)) \in L \int \mathcal{C}^\otimes$. We need to find an object F in the fiber over $*$, that is in LC , such that for any morphism $f : (\{1, 2\}, (E_1, E_2)) \rightarrow (S, (G_s)_{s \in S})$, with $(S, (G_s)_{s \in S})$ in the fiber over S , and every commutative diagram

$$\begin{array}{ccc} & & * \\ & \nearrow \nabla & \downarrow \\ \{1, 2\} & \longrightarrow & S \end{array}$$

such that f covers $\{1, 2\} \rightarrow S$, there exists an essentially unique morphism $(*, F) \rightarrow (S, (G_s)_{s \in S})$. But $\{1, 2\} \rightarrow S$ factors through $\{1, 2\} \rightarrow *$. Thus it suffices to show that for any $T \in LC$ there exists an equivalence

$$\text{Map}_{\mathcal{C}}(E_1 \otimes E_2, T) \simeq \text{Map}_{LC}(F, T).$$

But $LC \hookrightarrow \mathcal{C}$ admits a left adjoint $L : \mathcal{C} \rightarrow LC$. Thus we can take $F = L(E_1 \otimes E_2)$.

Thus we obtain a symmetric monoidal structure on LC . Our next goal is to construct the symmetric monoidal localization functor. For this, remember that the inclusion $(LC)^S \hookrightarrow \mathcal{C}^S$ also admits a left adjoint L^S . Let now $\alpha : S \rightarrow S' \in \text{Span}(\text{Fin})$ be a general morphism. We claim that for $E \in \mathcal{C}^S$ we have

$$L^{S'} \mathcal{C}^\otimes(\alpha)(E) \simeq (LC^\otimes)(\alpha)(L^S E) \simeq L^{S'} \mathcal{C}^\otimes(\alpha)(L^S E).$$

Indeed, For this we need that, if $\alpha, \beta \in \bar{W}$, then so is $\alpha \otimes \beta$. Indeed, as $\alpha \otimes \beta \simeq (\alpha \otimes \text{id}) \circ (\text{id} \otimes \beta)$, it suffices to prove that if $\alpha \in \bar{W}$, then so is $\alpha \otimes \text{id}$. Now α is in the strong saturation of W . As $\otimes$ preserves colimits in both variables, we can reduce to the case of α being in equivalence, in which case the statement is clear, or α being a composition or factor of morphisms in W . Then the claim follows from the assumption. With this, we may apply Lemma 7.6.2, using that the inclusion preserves cocartesian edges, to obtain a relative adjunction

$$\begin{array}{ccc} L \int \mathcal{C}^\otimes & \begin{array}{c} \xleftarrow{\exists} \\ \xrightarrow{\quad} \end{array} & \int \mathcal{C}^\otimes \\ & \searrow \quad \swarrow & \\ & \text{Span}(\text{Fin}) & \end{array}$$

finishing the claim. $\square$

Remark 7.6.5. One can show that $LC^\otimes$ satisfies the usual universal property in $\text{CAlg}(\text{Pr}^{\text{L}})$, that is, it is the initial object under $\mathcal{C}^\otimes$ in which all morphisms in W are invertible.

Example 7.6.6. Let S be a qcqs scheme. Consider $\mathcal{P}(\text{Sm}_S)_*$ with the smash product symmetric monoidal structure. The class W_{mot} in $\mathcal{P}(\text{Sm}_S)_*$ is generated by $(X \times \mathbb{A}^1)_+ \rightarrow X_+$ and $\check{C}(\text{Nis.} - \text{cover of } X)_+ \rightarrow X_+$. This class satisfies the assumptions of Proposition 7.6.4. For example, we have

$$(X \times \mathbb{A}^1)_+ \wedge Y_+ \simeq (X \times Y \times \mathbb{A}^1)_+ \rightarrow (X \times Y)_+ \simeq X_+ \wedge Y_+.$$

Thus we obtain that $\text{Spc}(S) \simeq L_{W_{\text{mot}}} \mathcal{P}(\text{Sm}_S)$ inherits a unique symmetric monoidal structure, which we call the smash product structure.

The general proof.

Extend Toms example

7.7 Symmetric monoidal stabilization

Our next goal is to extend the symmetric monoidal structure on $\mathrm{Spc}(S)$ from Example 7.6.6 to motivic spectra. For this we need to discuss how to formally “invert” objects in a presentably monoidal category.

Definition 7.7.1. Let $\mathcal{C} \in \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$ be a presentably symmetric monoidal category. We call an object $X \in \mathcal{C}$ *invertible*, if there exists some $Y \in \mathcal{C}$ and an equivalence $X \otimes Y \simeq \mathbf{1}_{\mathcal{C}}$.

Theorem 7.7.2 (Robalo). *Let $\mathcal{C} \in \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$ be a presentably symmetric monoidal category, and let $X \in \mathcal{C}$ be any object. Then there exists an initial object in the full subcategory of $\mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})_{\mathcal{C}/}$ consisting of those $F : \mathcal{C} \rightarrow \mathcal{D}$ in which $F(X) \in \mathcal{D}$ is invertible. We denote this initial object by $\mathcal{C}[X^{-1}]$.*

In particular, a colimit preserving, symmetric monoidal functor $F : \mathcal{C} \rightarrow \mathcal{D} \in \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$ can be extended over $\mathcal{C}[X^{-1}]$ if and only if $F(X)$ is invertible in $\mathcal{D}$, and in this case this extension is essentially unique.

Remark 7.7.3 (Picard groupoid). Let $\mathcal{D} \in \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$ be a presentably symmetric monoidal category. Denote by $\mathrm{Pic}(\mathcal{D})$ the “full subspace” of invertible objects of $\mathcal{D}$. One can show, that is actually small and lives in $\mathrm{CMon}(\mathrm{Ani})$, see [ABG18, Section 7]. We then obtain a cartesian square

$$\begin{array}{ccc} \mathrm{Fun}^{L,\otimes}(\mathcal{C}[X^{-1}], \mathcal{D}) & \hookrightarrow & \mathrm{Fun}^{L,\otimes}(\mathcal{C}, \mathcal{D}) \\ \downarrow & \lrcorner & \downarrow \mathrm{ev}_X \\ \mathrm{Pic}(\mathcal{D}) & \hookrightarrow & \mathcal{D} \end{array}$$

Here $\mathrm{Fun}^{L,\otimes}(-, -)$ denotes the full subcategory of colimit preserving, symmetric monoidal functors.

Definition 7.7.4. Let $\mathcal{C} \in \mathrm{CA}\uparrow\downarrow(\mathrm{Pr}^{\mathrm{L}})$ be a symmetric monoidal category. An object $X \in \mathcal{C}$ is called *symmetric* if there exists some $n \geq 2$ such that the cyclic permutation on $X^{\otimes n}$ is homotopic to the identity.

Theorem 7.7.5 (Robalo). *Let $\mathcal{C} \in \mathrm{CA}\uparrow\downarrow(\mathrm{Pr}^{\mathrm{L}})$ be a presentably symmetric monoidal category and X a symmetric object. Then the underlying category of $\mathcal{C}[X^{-1}]$ is given by*

$$\lim(\dots \xrightarrow{\omega_X} \mathcal{C} \xrightarrow{\Omega_X} \mathcal{C}) \simeq \mathrm{colim}^{\mathrm{Pr}^{\mathrm{L}}}(\mathcal{C} \xrightarrow{-\otimes X} \mathcal{C} \xrightarrow{-\otimes X} \dots).$$

Here $\Omega_X := \mathrm{Hom}(X, -)$, with Hom the internal mapping object in $\mathcal{C}$.

Example 7.7.6. $S^1 \in \mathrm{Ani}_*$, with an arbitrary base point, is symmetric. We have $S^1 \otimes S^1 = S^1 \wedge S^1 \simeq S^2$ we have that

$$\pi_0 \mathrm{Map}_{\mathrm{Ani}_*}(S^1 \otimes S^1, S^1 \otimes S^1) \cong [S^2, S^2] \cong \pi_2(S^2) \cong \mathbb{Z}$$

and the cyclic permutation swap corresponds to the morphism interchanging the coordinates, hence being a morphism of degree -1 . So $n = 2$ is not sufficient. However, for $n = 3$ we obtain

$$\pi_0 \mathrm{Map}_{\mathrm{Ani}_*}(S^1 \otimes S^1 \otimes S^1, S^1 \otimes S^1 \otimes S^1) \cong [S^3, S^3] \cong \pi_3(S^3) \cong \mathbb{Z}$$

where the cyclic permutation permutes the coordinates cyclic, which corresponds to two transpositions, hence to $\mathrm{swap} \circ \mathrm{swap}$, which is mapped under the above isomorphism to $(-1)(-1) = 1$, hence is homotopic to the identity.

It follows that $\mathrm{Sp} = \mathrm{Ani}[(S^1)^{-1}]$ inherits a symmetric monoidal structure such that Σ^∞ is symmetric monoidal.

Having Definition 7.1.1 in mind, we would like $\mathbb{P}^1$ to be symmetric. This is in fact true.

Lemma 7.7.7. $\mathbb{P}^1 \in \mathrm{Spc}(S)$ is symmetric.

Proof. being inspired from Example 7.7.6, we consider $(\mathbb{P}^1)^{\otimes 3} \simeq \mathbb{A}^3/(\mathbb{A}^3 \setminus 0)$ where we used that $\otimes$ preserves colimits in both variables. The cyclic permutation is then induced from the cyclic permutation on $\mathbb{A}^3$, which can be represented by the matrix

$$\text{swap} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

This matrix can be rewritten in terms of elementary row and column operations as

$$\text{swap} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Now, any elementary row or column operations is $\mathbb{A}^1$ -homotopic to the identity. For example

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \stackrel{H}{\sim} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

where

$$H = \begin{pmatrix} 1 & t & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

□

Corollary 7.7.8. Let S be a qcqs scheme. The category of motivic spectra $\mathrm{SH}(S) \simeq \mathrm{Spc}(S)_*[(\mathbb{P}^1)^{-1}]$ admits a symmetric monoidal structure such that the functor $\Sigma^\infty : \mathrm{Spc}(S)_* \rightarrow \mathrm{SH}(S)$ is symmetric monoidal.

Remark 7.7.9. Thus, a colimit preserving, symmetric monoidal functor $\mathrm{SH}(S) \rightarrow \mathcal{D} \in \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$ is the same data as a functor $\mathrm{Sm}_S \xrightarrow{F} \mathcal{D}$ such that

- F is symmetric monoidal;
- $F(\mathbb{A}^1) \simeq \mathbb{1}_{\mathcal{D}}$ is the monoidal unit;
- $F(\text{Nis. square})$ is a pullback square;
- $F(\mathbb{P}^1) \in \mathcal{D}$ is invertible.

Notice that the first condition implies that $F(\emptyset) \simeq *$. Indeed, for any $X \in \mathrm{Sm}_S$ we have

$$F(\emptyset) \simeq F(X \times \emptyset) \simeq F(X) \otimes F(\emptyset)$$

which is only true for $F(\emptyset) \simeq *$. Thus the first and the third condition together imply that F is a Nisnevich sheaf by Theorem 3.2.4. Furthermore, the first and second condition together imply the F is $\mathbb{A}^1$ -local. Indeed, for any $X \in \mathrm{Sm}_S$ we have

$$F(X \times \mathbb{A}^1) \simeq F(X) \otimes F(\mathbb{A}^1) \simeq F(X) \otimes \mathbb{1}_{\mathcal{D}} \simeq F(X).$$

This ends our discussion of the symmetric monoidal structure on $\mathrm{SH}(S)$, and thus our discussion of the fundamental properties of motivic spectra. We will now turn to more advanced properties of motivic spectra.

Is this true?

Chapter 8

The $6(+\varepsilon)$ functor formalism on $\mathrm{SH}(-)$

In this chapter we discuss the six functor formalism on motivic spectra. The six functor formalism encodes a lot of structure, like Grothendieck-Verdier duality, or even the definition of cohomology. We will assume every scheme in this chapter to be qcqs.

8.1 The pullback

Let $f : S' \rightarrow S \in \mathrm{Sch}$ be a morphism of schemes. We obtain a functor

$$\begin{aligned} f^* : \mathrm{Sm}_S &\rightarrow \mathrm{Sm}_{S'} \\ X &\mapsto S' \times_S X \end{aligned}$$

Using the left Kan extension of the composition $\mathrm{Sm}_S \xrightarrow{f^*} \mathrm{Sm}_{S'} \xrightarrow{y} \mathcal{P}(\mathrm{Sm}_{S'})$ along the Yoneda embedding $y : \mathrm{Sm}_S \hookrightarrow \mathcal{P}(\mathrm{Sm}_S)$, we obtain an adjunction

$$f^* : \mathcal{P}(\mathrm{Sm}_S) \xrightleftharpoons[\perp]{} \mathcal{P}(\mathrm{Sm}_{S'}) : f_*$$

where $f_*F(X) \simeq F(f^*X)$ for any $F \in \mathcal{P}(\mathrm{Sm}_{S'})$ and $X \in \mathrm{Sm}_S$. Furthermore, as f^* is the left Kan extension along the fully faithful functor y , the following diagram commutes

$$\begin{array}{ccc} \mathcal{P}(\mathrm{Sm}_S) & \xrightarrow{\exists! f^*} & \mathcal{P}(\mathrm{Sm}_{S'}) \\ y \uparrow & & \uparrow y \\ \mathrm{Sm}_S & \xrightarrow{f^*} & \mathrm{Sm}_{S'} \end{array}$$

We want to extend f^* first to motivic spaces, then to motivic spectra. The first step is achieved by the following lemma.

Lemma 8.1.1. *Let $f : S \rightarrow S' \in \mathrm{Sch}$ be a morphism of schemes. The induced functor $f^* : \mathcal{P}(\mathrm{Sm}_S) \rightarrow \mathcal{P}(\mathrm{Sm}_{S'})$ preserves*

- (1) *finite products;*
- (2) *Nisnevich equivalences;*
- (3) $\mathbb{A}^1$ -*equivalences;*
- (4) *motivic equivalences.*

Proof. We start with (1). First notice, that on $\mathcal{P}(-)$ limits and colimits are computed section wise. As $- \times - : \mathbf{Ani} \times \mathbf{Ani} \rightarrow \mathbf{Ani}$ commutes with colimits in both variables, it follows that $- \times - : \mathcal{P}(-) \times \mathcal{P}(-) \rightarrow \mathcal{P}(-)$ commutes with colimits in both variables. Furthermore, $y : \mathbf{Sm}_S \rightarrow \mathcal{P}(\mathbf{Sm}_S)$ preserves products.

We can now start with the prove itself. As the empty product in $\mathcal{P}(\mathbf{Sm}_S)$ is given by $* = S$, we have

$$f^*(*) = f^*(S) \simeq S \times_S S' \simeq S' \simeq * \in \mathcal{P}(\mathbf{Sm}_{S'}).$$

Here we used in the first equivalence that the Yoneda embedding commutes with f^* . Thus, it is enough to prove that binary products are preserved by f^* . For this, let $G, F \in \mathcal{P}(\mathbf{Sm}_S)$ be two arbitrary presheaves. Using the Yoneda density, we rewrite $F \simeq \text{colim}_{X \rightarrow F} X$ and $G \simeq \text{colim}_{Y \rightarrow G} Y$, where the diagram is indexed by smooth schemes mapping into F and G respectively. Then, as f^* is left adjoint, and by the discussion at the beginning of the prove, we have

$$\begin{aligned} f^*(F \times G) &\simeq f^*((\text{colim}_{X \rightarrow F} X) \times (\text{colim}_{Y \rightarrow G} Y)) \\ &\simeq f^*(\text{colim}_{X \rightarrow F} \text{colim}_{Y \rightarrow G} (X \times Y)) \\ &\simeq \text{colim}_{X \rightarrow F} \text{colim}_{Y \rightarrow G} f^*(X \times Y) \\ &\simeq \text{colim}_{X \rightarrow F} \text{colim}_{Y \rightarrow G} f^*(X) \times f^*(Y) \\ &\simeq f^*(F) \times f^*(G). \end{aligned}$$

Here we also used

$$\begin{aligned} f^*(X \times Y) &= f^*(X \times_S Y) \\ &\cong S' \times_S (X \times_S Y) \\ &\cong S' \times_S ((X \times_{S'} S') \times_S Y) \\ &\cong (S' \times_S X) \times_{S'} (S' \times_S Y) \\ &\cong f^*(X) \times f^*(Y). \end{aligned}$$

This shows (1).

For (2), as f^* is colimit preserving, it suffices to prove that f^* preserves generating Nisnevich equivalences, that is, Čech nerves of Nisnevich covers. As f^* preserves colimits and those fiber products of representables that exist in $\mathbf{Sm}_S$, it suffices to prove that $f^* : \mathbf{Sm}_S \rightarrow \mathbf{Sm}_{S'}$ preserves Nisnevich covers. But this is clear, as étale morphisms are stable under pullbacks, and the Nisnevich condition for the residue fields is so too.

For (3) it suffices to prove that f^* preserves the generating $\mathbb{A}^1$ -equivalences, that is the morphisms $X \times \mathbb{A}^1 \rightarrow X$ for $X \in \mathbf{Sm}_S$. But this is clear, as f^* preserves finite products and representables, and as

$$f^*(\mathbb{A}_S^1) \cong S' \times_S \mathbb{A}_S^1 \cong \mathbb{A}_{S'}^1$$

by definition.

(4) is now the combination of (2) and (3). □

Thus, the adjunction of presheaves descends to an adjunction on motivic spaces.

Corollary 8.1.2. *There exists an adjunction*

$$f^* : \mathbf{Spc}(S) \xrightleftharpoons[\perp]{} \mathbf{Spc}(S') : f_*$$

These adjunction have the property, that the following squares commute

$$\begin{array}{ccc} \mathcal{P}(\mathbf{Sm}_S) & \xrightarrow{f^*} & \mathcal{P}(\mathbf{Sm}_{S'}) \\ \text{L}_{\text{mot}} \downarrow & & \downarrow \text{L}_{\text{mot}} \\ \mathbf{Spc}(S) & \xrightarrow{f^*} & \mathbf{Spc}(S') \end{array} \quad \begin{array}{ccc} \mathcal{P}(\mathbf{Sm}_S) & \xleftarrow{f_*} & \mathcal{P}(\mathbf{Sm}_{S'}) \\ \uparrow & & \uparrow \\ \mathbf{Spc}(S) & \xleftarrow{f_*} & \mathbf{Spc}(S') \end{array}$$

Proof. The right adjoint exists as f^* is colimit preserving, and since $\mathrm{Spc}(S), \mathrm{Spc}(S')$ are presentable categories, hence we can apply the adjoint functor theorem. Further, the left hand square commutes by construction of $f^* : \mathrm{Spc}(S) \rightarrow \mathrm{Spc}(S')$, while commutativity of the right hand square follows from the commutativity of the left hand square by going over to right adjoint. $\square$

To extend the adjunction to motivic spectra, we first need to extend it to pointed motivic spaces.

Corollary 8.1.3. *The adjunction from Corollary 8.1.2 induces an adjunction*

$$f^* : \mathrm{Fun}(\Delta^1, \mathrm{Spc}(S)) \xrightleftharpoons[\perp]{} \mathrm{Fun}(\Delta^1, \mathrm{Spc}(S')) : f_*$$

which restricts to an adjunction

$$f^* : \mathrm{Spc}(S)_* \xrightleftharpoons[\perp]{} \mathrm{Spc}(S')_* : f_*$$

Proof. The first adjunction is general non-sense that an adjunction induces adjunctions on functor categories. The second adjunction follows from $f^*(*) \simeq *$, as $f^* : \mathrm{Spc}(S) \rightarrow \mathrm{Spc}(S')$ preserves finite products, and $f_*(*) \simeq *$, as f_* is a right adjoint. $\square$

Remark 8.1.4. Let $U : \mathrm{Spc}(S)_* \rightarrow \mathrm{Spc}(S)$ denote the forgetful functor. Then, by construction, we obtain two commutative squares

$$\begin{array}{ccc} \mathrm{Spc}(S)_* & \xrightarrow{f^*} & \mathrm{Spc}(S')_* \\ U \downarrow & & \downarrow U \\ \mathrm{Spc}(S) & \xrightarrow{f^*} & \mathrm{Spc}(S') \end{array} \quad \begin{array}{ccc} \mathrm{Spc}(S)_* & \xleftarrow{f_*} & \mathrm{Spc}(S')_* \\ U \downarrow & & \downarrow U \\ \mathrm{Spc}(S) & \xleftarrow{f_*} & \mathrm{Spc}(S') \end{array}$$

Considering the right hand square and passing to left adjoints, we obtain a third commutative square

$$\begin{array}{ccc} \mathrm{Spc}(S)_* & \xrightarrow{f^*} & \mathrm{Spc}(S')_* \\ (-)_+ \uparrow & & \uparrow (-)_+ \\ \mathrm{Spc}(S) & \xrightarrow{f^*} & \mathrm{Spc}(S') \end{array}$$

To extend the adjunction to motivic spectra, we now also need that f^* is symmetric monoidal, and that $f^*(\mathbb{P}_S^1) \simeq \mathbb{P}_{S'}^1$. The second requirement is clear. For the first, this follows from $f^* : \mathrm{Spc}(S) \rightarrow \mathrm{Spc}(S')$ preserving finite products, and hence, by the naturality of the smash product symmetric monoidal structure on pointed spaces, $f^* : \mathrm{Spc}(S)_* \rightarrow \mathrm{Spc}(S')_*$ is symmetric monoidal for the smash product.

Corollary 8.1.5. *The adjunction from Corollary 8.1.3 induces an adjunction*

$$f^* : \mathrm{SH}(S) \xrightleftharpoons[\perp]{} \mathrm{SH}(S') : f_*$$

such that the following diagram commutes

$$\begin{array}{ccc} \mathrm{Spc}(S)_* & \xrightarrow{f^*} & \mathrm{Spc}(S')_* \\ \Sigma^\infty \downarrow & & \downarrow \Sigma^\infty \\ \mathrm{SH}(S) & \xrightarrow{f^*} & \mathrm{SH}(S') \end{array}$$

Proof. This follows directly from the universal property of $\mathrm{SH}(S')$ as the initial presentably symmetric monoidal stable category in which $\mathbb{P}_{S'}^1$ is invertible. $\square$

Remark 8.1.6. Just as before, passing to right adjoints, we obtain another commutative square

$$\begin{array}{ccc} \mathrm{Spc}(S)_* & \xleftarrow{f_*} & \mathrm{Spc}(S')_* \\ \Omega^\infty \uparrow & & \uparrow \Omega^\infty \\ \mathrm{SH}(S) & \xleftarrow{f_*} & \mathrm{SH}(S') \end{array}$$

With this we have constructed the first two out of our six functors. We already now the existence of the next two, $(\otimes, \mathrm{Hom})$, with Hom the internal mapping object. To see how these four functors related to one another, we are going to make a short digression.

8.2 Interlude: Invertible and strongly dualizable objects

Definition 8.2.1. Let $\mathcal{C}$ be a symmetric monoidal category. A *duality* between two objects $A, A^\vee \in \mathcal{C}$ consists of two morphisms

- $\mathrm{ev}_A : A^\vee \otimes A \rightarrow \mathbb{1}_{\mathcal{C}}$;
- $\mathrm{coev}_A : \mathbb{1}_{\mathcal{C}} \rightarrow A \otimes A^\vee$.

such that the two compositions

$$\begin{aligned} A^\vee &\simeq A^\vee \otimes \mathbb{1}_{\mathcal{C}} \xrightarrow{\mathrm{coev}_A} (A^\vee \otimes A) \otimes A^\vee \xrightarrow{\mathrm{ev}_A} \mathbb{1}_{\mathcal{C}} \otimes A^\vee \simeq A^\vee, \\ A &\simeq \mathbb{1}_{\mathcal{C}} \otimes A \xrightarrow{\mathrm{coev}_A} A \otimes (A^\vee \otimes A) \xrightarrow{\mathrm{ev}_A} A \otimes \mathbb{1}_{\mathcal{C}} \simeq A \end{aligned}$$

are homotopic to the identity. These compositions are called the *triangle identities*.

We call A *strongly dualizable*, if there exists an $A^\vee$ and a duality between A and $A^\vee$. In this case we call $A^\vee$ a *strong dual* of A .

Remark 8.2.2. • A duality between A and $A^\vee$ induces an adjunction $(A \otimes -, - \otimes A^\vee)$. However, it is often relatively easy, in comparison to writing down an adjunction, to write down a duality.

- As $\mathcal{C}$ is symmetric, a duality between A and $A^\vee$ induces a duality between $A^\vee$ and A , so that also $(- \otimes A^\vee, A \otimes -)$ is an adjunction.
- Furthermore, as any two right adjoint of the same functor are equivalent, if $A^\vee$ and A' are both strong duals of A , then $- \otimes A^\vee \simeq - \otimes A'$, and hence, by plugging in $\mathbb{1}_{\mathcal{C}}$, we obtain an equivalence $A^\vee \simeq A'$. Thus, strong duals, if they exists, are unique.
- dualities, and thus strong duals, are preserved by symmetric monoidal functors.

Example 8.2.3. Suppose that $\mathcal{C}$ is closed symmetric monoidal, that is, $\mathcal{C}$ admits an internal hom-object $\mathrm{Hom}(-, -)$. Let $X \in \mathcal{C}$ be strongly dualizable, and let $X^\vee$ be a strong dual. Then both $\mathrm{Hom}(X, -)$ and $X^\vee \otimes -$ are right adjoint to $- \otimes X$, hence $\mathrm{Hom}(X, -) \simeq X^\vee$, and hence, by evaluating at $\mathbb{1}_{\mathcal{C}}$, $X^\vee \simeq \mathrm{hom}(X, \mathbb{1}_{\mathcal{C}})$.

Notice however, that although $\mathrm{Hom}(X, \mathbb{1}_{\mathcal{C}})$ does exist for all $X \in \mathcal{C}$, it is in general not a strong dual. Indeed, if this would be the case, then $\mathrm{Hom}(X, \mathbb{1}) \otimes - \simeq \mathrm{Hom}(X, -)$ would be left adjoint to $- \otimes X$. Hence $\mathrm{Hom}(X, -)$ would always preserve colimits, which is not true in general.

Counter
example

We can strengthen the first property of Remark 8.2.2 as follows

Lemma 8.2.4. *Let $\mathcal{C}$ be a symmetric monoidal category, and let $A, A^\vee \in \mathcal{C}$ be objects together with a morphism $\text{ev} : A^\vee \otimes A \rightarrow \mathbf{1}$. Then A is strongly dualizable with strong dual $A^\vee$ and evaluation map $\text{ev} : A \otimes A^\vee \rightarrow \mathbf{1}$ if and only if the following composition is an equivalence for all $X, Y \in \mathcal{C}$*

$$\text{Map}_{\mathcal{C}}(X, Y \otimes A^\vee) \xrightarrow{\otimes A} \text{Map}_{\mathcal{C}}(X \otimes A, Y \otimes A^\vee \otimes A) \xrightarrow{(\text{id} \otimes \text{ev})_*} \text{Map}_{\mathcal{C}}(X \otimes A, Y).$$

Proof. We first assume the composition to be an equivalence. This implies that there is an adjunction $(-\otimes A, -\otimes A^\vee)$ with counit given by $\text{id} \otimes \text{ev}$. Let $u : \text{id} \Rightarrow -\otimes A^\vee \otimes A$ denote the unit. We obtain the duality datum for A and $A^\vee$, that is, the evaluation and coevaluation, by considering the the component of the counit and unit at the monoidal unit respectively. The triangle identities follow from the triangle identities of the adjunction.

Let us now assume that A is strongly dualizable with string dual A . Denote the coevaluation by coev . We show that the composition is an equivalence by constructing an inverse. Indeed, consider the composition

$$\begin{aligned} \text{Map}_{\mathcal{C}}(X, Y \otimes A^\vee) &\xrightarrow{\otimes A} \text{Map}_{\mathcal{C}}(X \otimes A, Y \otimes A^\vee \otimes A) \xrightarrow{(\text{id} \otimes \text{ev})_*} \text{Map}_{\mathcal{C}}(X \otimes A, Y) \\ &\xrightarrow{-\otimes A^\vee} \text{Map}_{\mathcal{C}}(X \otimes A \otimes A^\vee, Y \otimes A^\vee) \xrightarrow{\text{coev}_*} \text{Map}_{\mathcal{C}}(X, Y \otimes A^\vee). \end{aligned}$$

By Yoneda, this composition is induced by postcomposition by a essentially unique morphism. To see from which morphism, we consider the action of the composition on a single morphism.

$$\begin{aligned} (X \xrightarrow{f} Y \otimes A^\vee) &\mapsto (X \otimes A \xrightarrow{f \otimes \text{id}} Y \otimes A^\vee \otimes A) \\ &\mapsto (X \otimes A \xrightarrow{f \otimes \text{id}} Y \otimes A^\vee \otimes A \xrightarrow{\text{id} \otimes \text{ev}}) \\ &\mapsto (X \otimes A \otimes A^\vee \xrightarrow{f \otimes \text{id} \otimes \text{id}} Y \otimes A^\vee \otimes A \otimes A^\vee \xrightarrow{\text{id} \otimes \text{ev} \otimes \text{id}} Y \otimes A^\vee) \\ &\mapsto (X \xrightarrow{\text{id} \otimes \text{coev}} X \otimes A \otimes A^\vee \xrightarrow{f \otimes \text{id} \otimes \text{id}} Y \otimes A^\vee \otimes A \otimes A^\vee \xrightarrow{\text{id} \otimes \text{ev} \otimes \text{id}} Y \otimes A^\vee). \end{aligned}$$

using functoriality of the tensor product, we might interchange the first two morphisms to obtain

$$\begin{aligned} (X \xrightarrow{f} Y \otimes A^\vee &\xrightarrow{\text{id} \otimes \text{id} \otimes \text{coev}} Y \otimes A^\vee \otimes A \otimes A^\vee \xrightarrow{\text{id} \otimes \text{ev} \otimes \text{id}} Y \otimes A^\vee) \\ &\simeq (X \xrightarrow{f} Y \otimes A^\vee \xrightarrow{\text{id} \otimes ((\text{id} \otimes \text{coev}) \circ (\text{ev} \otimes \text{id}))} Y \otimes A^\vee). \end{aligned}$$

Now the last morphism is homotopic to the identity by the duality datum of A and $A^\vee$. Thus the composition is homotopic to the identity, and hence the composition on morphism spaces had to be induced by the identity, proving the claim. $\square$

Remember from Corollary 7.6.3, that if we are given an adjunction

$$L : \mathcal{C} \xrightleftharpoons[\perp]{} \mathcal{D} : R$$

where $\mathcal{C}$ and $\mathcal{D}$ are both symmetric monoidal categories, and L is a symmetric monoidal functor, then R is lax symmetric monoidal. In particular, there exists a canonical morphism $\text{can}_{X,Y} : RX \otimes RY \rightarrow R(X \otimes Y)$ for any $X, Y \in \mathcal{D}$, given as the composition

$$RX \otimes RY \xrightarrow{u} RL(RX \otimes RY) \simeq R(LRX \otimes LRY) \xrightarrow{R(c \otimes c)} R(X \otimes Y)$$

where $u : \text{id}_{\mathcal{C}} \Rightarrow RL$ is the unit of the adjunction, and $c : LR \Rightarrow \text{id}_{\mathcal{D}}$ is the counit. The middle equivalence uses the symmetric monoidality of L .

Lemma 8.2.5. *In the situation of the discussion directly above, let $X \in \mathcal{C}$ be a strongly dualizable object. Then, for $Y \in \mathcal{D}$ the composition*

$$X \otimes RY \xrightarrow{u \otimes \text{id}} RLX \otimes RY \xrightarrow{\text{can}_{X,Y}} R(LX \otimes Y)$$

is an equivalence.

Proof. Let $T \in \mathcal{C}$ and let $X^\vee$ be a strong dual of X . Then we have a chain of equivalences

$$\begin{aligned} \text{Map}_{\mathcal{C}}(T, X \otimes RY) &\simeq \text{Map}_{\mathcal{C}}(T \otimes X^\vee, RY) \\ &\simeq \text{Map}_{\mathcal{D}}(L(T \otimes X), Y) \\ &\simeq \text{Map}_{\mathcal{D}}(LT \otimes L(X^\vee), Y) \\ &\simeq \text{Map}_{\mathcal{D}}(LT \otimes (LX)^\vee, Y) \\ &\simeq \text{Map}_{\mathcal{D}}(LT, LX \times Y) \\ &\simeq \text{Map}_{\mathcal{C}}(T, R(LX \otimes Y)). \end{aligned}$$

This chain of equivalences is indeed induced by the claimed morphism. To see this, we have to write out the action of the the equivalences on morphisms:

$$\begin{aligned} (T \xrightarrow{f} X \otimes RY) &\mapsto (T \otimes X^\vee \xrightarrow{f \otimes \text{id}} RY \otimes X \otimes X^\vee \xrightarrow{\text{id} \otimes \text{ev}_X} RY) \\ &\mapsto (LT \otimes LX^\vee \xrightarrow{Lf \otimes \text{id}} L(RY \otimes X) \otimes LX^\vee \\ &\quad \simeq L(RY \otimes X \otimes X^\vee) \xrightarrow{L(\text{id} \otimes \text{ev}_X)} LRY \xrightarrow{c_Y} Y) \\ &\mapsto (LT \xrightarrow{L(\text{id} \otimes \text{coev}_X)} L(T \otimes X^\vee \otimes X) \xrightarrow{L(f \otimes \text{id} \otimes \text{id})} L(RY \otimes X \otimes X^\vee \otimes X) \\ &\quad \xrightarrow{L(\text{id} \otimes \text{ev}_X \otimes \text{id})} L(RY \otimes X) \simeq LRY \otimes LX \xrightarrow{c_Y \otimes \text{id}} Y \otimes LX) \\ &\mapsto (T \xrightarrow{u_T} RLT \xrightarrow{RL(\text{id} \otimes \text{coev}_X)} RL(T \otimes X^\vee \otimes X) \\ &\quad \xrightarrow{RL(f \otimes \text{id} \otimes \text{id})} RL(RY \otimes X \otimes X^\vee \otimes X) \xrightarrow{RL(\text{id} \otimes \text{ev}_X \otimes \text{id})} RL(RY \otimes X) \\ &\quad \simeq R(LRY \otimes LX) \xrightarrow{R(c_Y \otimes \text{id})} R(Y \otimes LX)) \end{aligned}$$

We now want to rewrite the last expression. Using naturality of the unit $u_T : T \rightarrow RLT$, we can push this to the very end to obtain

$$\begin{aligned} T &\xrightarrow{\text{id} \otimes \text{coev}_X} T \otimes X^\vee \otimes X \xrightarrow{f \otimes \text{id} \otimes \text{id}} RY \otimes X \otimes X^\vee \otimes X \\ &\xrightarrow{\text{id} \otimes \text{ev}_X \otimes \text{id}} RY \otimes X \xrightarrow{u_{RY \otimes X}} R(LRY \otimes LX) \xrightarrow{R(c_Y \otimes \text{id})} R(Y \otimes LX). \end{aligned}$$

using functoriality of the tensor product, we can exchange the first two morphisms

$$\begin{aligned} T &\xrightarrow{f} RY \otimes X \xrightarrow{\text{id} \otimes \text{coev}_X} RY \otimes X \otimes X^\vee \otimes X \\ &\xrightarrow{\text{id} \otimes \text{ev}_X \otimes \text{id}} RY \otimes X \xrightarrow{u_{RY \otimes X}} R(LRY \otimes LX) \xrightarrow{R(c_Y \otimes \text{id})} R(Y \otimes LX). \end{aligned}$$

Using the duality data and functoriality of the tensor product, we see that the composition of the second and third map is homotopic to the identity, leaving us with the composition

$$T \xrightarrow{f} RY \otimes X \xrightarrow{u_{RY \otimes X}} R(LRY \otimes LX) \xrightarrow{R(c_Y \otimes \text{id})} R(Y \otimes LX)$$

Now, using the triangle identity for LX and functoriality of R and the tensor product, we can rewrite this again as

$$T \xrightarrow{f} RY \otimes X \xrightarrow{u_{RY \otimes X}} R(LRY \otimes LX) \xrightarrow{R(\text{id} \otimes Lu_X)} R(LRY \otimes LRLX) \xrightarrow{R(c_Y \otimes c_{LX})} R(Y \otimes LX).$$

Using the symmetric monoidality of L , we can pullout the L from the middle two morphism, and hence use naturality of the unit to rewrite this again as

$$T \xrightarrow{f} RY \otimes X \xrightarrow{\text{id} \otimes u_X} RY \otimes RLX \xrightarrow{u_{RY \otimes RLX}} R(LRY \otimes LRLX) \xrightarrow{R(c_Y \otimes c_{LX})} R(Y \otimes LX).$$

But the composition of the last two morphisms is the canonical map, hence

$$T \xrightarrow{f} RY \otimes X \xrightarrow{\text{id} \otimes u_X} RY \otimes RLX \xrightarrow{\text{can}_{X,Y}} R(Y \otimes LX).$$

Hence, the chain of equivalences is homotopic to post composition with the morphism from the claim. By Yoneda, this implies that the chain of natural transformations is equivalent to the post composition by the claimed morphism, and hence implies the claim as such. $\square$

The formula from Lemma 8.2.5 is known as the *projection formula*, sometimes also phrased as that R is a $\mathcal{C}$ -linear functor.

Example 8.2.6. Let S be a qcqs scheme and consider $\Sigma^\infty \mathbb{P}_S^1 \in \text{SH}(S)$, where we consider $\mathbb{P}_S^1$ as being pointed at infinity. By definition $\Sigma_{\mathbb{P}^1} = - \otimes \Sigma^\infty \mathbb{P}_S^1$, and $\Sigma^\infty \mathbb{P}_S^1$ is by construction of $\text{SH}(S)$ invertible, in particular strongly dualizable. Let now $f : S' \rightarrow S$ be a morphism of qcqs schemes. Then we have

$$f_* \Sigma_{\mathbb{P}^1} = f_* (- \otimes \Sigma^\infty \mathbb{P}_{S'}^1) \simeq f_* (- \otimes \Sigma^\infty f^* \mathbb{P}_S^1) \simeq f_* (- \otimes f^* \Sigma^\infty \mathbb{P}_S^1) \simeq f^* (-) \otimes \Sigma^\infty \mathbb{P}_S^1 = \Sigma_{\mathbb{P}^1} f_*.$$

Here we used in order the definition of $\Sigma_{\mathbb{P}^1}$, that $f^* \mathbb{P}_S^1 \simeq \mathbb{P}_{S'}^1 \times_S S'$, as $\mathbb{P}_S^1$ is a representable, that f^* commutes with Σ^∞ by Corollary 8.1.5, the projection formula Lemma 8.2.5, and again the definition of $\Sigma_{\mathbb{P}^1}$.

Let now $E = (E^0, E^1, \dots) \in \text{SH}(S')$ be a motivic spectrum. By definition, one has $E^i = \Omega^\infty \Sigma_{\mathbb{P}^1}^i E$, together with equivalences $E^i \simeq \Omega_{\mathbb{P}^1} E^{i+1}$. From this it follows that $E^i \simeq \Omega^\infty \Sigma_{\mathbb{P}^1}^i E$. Indeed, we know that $E^0 \simeq \Omega^\infty E$. Now we can calculate inductively. In particular, we see that

$$f_*(E^i) \simeq f_* \Omega^\infty \Sigma_{\mathbb{P}^1}^i E \simeq \Omega^\infty f_* \Sigma_{\mathbb{P}^1}^i E \simeq \Omega^\infty \Sigma_{\mathbb{P}^1}^i f_* E = (f_*)^i.$$

Thus $f_* E = (f_* E^0, f_* E^1, \dots)$.

Klaus was unsure, look up how to do this

8.3 The left adjoint of the pullback

Let $f : S' \rightarrow S$ be a smooth morphism of qcqs scheme. It hence induces a left adjoint

$$f_\# : \text{Sm}_{S'} \xrightarrow{\quad \xrightarrow{\quad} \quad} \text{Sm}_S : f^*$$

given by sending $X \rightarrow S'$ to $f_\# X = X \rightarrow S' \xrightarrow{f} S$. Left Kan extending the composition $\text{Sm}_{S'} \xrightarrow{f_\#} \text{Sm}_S \xrightarrow{y} \mathcal{P}(\text{Sm}_S)$ along the Yoneade embedding, we obtain an induced adjunction

$$f_\# : \mathcal{P}(\text{Sm}_{S'}) \xrightarrow{\quad \xrightarrow{\quad} \quad} \mathcal{P}(\text{Sm}_S) : R$$

where the right adjoint is explicitly given as the restriction along $f_\#$, that is $(RF)(X) \simeq F(f_\# X)$ for $F \in \mathcal{P}(\text{Sm}_S)$ and $X \in \text{Sm}_{S'}$. As colimits of presheaves are calculated point wise, it follows that R preserves colimits. By the universal property of $\mathcal{P}(-)$, the functor

R is the unique colimit preserving extension of its restriction to representables. Thus let $X \in \mathbf{Sm}_S$ and $X' \in \mathbf{Sm}_{S'}$. Then we have

$$\begin{aligned} \mathrm{Map}_{\mathcal{P}(\mathbf{Sm}_{S'})}(X', RX) &\simeq (RX)(X') \simeq X(f_{\#}X') \\ &\simeq \mathrm{Map}_{\mathcal{P}(\mathbf{Sm}_S)}(f_{\#}X', X) \\ &\simeq \mathrm{Map}_{\mathbf{Sm}_S}(f_{\#}X', X) \\ &\simeq \mathrm{Map}_{\mathbf{Sm}_{S'}}(X', f^*X) \\ &\simeq \mathrm{Map}_{\mathcal{P}(\mathbf{Sm}_{S'})}(X', f^*X). \end{aligned}$$

It follows from Yoneda that $RX \simeq f^*X$ for all $X' \in \mathbf{Sm}_S$, so $R \simeq f^* : \mathcal{P}(\mathbf{Sm}_S) \rightarrow \mathbf{Sm}_{S'}$, where f^* is the pullback functor from Section 8.1, as f^* is colimit preserving as a left adjoint. In particular, we obtain a triple of adjoint functors $f_{\#} \dashv f^* \dashv f_*$.

Lemma 8.3.1. *Let $f : S' \rightarrow S$ be a smooth morphism of qcqs schemes. Then*

- (1) *the pullback functor $f^* : \mathcal{P}(\mathbf{Sm}_S) \rightarrow \mathcal{P}(\mathbf{Sm}_{S'})$ admits a left adjoint $f_{\#} : \mathcal{P}(\mathbf{Sm}_S) \rightarrow \mathcal{P}(\mathbf{Sm}_{S'})$;*
- (2) *the lower hash functor $f_{\#}$ preserves Nisnevich, $\mathbb{A}^1$, and motivic equivalences.*

Proof. (1) is exactly the discussion before the lemma, so we only have to prove (2). As $f_{\#}$ is a left adjoint, it commutes with colimits. Hence it suffices to prove that $f_{\#} : \mathbf{Sm}_{S'} \rightarrow \mathbf{Sm}_S$ preserves generating Nisnevich and $\mathbb{A}^1$ equivalences. For the Nisnevich equivalences, it suffices to remark that $f_{\#}$ preserves pullbacks of smooth schemes and Nisnevich covers for trivial reasons:

$$\begin{array}{ccccc} W & \longrightarrow & Z & \xrightarrow{f_{\#}} & W & \longrightarrow & Z \\ \downarrow & \lrcorner & \downarrow & \searrow & \downarrow & \searrow & \downarrow \\ X & \longrightarrow & Y & \searrow & X & \longrightarrow & Y \\ & \searrow & \searrow & \searrow & \searrow & \searrow & \searrow \\ & & S' & & S' & \xrightarrow{f} & S \end{array}$$

For $\mathbb{A}^1$ equivalences, this follows from the calculation

$$\begin{aligned} f_{\#}(X \times_{S'} \mathbb{A}_{S'}^1 \rightarrow S') &= (X \times_{S'} \mathbb{A}_{S'}^1 \rightarrow S' \xrightarrow{f} S) \\ &\cong (X \times_{S'} S' \times_S \mathbb{A}_S^1 \rightarrow S' \xrightarrow{f} S) \\ &\cong (X \times_S \mathbb{A}_S^1 \rightarrow S). \end{aligned}$$

□

Lemma 8.3.2 (Projection formula). *Let $f : S' \rightarrow S$ be a smooth morphism of schemes. Let $X \in \mathcal{P}(\mathbf{Sm}_S)$ and $Y \in \mathcal{P}(\mathbf{Sm}_{S'})$ be two presheaves. Then the dashed composition in the diagram*

$$\begin{array}{ccc} f_{\#}(X \times f^*Y) & \dashrightarrow & f_{\#}X \times Y \\ f_{\#}(u_X \times \mathrm{id}) \downarrow & & \uparrow c_{f_{\#}X \times Y} \\ f_{\#}((f^*f_{\#}X) \times f^*Y) & \xrightarrow{\simeq} & f_{\#}f^*(f_{\#}X \times Y) \end{array}$$

is an equivalence.

Proof. All functors appearing preserve colimits, in the case of $\times$ in both variables separately. Thus we may assume $X \in \mathbf{Sm}_{S'}$ and $Y \in \mathbf{Sm}_S$ to be representable. But in this case we can explicitly calculate the composition to be the isomorphism

$$X \times_{S'} (S' \times_S Y) \xrightarrow{\cong} X \times_S Y.$$

□

We thus get an induced adjunction

$$f_{\#} : \mathbf{Spc}(S') \xrightleftharpoons[\perp]{} \mathbf{SH}(S) : f^*$$

such that

$$\begin{array}{ccc} \mathcal{P}(\mathbf{Sm}_{S'}) & \xrightarrow{f_{\#}} & \mathcal{P}(\mathbf{Sm}_S) \\ \text{L}_{\text{mot}} \downarrow & & \downarrow \text{L}_{\text{mot}} \\ \mathbf{Spc}(S') & \xrightarrow{f_{\#}} & \mathbf{Spc}(S) \end{array}$$

and such that the projection formula still holds.

We again want to extend this again to pointed spaces. But $f_{\#}$ does not preserve the terminal object:

$$f_{\#}(*) = f_{\#}(S' \xrightarrow{\text{id}} S') = (S' \xrightarrow{f} S) \neq *.$$

However, consider $(* \rightarrow X) \in \mathbf{Spc}(S')_*$. We obtain a morphism $(f_{\#}(*)f_{\#}X) \in \mathbf{Spc}(S)$. Consider the commutative diagram in $\mathbf{Spc}(S)$

$$\begin{array}{ccc} f_{\#}(*) & \xrightarrow{\text{id}} & f_{\#}(*) \\ \text{id} \downarrow & & \downarrow \\ f_{\#}(*) & \longrightarrow & f_{\#}X \end{array}$$

Taking horizontal cofibers, we obtain an object $(* \rightarrow f_{\#}X/f_{\#}(*)) \in \mathbf{Spc}(S)_*$. We now calculate

$$\begin{aligned} \text{Map}_{\mathbf{Spc}(S)_*}(f_{\#}X/f_{\#}(*), Y) &\simeq \text{Map}_{\mathbf{Spc}(S)^{\Delta^1}}((f_{\#}(*) \rightarrow f_{\#}X), (* \rightarrow Y)) \\ &\simeq \text{Map}_{\mathbf{Spc}(S')^{\Delta^1}}((* \rightarrow X), (f^* * \rightarrow f^* Y)) \\ &\simeq \text{Map}_{\mathbf{Spc}(S')^{\Delta^1}}((* \rightarrow X), (* \rightarrow f^* Y)) \\ &\simeq \text{Map}_{\mathbf{Spc}(S')_*}(X, f^* Y). \end{aligned}$$

Thus $f^* : \mathbf{Spc}(S)_* \rightarrow \mathbf{Spc}(S')_*$ admits a left adjoint $f_{\#} : \mathbf{Spc}(S')_* \rightarrow \mathbf{Spc}(S)_*$ given explicitly by $f_{\#}X := f_{\#}X/f_{\#}(*) \in \mathbf{Spc}(S)_*$, pointed at the canonical base point. In particular, the following diagram commutes

$$\begin{array}{ccc} \mathbf{Spc}(S') & \xrightarrow{f_{\#}} & \mathbf{Spc}(S) \\ (-)_+ \downarrow & & \downarrow (-)_+ \\ \mathbf{Spc}(S')_* & \xrightarrow{f_{\#}} & \mathbf{Spc}(S)_* \end{array}$$

as this is the left adjoint diagram to the diagram from Remark 8.1.4

$$\begin{array}{ccc} \mathbf{Spc}(S') & \xleftarrow{f^*} & \mathbf{Spc}(S) \\ U \uparrow & & \uparrow U \\ \mathbf{Spc}(S')_* & \xleftarrow{f^*} & \mathbf{Spc}(S)_* \end{array}$$

where U is the forgetful functor.

Lemma 8.3.3. *Let $f : S' \rightarrow S$ be a smooth morphism of qcqs schemes. The following (dual) projection formulas hold for $X \in \mathrm{Spc}(S')_*$ and $Y \in \mathrm{Spc}(S)_*$.*

- (1) $f_{\#}(X \otimes f^*Y) \simeq f_{\#} \otimes Y$;
- (2) $f^*\mathrm{Hom}(Y, Z) \simeq \mathrm{Hom}(f^*Y, f^*Z)$.

Here, $\mathrm{Hom}(-, -)$ is the internal mapping object in $\mathrm{Spc}(-)$.

Proof. For (1), notice that every object $X \in \mathrm{Spc}(S)_*$ can be written as a colimit of objects in the essential image of $(-)_+ : \mathrm{Spc}(S) \rightarrow \mathrm{Spc}(S)_*$. Indeed, we have the following pushout in $\mathrm{Spc}(S)_*$

$$\begin{array}{ccc} (*)_+ & \longrightarrow & (\emptyset)_+ \\ \downarrow & \lrcorner & \downarrow \\ X_+ & \longrightarrow & X \end{array}$$

where the left vertical morphism is induced from the base point of X . As all functors in (1) preserve colimits, we may assume $X = \tilde{X}_+$ and $Y = \tilde{Y}_+$. But then we can use that the smash product monoidal structure is symmetric monoidal to calculate

$$\begin{aligned} f_{\#}(\tilde{X}_+ \otimes f^*(\tilde{Y}_+)) &\simeq f_{\#}(\tilde{X}_+ \otimes (f^*\tilde{Y})_+) \\ &\simeq f_{\#}((\tilde{X} \times f^*\tilde{Y})_+) \\ &\simeq (f_{\#}(\tilde{X} \times f^*\tilde{Y}))_+ \\ &\simeq (f_{\#}\tilde{X} \times \tilde{Y})_+ \\ &\simeq f_{\#}(\tilde{X}_+) \otimes \tilde{Y}_+. \end{aligned}$$

Here we used the projection formula in the fourth equivalence.

For (2), we calculate for an arbitrary $T \in \mathrm{Spc}(S')_*$

$$\begin{aligned} \mathrm{Map}_{\mathrm{Spc}(S')_*}(T, f^*\mathrm{Hom}(Y, Z)) &\simeq \mathrm{Map}_{\mathrm{Spc}(S)_*}(f_{\#}T, \mathrm{Hom}(Y, Z)) \\ &\simeq \mathrm{Map}_{\mathrm{Spc}(S)_*}(f_{\#}T \otimes Y, Z) \\ &\simeq \mathrm{Map}_{\mathrm{Spc}(S)_*}(f_{\#}(T \otimes f^*Y), Z) \\ &\simeq \mathrm{Map}_{\mathrm{Spc}(S')_*}(T \otimes f^*Y, f^*Z) \\ &\simeq \mathrm{Map}_{\mathrm{Spc}(S')_*}(T, f^*\mathrm{Hom}(f^*Y, f^*Z)). \end{aligned}$$

Here we used (1) in the third equivalence. The claim now follows from Yoneda. $\square$

Proposition 8.3.4. *Let $f : S' \rightarrow S$ be a smooth morphism of qcqs schemes. Then $f^* : \mathrm{SH}(S) \rightarrow \mathrm{SH}(S')$ admits a left adjoint $f_{\#} : \mathrm{SH}(S') \rightarrow \mathrm{SH}(S)$, such that*

$$\begin{array}{ccc} \mathrm{Spc}(S')_* & \xrightarrow{f_{\#}} & \mathrm{Spc}(S)_* \\ \Sigma^{\infty} \downarrow & & \downarrow \Sigma^{\infty} \\ \mathrm{SH}(S') & \xrightarrow{f_{\#}} & \mathrm{SH}(S) \end{array}$$

commutes and such that $f_{\#}$ satisfies the projection formula

$$f_{\#}(X \otimes f^*Y) \simeq f_{\#} \otimes Y$$

for $X \in \mathrm{SH}(S')$ and $Y \in \mathrm{SH}(S)$.

Proof. For $E = (E^0, E^1, \dots) \in \mathrm{SH}(S)$ we have that $f^*E = (f^*E^0, f^*E^1, \dots)$. Indeed we have for $X \in \mathrm{Spc}(S)_*$, using the projection formula Lemma 8.3.3 (2)

$$f^*\Omega_{\mathbb{P}^1}X \simeq f^*\mathrm{Hom}(\mathbb{P}_S^1, X) \simeq \mathrm{Hom}(f^*\mathbb{P}_S^1, f^*X) \simeq \mathrm{Hom}(\mathbb{P}_{S'}^1, f^*X) \simeq \Omega_{\mathbb{P}^1}f^*X.$$

Thus, the functor $f^* : \mathrm{Spc}(S)_* \rightarrow \mathrm{Spc}(S')_*$ induces a functor $f_2^* : \mathrm{SH}(S) \rightarrow \mathrm{SH}(S')$ on the limit, left adjoint to f_* , as both are computed level wise. But, as f^* is already left adjoint to f_* , we obtain $f_2^* \simeq f^*$ by uniqueness of adjoints. But limits and colimits are computed level wise, and $f^* : \mathrm{Spc}(S)_* \rightarrow \mathrm{Spc}(S')_*$ is both a left and a right adjoint, hence commutes with limits and colimits. It follows that $f^* : \mathrm{SH}(S) \rightarrow \mathrm{SH}(S')$ is a left exact, accessible functor between presentable categories, thereby there exists a left adjoint $f_{\#} : \mathrm{SH}(S') \rightarrow \mathrm{SH}(S)$.

Now the diagram commutes, as it is the left adjoint of the diagram

$$\begin{array}{ccc} \mathrm{Spc}(S')_* & \xleftarrow{f^*} & \mathrm{Spc}(S)_* \\ \Omega^\infty \uparrow & & \uparrow \Omega^\infty \\ \mathrm{SH}(S') & \xleftarrow{f_*} & \mathrm{SH}(S) \end{array}$$

which commutes as f^* is calculated point wise.

For (2), notice that the projection formula holds for strongly dualizable objects by Lemma 8.2.5. In particular, it holds for $\Sigma_{\mathbb{P}^1}^n \mathbb{1}$ for any $n \in \mathbb{Z}$. But any $E \in \mathrm{SH}(S)$ is a colimit of the tower $\Sigma_{\mathbb{P}^1}^{-n} \Sigma^\infty E^n \simeq \Sigma_{\mathbb{P}^1}^{-n} \mathbb{1} \otimes \Sigma^\infty E^n$. As $\otimes$ preserves colimits in both variables, we may reduce to objects in the essential image of $\Sigma^\infty : \mathrm{Spc}(S)_* \rightarrow \mathrm{SH}(S)$. But then, for $X \in \mathrm{Spc}(S')_*$ and $Y \in \mathrm{Spc}(S)_*$

$$\begin{aligned} f_{\#}(\Sigma^\infty X \otimes f^*(\Sigma^\infty Y)) &\simeq f_{\#}(\Sigma^\infty X \otimes \Sigma^\infty(f^*Y)) \\ &\simeq f_{\#}\Sigma^\infty(X \otimes f^*Y) \\ &\simeq \Sigma^\infty f_{\#}(X \otimes f^*Y) \\ &\simeq \Sigma^\infty(f_{\#}X \otimes Y) \\ &\simeq f_{\#}(\Sigma^\infty X) \otimes \Sigma^\infty Y. \end{aligned}$$

Here we used the symmetric monoidality and the projection formula for pointed motivic spaces Lemma 8.3.3. $\square$

8.4 Descent

In this section we want to study the descent behavior for $\mathrm{Spc}(-)$, $\mathrm{Spc}(-)_*$, and $\mathrm{SH}(-)$.

Lemma 8.4.1. *Let $g : S' \rightarrow S$ be an étale morphism of schemes. Then there exists an equivalence*

$$\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S)_{S'} \simeq \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_{S'}).$$

Proof. We have that, in the following commutative diagram of schemes

$$\begin{array}{ccc} X & \xrightarrow{h} & S \\ & \searrow f & \nearrow g \\ & S' & \end{array}$$

if g is étale and h is smooth, then g was already smooth. This can be seen as follows. Considered the graph

$$\begin{array}{ccccc}
 & & \text{id} & & \\
 & \nearrow & & \searrow & \\
 X & \xrightarrow{\Gamma_f} & X \times_S S' & \longrightarrow & X \\
 & \searrow f & \downarrow & \lrcorner & \downarrow h \\
 & & S & \xrightarrow{g} & S'
 \end{array}$$

It follows that the projection is smooth, as the pullback of a smooth morphism. We further have the pullback square

$$\begin{array}{ccc}
 X & \xrightarrow{\Gamma_f} & X \times_S S' \\
 f \downarrow & \lrcorner & \downarrow \\
 S' & \xrightarrow{\Delta} & S' \times_S S'
 \end{array}$$

As $S' \rightarrow S$ is étale, hence unramified. It follows, that the diagonal $\Delta : S' \times_S S' \rightarrow S'$ is an open immersion, see [Sta25, Tag 02GE]. But open immersions are étale as local isomorphisms of schemes. Thus Γ_f is étale. It follows that $f = \text{pr} \circ \Gamma_f$ is the composition of a smooth and an étale morphism, hence smooth.

Notice that we really needed $S' \rightarrow S$ to be étale for this argument to work, as we essentially used the unramifiedness to conclude that Γ_f is smooth.

It follows from this cancellation property, that there is an equivalence of categories

$$(\text{Sm}_{S'})_{/S} \simeq \text{Sm}_S.$$

Using the adjunction

$$L : \mathcal{P}(\text{Sm}_{S'}) \xrightleftharpoons[\perp]{} \text{Shv}_{\text{Nis}}(\text{Sm}_{S'}) : \iota$$

and the fact that the Nisnevich topology is subcanonical, implying that S is a sheaf, we obtain an adjunction

$$L' : \mathcal{P}(\text{Sm}_{S'})_{/S} \xrightleftharpoons[\perp]{} \text{Shv}_{\text{Nis}}(\text{Sm}_{S'})_{/S} : \iota'$$

such that the right adjoint is fully faithful. Using that limits in slice categories are computed underlying, we obtain that L preserves small limits, hence is a left exact left adjoint. By [Lur09, Corollary 5.1.6.12], we further have an equivalence

$$\mathcal{P}(\text{Sm}_{S'})_{/S} \simeq \mathcal{P}((\text{Sm}_{S'})_{/S}).$$

In summary, we obtain that $\text{Shv}_{\text{Nis}}(\text{Sm}_{S'})_{/S}$ is a reflexive subcategory of $\mathcal{P}((\text{Sm}_{S'})_{/S})$, hence is the topos of some site on $(\text{Sm}_{S'})_{/S}$. But L inverts precisely those morphisms, which are inverted under L after forgetting to $\mathcal{P}(\text{Sm}_{S'})$, we obtain that

$$\text{Shv}_{\text{Nis}}(\text{Sm}_{S'})_{/S} \simeq \text{Shv}_{\text{Nis}}((\text{Sm}_{S'})_{/S}) \simeq \text{Shv}_{\text{Nis}}(\text{Sm}_S).$$

□

Let now $S' \rightarrow S$ be a Nisnevich cover and consider the simplicial étale scheme $S'_\bullet$ induced by the Čech nerve $\check{C}(S' \rightarrow S)$.

Proposition 8.4.2. *The functors*

$$\text{Spc}(-), \text{Spc}(-)_*, \text{SH}(-) : \text{Sch}^{\text{qcqs}} \rightarrow \text{Cat}$$

are all Nisnevich sheaves, that is $F(S) \simeq \lim_{\Delta} F(S'_\bullet)$ for $F \in \{\text{Spc}(-), \text{Spc}(-)_*, \text{SH}(-)\}$ and for every Nisnevich cover $S' \rightarrow S$.

Proof. The descend property for $\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_-) \simeq \mathrm{Shv}_{\mathrm{Nis}}((\mathrm{Sm}_-)/_-)$ follows from descend for topoi, see Section 2.1.

We can consider $F(-) = \mathrm{Spc}(-)$. Let $p : A \rightarrow B$ be an étale morphism of schemes, $X \in \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_B)$ an $\mathbb{A}^1$ -invariant Nisnevich sheaf. Then, p^*X is $\mathbb{A}^1$ -invariant. Indeed, we have for any $Y \in \mathrm{Sm}_A$

$$\begin{aligned} (p^*X)(\mathbb{A}_A^1 \times Y) &\simeq X(p_{\#}(\mathbb{A}_A^1 \times Y)) \\ &\simeq X(p_{\#}(p^*\mathbb{A}_B^1 \times Y)) \\ &\simeq X(\mathbb{A}^1 \times p_{\#}Y) \simeq X(p_{\#}Y) \\ &\simeq (p^*X)(Y). \end{aligned}$$

Here we used Lemma 8.3.2 in the third equivalence. It follows that the composition $\mathrm{Spc}(B) \hookrightarrow \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_B) \xrightarrow{p^*} \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_A)$ factors essentially uniquely over $\mathrm{Spc}(A)$. Checking on representables, using a cocontinuity argument again, we see the factorization is given by the pullback functor from Section 8.1

$$\begin{array}{ccc} \mathrm{Spc}(B) & \hookrightarrow & \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_B) \\ p^* \downarrow & & \downarrow p^* \\ \mathrm{Spc}(A) & \hookrightarrow & \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_A) \end{array}$$

By the essential uniqueness of the factorization, we obtain a natural transformation $\mathrm{Spc}(-) \hookrightarrow \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_-)$, which is point wise fully faithful.

Recall now, that we can describe the limit over Δ of a Cat -valued functor F explicitly as

$$\lim_{\Delta} F \simeq \mathrm{sect}_{\mathrm{cart}} \int F \simeq \{(X_i \in F(i))_{i \in \Delta} \mid \forall \alpha : i \rightarrow j \in \Delta, \alpha^* X_j \simeq X_i\}.$$

As limits preserve fully faithful inclusions, we obtain a commutative diagram

$$\begin{array}{ccc} \lim_{\Delta} \mathrm{Spc}(S'_{\bullet}) & \hookrightarrow & \lim_{\Delta} \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_{S'_{\bullet}}) \\ \uparrow & & \uparrow \simeq \\ \mathrm{Spc}(S) & \hookrightarrow & \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S) \end{array}$$

By the above, the essential image of the upper horizontal morphism is given by these $(X_i)_{i \in \Delta}$, such that all X_i are $\mathbb{A}^1$ -invariant. So, to show the claim, it suffices to show, by the right vertical equivalence, that if $X|_{S'_n} \simeq (S'_n \rightarrow S)^* X$ is $\mathbb{A}^1$ -invariant for all n , where $S'_n = (S')^{\times_{S^n}}$ is the n -th step of the Čech nerve, then X was already $\mathbb{A}^1$ -invariant. This says, that if $X|_{S'_n}$ lies in the essential image of the inclusion $\lim_{\Delta} \mathrm{Spc}(S'_{\bullet}) \hookrightarrow \lim_{\Delta} \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_{S'_{\bullet}})$, it was already in the essential image of $\mathrm{Spc}(S) \hookrightarrow \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S)$. Phrased differently, the left vertical fully faithful inclusion is essentially surjective, and hence an equivalence.

As we already know that $\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_-)$ is a Nisnevich sheaf, we obtain an equivalence

$$S \simeq \mathrm{colim}_{\Delta} S'_{\bullet} \in \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S).$$

Let now $Y \in \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S)$ be a Nisnevich sheaf. As products preserve colimits, we in particular obtain an equivalence

$$Y \simeq \mathrm{colim}_{\Delta} (Y \times S'_{\bullet}) \in \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S).$$

But this implies that, for any $X \in \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S)$,

$$\begin{aligned} \mathrm{Map}_{\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S)}(Y, X) &\simeq \lim_{\Delta} \mathrm{Map}_{\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S)}(Y \times S'_{\bullet}, X) \\ &\simeq \mathrm{Map}_{\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S)}(Y, \mathrm{Hom}(\mathbb{A}^1, X)) \\ &\simeq \mathrm{Map}_{\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S)}(Y \times \mathbb{A}^1, X) \\ &\simeq \lim_{\Delta} \mathrm{Map}_{\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S)}(Y \times S'_{\bullet} \times \mathbb{A}^1, X) \end{aligned}$$

Hence it suffices to check $\mathbb{A}^1$ -invariance of X by mapping out of objects of the form $Y \times S'_n$. However, $Y \times S'_n$ is in the essential image of $(p_n)_\# := (S'_n \rightarrow S)_\# : \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_{S'_n}) \rightarrow \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S)$, so that

$$\begin{aligned}
\mathrm{Map}_{\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S)} \left((Y \times S'_n, \mathrm{Hom}(\mathbb{A}^1, X)) \right) &\simeq \mathrm{Map}_{\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S)} \left((Y \times S'_n \times \mathbb{A}^1, X) \right) \\
&\simeq \mathrm{Map}_{\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S)} \left((p_n)_\#((Y \times \mathbb{A}^1) \times S'_n), X \right) \\
&\simeq \mathrm{Map}_{\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S)} \left((p_n)_\#((Y \times \mathbb{A}^1) \times S'_n), X \right) \\
&\simeq \mathrm{Map}_{\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_{S'_n})} \left(Y \times S'_n \times \mathbb{A}^1, p_n^* X \right) \\
&\simeq \mathrm{Map}_{\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_{S'_n})} \left(Y \times S'_n, \mathrm{Hom}(\mathbb{A}^1, p_n^* X) \right) \\
&\simeq \mathrm{Map}_{\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_{S'_n})} \left(Y \times S'_n, p_n^* X \right) \\
&\simeq \mathrm{Map}_{\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S)} \left(Y \times S'_n, X \right).
\end{aligned}$$

This shows the proposition for $\mathrm{Spc}(-)$. For $\mathrm{Spc}(-)_*$, we first note that, since $\mathrm{Fun}(\Delta^1, -)$, aa right adjoint, preserves limits. Thus, $\mathrm{Fun}(\Delta^1, \mathrm{Spc}(-))$ is a Nisnevich sheaf. Now $\mathrm{Spc}(S)_*$ is given as the pullback

$$\begin{array}{ccc}
\mathrm{Spc}(S)_* & \hookrightarrow & \mathrm{Fun}(\Delta^1, \mathrm{Spc}(S)) \\
\downarrow & \lrcorner & \downarrow \mathrm{ev}_0 \\
* & \xrightarrow{S} & \mathrm{Spc}(S)
\end{array}$$

In particular, as limits preserve fully faithful inclusions,

$$\lim_{\Delta} \mathrm{Spc}(S'_\bullet)_* \hookrightarrow \lim_{\Delta} \mathrm{Fun}(\Delta^1, \mathrm{Spc}(S'_\bullet)) \simeq \mathrm{Fun}(\Delta^1, \mathrm{Spc}(S))$$

with essential image those $A \rightarrow B \in \mathrm{Fun}(\Delta^1, \mathrm{Spc}(S))$, such that $A|_{S'_n} \simeq (S'_n \rightarrow S)^* A \simeq *$ for all n . Now we can use that we already have proven that $\mathrm{Spc}(-)$ is a Nisnevich sheaf. Thus

$$A \simeq \lim_{\Delta} A|_{S'_\bullet} \simeq \lim_{\Delta} * \simeq *.$$

Thus, the image is exactly $\mathrm{Spc}(-)_*$, showing the claim.

Lastly, we consider $\mathrm{SH}(-)$. However, as f^* commutes with $\Omega_{\mathbb{P}^1}$, we obtain a diagram in $\mathrm{Fun}(\Delta, \mathrm{Cat})$

$$\dots \mathrm{Spc}(S'_\bullet)_* \xrightarrow{\Omega_{\mathbb{P}^1}} \mathrm{Spc}(S'_\bullet)_* \xrightarrow{\Omega_{\mathbb{P}^1}} \mathrm{Spc}(S'_\bullet)_*.$$

As the order of taking limits is irrelevant for the result, we obtain an equivalence, using that we already know that $\mathrm{Spc}(-)_*$ is a Nisnevich sheaf

$$\mathrm{SH}(S) \simeq \lim_{\Delta} \mathrm{SH}(S'_\bullet),$$

finishing the proposition. □

Remark 8.4.3. As a very special case we get, for any Nisnevich cover $S' \rightarrow S$ that $F(S) \rightarrow F(S')$ is conservative for any choice of $F(-) = \{\mathrm{Spc}(-), \mathrm{Spc}(-)_*, \mathrm{SH}(-)\}$. Indeed, if $f \in F(S)$ is such that it becomes an equivalence in $F(S')$, it also becomes an equivalence in all of the $F(S'_n)$, as functors preserve equivalences. In particular, f is an equivalence in $\lim_{\Delta} F(S'_\bullet) \simeq F(S)$.

Lemma 8.4.4. *Let $f : S' \rightarrow S$ be a Nisnevich cover, $E \in \mathrm{SH}(S)$ a motivic spectrum. Then E is invertible, resp. strongly dualizable, if and only if f^*E is invertible, resp. strongly dualizable.*

Proof. As f^* is symmetric monoidal, it preserves invertible and dualizable objects. Further, E is invertible if and only if $E \otimes \mathrm{Hom}(E, \mathbb{1}) \xrightarrow{\sim} \mathbb{1}$ is an equivalence. But f^* is conservative, symmetric monoidal, and commutes with Hom . Lastly, let $E \in \mathrm{SH}(S)$ be such that $f^*E \in \mathrm{SH}(S')$ is strongly dualizable. Let $F, G \in \mathrm{SH}(S)$ be two motivic spectra, and considered the composition

$$\begin{aligned} \mathrm{Map}_{\mathrm{SH}(S)}(F, G \otimes \mathrm{Hom}(E, \mathbb{1})) &\xrightarrow{- \otimes E} \mathrm{Map}_{\mathrm{SH}(S)}(F \otimes E, G \otimes \mathrm{Hom}(E, \mathbb{1}) \otimes E) \\ &\xrightarrow{(u_G)_*} \mathrm{Map}_{\mathrm{SH}(S)}(F \otimes E, G). \end{aligned}$$

By Lemma 8.2.4, it suffices to show that this is an equivalence. But, as f^*E is dualizable, this becomes an equivalence after applying f^* . Thereby, it becomes an equivalence after applying $f_n^* : \mathrm{SH}(S) \rightarrow \mathrm{SH}(S'_n)$ for all n . As

$$\mathrm{Map}_{\mathrm{SH}(S)}(F, G) \simeq \lim_{\Delta} \mathrm{Map}_{\mathrm{SH}(S'_n)}(f_n^*F, f_n^*G)$$

the claim follows. $\square$

Remark 8.4.5. Let V/S be a vector bundle. Then $\Sigma^\infty \mathrm{Th}(V) \in \mathrm{SH}(S)$ is invertible. Indeed, choosing a trivializing Zariski cover $f : S' \rightarrow S$, which is in particular a Nisnevich cover, we get that $f^* \Sigma^\infty \mathrm{Th}(V)$ is a sum of spheres, hence invertible, hence we conclude by Lemma 8.4.4.

8.5 The exceptional functor

In this section we are going to define the missing two functors for our six functor formalism¹. These are given by the so-called exceptional functors.

Let $f : S' \rightarrow S$ be a separable morphism of finite type of qcqs schemes. By the Nagata compactification theorem, see [Sta25, Tag 0F41], we obtain a factorization

$$\begin{array}{ccc} S' & \xrightarrow{j} & X \\ & \searrow f & \downarrow p \\ & & S \end{array}$$

where j is an open immersion and p is proper. We now define a new functor as

$$f_! := p_* \circ j_\# : \mathrm{SH}(S') \rightarrow \mathrm{SH}(S).$$

Theorem 8.5.1. *The lower shriek functor satisfies the following properties*

- $f_!$ is independent of the chosen factorization, that is, given any other factorization $f = p' \circ j'$, the functor $p'_* \circ j'_\#$ is homotopic to $f_!$, and all homotopies are coherent.
- $f_!$ admits a right adjoint $f^!$.
- There is an equivalence $f_! \simeq f_* : \mathrm{SH}(S') \rightarrow \mathrm{SH}(S)$ for f proper.
- There is an equivalence $f^! \simeq ((-) \otimes \Sigma^\infty \mathrm{Th} \Omega_f^1) \circ f^*$ for f smooth.

Proof. This is a long proof, so we instead refer to [Hoy17]. There everything is proven in the equivariant setting, but we obtain the above result simply by letting $G \simeq \{e\}$. $\square$

We will now use the above theorem to prove some incarnation of Atiyah duality.

¹Notice that $f_\#$ does not count as one of the six functors.

Proposition 8.5.2. *Let $f : S' \rightarrow S$ be a smooth and proper morphism of qcqs schemes. Then $\Sigma_+^\infty S' \in \mathrm{SH}(S)$ is strongly dualizable with dual $\mathrm{Th}_S(-\Omega_{S'/S}^1) := f_\#((\Sigma^\infty \mathrm{Th} \Omega_f^1)^{-1})$.*

Notice that $\Omega_{S'/S}^1 = \Omega_f^1$ is a vector bundle over S , hence $\Sigma^\infty \mathrm{Th} \Omega_f^1$ is invertible in $\mathrm{SH}(S)$ by Remark 8.4.5, so the claim does make sense.

Proof. As f is smooth and proper, we obtain an adjunction from Theorem 8.5.1

$$f_* \simeq f_! \dashv f^! \simeq ((-) \otimes \Sigma^\infty \mathrm{Th} \Omega_f^1) \circ f^*.$$

As f^* admits a left adjoint $f_\#$, as f is smooth, and $\Sigma^\infty \mathrm{Th} \Omega_f^1$ is invertible, we obtain an equivalence

$$f_* \simeq f_\#((-) \otimes (\Sigma^\infty \mathrm{Th} \Omega_f^1)^{-1})$$

Further, we have by the projection formula Proposition 8.3.4

$$(\Sigma_+^\infty S') \otimes (-) \simeq (f_\#(\mathbb{1}_{\mathrm{SH}(S')}) \otimes (-) \simeq f_\#(\mathbb{1}_{\mathrm{SH}(S')} \otimes f^*) \simeq f_\# f^*.$$

Both functors admit right adjoint, and hence

$$\begin{aligned} (\Sigma_+^\infty S') \otimes (-) \dashv f_* f^* &\simeq f_\#(f^*(-) \otimes (\Sigma^\infty \mathrm{Th} \Omega_f^1)^{-1}) \\ &\simeq f_\#((\Sigma^\infty \mathrm{Th} \Omega_f^1)^{-1}) \otimes (-) \\ &= \mathrm{Th}_S(-\Omega_{S'/S}^1) \otimes (-). \end{aligned}$$

Here we used again the projection formula from Proposition 8.3.4 in the second equivalence. Thus $(\Sigma_+^\infty S') \otimes (-)$ admits a right adjoint which is also given by tensoring with an object, and hence, by Lemma 8.2.4, the claim follows. $\square$

8.6 An application

Theorem 8.6.1. *Let k be a field which has resolution of singularities. Let $U \in \mathrm{Sm}_k$. Then $\Sigma_+^\infty U \in \mathrm{SH}(k)$ is strongly dualizable.*

Remark 8.6.2. As the tensor product of strongly dualizable objects is again strongly dualizable, the theorem implies that $\mathrm{SH}(k)$, which is generated by twists of the suspension spectra of schemes, is generated by strongly dualizable objects. Furthermore, the subcategory of $\mathrm{SH}(k)$ consisting of dualizable objects is stable and closed under taking summands.² It follows, that an object in $\mathrm{SH}(k)$ is compact if and only if it is dualizable. Indeed, as the monoidal unit is compact, every strongly dualizable object is compact by the following calculation

$$\begin{aligned} \mathrm{Map}_{\mathrm{SH}(k)}(A, \mathrm{colim}_i X_i) &\simeq \mathrm{Map}_{\mathrm{SH}(k)}(\mathbb{1} \otimes A, \mathrm{colim}_i X_i) \\ &\simeq \mathrm{Map}_{\mathrm{SH}(k)}(\mathbb{1}, A^\vee \otimes (\mathrm{colim}_i X_i)) \\ &\simeq \mathrm{colim}_i \mathrm{Map}_{\mathrm{SH}(k)}(\mathbb{1}, A^\vee \otimes X_i) \\ &\simeq \mathrm{colim}_i \mathrm{Map}_{\mathrm{SH}(k)}(A, X_i), \end{aligned}$$

where we used that $- \otimes A^\vee$ commutes with colimits as a left adjoint. On the other hand, as $\mathrm{SH}(k)$ is compactly generated, every compact object is a summand of finite iterated colimits of compact generators. As the compact generators are strongly dualizable, and the subcategory of strongly dualizable objects is stable and closed under taking summands, this shows the claim.

²This is true in more generality for general symmetric monoidal categories.

Remark 8.6.3. As a reminder, a field k has resolution of singularities if for all finite-type k -schemes X there exists a proper birational morphism $X' \xrightarrow{\varphi} X$ with X' smooth and $\varphi^{-1}(X) \xrightarrow{\cong} X_{\text{reg}}$. This implies that k has to be perfect. Indeed, the condition implies that X has to be generically smooth, and if k would not be perfect, you could construct a variety which is singular everywhere.

it is known that every field of characteristic 0 has resolution of singularities, this is a deep theorem by Hironaka.

Proof. With out loss of generality, using that the subcategory of strongly dualizable objects is closed under coproducts, we may assume U to be connected. We prove the theorem via induction on the dimension of the scheme U . If $\dim U = 0$, then U is finite étale over k and hence smooth and proper, in particular strongly dualizable by Proposition 8.5.2.

Now assume that U has dimension d . By the Nagata compactification, there exists an open immersion $U \hookrightarrow \bar{U}$ over k with $\bar{U}$ proper over k . Replacing $\bar{U}$ by the closure of U inside $\bar{U}$, which is still proper as a closed subscheme of a proper scheme, we may assume U to be dense in $\bar{U}$. In particular, $\dim(\bar{U} \setminus U) < d$. Using resolution of singularities, we may further assume that $\bar{U}$ is smooth and proper over k . Here we use that as the resolution of singularities of $\bar{U}$ is an isomorphism over the regular locus of $\bar{U}$, which contains U since U is smooth and open subscheme of $\bar{U}$, the embedding $U \hookrightarrow \bar{U}$ factors over the resolution of singularities.

Define $Z_0 := \bar{U} \setminus U$ and inductively $Z_{i+1} := (Z_i)_{\text{Sing}}$. It follows that $Z_i \setminus Z_{i+1}$ is smooth, and, as we assume resolution of singularities, there exists generic smoothness, in every step the dimension decreases by one, and hence $Z_d = \emptyset$. We define further $U_i := \bar{U} \setminus Z_i = U_{i+1} \setminus (Z_i \setminus Z_{i+1})$, hence an open subset of $\bar{U}$ where we removed a smooth closed subscheme of dimension $< d$. We obtain an decreasing sequence of schemes

$$\bar{U} = U_d \subseteq U_{d-1} \subseteq U_{d-2} \subseteq \dots \subseteq U_0 = U$$

where, in every step, we remove a smooth subscheme.

Since $U_0 = U$ and $U_d = \bar{U}$, it suffices to show that, if $V, W \in \text{Sm}_k$ with $W \hookrightarrow V$ a closed subscheme of V of dimension less than d , $\Sigma_+^\infty V$ strongly dualizable, then also $\Sigma_+^\infty (V \setminus W)$ is strongly dualizable. Indeed, if this claim is true, then, as $\bar{U} = U_d$ is dualizable as a smooth proper k scheme, we get by a downward induction on i that U_i is strongly dualizable for all $i \leq d$, in particular for $i = d$, which shows the proposition.

To prove the claim, consider the cofiber sequence

$$(V \setminus W)_+ \rightarrow V_+ \rightarrow \text{Th} N_{W/V},$$

where we used purity Theorem 5.0.1 to identify the cofiber with the Thom space of the normal bundle. Then, by stability of the subcategory of strongly dualizable objects, it suffices to prove that $\Sigma^\infty \text{Th} N_{W/V}$ is strongly dualizable. If $N_{W/V}$ is trivial, then

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$$\Sigma^\infty \text{Th} N_{W/V} \simeq \Sigma_+^\infty W \otimes (\Sigma_+^\infty \mathbb{P}^1)^{\otimes c},$$

with c the dimension of $N_{W/V}$, is a tensor product of strongly dualizable objects, as W is of dimension less than d , hence strongly dualizable by the induction hypothesis of the outer induction, and $\Sigma^\infty \mathbb{P}^1$ is even invertible.

In general, we can choose a finite, as W is smooth hence of finite type, hence quasi-compact, open cover $W = \bigcup_{i=1}^n W_i$ with $W_i \subseteq W$ open subschemes, such that $N_{W/V}|_{W_i}$ is trivial. By induction on N , the case $N = 1$ already proven above, it suffices to prove that if $W = W_1 \cup W_2$ is an open cover such that $\Sigma^\infty \text{Th}(N_{W/V}|_{W_1})$ is strongly dualizable, and $N_{W/V}|_{W_2}$ is trivial and hence also dualizable, then $\Sigma^\infty \text{Th}(N_{W/V})$ is dualizable. But this

follows from the pushout square

$$\begin{array}{ccc}
\mathrm{Th}(N_{W/V}|_{W_1 \cap W_2}) & \longrightarrow & \mathrm{Th}(N_{W/V}|_{W_1}) \\
\downarrow & \lrcorner & \downarrow \\
\mathrm{Th}(N_{W/V}|_{W_2}) & \longrightarrow & \mathrm{Th} N_{W/V}
\end{array}$$

in $\mathrm{Spc}(k)_*$, as Σ^∞ preserves pushouts, and by the stability of the subcategory of strongly dualizable objects. $\square$

This finishes the discussion of the 6-functors. There is a lot more to say in this regard, but we want, on the last few pages, include a preview for the general theory of motives, as well as the prove of the Steinberg relations in the homotopy groups of the sphere.

Chapter 9

Motives

The theory of motives was sparked by an idea of Grothendieck, that there should exist a universal Weil cohomology theory encoding all the arithmetic properties of a scheme. We will discuss shortly the general framework for motives, and afterwards we give an idea of the theory of motivic cohomology.

9.1 Motivic formalism

Fix a field k and consider a functor $\mu : \mathrm{Sm}_k \rightarrow \mathcal{C}$ where $\mathcal{C}$ is a symmetric monoidal, pointed category admitting finite coproducts, and such that the tensor product preserves finite coproducts in both variables, and where μ is a symmetric monoidal, essentially surjective functor, which preserves coproducts.

Remark 9.1.1. We recall the existence of the category $\mathcal{P}_\Sigma(\mathcal{C}) := \mathrm{Fun}^\times(\mathcal{C}, \mathrm{Ani})$ of product preserving presheaves, sometimes also called spherical presheaves. This category carries a symmetric monoidal structure, called the *Day convolution*, such that the Yoneda embedding $\mathcal{C} \hookrightarrow \mathcal{P}_\Sigma(\mathcal{C})$ is symmetric monoidal and such that the Day convolution product preserves colimits separately in each variable, see [Lur17b, Proposition 4.8.1.10].

By the universal property of the presheaf category, μ extends uniquely to a symmetric monoidal functor $\mu : \mathcal{P}_\Sigma(\mathrm{Sm}_k) \rightarrow \mathcal{P}_\Sigma(\mathcal{C})$. Write $W_{\mathrm{mot}} \subseteq \mathrm{Ar}(\mathcal{P}_\Sigma(\mathrm{Sm}_k))$ for the strongly saturated class of motivic equivalences generated by the projections $X \times \mathbb{A}^1 \rightarrow X$ and the Čech nerves of Nisnevich covers. Further write $W_{\mathrm{mot}}^\mathcal{C} \subseteq \mathrm{Ar}(\mathcal{P}_\Sigma(\mathcal{C}))$ for the strong saturation of the image of W_{mot} under μ . Notice that this is in particular of small generation.

Definition 9.1.2. We define the *category of unstable $\mathcal{C}$ -motives* as the symmetric monoidal Bousfield localization

$$\mathrm{Spc}^\mathcal{C}(k) := \mathcal{P}_\Sigma(\mathcal{C})[(W_{\mathrm{mot}}^\mathcal{C})^{-1}].$$

We further define the *stable category of $\mathcal{C}$ -motives* as the symmetric monoidal stabilization

$$\mathrm{SH}^\mathcal{C}(k) := \mathrm{Spc}^\mathcal{C}(k)[P^{-1}]$$

where $P := \mathrm{cofib}(\mu(*) \rightarrow \mu(\mathbb{P}^1))$.

Remark 9.1.3. Notice that if we would take μ to be the identity functor, which is not allowed in our setting above, $\mathrm{Spc}^\mathcal{C}(k) = \mathcal{P}_\Sigma(\mathrm{Sm}_k)[(W_{\mathrm{mot}})^{-1}] \simeq \mathrm{Spc}(k)$. indeed, we have that $\mathcal{P}_\Sigma(\mathrm{Sm}_k) \simeq \mathcal{P}(\mathrm{Sm}_k)[W_\Sigma^{-1}]$, where W_Σ is the strongly saturated class generated by $y(X) \amalg y(Y) \rightarrow y(X \amalg Y)$ for smooth schemes $X, Y \in \mathrm{Sm}_k$, and $y(\emptyset) \rightarrow \emptyset$. But $W_\Sigma \subseteq W_{\mathrm{mot}}$.

In particular, we obtain adjunctions induced by the functor $\mu : \mathcal{P}_\Sigma(\mathrm{Sm}_k) \rightarrow \mathcal{P}_\Sigma(\mathcal{C})$

$$\mathrm{Spc}(k) \xrightleftharpoons{\mu} \mathrm{Spc}^{\mathcal{C}}(k)$$

$$\mathrm{SH}(k) \xrightleftharpoons{\mu} \mathrm{SH}^{\mathcal{C}}(k)$$

Example 9.1.4. Consider the category of smooth k schemes with a disjoint basepoint, that is the category $\mathcal{C} = \mathrm{Sm}_{k,+} \subseteq \mathrm{Sm}_{k,*}$ given by the on the nose image of $(-)_+ : \mathrm{Sm}_k \rightarrow \mathrm{Sm}_{k,*}$. In formulas $\mathrm{Sm}_{k,+} := \{X \amalg * \mid X \in \mathrm{Sm}_k\}$. The monoidal structure is given by $(X_+ \otimes Y_+) \simeq (X \times Y)_+$ and is induced by the restriction of the smash product. Let $\mu = (-)_+ : \mathrm{Sm}_k \rightarrow \mathrm{Sm}_{k,+}$. As $\mathcal{P}_\Sigma(\mathrm{Sm}_{k,+}) \simeq \mathcal{P}_\Sigma(\mathrm{Sm}_k)_*$, see [BH21, Lemma 2.1], we obtain

$$\mathrm{Spc}^{\mathcal{C}}(k) \simeq \mathrm{Spc}(k).$$

Example 9.1.5. A more interesting example can be obtained by considering the subcategory $\mathcal{C} \subseteq \mathrm{Span}(\mathrm{Sch}_k)$ with objects Sm_k and morphisms those spans $(X \xleftarrow{f} Z \xrightarrow{g} Y)$ such that g is arbitrary and f is finite locally free morphisms, also known as finite flat morphisms. This really defines a subcategory, as finite locally free morphisms are stable under composition and basechange. Further, this category is symmetric monoidal if tensor product given by the product of schemes, finite products and coproducts exists and agree, in particular, $\mathcal{C}$ is pointed. We take $\mu : \mathrm{Sm}_k \rightarrow \mathcal{C}$ to be the inclusion of the right pointing morphisms. We obtain categories

$$\Sigma_+^\infty : \mathrm{Spc}^{\mathrm{ff}}(k) \xrightleftharpoons{\mu} \mathrm{SH}^{\mathrm{ff}}(k) : \Omega^\infty$$

These categories are known as the unstable and stable motives with finite locally free transfers.

This example allows for more variation. For one we can take trivial variations, replacing for example “finite locally free” with “finite étale” or “finite syntomic” to obtain categories $\mathrm{SH}^{\mathrm{ét}}(k)$ and $\mathrm{SH}^{\mathrm{syn}}(k)$ of motives with finite étale and finite syntomic transfers respectively.

As a not-so-trivial variation, we can also take as $\mathcal{C} = \mathrm{Cor}_k^{\mathrm{fr}}$ the category of framed correspondences. Roughly, objects in $\mathrm{Cor}_k^{\mathrm{fr}}$ are the smooth k schemes Sm_k as before, but the morphisms are instead given by spans $(X \xleftarrow{f} Z \xrightarrow{g} Y)$, such that g is arbitrary and f is finite syntomic, together with a specified zero-homotopy $[L_f] \simeq 0 \in K(Z)$ of the class of the cotangent complex of f in the K -theory space of Z . The composition becomes a lot harder in this context. We refer to [EHK⁺21]. In this case we obtain the category $\mathrm{SH}^{\mathrm{fr}}(k)$ of motives with framed transfers.

Example 9.1.6. Another example in the spirit of the precious one is given by the category of finite correspondences of Voevodsky $\mathcal{C} = \mathrm{Cor}_k$, which is a classical, discrete category. Its objects are again the smooth k -schemes, while the Hom-sets $\mathrm{Map}_{\mathrm{Cor}_k}((X), Y)$ are given as the free abelian group on the elementary correspondences from X to Y , that is an integral closed subscheme $Z \subseteq X \times Y$ such that the projection $Z \rightarrow X$ is finite and surjective over one component. The composition is defined using intersection theory. From this we obtain the category of Voevodsky motives $\mathcal{DM}(k) := \mathcal{DM}(k, \mathbb{Z}) := \mathrm{SH}^{\mathrm{Cor}_k}(k)$. As the notation indicates, we can also vary the ring in the definition of the Hom-sets to obtain as a variation Voevodsky motives with coefficients in A $\mathcal{D}(k, A)$.

Remark 9.1.7. The notation $\mathcal{DM}$ hints to the fact, that the category of Voevodsky motives can be seen as something like a derived category of motives. It is one of the goals of motivic homotopy theory to define a t -structure on $\mathcal{DM}$ such that the heart of this t -structure defines the “real motives”, and not only the derived ones.

Remark 9.1.8. There are functors, not necessarily obvious ones

$$\mathrm{Cor}_k^{\mathrm{fét}} \rightarrow \mathrm{Cor}_k^{\mathrm{fr}} \rightarrow \mathrm{Cor}_k^{\mathrm{fsyn}} \rightarrow \mathrm{Cor}_k$$

which in turn induce functors

$$\begin{array}{ccccccc} \mathrm{SH}^{\mathrm{fét}}(k) & \longrightarrow & \mathrm{SH}^{\mathrm{fr}}(k) & \longrightarrow & \mathrm{SH}^{\mathrm{fsyn}}(k) & \longrightarrow & \mathcal{DM}(k) \\ \uparrow & & & & & & \\ \mathrm{SH}(k) & & & & & & \end{array}$$

It is a big theorem that $\mathrm{SH}(k) \simeq \mathrm{SH}^{\mathrm{fr}}(k)$, even over a general base. It is also true, but less useful, that $\mathrm{SH}(k) \simeq \mathrm{SH}^{\mathrm{fét}}$.

9.2 Module categories

We want to describe the category of motives with transfer in another way. For this we need the language of categories of modules over algebras. We will not define them here, as this would need to introduce the language of operads, but only state some of their properties. We refer to [Cno26, Section II] for a first introduction.

Proposition 9.2.1. *Let $\mathcal{C}$ be a symmetric monoidal category and let $A \in \mathrm{CAlg}(\mathcal{C})$ be a commutative algebra object. Then there exists a category of A -modules in $\mathcal{C}$ which we denote by $\mathrm{Mod}_A(\mathcal{C})$. They satisfy the following properties.*

- *There exists a forgetful functor $U : \mathrm{Mod}_A(\mathcal{C}) \rightarrow \mathcal{C}$. This forgetful functor is conservative and admits a left adjoint, the free A -module functor $FC \rightarrow \mathrm{Mod}_A(\mathcal{C})$ such that the underlying object of a free module is given by $UF X \simeq A \otimes X$ for any $X \in \mathcal{C}$.*
- *Suppose that $\mathcal{C}$ is presentably symmetric monoidal. Then U preserves all sifted colimits and $\mathrm{Mod}_A(\mathcal{C})$ admits a symmetric monoidal structure, such that the free functor $F : \mathcal{C} \rightarrow \mathrm{Mod}_A(\mathcal{C})$ is symmetric monoidal. The tensor product is given as the relative tensor product over A , that is*

$$M_1 \otimes_A M_2 \simeq |M_1 \otimes M_2 \rightrightarrows M_1 \otimes A \otimes M_2 \rightrightarrows M_1 \otimes A \otimes M_2 \rightrightarrows \dots|$$

If $\mathcal{C}$ is moreover stable, U preserves all colimits.

- *Let $F : \mathcal{C} \rightleftarrows \mathcal{D} : G$ by an adjunction with $\mathcal{C}$ and $\mathcal{D}$ presentably symmetric monoidal, and the left adjoint F symmetric monoidal. By Corollary 7.6.3 we know that G is lax symmetric monoidal and hence induces*

$$\begin{array}{ccc} \mathrm{CAlg}(\mathcal{D}) & \xrightarrow{G} & \mathrm{CAlg}(\mathcal{C}) \\ \mathbb{1}_{\mathcal{D}} & \longmapsto & G(\mathbb{1}_{\mathcal{D}}) \end{array}$$

Furthermore there exists a canonical factorization

$$G : \mathcal{D} \xrightarrow{\bar{G}} \mathrm{Mod}_{G(\mathbb{1}_{\mathcal{D}})}(\mathcal{C}) \xrightarrow{U} \mathcal{C}$$

as lax symmetric monoidal functors.

Proof. This is contained in [Lur17b, Sections 7.2 to 7.5]. □

Lemma 9.2.2. *Let $F : \mathcal{C} \rightleftarrows \mathcal{D} : G$ by an adjunction with $\mathcal{C}$ and $\mathcal{D}$ presentably symmetric monoidal. Assume further that*

- $\mathcal{C}$ and $\mathcal{D}$ are stable;
- $\mathcal{C}$ is compact-rigidly generated, that is, there exists a family $\{S_\lambda \in \mathcal{C}\}_{\lambda \in \Lambda}$, such that each S_λ is compact and strongly dualizable, and the smallest stable subcategory of $\mathcal{C}$ closed under colimits and containing all the S_λ is $\mathcal{C}$ itself;
- $F(S_\lambda)$ is compact for all $\lambda \in \Lambda$;
- The $F(S_\lambda)$ generate $\mathcal{D}$.

which type of generation?

Then there is an equivalence $\overline{G} : \mathcal{D} \simeq \text{Mod}_{\mathbb{1}_{\mathcal{D}}}(\mathcal{C})$.

Proof. Consider the family $\{G(\mathbb{1}_{\mathcal{D}}) \otimes S_\lambda \in \text{Mod}_{G(\mathbb{1}_{\mathcal{D}})}(\mathcal{C})\}_{\lambda \in \Lambda}$. This forms a compact rigid generating family. Indeed, they are strongly dualizable as a tensor product of strongly dualizable objects, they are compact as U preserves filtered colimits, and they generate, as $\{\text{Hom}(G(\mathbb{1}_{\mathcal{D}}) \otimes S_\lambda, -)\}$ forms a jointly conservative family .

details

Hence we may replace $\mathcal{C}$ by $\text{Mod}_{G(\mathbb{1}_{\mathcal{D}})}(\mathcal{C})$ and thereby assume $G(\mathbb{1}_{\mathcal{D}}) \simeq \mathbb{1}_{\mathcal{C}}$. Since $F(\mathcal{C})$ generated $\mathcal{D}$ under colimits, G is conservative. Indeed, let $\alpha \in \text{Fun}(\Delta^1, \mathcal{C})$ be a morphism. Then $G(\alpha)$ is an equivalence if and only if $\text{Map}_{\mathcal{C}}(X, G(\alpha))$ is an equivalence for all $X \in \mathcal{C}$. This in turn is equivalent by adjunction to $\text{Map}_{\mathcal{D}}(FX, \alpha)$ is an equivalence for all $X \in \mathcal{C}$. By generation under colimits, this is equivalent to $\text{Map}_{\mathcal{D}}(Y, \alpha)$ being an equivalence for all $Y \in \mathcal{D}$, which is finally equivalent to α being an equivalence.

Hence it suffices to prove that F is fully faithful. Let $\mathcal{K} \subseteq \mathcal{C}$ be the full subcategory on those $X \in \mathcal{C}$ such that the unit induces an equivalence $X \xrightarrow{\simeq} GFX$. Then, by assumption

- $\mathbb{1} \in \mathcal{K}$, as F is symmetric monoidal and we assumed $G(\mathbb{1}_{\mathcal{D}}) \simeq \mathbb{1}_{\mathcal{C}}$.
- $S_\lambda \in \mathcal{K}$. Indeed, for all λ we have

$$GF(S_\lambda) \simeq G(F(S_\lambda) \otimes \mathbb{1}_{\mathcal{D}}) \simeq S_\lambda \otimes G(\mathbb{1}_{\mathcal{D}}) \simeq S_\lambda$$

where we used Lemma 8.2.5 in the third step. That this equivalence is homotopic to the unit follows from the explicit description of the projection formula.

- $\mathcal{K}$ is closed under desuspensions. Indeed, $\mathcal{C}$ and $\mathcal{D}$ are stable, hence F and G are exact, in particular commute with suspensions and desuspensions.
- $\mathcal{K}$ is closed under colimits. From the point before, we already know that F preserves all colimits, while G preserves finite colimits. Hence it suffices to show that G preserves filtered colimits. It suffices to check on generators, so we compute

$$\begin{aligned} \text{Hom}(S_\lambda, G(\text{colim}_i X_i)) &\simeq \text{Hom}(F(S_\lambda), \text{colim}_i X_i) \\ &\simeq \text{colim}_i \text{Hom}(F(S_\lambda), X_i) \\ &\simeq \text{colim}_i \text{Hom}(S_\lambda, G(X_i)) \\ &\simeq \text{Hom}(S_\lambda, \text{colim}_i G(X_i)). \end{aligned}$$

Here we used that the $F(S_\lambda)$ are compact by assumption.

Hence $\mathcal{K}$ is a stable subcategory of $\mathcal{C}$ which contains all S_λ and the monoidal unit, hence $\mathcal{K} \simeq \mathcal{C}$. $\square$

We obtain the following reformulation of the category of motives with transfer

Corollary 9.2.3. *Let k be a field which has resolution of singularities. Then*

$$\mathrm{SH}^{\mathcal{C}}(k) \simeq \mathrm{Mod}_{R^{\mathcal{C}}}(\mathrm{SH}(k))$$

where $R^{\mathcal{C}}$ is the image of the unit $1_{\mathrm{SH}^{\mathcal{C}}(k)}$ under the canonical morphism $\mu^* : \mathrm{SH}^{\mathcal{C}}(k) \rightarrow \mathrm{SH}(k)$, which is the right adjoint to the functor induced by μ .

Proof. Combine Theorem 8.6.1 and Lemma 9.2.2. □

Remark 9.2.4. The ring spectrum $R^{\mathcal{C}}$ for the examples of motives with transfers have all been identified/given names:

- $R^{\mathrm{fét}} \simeq 1_{\mathrm{SH}(k)} \simeq R^{\mathrm{fr}}$ the sphere spectrum;
- $R^{\mathrm{fsyn}} \simeq MGL$ the algebraic cobordism spectrum;
- $R^{\mathrm{ff}} \simeq kgl$ the effective K -theory spectrum;
- $R^{\mathrm{Cor}_k} =: \mathrm{HZ}$ the motivic cohomology spectrum;
- More generally $R^{\mathrm{Cor}_k^A} =: \mathrm{HA}$ the motivic cohomology spectrum with values in A .

In most of the cases, the objects appearing on the right hand side had already been defined, in which cases the equivalences are highly nontrivial theorem. We refer to [EHK⁺21] and the followup papers [EHK⁺20] and [HJN⁺25].

9.3 Properties of motives

Definition 9.3.1. Let $\mathcal{C}$ be a category for which we can define a motivic context, see the beginning of 9.1. We call $\mathcal{C}$ *compatible with the Nisnevich topology*, if for every Nisnevich cover $f : U \rightarrow X \in \mathrm{Sm}_k$ the morphism $\mu^* \mu(\check{C}(f) \rightarrow X) \in \mathcal{P}_{\Sigma}(\mathrm{Sm}_k)$ is a Nisnevich equivalence.

Here $\mu^* : \mathcal{P}_{\Sigma}(\mathcal{C}) \rightarrow \mathcal{P}_{\Sigma}(\mathrm{Sm}_k)$ is the right adjoint to μ .

Lemma 9.3.2. $\mathcal{C} = \mathrm{Cor}_k^{\mathrm{ff}}$ is compatible with the Nisnevich topology.

Proof. The right adjoint μ^* limits, so in particular fiber products, and sifted colimits as a functor between spherical presheaves. hence $\mu^* \mu \check{C}(f) \simeq \check{C}(\mu^* \mu f)$. As $\mu^* \mu X$ is 1-truncated, it suffices to show that $\mu^* \mu f(\mathcal{O})$ is surjective for all henselian local ring $\mathcal{O}$, that is, for every span as bellow, we find a dashed factorization

$$\begin{array}{ccc} & & U \\ & \nearrow \text{dashed} & \downarrow \\ & Z & \\ \swarrow \text{ff} & & \searrow \\ \mathcal{O} & & X \end{array}$$

However, as $Z \rightarrow \mathcal{O}$ is finite locally free, it follows from Proposition 3.1.4 that $Z = Z_1 \amalg \dots \amalg Z_n$, where the Z_i are henselian local and there exists a surjection $U(Z_i) \twoheadrightarrow X(Z_i)$ for all i . □

Remark 9.3.3. Similar arguments work in all the other examples.

Lemma 9.3.4. *Let $\mathcal{C}$ be a category for which we can define a motivic context, and assume that $\mathcal{C}$ is compatible with the Nisnevich topology. Then $\mu^* : \mathcal{P}_{\Sigma}(\mathcal{C}) \rightarrow \mathcal{P}_{\Sigma}(\mathrm{Sm}_k)$ commutes with $L_{\mathbb{A}^1}$, L_{Nis} , and L_{mot} .*

Find reference

Proof. By construction, μ^* already preserves Nisnevich, respectively $\mathbb{A}^1$, respectively motivic, local objects, hence it suffices to prove that μ^* preserves local equivalences. We know use the fact that if $W \subseteq \text{Fun}(\Delta^1, \mathcal{C})$ is a class of morphism, then the strong saturated class $\overline{W}$ generated by W agrees with the class generated under sifted colimits and 2-out-of-3 from the class $\{f \amalg \text{id}_Y \mid f \in W, Y \in \mathcal{C}\}$. We now set

- $W_1 := \{\check{C}(f) \rightarrow X \mid f : U \rightarrow X \text{ Nisnevich cover}\};$
- $W_2 := \{\mathbb{A}^1\text{-homotopy equivalences}\}.$

Then W_1 and W_2 are closed under $-\amalg \text{id}_{(-)}$. Since $\mu^*\mu$ preserves sifted colimits, see the proof of Lemma 9.2.2, it follows that for L_{Nis} it suffices to prove that $\mu^*\mu(W_1)$ consists of Nisnevich equivalences, which is true by assumption. For $L_{\mathbb{A}^1}$ it suffices to prove that $\mu^*\mu(W_2)$ consists of $\mathbb{A}^1$ -homotopy equivalences, but this follows from

$$\mu^*\mu(X) \times \mathbb{A}^1 \xrightarrow{\text{id} \times u} \mu^*\mu(X) \times \mu^*\mu(\mathbb{A}^1) \rightarrow \mu^*\mu(X \times \mathbb{A}^1)$$

where the first morphism is induced by the unit, and we used the symmetric monoidality of μ and the lax symmetric monoidality of μ^* .

For L_{mot} we simply take $W_1 \cap W_2$ and conclude. $\square$

Remark 9.3.5. Let $\mathcal{C}$ be a category for which we can define a motivic context. We mention two other deep properties.

- (1) We say that $\mathcal{C}$ is *strict $\mathbb{A}^1$ -invariant*, if for a Nisnevich local presheaf $E \in \mathcal{P}_{\Sigma}(\mathcal{C})$, there is an equivalence $L_{\text{mot}}E \simeq L_{\text{Nis}} \circ L_{\mathbb{A}^1}E$.
- (2) We say that $\mathcal{C}$ satisfies *cancellation*, if the functor $\text{Spc}^{\mathcal{C}}(k) \rightarrow \text{SH}^{\mathcal{C}}(k)$ is fully faithful on connected objects.

These properties “usually” hold, for example, they are true for $\text{Cor}^{\text{fr}}, \text{Cor}^{\text{fsyn}}, \text{Cor}^{\text{ff}},$ and Cor . It is however not known, whether they are true for $\text{Cor}^{\text{fét}}$.

Remark 9.3.6. As we did not introduce these notions, we only mention this as a remark. Usually, $\mathcal{C}$ is “semiadditive”, in which case one can write “grouplike” in place of “connected”.

9.4 Motivic (co)homology

In this section we give a short account of motivic cohomology. Remember that we denote the image of the unit of $\mathcal{DM}(k, R)$ under the functor $\mathcal{DM}(k, R) \rightarrow \text{SH}(k)$ as HR .

Definition 9.4.1. Let $X \in \text{Spc}(k)$. We define the *motivic homology of X with values in R* as

$$HR_{**}(X) := H_{**}(X, R) := \pi_{**}(HR \otimes \Sigma_+^{\infty} X).$$

We define the *motivic cohomology of X with values in R* as

$$HR^{**}X := H^{**}(X, R) := H^*(X, R(*)) = \pi_0 \text{Map}_{\text{SH}(k)}(\Sigma_+^{\infty} X, \Sigma^{**} HR).$$

Lastly, we define the *motive of X* as the image of $\Sigma_+^{\infty} X$ under the free HR module functor, that is, the motive of X is given by $HR \otimes \Sigma_+^{\infty} X \in \text{Mod}_{HR}(\text{SH}(k)) \simeq \mathcal{DM}(k, R)$.

Note in particular that cohomology forms a ring, as HR is a commutative algebra. Indeed, just as in topology, if $X \in \mathrm{Spc}(k)$ and we are given two elements $\alpha \in H^{p,q}(X, R)$ and $\beta \in H^{\alpha', \beta'}(X, R)$, we can represent them by morphisms

$$\begin{aligned}\alpha &: \Sigma_+^\infty X \rightarrow \Sigma^{p,q} HR \\ \beta &: \Sigma_+^\infty X \rightarrow \Sigma^{p',q'} HR\end{aligned}$$

and hence the corresponding multiplication as

$$\Sigma_+^\infty X \xrightarrow{\Delta} \Sigma_+^\infty (X \times X) \simeq \Sigma_+^\infty X \otimes \Sigma_+^\infty X \xrightarrow{\alpha \otimes \beta} \Sigma^{p,q} HR \otimes \Sigma^{p',q'} HR \xrightarrow{\mu} \Sigma^{p+p', q+q'} HR$$

and hence we obtain an element $\alpha \cdot \beta \in H^{p+p', q+q'}(X, R)$. One can show that this ring is graded commutative in the first degree, that is

$$\alpha \cdot \beta = (-1)^{pp'} \beta \cdot \alpha \in H^{p+p', q+q'}(X, R).$$

We elect not to do this and incited refer to

Setting $HR^{**} := HR_{**} := H^{**}(\mathrm{Spec} k, R) \simeq \pi_{**} HR$, which is in particular a ring, we see that, for every $X \in \mathrm{Spc}(X)$, the cohomology $H^{**}(X, R)$ is an HR^{**} -algebra via the terminal morphism $X \rightarrow *$, while $H_{**}(X, R)$ is an HR_{**} -module. The spaces $K(R(i), j) = HR^{j,i} = \Omega^\infty \Sigma^{j,i} HR \mathrm{Spc}(k)$ are called the *motivic Eilenberg-Mac Lane spaces*. By construction, they are the representing objects for Motivic cohomology.

We want to discuss a bit the relationship with symmetric powers. While all of the above was more or less applicable for all $\mathcal{C}$ for which a motivic context can be defined, the next part is special to motivic cohomology, and was historically the starting point for the theory of motivic cohomology in the sense Voevodsky and Suslin.

Remark 9.4.2. We first recall the relation in the topological situation. For this let X be a topological spaces. Define $\mathrm{Sp}^\infty(X) = \mathrm{colim}_N (\prod_{i=1}^N X) / \Sigma_N$. This is closely related to the free commutative monoid on the X viewed as a topological space, where multiplication is given by concatenation of sequences unordered tuples of points in X .

The *Dold-Thom theorem* now says that, if X is connected, then

$$\mathrm{Sp}^\infty(X) \simeq \mathbb{Z}\{X\} \in \mathrm{Ab}_\Delta$$

where, on the right hand side we view X as a simplicial set, take the free simplicial group on X . That is, the (Kan complex) $\mathrm{Sp}^\infty(X)$ is simplicially homotopy equivalent to the simplicial abelian group underlying the simplicial chain complex. In particular

$$\pi_i(\mathrm{Sp}^\infty(X)) \cong \tilde{H}_i(X)$$

where on the left hand side we may take any base point.

It was (perhaps) an Idea of Suslin, that one can take this as a definition for motivic cohomology. That is, given a smooth, quasi-projective scheme, one can define the *n-th symmetric power of X*

$$\mathrm{SP}^n(X) := X^n / \Sigma_n.$$

This is a quasi-projective scheme, but it is usually not smooth, making it hard to handle. However, in [Voe10] Voevodsky defined

$$\mathrm{SP}^n : \mathcal{P}_\Sigma(\mathrm{SmQP}_k) \rightarrow \mathcal{P}_\Sigma(\mathrm{SmQP}_k)$$

as the unique colimit preserving extension of

$$\begin{aligned}\mathrm{SP}^n : \mathrm{SmQP}_k &\rightarrow \mathcal{P}_\Sigma(\mathrm{SmQP}_k), \\ X &\mapsto y(\mathrm{SP}^n).\end{aligned}$$

find reference

and managed to show, that this functor preserves Nisnevich, $\mathbb{A}^1$, and hence, motivic equivalences, and thereby descends to a functor

$$\mathrm{Sp}^n : \mathrm{Spc}(k) \simeq \mathrm{L}_{\mathrm{mot}} \mathcal{P}_{\Sigma}(\mathrm{SmQP}_k) \rightarrow \mathrm{Spc}(k).$$

Here the first equivalence is apriori non-obvious. It uses that every smooth k -scheme admits a cover by open, quasi-projective k -schemes, see [BH21, Section 6.1].

Now, by construction, we have that $\mathrm{Sp}^n(*) \simeq *$ for all n , and hence, the n -th symmetric power descends to a functor

$$\mathrm{SP}^n : \mathrm{Spc}(k)_* \rightarrow \mathrm{Spc}(k)_*.$$

With a bit more work than what we are willing to do at this point, one can construct natural equivalences

$$\begin{aligned} \mathrm{Sp}^n &\rightarrow \mathrm{SP}^{n+1} \in \mathrm{Fun}(\mathrm{Spc}(k), \mathrm{Spc}(k)), \\ (x_1, \dots, x_n) &\mapsto (x_1, \dots, x_n, *). \end{aligned}$$

We then define the *motivic infinite symmetric powers* as $\mathrm{SP}^{\infty} := \mathrm{colim}_n \mathrm{SP}^n$. Voevodsky then showed the following.

Theorem 9.4.3 (Motivic Dold-Thom theorem). *With the notation from above, there are equivalences for $n \geq 1$*

$$K(\mathbb{Z}(n), 2n) \simeq \mathrm{SP}^{\infty}(S^{2n,n})$$

between the motivic Eilenberg-Mac Lane spaces and the motivic infinite symmetric powers of the sphere.

Example 9.4.4. • For $n = 0$ we have that $\mathrm{SP}^{\infty}(S^0) \simeq \mathbb{N}$, while one can show that $K(\mathbb{Z}(0), 0) \simeq \mathbb{Z}$.

- For $n = 1$ we have that $S^{2,1} \simeq \mathbb{P}^1$. Further, there is an isomorphism of schemes $\mathrm{SP}^n(\mathbb{P}^1) \cong \mathbb{P}^n$, by mapping a point on the right hand side, given by homogeneous coordinates $[x_0 : \dots : x_n]$ to the unordered n -tuple of roots of the homogeneous polynomial $x_0 U^n + x_1 U^{n-1} T + \dots x_n T^n$. One can show that $K(\mathbb{Z}(1), 2) \simeq \mathbb{P}^{\infty}$.

In algebraic topology, integral coefficients for invariants are usually hard, while considering rational coefficients is a lot easier. Something similar happens in motivic homotopy theory. Indeed

Theorem 9.4.5 (Voevodsky). *Let $X \in \mathrm{Spc}(X)_*$. Then the Σ_r -invariant morphism $X^r \rightarrow \mathrm{SP}^r(X)$ induces a factorization*

$$\begin{array}{ccc} X^r & \longrightarrow & \mathrm{SP}^r(X) \\ \downarrow & \nearrow \text{dashed} & \\ X_{h\Sigma_r}^r & & \end{array}$$

which becomes an equivalence in $\mathcal{DM}(k, \mathbb{Q})$.

Here $(-)_h\Sigma_r$ denotes the *homotopy orbits* of the Σ_r action,

Remark 9.4.6. Let G be a discrete group. Then one can define the discrete category BG , which has one object and as morphisms the elements of G , while composition is given by the group law of G . Let $\mathcal{C}$ be a category. Then, the *category of G objects in $\mathcal{C}$* is defined as $\mathrm{Fun}(\mathrm{BG}, \mathcal{C})$. Given a G -object X in $\mathcal{C}$, that is, $X \in \mathrm{Fun}(\mathrm{BG}, \mathcal{C})$ we define the *homotopy fixed points* as

$$X^{hG} := \lim_{\mathrm{BG}} X,$$

and the *homotopy orbits* as

$$X_{hG} := \mathrm{colim}_{\mathrm{BG}} X.$$

Proposition 9.4.7. *Let $\mathcal{C}$ be a stable and idem-complete category, such that $\pi_0 \text{Map}_{\mathcal{C}}(X, Y) \in \text{Vect}_{\mathbb{Q}} \subseteq \text{Ab}$ for all $X, Y \in \mathcal{C}$. Then for $X \in \text{Fun}(\text{BG}, \mathcal{C})$ and $|G| < \infty$, we have that both homotopy orbits and colimits exist, and the composition*

$$X^{hG} \longrightarrow X \longrightarrow X_{hG}$$

is an equivalence. In particular, both homotopy orbits and fixpoints are summands of X , and the projector on X is given by $\frac{1}{|G|} \sum_{g \in G} g$.

Proof.

□ todo

We want to compute $(\Sigma_+^{\infty} \text{SP}^{\infty}(S^{2n,n})) \otimes H\mathbb{Q}$. Using compatibility of the tensor product with colimits, we may start with calculating the finite symmetric powers. Using Theorem 9.4.5 together with the compatibility of $\otimes$ and Σ_+^{∞} with colimits, we obtain

$$M(\text{SP}^r(S^{2n,n})) := (\Sigma_+^{\infty} \text{SP}^r(S^{2n,n})) \otimes H\mathbb{Q} \simeq ((\Sigma_+^{\infty}(S^{2n,n})^r) \otimes H\mathbb{Q})_{h\Sigma_r} \in \mathcal{DM}(k, \mathbb{Q})$$

where we set $M : \text{Spc}(k) \rightarrow \mathcal{DM}(k, \mathbb{Q})$, $X \mapsto M(X) := (\Sigma_+^{\infty} X) \otimes H\mathbb{Q}$ for the *motivic functor*, assigning a motivic space its motive.

By construction, M , as a composition of symmetric monoidal functors, is symmetric monoidal and preserves coproducts. From the compatibility with coproducts, we obtain a stable splitting

$$M(S^{2n,n}) := (\Sigma_+^{\infty}(S^{2n,n})^r) \otimes H\mathbb{Q} \simeq (\Sigma^{\infty} S^{0,0} \oplus \Sigma^{\infty} S^{2n,n}) \otimes H\mathbb{Q} \simeq H\mathbb{Q} \oplus \Sigma^{2n,n} H\mathbb{Q}.$$

Using the symmetric monoidality, we hence obtain

$$\begin{aligned} M((S^{2n,n})^{\otimes r}) &\simeq (H\mathbb{Q} \oplus \Sigma^{2n,n} H\mathbb{Q})^{\otimes r} \\ &\simeq H\mathbb{Q} \oplus (\Sigma^{2n,n} H\mathbb{Q})^{\oplus r} \oplus \dots \oplus (\Sigma^{2n,n} H\mathbb{Q})^{\otimes r}. \end{aligned}$$

In the last line we silently already identified each term of the form $1 \otimes X = H\mathbb{Q} \otimes X \simeq X$. To understand the Σ^r action, it is however important to remember where the monoidal units appeared. First of all Σ_r acts transitively on each summand separately. Furthermore, as Σ_r is given by permuting the factors in the tensor products, we see that it acts on each summand transitively. Furthermore, the swap map on $\Sigma^{2n,n} H\mathbb{Q} \otimes \Sigma^{2n,n} H\mathbb{Q}$ is given by the identity, see also the formula for the product in motivic cohomology. This is related to the product formula, where we would obtain a factor of $(-1)^{4n^2} = 1$. We hence obtain from Proposition 9.4.7

$$M((S^{2n,n})^r)_{h\Sigma_r} \simeq H\mathbb{Q} \oplus \Sigma^{2n,n} H\mathbb{Q} \oplus \Sigma^{4n,2n} H\mathbb{Q} \oplus \dots \oplus \Sigma^{2nr,nr} H\mathbb{Q}.$$

Being more careful, one may prove

Proposition 9.4.8. *$M(\text{SP}^r(S^{2n,n})) \in \mathcal{DM}(k, \mathbb{Q})$ splits into a finite direct sum of terms of $\Sigma^{2*,*} H\mathbb{Q}$. Moreover, we have*

$$H\mathbb{Q}^{**}(\text{SP}^r(S^{2n,n})) \simeq H\mathbb{Q}^{**}[t_n]/t_n^{r+1},$$

where $|t_n| = (2n, n)$.

Remark 9.4.9. Under the canonical map $S^{2n,n} \simeq \text{SP}^1(S^{2n,n}) \rightarrow \text{SP}^r(S^{2n,n})$, the class $t_n \in H^{**}(\text{SP}^r(S^{2n,n}), \mathbb{Q})$ is sent to $x_n \in H^{2n,n}(S^{2n,n}, \mathbb{Q})$, and, in fact, t_n is sent to the unique preimage of x_n . This is, however, non-obvious, and more or less part of the above proposition.

It is furthermore worth to remark that knowing the cohomology determines the motive, as this is a finite sum of shifts of the unit $H\mathbb{Q}$, as we “only” have to label the generating elements and understand their behavior, which is what happens in Proposition 9.4.8.

Corollary 9.4.10. *There is an equivalence $H^{**}(K(\mathbb{Z}(n), 2n), \mathbb{Q}) \cong^{**} [t_n]$. Furthermore, there are equivalences*

$$H\mathbb{Z} \otimes H\mathbb{Q} \simeq H\mathbb{Q} \otimes H\mathbb{Q} \simeq H\mathbb{Q} \in \mathrm{SH}(k).$$

Proof. As $K(\mathbb{Z}(n), 2n) \simeq \mathrm{colim}_r \mathrm{SP}^R(S^{2n,n})$, the first part follows as usual from taking the limits, and realizing that the $\lim^1$ term vanishes by the Mittag-Leffler condition, and the explicit calculation from Proposition 9.4.8.

For the second part, by definition of the motivic Eilenberg-Mac Lane spaces, there is an equivalence

$$H\mathbb{Z} \simeq \mathrm{colim}_n \Sigma^{-2n,n} \Sigma^\infty K(\mathbb{Z}(n), 2n).$$

Using compatibility of the tensor product with colimits, we obtain

$$\begin{aligned} H\mathbb{Z} \otimes H\mathbb{Q} &\simeq \mathrm{colim}_n \Sigma^{-2n, -n}(H\mathbb{Q} \otimes \Sigma^\infty K(\mathbb{Z}(n), 2n)) \\ &\simeq \mathrm{colim}_n (\Sigma^{-2n, -n}(\Sigma^{2,1} H\mathbb{Q} \oplus \Sigma^{4,2} H\mathbb{Q} \oplus \dots)) \rightarrow \\ &\Sigma^{-4, -2}(\Sigma^{4,2} H\mathbb{Q} \oplus \Sigma^{8,4} H\mathbb{Q} \oplus \dots) \rightarrow \\ &\Sigma^{-6, -3}(\Sigma^{6,3} H\mathbb{Q} \oplus \Sigma^{12,6} H\mathbb{Q} \oplus \dots) \rightarrow \dots \end{aligned}$$

To calculate the colimit we have to determine the morphisms. These morphisms are, up to desuspensions, dual to the morphisms

$$H\mathbb{Q}^{**}(\Sigma^{2,1} K(\mathbb{Z}(n), 2n) \rightarrow K(\mathbb{Z}(n+1), 2n+2)) \simeq (H\mathbb{Q}^{**}[t_{n+1}] \rightarrow \Sigma^{2,1} H\mathbb{Q}[t_n] \simeq H\mathbb{Q}\{1, \sigma 1, \sigma t_n, \dots\}),$$

where σ is the action of the suspension isomorphism on the $H\mathbb{Q}^{**}$ module generators of $H\mathbb{Q}[t_n]$. For degree reasons, t_{n+1} can only be mapped to 0 or to σt_n . By more formal reasons, which we cannot state here due to our lack of discussion, $t_{n+1} \mapsto \sigma t_n$. Now we know that the ring structure on the cohomology of the suspension of a space is trivial, so that $t_{n+1}^i \mapsto 0$ for $i \geq 1$. From this we obtain that the morphisms in the above colimit diagram are given by the identity on the first summand, and by zero on all others, so that the colimit degenerates to a constant sequential diagrams of $H\mathbb{Q}$ and hence is equivalent to $H\mathbb{Q}$.

For the last part, we note that $\mathbb{Q} \cong \mathrm{colim}(\mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{3} \mathbb{Z} \xrightarrow{4} \dots)$. Using functoriality of $H : \mathrm{CRing} \rightarrow \mathrm{DM}(k)$, we obtain

$$H\mathbb{Q} \simeq \mathrm{colim}(H\mathbb{Z} \xrightarrow{2} H\mathbb{Z} \xrightarrow{3} H\mathbb{Z} \xrightarrow{4} \dots),$$

and hence

$$H\mathbb{Q} \otimes H\mathbb{Q} \simeq \mathrm{colim}(H\mathbb{Q} \xrightarrow{2} H\mathbb{Q} \xrightarrow{3} H\mathbb{Q} \xrightarrow{4} \dots) \simeq H\mathbb{Q}.$$

□

The last thing we want to mention regarding motivic cohomology, is the motivic Steenrod algebra. For this, consider first an odd prime l . There are natural operations for $X \in \mathrm{Spc}(k)$ and $p, q, i \in \mathbb{Z}$

$$P^i : H^{p,q}(X, \mathbb{F}_l) \rightarrow H^{p+2i(l-1), q+i(l-1)}(X, \mathbb{F}_l).$$

We further have the Bockstein homomorphism

$$\beta : H^{p,q}(X, \mathbb{F}_l) \rightarrow H^{(p+1), q}(X, \mathbb{F}_l)$$

defined as the boundary morphism of the long exact sequence associated to the short exact sequence of coefficient groups $0 \rightarrow \mathbb{F}_l \rightarrow \mathbb{Z}/l^2 \rightarrow \mathbb{F}_l \rightarrow 0$.

These operations satisfy the following properties.

work out argument. Idea: due to suspension need to involve diagonal $S1$ to $S1 \wedge S1$, which is automatically null

- They are natural in X ;
- $P^i = 0$ for all $i < 0$, and $P^0 = \text{id}$;
- They satisfy Cartan-type formulas $P^i(a \cdot b) = \sum_{i=j+k} P^j(a) \cdot P^k(b)$, and $\beta(a \cdot b) = \beta(a)b + a\beta(b)$;
- They are bistable, that is, the following two diagrams commute

$$\begin{array}{ccc}
H^{**}(\Sigma X, \mathbb{F}_l) & \longrightarrow & H^{\bullet\bullet}(\Sigma X, \mathbb{F}_l) \\
\cong \downarrow & & \downarrow \cong \\
H^{*-1,*}(X, \mathbb{F}_l) & \longrightarrow & H^{\bullet-1,\bullet}(X, \mathbb{F}_l)
\end{array}
\qquad
\begin{array}{ccc}
H^{**}(\mathbb{G}_m \wedge X, \mathbb{F}_l) & \longrightarrow & H^{\bullet\bullet}(\mathbb{G}_m \wedge X, \mathbb{F}_l) \\
\cong \downarrow & & \downarrow \cong \\
H^{*-1,*-1}(X, \mathbb{F}_l) & \longrightarrow & H^{\bullet-1,\bullet-1}(X, \mathbb{F}_l)
\end{array}$$

where the horizontal arrows can be filled with either P^i or β ;

- If $u \in H^{2n,n}(X, \mathbb{F}_l)$ then $P^n u = u^l$;
- $u \in H^{p,q}(X, \mathbb{F}_l)$ such that $n \geq p - q$ and $n \geq q$, then $P^n u = 0$.
- They satisfy an analog of the Adem relation. The explicit formula is too complicated to write it down here.

Formula

Let now $l = 2$. In this case we have that $K(\mathbb{Z}(1), 2) \simeq \mathbb{P}^1 \simeq B\mathbb{G}_m$, see Example 9.4.4. Due to this, one can identify motivic cohomology with classical cohomology with values in $\mathbb{G}_m$

$$H^{*,1}(X, \mathbb{Z}) \cong H_{\text{Zar}}^{*-1}(X, \mathbb{G}_m).$$

In particular, one has for $X \in \text{Sm}_k$

$$\begin{aligned}
H^{0,1}(X, \mathbb{Z}) &= 0 \\
H^{1,1}(X, \mathbb{Z}) &\cong \mathcal{O}^\times(X) \\
H^{2,1}(X, \mathbb{Z}) &\cong \text{Pic}(X)
\end{aligned}$$

We can now use the long exact sequence of cohomology groups induced by the short exact sequence of coefficients $0 \rightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \rightarrow \mathbb{Z}/2$ to calculate the motivic cohomology of $\text{Spec} k$ with $\mathbb{Z}/2$ coefficients:

$$\begin{array}{ccccccc}
H^{0,1}(k, \mathbb{Z}) & \longrightarrow & H^{0,1}(k, \mathbb{Z}/2) & \xrightarrow{\partial} & H^{1,1}(k, \mathbb{Z}) & \xrightarrow{2} & H^{1,1}(k, \mathbb{Z}) \longrightarrow H^{1,1}(k, \mathbb{Z}/2) \longrightarrow H^{2,1}(k, \mathbb{Z}) \\
\cong \downarrow & & & & \downarrow \cong & & \downarrow \cong \\
0 & & & & k^\times & & k^\times \longrightarrow 0
\end{array}$$

It follows that $H^{0,1}(k, \mathbb{Z}/2) \cong \ker(k^\times \xrightarrow{2} k^\times)$, while $H^{1,1}(k, \mathbb{Z}/2) \cong \text{coker}(k^\times \xrightarrow{2} k^\times)$. Assume now, that $\text{char}(k) \neq 2$. Then

$$\begin{aligned}
H^{0,1}(k, \mathbb{Z}/2) &\cong \{\pm 1\} \cong \mathbb{Z}/2\{\tau\}, \\
-1 &\mapsto \tau
\end{aligned}$$

while

$$H^{1,1}(k, \mathbb{Z}/2) \cong k^\times / 2.$$

We denote the preimage of the class of (-1) under the above isomorphism by ρ . Then, by the usual properties of the Bockstein homomorphism, we see that

$$\beta(\tau) = \rho.$$

With these notions, there exists natural operations, the *motivic Steenrod operations*

$$\begin{aligned}\mathrm{Sq}^{2i} &:= P^i : H^{**}(X, \mathbb{F}_2) \rightarrow H^{*+2i, *+i}(X, \mathbb{F}_2), \\ \mathrm{Sq}^{2i+1} &:= \beta \mathrm{Sq}^{2i}.\end{aligned}$$

These operations satisfy naturality, bistability, the squaring property, and the Cartan formula as before, but this time, we can explicitly state the Adem relations, although only for a, b even:

Let $0 < a < 2b$ and a, b even. Then

$$\mathrm{Sq}^a \mathrm{Sq}^b = \sum_{t=0}^{\lfloor \frac{a}{b} \rfloor} \tau^{c_t} \binom{b-t-1}{a-2t} \mathrm{Sq}^{a+b-t} \mathrm{Sq}^t$$

where $c_t \equiv t \pmod{2}$ such that the grading works out. If a and b are not even, we get extra terms involving ρ .

Chapter 10

The sphere spectrum

In this chapter we give an overview of some theorems involving the motivic sphere spectrum. Our ultimate goal will be to prove that the Steinberg relations hold in the homotopy groups of the motivic sphere spectrum.

10.1 Milnor-Witt K -theory

We begin with an ad-hoc definition of a variant of K -theory, which plays a mayor role in motivic cohomology theory.

Definition 10.1.1. Let F be a field. Denote by A_F the free associative, graded ring on the generator η in degree (-1) and for all $a \in F^\times$ a generator $[a]$ in degree 1. We define the *Milnor-Witt K -groups of F* as

$$K_*^{\text{MW}}(F) = A_F/I$$

where I is the two sided ideal generated by the four relations

- $[ab] = [a] + [b] + \eta[a][b]$ for all $a, b \in F^\times$;
- $[a][1 - a] = 0$ for all $a \in F \setminus \{0, 1\}$, these are known as the *Steinberg relations*;
- $\eta[a] = [a]\eta$;
- $\eta(2 + \eta[-1])$.

Remark 10.1.2. It is clear from the definition that the quotient K_*^{MW}/η is given by Milnor K -theory K_*^{M} , see for example [Lam05] for the classical definition of Milnor K theory as the free associated graded ring on generators the units of the field F , satisfying the Steinberg relations.

The next theorems are deep results.

Theorem 10.1.3 (Morel). *For any field F there is an isomorphism*

$$\pi_0(\mathbb{1})_* := \pi_0(\mathbb{1})_*(F) \cong K_*^{\text{MW}}(F)$$

where $\mathbb{1} \in \text{SH}(F)$ denotes the monoidal unit, that is, the sphere spectrum.

Definition 10.1.4. We define the *unramified Milnor-Witt K -groups* as

$$\underline{K}_*^{\text{MW}} := \pi_0(\mathbb{1})_* \in \text{HI}_*(F)$$

We further define the *unramified Milnor K -groups* as

$$\underline{K}_*^{\text{M}} := \underline{K}_*^{\text{MW}}/\eta.$$

Theorem 10.1.5 (Nesterenko-Suslin, Totaro). *The unit map $\mathbb{1} \rightarrow H\mathbb{Z} \in \mathrm{SH}(F)$ induces an equivalence*

$$\pi_0(H\mathbb{Z})_* \simeq \underline{K}_*^{\mathrm{M}}.$$

Furthermore, $\pi_i(H\mathbb{Z})_* = 0$ for all $i < 0$.

We are not going to prove these theorems, we will, however, explain a bit what is going on. This will be the context of the next section.

10.2 Glimpses of the theorems

In this section we want to construct the morphism from Milnor-Witt K -theory into the global sections of the degree 0 homotopy sheaf of the unit. For this we need to construct images of the generators $[a]$ for all $a \in F^\times$, and the generator η . We start with the generators $[a]$.

For this, consider the composition

$$\begin{aligned} \mathcal{O}^\times(F) &\xrightarrow{\simeq} \pi_0 \mathrm{Map}_{\mathrm{Spc}(F)}(*, \mathbb{G}_m) \xrightarrow{\simeq} \pi_0 \mathrm{Map}_{\mathrm{Spc}(A_n)}(S^0, \mathbb{G}_m) \xrightarrow{\Sigma^\infty} \pi_0 \mathrm{Map}_{\mathrm{SH}(F)}(\mathbb{1}, \Sigma^{1,1}\mathbb{1}) \\ &\simeq \pi_0(\mathbb{1})_1. \end{aligned}$$

We define the images of $a \in \mathcal{O}^\times(F) \cong F^\times$ as the elements $[a]$.

For η consider the Hopf morphism $H : S^1 \otimes \mathbb{G}_m \otimes \mathbb{G}_m \simeq (\mathbb{A}^2 \setminus 0) \rightarrow \mathbb{P}^1 \simeq S^1 \otimes \mathbb{G}_m \in \mathrm{Spc}(F)_*$. Stabilizing and desuspending accordingly, we obtain

$$\eta := \Sigma^{-3,-2} \Sigma^\infty H : \mathbb{1} \rightarrow \Sigma^{-1,-1}\mathbb{1}.$$

Hence we obtain an element $\eta \in \pi_0(\mathbb{1})_{-1}$. One would now need to check that these defined objects satisfy all the relations of Milnor-Witt K theory. We will check some of the relations later on. But first, we want to discuss the meaning of $\eta = 0$ in $H\mathbb{Z}$.

For this, recall that $\mathbb{P}^2 \setminus \mathbb{P}^1 \cong \mathbb{A}^2 \in \mathrm{Sm}_F$ where we embed $\mathbb{P}^1 \hookrightarrow \mathbb{P}^2$ as the line at infinity. This implies that we have the following sequence of equivalences in $\mathrm{Spc}(F)$

$$\begin{aligned} \mathbb{P}^2 &\simeq \mathbb{P}^2 / \mathbb{A}^2 \simeq \mathbb{P}^2 / (\mathbb{P}^2 \setminus \mathbb{P}^1) \\ &\simeq \mathrm{Th} N_{\mathbb{P}^1 / \mathbb{P}^2} \\ &= N_{\mathbb{P}^1 / \mathbb{P}^2} / (N_{\mathbb{P}^1 / \mathbb{P}^2} \setminus 0) \\ &\simeq \mathbb{P}^1 / (\mathbb{A}^2 \setminus 0) \\ &\simeq \mathrm{cofib} H. \end{aligned}$$

Here in the second line we used Purity Theorem 5.0.1, to identify the quotient with the normal bundle of the inclusion, which is given by the canonical bundle $\mathcal{O}(1)$. Phrased differently, there is a cofiber sequence in $\mathrm{Spc}(F)$

$$\mathbb{A}^2 \setminus 0 \simeq S^{3,2} \xrightarrow{\eta''} \mathbb{P}^1 \simeq S^{2,1} \rightarrow \mathbb{P}^2$$

Hence we see that $\eta = 0$ in $\mathcal{DM}(F)$ if and only if the motive of $\mathbb{P}^2$ splits as $M(\mathbb{P}^2) \simeq \mathbb{1} \oplus S^{2,1} \oplus S^{4,2}$.

This is an incarnation of the projective bundle formula, a property, which the motive functor processes. Indeed, we have

$$M(\mathbb{P}^n) \simeq \mathbb{1} \oplus S^{2,1} \oplus \dots \oplus S^{2n,n}.$$

We now construct a further element in $\pi_0(\mathbb{1})_0$, which measures the failure of commutativity in $\pi_0(\mathbb{1})_0$:

We denote by $\varepsilon \in \pi_0(\mathbb{1})_0$ the image of $u \mapsto u^{-1} \in F^\times$ under the composition

$$\pi_0 \mathrm{Map}_{\mathrm{Spc}(F)}(\mathbb{G}_m, \mathbb{G}_m) \xrightarrow{\Sigma^\infty} \pi_0 \mathrm{Map}_{\mathrm{SH}(F)}(\Sigma^\infty \mathbb{G}_m, \Sigma^\infty \mathbb{G}_m) \simeq \pi_0 \mathrm{Map}_{\mathrm{SH}(F)}(\mathbb{1}, \mathbb{1}).$$

Why?

Lemma 10.2.1. *The element ε satisfies the following properties.*

- *The pointed morphism*

$$\begin{aligned} f : (\mathbb{P}^1, [1 : 1]) &\rightarrow (\mathbb{P}^1, [1 : 1]) \\ [x : y] &\rightarrow [y : x] \end{aligned}$$

corresponds to

$$\begin{aligned} -\varepsilon &\in \pi_0(\mathbb{1})_0 \simeq \pi_0 \mathrm{Map}_{\mathrm{SH}(F)}(\mathbb{1}, \mathbb{1}) \\ &\simeq \pi_0 \mathrm{Map}_{\mathrm{SH}(F)}(\Sigma^{2,1}\mathbb{1}, \Sigma^{2,1}\mathbb{1}) \\ &\simeq \pi_0 \mathrm{Map}_{\mathrm{SH}(F)}(\Sigma^\infty \mathbb{P}^1, \Sigma^\infty \mathbb{P}^1). \end{aligned}$$

- *The morphism*

$$\mathrm{switch} : \mathbb{G}_m \otimes \mathbb{G}_m \rightarrow \mathbb{G}_m \otimes \mathbb{G}_m$$

corresponds to

$$\begin{aligned} \varepsilon &\in \pi_0(\mathbb{1})_0 \simeq \pi_0 \mathrm{Map}_{\mathrm{SH}(F)}(\mathbb{1}, \mathbb{1}) \\ &\simeq \pi_0 \mathrm{Map}_{\mathrm{SH}(F)}(\Sigma^{2,2}\mathbb{1}, \Sigma^{2,2}\mathbb{1}) \\ &\simeq \pi_0 \mathrm{Map}_{\mathrm{SH}(F)}(\mathbb{G}_m \otimes \mathbb{G}_m, \mathbb{G}_m \otimes \mathbb{G}_m). \end{aligned}$$

Proof. Consider the following pushout diagram

$$\begin{array}{ccc} \mathbb{G}_m & \longrightarrow & \mathbb{A}^1 \\ \downarrow & & \downarrow \\ \mathbb{A}^1 & \longrightarrow & \mathbb{P}^1 \end{array} \quad \lrcorner$$

The automorphism f of $\mathbb{P}^1$ corresponds to the automorphism of $\mathbb{G}_m$ given by $u \mapsto u^{-1}$, which corresponds to ε . However, to make this into an automorphism of the diagram, we cannot simply act on $\mathbb{G}_m$ and $\mathbb{P}^1$, as f interchanges the coordinates. Hence we also have to swap the corners of the diagram. As $\mathbb{A}^1 \simeq * \in \mathrm{Spc}(F)$, the swapping of the coordinates corresponds to the swapping of the suspension coordinates, which in turn corresponds to an additional (-1) in the group structure of $\pi_0(\mathbb{1})_0$. This shows the first point.

To prove the second point, we use that S^1 is invertible in $\mathrm{SH}(F)$ and consider instead $\mathbb{P}^1 \simeq S^1 \otimes \mathbb{G}_m$. The switch morphism on S^1 corresponds again to (-1) in the group structure, hence it suffices to show that $\mathrm{swap} : \mathbb{P}^1 \otimes \mathbb{P}^1 \rightarrow \mathbb{P}^1 \otimes \mathbb{P}^1$ corresponds to $(-\varepsilon)$. We have

$$\mathbb{P}^1 \otimes \mathbb{P}^1 \simeq \mathbb{A}^2 / (\mathbb{A}^2 \setminus 0) \in \mathrm{Spc}(F)$$

and the switch morphism on $\mathbb{P}^1 \otimes \mathbb{P}^1$ is represented under the above equivalence by the matrix

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

But the second matrix in the decomposition is an element of the special linear group $\mathrm{SL}_2(\mathbb{Z})$, which is generated by the elementary matrices

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

which are both $\mathbb{A}^1$ -invariant to the identity, see also the proof of Lemma 7.7.7. Hence, the switch map on $\mathbb{P}^1$ is homotopic to the morphism represented by

$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

But this corresponds to the morphism $h \otimes \text{id} : \mathbb{P}^1 \otimes \mathbb{P}^1 \rightarrow \mathbb{P}^1 \otimes \mathbb{P}^1$, where

$$\begin{aligned} h : \mathbb{P}^1 &\rightarrow \mathbb{P}^1 \\ [x : y] &\mapsto [-x : y] \end{aligned}$$

which is represented by the matrix

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

This matrix is now $\mathbb{A}^1$ -homotopic to

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

which in turn represents the automorphism

$$\begin{aligned} \mathbb{P}^1 &\rightarrow \mathbb{P}^1 \\ [x : y] &\mapsto [y : x]. \end{aligned}$$

But this is the morphism from the first part of the lemma, which we proved to represent $(-\varepsilon)$. $\square$

We now return to the element $\eta \in \pi_0(\mathbb{1})_{-1}$. Using Lemma 7.2.1, we see that the multiplication map

$$\mu : \mathbb{G}_m \times \mathbb{G}_m \rightarrow \mathbb{G}_m \in \text{Sm}_F$$

admits a stable splitting

$$\Sigma^\infty \mu \simeq \mu_1 \oplus \mu_2 \oplus \mu_3 : \Sigma^\infty \mathbb{G}_m \oplus \Sigma^\infty \mathbb{G}_m \oplus \Sigma^\infty (\mathbb{G}_m \otimes \mathbb{G}_m).$$

Lemma 10.2.2. *The stable splitting of the multiplication map is given by*

$$\begin{aligned} \mu_1 = \mu_2 = \text{id}_{\Sigma^\infty \mathbb{G}_m} : \Sigma^\infty \mathbb{G}_m &\rightarrow \Sigma^\infty \mathbb{G}_m, \\ \mu_3 = \eta = \eta\varepsilon : \Sigma^\infty (\mathbb{G}_m \otimes \mathbb{G}_m) &\rightarrow \Sigma^\infty \mathbb{G}_m. \end{aligned}$$

Proof. The first two identifications follow directly from the fact that $\mu(x, 1) = \mu(1, x) = x$. Indeed, this follows from μ_1 and μ_2 being those components of $\Sigma^\infty \mu$ which live on $\Sigma^\infty (\mathbb{G}_m \vee \mathbb{G}_m)$, that is, the elements $(x, 1)$ and $(1, x)$ in $\mathbb{G}_m$.

For the last identity, consider the morphism $H : \mathbb{A}^2 \setminus 0 \rightarrow \mathbb{P}^1$. Covering both $\mathbb{A}^2 \setminus 0$ and $\mathbb{P}^1$ by a two piece cover in the obvious compatible way, we obtain the following diagram

$$\begin{array}{ccccc} & & \mu \circ (\text{id} \times (u \mapsto u^{-1})) & & \\ & \searrow & & \searrow & \\ \mathbb{G}_m \times \mathbb{G}_m & \hookrightarrow & \mathbb{G}_m \times \mathbb{A}^1 & \xrightarrow{\quad} & \mathbb{G}_m \longrightarrow \mathbb{A}^1 \\ \downarrow & & \downarrow & & \downarrow \\ \mathbb{G}_m \times \mathbb{A}^1 & \hookrightarrow & \mathbb{A}^2 \setminus 0 & \xrightarrow{\quad} & \mathbb{A}^1 \longrightarrow \mathbb{P}^1 \\ & & & \searrow & \\ & & & & H \end{array}$$

Here the top map arises from tracing the Hopf map through the diagram to the top left. Considering this diagram in $\mathrm{SH}(F)$, we see that both the two corners of the left hand diagram cancel out μ_1 and μ_2 . Hence, we obtain a homotopy

$$\Sigma^\infty H \simeq \Sigma^{3,2} \mu_3(\mathrm{id} \otimes \varepsilon).$$

Notice that the additional $\Sigma^{2,2}$ comes from ε being defined on $\mathbb{1}$, and the equivalence $\mathbb{G}_m \otimes \mathbb{G}_m \simeq \Sigma^{1,1} \mathbb{1} \otimes \Sigma^{1,1} \mathbb{1} \simeq \Sigma^{2,2} \mathbb{1}$ inside μ_3 . Using that $\varepsilon^2 \simeq \mathrm{id}$ we obtain by precomposition $\eta\varepsilon = \mu_3$.

But we also have $\mu_3 \circ \varepsilon = \mu_3$, as ε corresponds to $\mathrm{flp} : \mathbb{G}_m \otimes \mathbb{G}_m \rightarrow \mathbb{G}_m \otimes \mathbb{G}_m$, and μ being commutative. $\square$

We want to deduce from this some of the relations that hold in $\pi_0(\mathbb{1})_*$.

Definition 10.2.3. For $u \in F^\times$, we define $\langle u \rangle := 1 + \eta[u] \in K_0^{\mathrm{MW}}(F)$, and similar in $\pi_0(\mathbb{1})_0$.

We obtain a multiplicative morphism $\langle - \rangle : F^\times \rightarrow K_0^{\mathrm{MW}}(F)$ and to $\pi_0(\mathbb{1})_0$. Indeed, we have

$$\langle ab \rangle = 1 + \eta[ab] = 1 + \eta([a] + [b] + \eta[a][b]) = 1 + \eta([a] + [b]) + \eta^2[a][b] = \langle a \rangle \langle b \rangle.$$

Note that for a fixed $u \in F^*$ the left multiplication, which is non-base point preserving $\mathbb{G}_m \rightarrow \mathbb{G}_m, x \mapsto ux$ can be factored as the composition

$$\mathbb{G}_m \simeq \mathbb{G}_m \times * \xrightarrow{\mathrm{id} \times \{u\}} \mathbb{G}_m \times \mathbb{G}_m \xrightarrow{\mu} \mathbb{G}_m.$$

From this one can deduce that $\langle u \rangle \in \pi_0(\mathbb{1})_*$ corresponds to the composition

details

$$\begin{aligned} \mathbb{P}^1 &\rightarrow \mathbb{P}^1, \\ [x : y] &\mapsto [ux : y] \end{aligned}$$

which is now base point preserving, taking $\mathbb{P}^1$ to be pointed at ∞ . By more inspection ,

details

this is the defining map for $1 + \eta[u] = \langle u \rangle \in \pi_0(\mathbb{1})_0$. But we also know from the proof of Lemma 10.2.1 (2), that $-\varepsilon$ corresponds to the morphism

$$\begin{aligned} h : \mathbb{P}^1 &\rightarrow \mathbb{P}^1 \\ [x : y] &\mapsto [-x : y] \end{aligned}$$

which is by the above discussion nothing else then $\langle -1 \rangle = 1 + \eta[-1]$. As $\eta = \eta\varepsilon$, which is true in $\pi_0(\mathbb{1})_0$ by Lemma 10.2.2, if and only if $\eta(1 - \varepsilon) = 0$, we obtain the relation

$$\eta(2 + \eta[-1]) = \eta(1 - \varepsilon) = 0 \in \pi_0(\mathbb{1})_0.$$

Another relation we see is that as $\mu : \mathbb{G}_m \times \mathbb{G}_m \rightarrow \mathbb{G}_m$ stabilizes to $\mathrm{id} + \mathrm{id} + \eta : \Sigma^\infty \mathbb{G}_m \oplus \Sigma^\infty \mathbb{G}_m \rightarrow \Sigma^\infty(\mathbb{G}_m \otimes \mathbb{G}_m)$, we obtain

$$[ab] = \Sigma^\infty \mu(a, b) = [a] + [b]_\eta [a][b].$$

Before we concern ourself with the proof of the Steinberg relations, we make a short digression discussing rational spectra.

10.3 Rational spectra

Let us first discuss how to invert numbers in spectra. Let $n \in \mathbb{Z}$ and $E \in \mathrm{SH}(F)$. Notice that we canonically have an endomorphism $n : E \rightarrow E$, using that $\mathrm{Map}_{\mathrm{SH}(F)}(E, E) \simeq \Omega^\infty \mathrm{Hom}(E, E)$ is an infinite loop space, and hence $\pi_0 \mathrm{Map}_{\mathrm{SH}(F)}(E, E)$ is an abelian group. We say that E is n -periodic, if this morphism is an equivalence. We write $\mathrm{SH}(F) [\frac{1}{n}]$ for the full subcategory of $\mathrm{SH}(F)$ consisting of n -periodic spectra. This category is closed under limits and colimits. The inclusion $\mathrm{SH}(F) [\frac{1}{n}] \hookrightarrow \mathrm{SH}(F)$ admits a left adjoint $E \mapsto \mathrm{colim}(E \xrightarrow{n} E \xrightarrow{n} \dots) =: E[\frac{1}{n}]$. See for example [Cno26, Chapter 8] for more details in the case of classical spectra, or [Lur25, Chapter 9.5.6] for the general theory.

Remark 10.3.1. One can also invert more than one number at the same time, for example

$$E_{\mathbb{Q}} := \mathrm{colim}(E \xrightarrow{2} E \xrightarrow{3} E \xrightarrow{4} \dots).$$

Remark 10.3.2. We have formulas like

$$\pi_0(\mathbb{1}[\frac{1}{n}])_0 = \pi_0 \mathrm{Map}_{\mathrm{SH}(F)}(\mathbb{1}[\frac{1}{n}], \mathbb{1}[\frac{1}{n}]) \cong \pi_0(\mathbb{1})_0[\frac{1}{n}],$$

using that we work in a stable category.

Consider now $E \in \mathrm{SH}(F) [\frac{1}{2}]$. We define

$$e_{\pm} := \frac{1 \mp \langle -1 \rangle}{2} \in K_0^{\mathrm{MW}}(F) [\frac{1}{2}].$$

Using multiplicativity of $\langle - \rangle$ one calculates that $e_{\pm}^2 = e_{\pm}$, $e_+ e_- = 0$, and $e_+ + e_- = 1$, that is, $e_{\pm}$ defines an idempotent split of the unit. Identifying $K_0^{\mathrm{MW}} \cong \pi_0(\mathbb{1})_0$, the $e_{\pm}$ act on motivic spectra $E \in \mathrm{SH}(F) [\frac{1}{2}]$.

Definition 10.3.3. We define the *positive and negative spectra* $\mathrm{SH}(F) [\frac{1}{2}]^{\pm}$ as the full subcategories of $\mathrm{SH}(F) [\frac{1}{2}]$ on those spectra $E \in \mathrm{SH}(F)$ on which $e_{\mp}$ acts by an equivalence.

Notice that if $e_{\mp}$ acts by an equivalence, then $e_{\pm}$ acts by 0, as $e_+ + e_- = 0$, and hence $e_{\mp}$ acts by id, using $e_+ + e_- = 1$. In particular, we see that $\langle -1 \rangle$ acts by ± 1 on $\mathrm{SH}(F) [\frac{1}{2}]^{\pm}$. For $E \in \mathrm{SH}(F) [\frac{1}{2}]$ we define $E^{\pm} := E[\frac{1}{e_{\mp}}] = \mathrm{colim}(E \xrightarrow{e_{\mp}} E \xrightarrow{e_{\mp}} E \xrightarrow{e_{\mp}} \dots)$. One then checks that $E^{\pm} \in \mathrm{SH}(F) [\frac{1}{2}]^{\pm}$ and that there is a splitting $E \simeq E^+ \oplus E^-$.

Example 10.3.4. Consider $\pi_0(\mathbb{1}[\frac{1}{2}])_* := \underline{K}_*^{\mathrm{MW}}[\frac{1}{2}]$. From the relations in Milnor-Witt K -theory we see that $\eta(1 + \langle -1 \rangle) = 0$ and $\langle -1 \rangle = 1 + \eta[-1]$. Hence $\langle -1 \rangle = 1$ if and only if $\eta = 0$. Hence

$$\underline{K}_*^{\mathrm{MW}}[\frac{1}{2}]^+ \simeq \underline{K}_*^{\mathrm{M}}[\frac{1}{2}].$$

On the other hand, if $\langle -1 \rangle = -1$, then $-2 = \eta[-1]$ is a unit, and hence

$$\underline{K}_*^{\mathrm{MW}}[\frac{1}{2}]^- \simeq \underline{K}_*^{\mathrm{MW}}[\frac{1}{2}][\eta^{-1}].$$

Lemma 10.3.5. The functor $- \otimes H\mathbb{Z} : \mathrm{SH}(F) [\frac{1}{2}]_{\geq 0}^+ \rightarrow \mathrm{SH}(F)$ is conservative.

Proof. Let $0 \neq E \in \mathrm{SH}(F) [\frac{1}{2}]_{\geq 0}^+$. Then there exists an $n \geq 0$ minimal with the property that $\pi_n(E)_* \neq 0$. We claim that $\pi_n(E \otimes H\mathbb{Z})_* \neq 0$. To see this, note that for $A, B \in \mathrm{SH}(F)_{\geq 0}$ we have that $\pi_0(A \otimes B)_* \cong \pi_0(\pi_0(A)_* \otimes \pi_0(B)_*)$. We then calculate

$$\begin{aligned} \pi_n(E \otimes H\mathbb{Z})_* &\cong \pi_0(\pi_n(E)_* \otimes \pi_0(H\mathbb{Z})_*) \\ &\cong \pi_0(\pi_n(E)_* \otimes \underline{K}_*^{\mathrm{M}}) \\ &\cong \pi_0(\pi_n(E)_* \otimes \underline{K}_*^{\mathrm{M}}[\frac{1}{2}]) \\ &= \pi_0(\pi_n(E)_* \otimes \pi_0(\mathbb{1}[\frac{1}{2}]^+)_*) \\ &\cong \pi_0(E \otimes \mathbb{1}[\frac{1}{2}]^+) \\ &\cong \pi_0(E)_*. \end{aligned}$$

□

The positive, rational spectra give a new description of rational motives.

Theorem 10.3.6. *The natural maps*

$$\begin{aligned} \mathbb{1}_{\mathbb{Q}}^+ &\rightarrow H\mathbb{Q} \in \mathrm{SH}(F), \\ \mathrm{SH}(F)_{\mathbb{Q}}^+ &\rightarrow \mathcal{DM}(F, \mathbb{Q}) \end{aligned}$$

are equivalences.

Proof. It suffices to prove the first part, as both sides in the second part are equivalent to module categories over their respective units.

For the first part, as both sides are connective, it suffices to prove by Lemma 10.3.5 that the map in the first part induces an equivalence

$$\mathbb{1}_{\mathbb{Q}}^+ \otimes H\mathbb{Z} \rightarrow H\mathbb{Q} \otimes H\mathbb{Z}.$$

But both sides are equivalent to $H\mathbb{Q}$. For the right hand side, this is Corollary 9.4.10, and for the left hand side, as $\mathbb{1}_{\mathbb{Q}}^+$ is rational, we may replace $H\mathbb{Z}$ with its rationalization $H\mathbb{Q}$. As $\eta = 0$ in $H\mathbb{Z}$, we immediately obtain $\mathbb{1}_{\mathbb{Q}}^+ \otimes H\mathbb{Q} \simeq H\mathbb{Q}$. Now the equivalence on the right hand side is essentially the multiplication, while the induced morphism is the unit morphism. Hence, the composition of the equivalences with the induced morphism is just the unit, hence the identity, hence an equivalence. $\square$

10.4 Steinberg relation

In this last section we discuss that the Steinberg relations hold in the global sections of the homotopy sheaf of the sphere spectrum. We follow the prove as given in [Hoy20].

Theorem 10.4.1. *Write $\mathrm{st} : (\mathbb{A}^1 \setminus \{0, 1\})_+ \rightarrow \mathbb{G}_m \oplus \mathbb{G}_m \in \mathrm{Spc}(F)_*$ for the morphism given by $x \mapsto (x, x - 1)$. Then $\Sigma(\mathrm{st})$ is null.*

Corollary 10.4.2. *In $\pi_0(\mathbb{1})_2$ we have that $[a][1 - a] = 0$ for all $a \in F \setminus \{0, 1\}$.*

Proof. This follows from $[a][1 - a]$ being represented by the infinite suspension of the composition

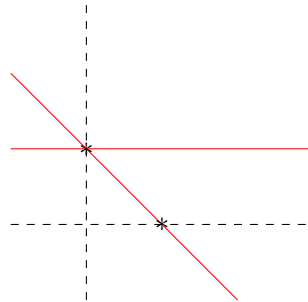
$$S^1 \simeq \Sigma^{1,0} S^0 \xrightarrow{\Sigma a} \Sigma(\mathbb{A}^1 \setminus \{0, 1\})_+ \xrightarrow{\Sigma \mathrm{st}} \Sigma(\mathbb{G}_m \oplus \mathbb{G}_m) \mathrm{sin} \mathrm{Spc}(F)_*.$$

Write the story of the proof. This is funny.

$\square$

Proof of the Theorem. The proof idea is to factor the suspension of the map representing $[a][1 - a] \in \pi_0(\mathbb{1})_2$ through something contractible.

To be precise, set $B = (\mathrm{Bl}_{\{(0,1),(1,0)\}} \mathbb{A}^2) \setminus ((\mathbb{A}^1 \times 0) \cup (0 \times \mathbb{A}^1))$. We visualize this with the following picture



Here, the diagonal red line is the image of $\mathfrak{st}$, while the dashed lines represent the two removed affine lines. The two stars represent the projective lines in the blow up, and lastley, the red horizontal line will be explained at a later stage of the proof.

Now set $U := B \setminus (1 \times \mathbb{A}^1) = (\mathrm{Bl}_{(0,1)}(\mathbb{A}^1 \times \mathbb{G}_m) \setminus (0 \times \mathbb{G}_m))$ and $U' := B \setminus (1 \times \mathbb{A}^1) = (\mathrm{Bl}_{(1,0)}(\mathbb{G}_m \times \mathbb{A}^1) \setminus (\mathbb{G}_m \times 0))$. Then we have $B = U \cup U'$, while $U \cap U' \cong \mathbb{G}_m \times \mathbb{G}_m$, as we removed exactly the coordinate cross¹.

Consider now the two morphisms

$$\begin{aligned} e &: (\mathbb{G}_m \times \mathbb{G}_m) \amalg_{\mathbb{G}_m \times 1} (\mathbb{A}^1 \times 1) \rightarrow U, \\ e' &: (\mathbb{G}_m \times \mathbb{G}_m) \amalg_{1 \times \mathbb{G}_m} (1 \times \mathbb{A}^1) \rightarrow U'. \end{aligned}$$

Note that $(\mathbb{G}_m \times \mathbb{A}^1) \amalg_{\mathbb{G}_m \times 1} (\mathbb{A}^1 \times 1) \simeq *$ is $\mathbb{A}^1$ -contractible, as we filled in the point at $(0, 1)$ into $\mathbb{A}^2 \setminus (0 \times \mathbb{A}^1)$. Furthermore, $(\mathrm{Bl}_{(0,1)}\mathbb{A}^2) \setminus (0 \times \mathbb{A}^1) \simeq \mathbb{A}^2 \simeq *$ is also $\mathbb{A}^1$ -contractible. It follows that Σe is equivalent to

$$(\mathbb{G}_m \times \mathbb{A}^1) \amalg_{\mathbb{G}_m \times 1} (\mathbb{A}^1 \times 1) \rightarrow ((\mathrm{Bl}_{(0,1)}\mathbb{A}^2) \setminus (0 \times \mathbb{A}^1))/U$$

The left-hand side is equivalent to

$$(\mathbb{G}_m \times \mathbb{A}^1)/(\mathbb{G}_m \times \mathbb{G}_m).$$

But now both open subsets that we quotient out are, in their respective spaces, the complements of the closed subscheme $\mathbb{G}_m \times 0$. So, by excision, Σe is an equivalence, and the same argumentation works for e' .

Let now $C \subseteq B$ be the union of the following three lines: the image of the Steinberg relation (i.e. the diagonal red line), the line $\mathbb{A}^1 \times 1$ (i.e. the horizontal red line) and the exceptional divisor at $(0, 1)$ (notice that we removed the point $(0, 1)$ in the exceptional divisor, hence it is an affine line again). As the intersections of these three lines is non-empty, C is $\mathbb{A}^1$ contractible. Consider now the diagram

$$\begin{array}{ccc} (\mathbb{A}^1 \setminus \{0, 1\})_+ & \xrightarrow{\quad \quad \quad} & \mathbb{G}_m \times \mathbb{G}_m \\ \downarrow & & \downarrow \\ & (\mathbb{A}^1 \times 1) \amalg_{\mathbb{G}_m \times 1} (\mathbb{G}_m \times \mathbb{G}_m) \amalg_{\mathbb{G}_m \times 1} (1 \times \mathbb{A}^1) & \\ & \downarrow e \amalg e' & \\ C & \xrightarrow{\quad \quad \quad} & U \amalg_{\mathbb{G}_m \times \mathbb{G}_m} U' \cong B \end{array}$$

We have

$$(\mathbb{A}^1 \times 1) \amalg_{\mathbb{G}_m \times 1} (\mathbb{G}_m \times \mathbb{G}_m) \amalg_{\mathbb{G}_m \times 1} (1 \times \mathbb{A}^1) \simeq (\mathbb{G}_m \times \mathbb{G}_m)/(\mathbb{G}_m \vee \mathbb{G}_m) = \mathbb{G}_m \otimes \mathbb{G}_m,$$

and the map $e \amalg e'$ becomes an equivalence after one suspension by what we discussed before.

The top horizontal morphism in the diagram is $\mathfrak{st}$, while the lower left corner is contractible. As $\Sigma(e \amalg e')$ is an equivalence, the image of $(\mathbb{A}^1 \setminus \{0, 1\})_+$ under $\mathfrak{st}$ in $\mathbb{G}_m \otimes \mathbb{G}_m$ becomes contractible. As the upper right vertical map is a monomorphism, the image of $(\mathbb{A}^1 \setminus \{0, 1\})_+$ under $\mathfrak{st}$ in $\mathbb{G}_m \times \mathbb{G}_m$ is contractible, which shows the theorem. $\square$

More details why the suspension is what I claim it is.

¹To see that these are really the union/intersection, it is easiest, in our opinion, to check on R points for any F -algebra R .

Bibliography

- [ABG18] Matthew Ando, Andrew Blumberg, and David Gepner. Parametrized spectra, multiplicative Thom spectra and the twisted Umkehr map. *Geometry & Topology*, 22(7):3761–3825, 2018.
- [Bac24] Tom Bachmann. Strongly $\mathbb{A}^1$ -invariant sheaves (after F. Morel), 2024.
- [Bar17] Clark Barwick. Spectral Mackey functors and equivariant algebraic K-theory (I). *Advances in Mathematics*, 304:646–727, 2017.
- [BH21] Tom Bachmann and Marc Hovey. *Norms in motivic homotopy theory*, volume 425. Société Mathématique de France, Paris, 2021.
- [CCNW26] Denis-Charles Cisinski, Bastiaan Cnossen, Kim Nguyen, and Tashi Walde. Synthetic category theory. In-progress book project, 2026.
- [CM21] Dustin Clausen and Akhil Mathew. Hyperdescent and étale k-theory. *Inventiones mathematicae*, 225(3):981–1076, 2021.
- [Cno26] Bastiaan Cnossen. Stable homotopy theory and higher algebra. In-progress textbook, available at <https://sites.google.com/view/bastiaan-cnossen/home>, 2026.
- [EHK⁺20] Elden Elmanto, Marc Hovey, Adeel A. Khan, Vladimir Sosnilo, and Maria Yakerson. Modules over algebraic cobordism. *Forum of Mathematics, Pi*, 8:e14, 2020.
- [EHK⁺21] Elden Elmanto, Marc Hovey, Adeel A. Khan, Vladimir Sosnilo, and Maria Yakerson. Motivic infinite loop spaces. *Cambridge Journal of Mathematics*, 9(2):431–549, 2021.
- [Ful98] William Fulton. *Intersection Theory*, volume 2 of *Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge. A Series of Modern Surveys in Mathematics*. Springer-Verlag, Berlin, 2nd edition, 1998.
- [HHLN23] Rune Haugseng, Fabian Hebestreit, Sil Linskens, and Joost Nuiten. Two-variable fibrations, factorisation systems and ∞ -categories of spans. *Forum of Mathematics, Sigma*, 11:e111, 2023.
- [HJN⁺25] Marc Hovey, Joachim Jelisiejew, Denis Nardin, Burt Totaro, and Maria Yakerson. The Hilbert scheme of infinite affine space and algebraic K-theory. *Journal of the European Mathematical Society*, 27(5):2037–2071, 2025.
- [Hoy17] Marc Hovey. The six operations in equivariant motivic homotopy theory. *Advances in Mathematics*, 305:197–279, 2017.
- [Hoy20] Marc Hovey. The Steinberg relation. Preprint, available at <https://hovey.app.uni-regensburg.de/papers/steinberg.pdf>, 2020.

- [JT07] André Joyal and Myles Tierney. Quasi-categories vs segal spaces. In *Categories in Algebra, Geometry and Mathematical Physics*, volume 431 of *Contemporary Mathematics*, pages 277–326. American Mathematical Society, Providence, RI, 2007.
- [Lam05] Tsit-Yuen Lam. *Introduction to Quadratic Forms over Fields*, volume 67 of *Graduate Studies in Mathematics*. American Mathematical Society, Providence, RI, 2005.
- [Lur09] Jacob Lurie. *Higher topos theory*. Princeton University Press, 2009.
- [Lur17a] Jacob Lurie. Higher algebra, 2017. Available at the author’s website.
- [Lur17b] Jacob Lurie. *Higher Algebra*. Preprint, 2017. Available at <https://www.math.ias.edu/~lurie/papers/HA.pdf>.
- [Lur25] Jacob Lurie. Kerodon. <https://kerodon.net>, 2025.
- [MM12] Saunders MacLane and Ieke Moerdijk. *Sheaves in geometry and logic: A first introduction to topos theory*. Springer Science & Business Media, 2012.
- [Ray06] Michel Raynaud. *Anneaux locaux henséliens*, volume 169. Springer, 2006.
- [Sta25] The Stacks project authors. The stacks project. <https://stacks.math.columbia.edu>, 2025.
- [Tam12] Günter Tamme. *Introduction to étale cohomology*. Springer Science & Business Media, 2012.
- [Tin] Tintin. Applications of homotopy purity theorem of morel-voevodsky. Math-Overflow. URL:<https://mathoverflow.net/q/217589> (version: 2016-05-17).
- [Voe10] Vladimir Voevodsky. Motivic eilenberg-macLane spaces. *Publications Mathématiques de l’IHÉS*, 112:1–99, 2010.
- [Wei94] Charles A. Weibel. *An Introduction to Homological Algebra*. Cambridge University Press, 1994.